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(54) **HUMAN PAPILLOMAVIRUS VACCINES AND USES OF THE SAME**

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None

See application file for complete search history.

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(57) **ABSTRACT**

Provided herein are engineered human papilloma virus (HPV) molecular vaccine constructs. Vaccine constructs can also include ligand-inducible engineered gene switch systems for modulating expression of heterologous genes, such as a cytokines, in host cells.

**12 Claims, 19 Drawing Sheets**

**Specification includes a Sequence Listing.**

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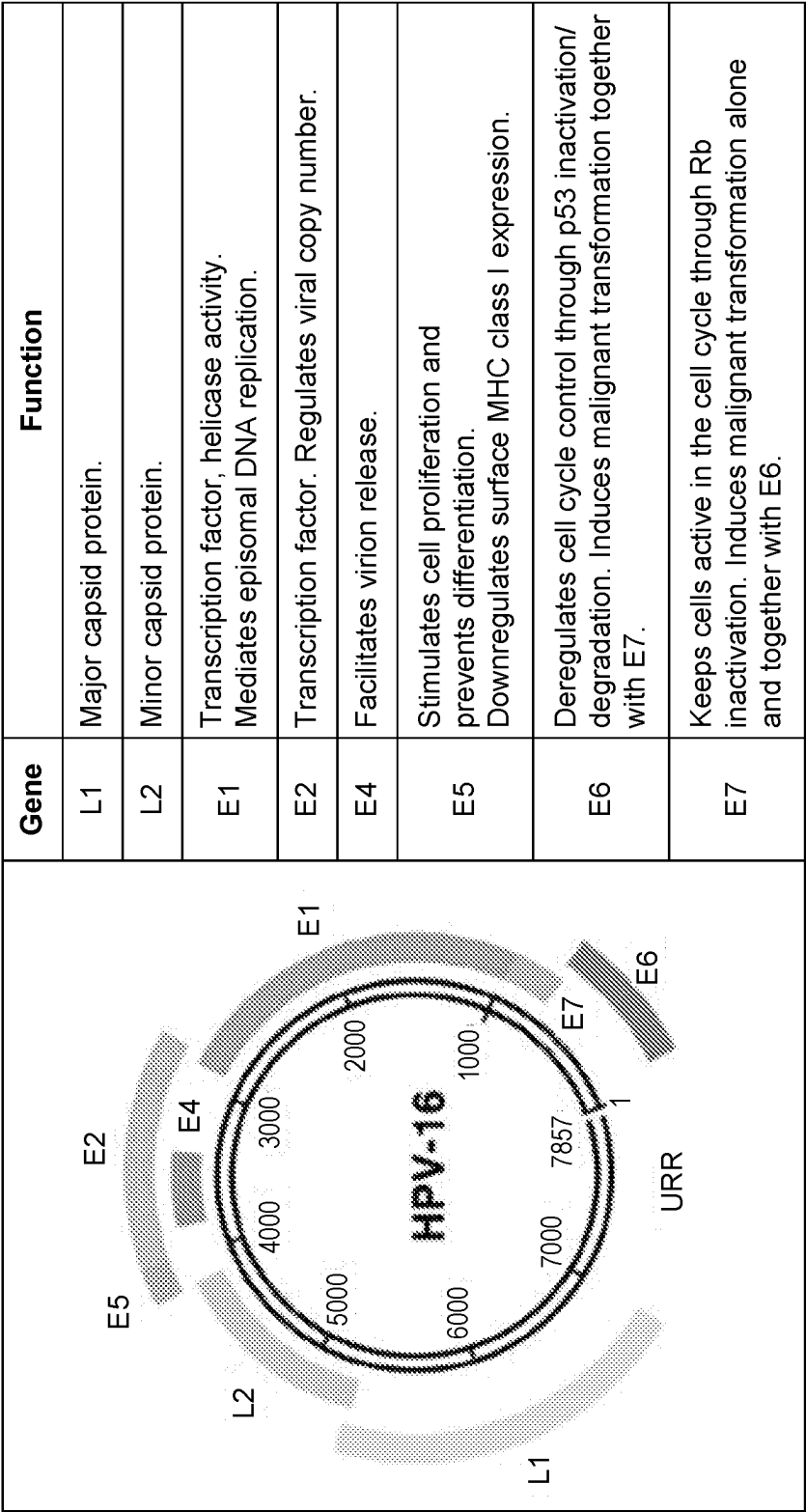
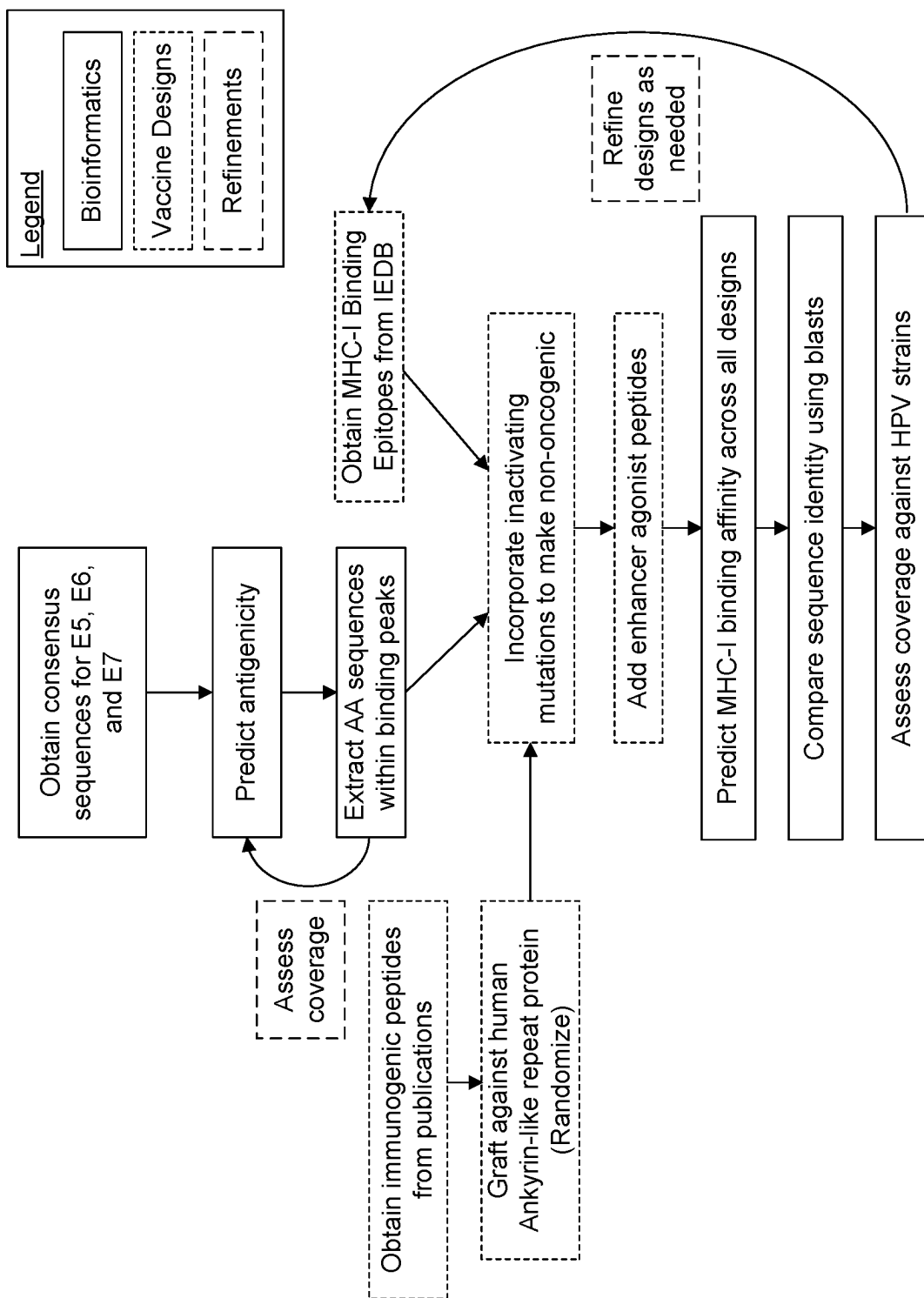
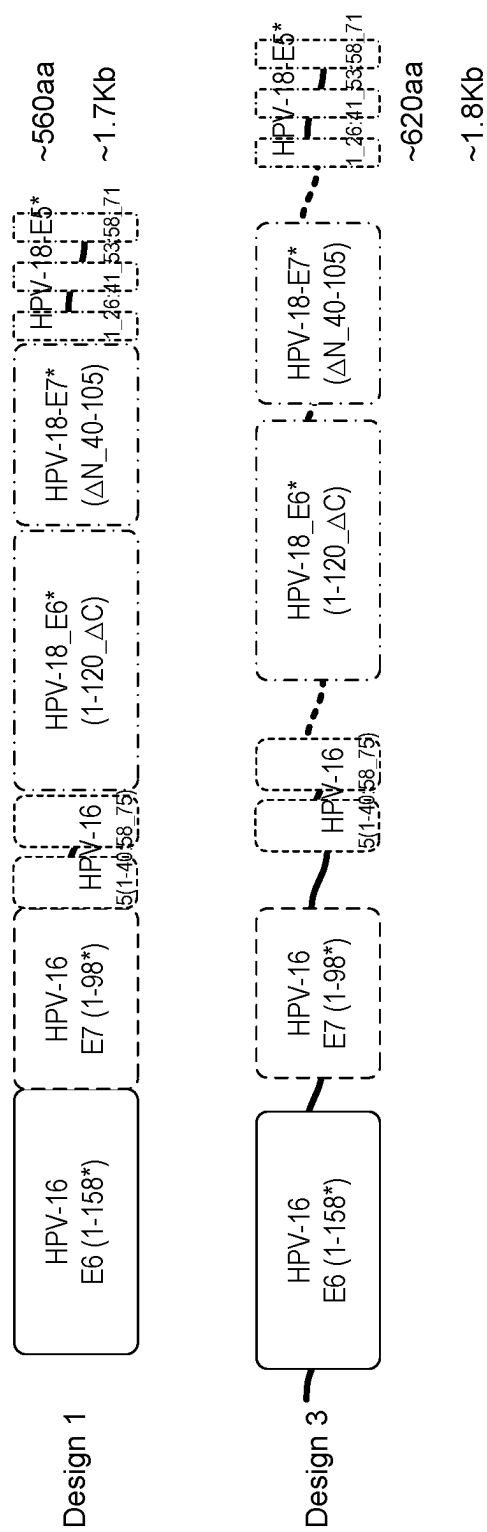


FIG. 1

**FIG. 2**



— Agonist peptides A1 and A3- E6 and A3 from E7 for HPV 16

\* Non oncogenic inactivation mutations in E6 and E7, **E5**

- - - - - Potential agonist peptides, identical to Schlom mutations for HPV-18

FIG. 3

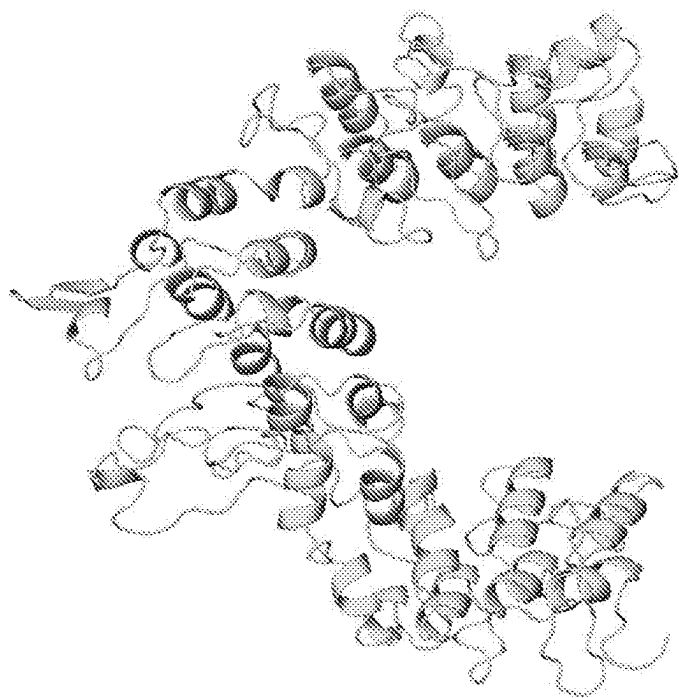


FIG. 4B

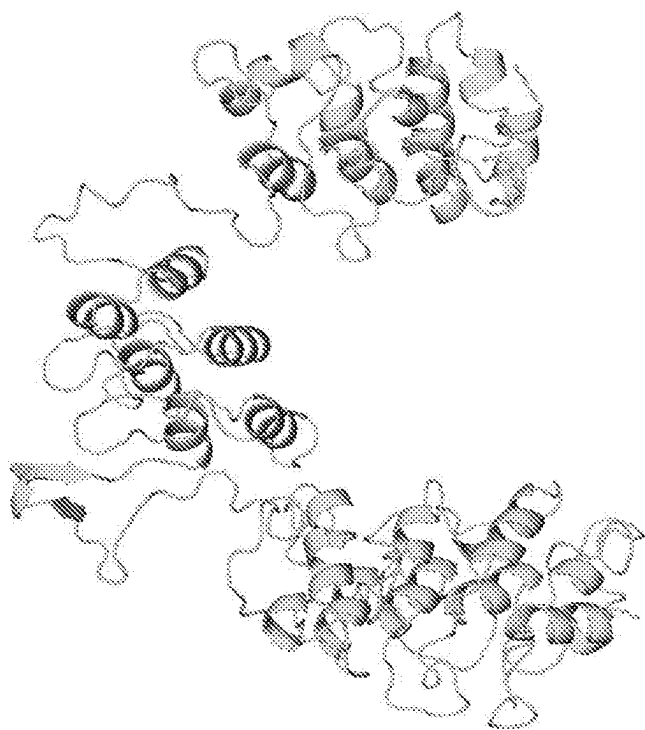


FIG. 4A

| Score          | Expect | Method                                                         | Identities   | Positives    | Gaps         |
|----------------|--------|----------------------------------------------------------------|--------------|--------------|--------------|
| 585 bits(1507) | 0.0    | Compositional matrix adjust.                                   | 365/566(64%) | 379/566(66%) | 134/566(23%) |
| Query 4        |        | RELTEVFEEFAARYSRSDYIIFVYIPLADAKLPQLCTEVDNMLMGTLGIVPLHAAVSADT   |              |              | 63           |
| Sbjct 2        |        | +ELTEVFEEFA +Y IIFVYIPL KLPQLCTEV LLMGTGLGIV T                 |              |              | 47           |
|                |        | RELTEVFEEFAKY-----IIFVYIPLK-KKLPQLCTEVKK-LLMGTLGIV-----KKT     |              |              |              |
| Query 64       |        | LHEYMLDLRNRAATDLSEENDEIDGVNHQHLPARRMHNDLLIRCLRCPLILAAARLAVLEL  |              |              | 123          |
| Sbjct 48       |        | LHEYMLDL+ SEENDEIDGVNHQHLPAR NLLIRCLRC LFL                     |              |              | 93           |
|                |        | LHEYMLDLKK-----SEENDEIDGVNHQHLPAR-KKNLLIRCLRC-----KKLFL        |              |              |              |
| Query 124      |        | NTLSFVCPWCASSHADVKCIDFYSRIDDLKLPDLCTELALHWAAVNNVQLFNLTLSFVG    |              |              | 183          |
| Sbjct 94       |        | NTLSFVCPWCAS KCIDFYSRI KLPDLCTEL QLFNLTLSFV                    |              |              | 139          |
|                |        | NTLSFVCPWCASKK---KCIDFYSRIKK-KLPDLCTEL-----KKQLFNLTLSFV-       |              |              |              |
| Query 184      |        | ANKFQQLFNLTNNRVYDFAFRDLPLFLAAREGSYEKLTNTGLYNLANRLCIVYRDGNPY    |              |              | 243          |
| Sbjct 140      |        | KFQQLFNLT +VYDFAFRDL +KLNTGLYNL +LCIVYRDGNPY                   |              |              | 186          |
|                |        | -KKFQQLFNLT--KKVYDFAFRDLK-----KKLTNTGLYNL-KKLCIVYRDGNPY        |              |              |              |
| Query 244      |        | AVCDDHMYMLDLQPETTPRDFAFRDLICIVYIVRLDLLEVSQTSKLTROTDLGIVCPIA    |              |              | 303          |
| Sbjct 187      |        | AVCD YMLDLQPETT + FAFRDLICIV +TLGIVCPI                         |              |              | 225          |
|                |        | AVCDKK-YMLDLQPETT-KKFAFRDLICIV-----KKTGLGIVCPIK                |              |              |              |
| Query 304      |        | ARYSRSDRTLEDLLMGVADATIHDIILECVDNMHLKKQRFHNIPLHAAVSADTLEQQY     |              |              | 363          |
| Sbjct 226      |        | RTLEDLLMGV TIHDIILECV HLDKKQRFHNI TTLEQQY                      |              |              | 270          |
|                |        | K-----RTLEDLLMGVK-KTIHDIILECVKK-HLDKKQRFHNI-----KKTILEQQY      |              |              |              |
| Query 364      |        | NKPLCDLLRNRAATDLCSTLRCLCVNHDRWTGRCMSCCPLILAAARLAVSLQDIEITCVSHA |              |              | 423          |
| Sbjct 271      |        | NKPLCDLL+ CDSTLRCLV RWTGRCMSCC SLQDIEITCV                      |              |              | 315          |
|                |        | NKPLCDLLK-----CDSTLRCLV-KRWWTGRCMSCC-----KKSQDIEITCVKK-        |              |              |              |
| Query 424      |        | DVLCVQSTHVDIDDLAHYNIIVTFCCLEHWAAVNNVDDLRAFCQLFNLTLSGANKISEYRH  |              |              | 483          |
| Sbjct 316      |        | LCVQSTHVDI AHYNIIVTFCC DDLRAFCQLFNLTLS KISEYRH                 |              |              | 362          |
|                |        | --LCVQSTHVDIKK-AHYNIIVTFCC-----KKDDLRAFCQLFNLTLS---KKISEYRH    |              |              |              |
| Query 484      |        | YCYNRRKLPQLCTELPLFLAAREGSYEKTVLELVEVANRRTLEDILMGTDHMFAPKDLFV   |              |              | 543          |
| Sbjct 363      |        | YCY +KLPQLCTEL +KTVLELVEV +RTLEDILMGTDHMFAPKDLFV               |              |              | 409          |
|                |        | YCY-KKKLPQLCTELK-----KKTVELELVEV-KKRTLEDILMGTDHMFAPKDLFV       |              |              |              |
| Query 544      |        | VPRDQLYNKPLCDVIVVTRTLDIVLHL 569                                |              |              |              |
| Sbjct 410      |        | V + QLYNKPLCDV +TLQDIVLHL 433                                  |              |              |              |
|                |        | V-KKQLYNKPLCDV-KKTLQDIVLHL                                     |              |              |              |

FIG. 5A

Design 2 Location of Binding Affinities

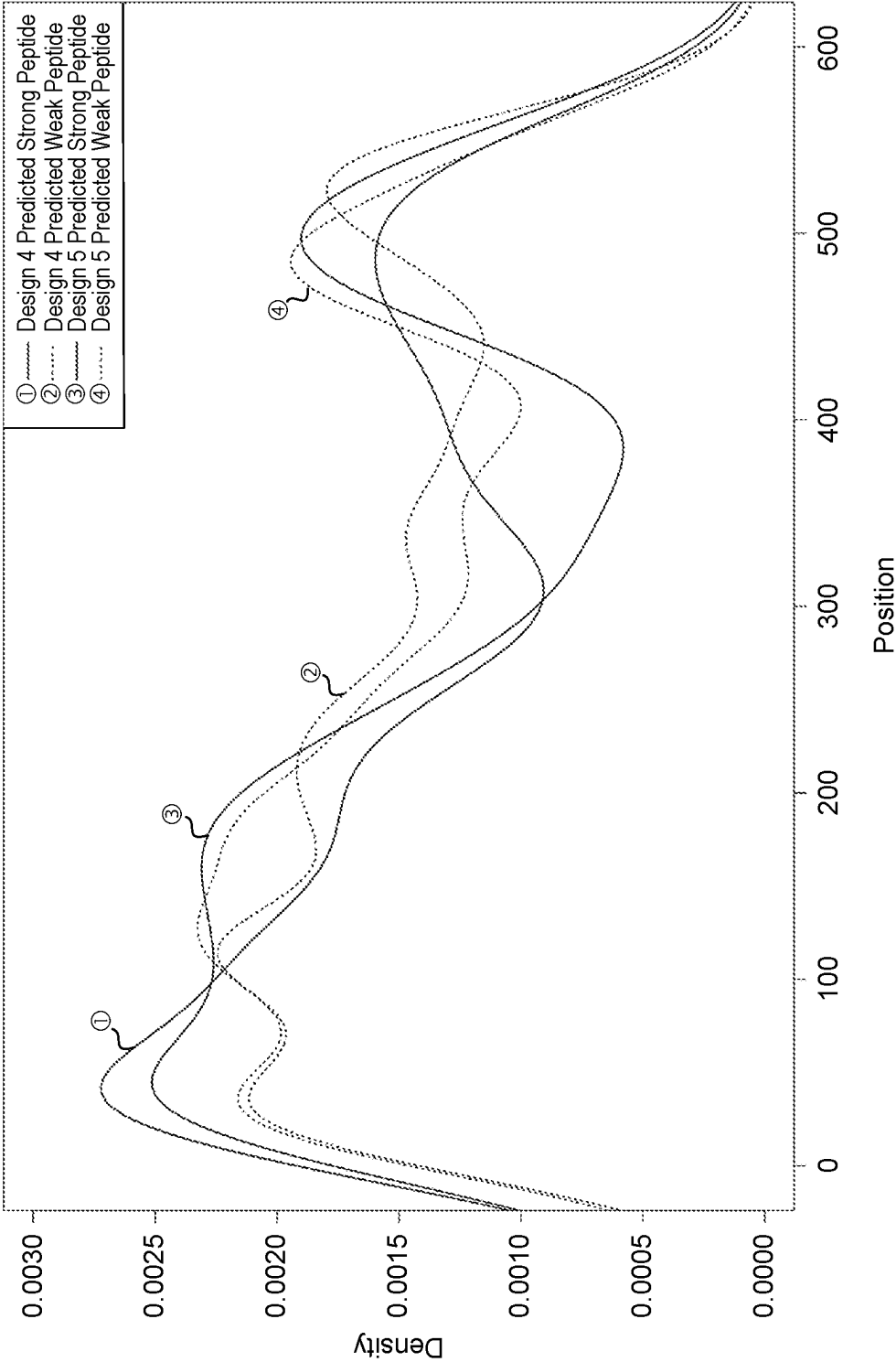


FIG. 5B

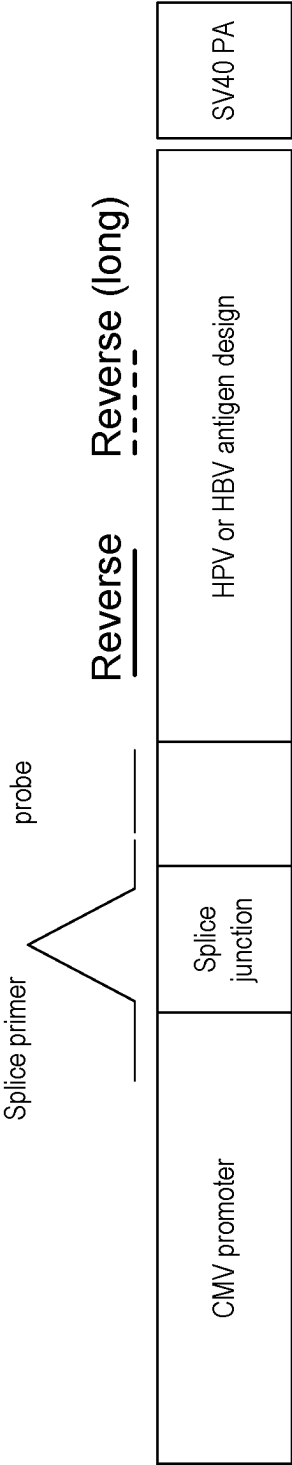


FIG. 6



E6 Location of Binding Affinities

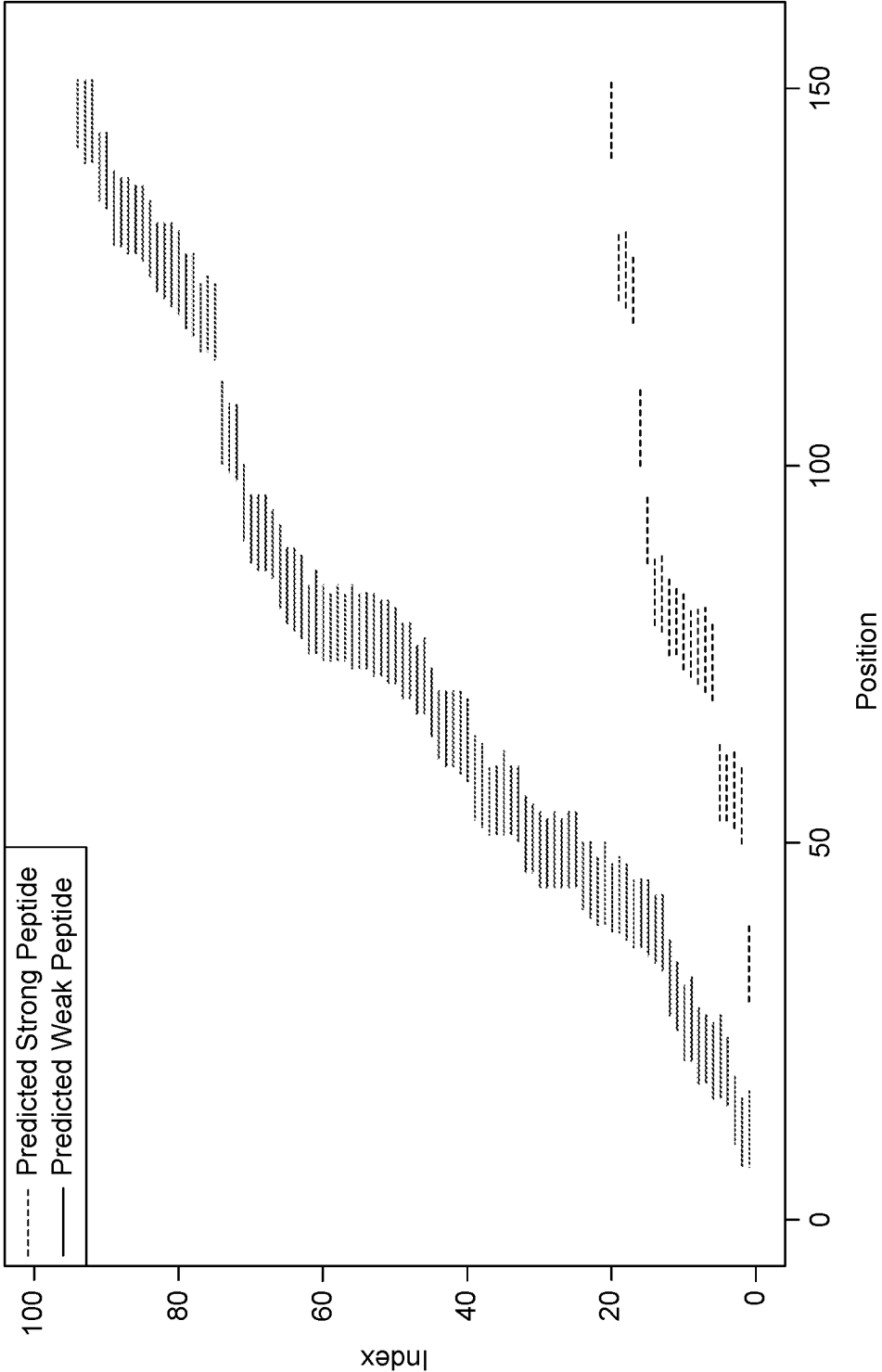
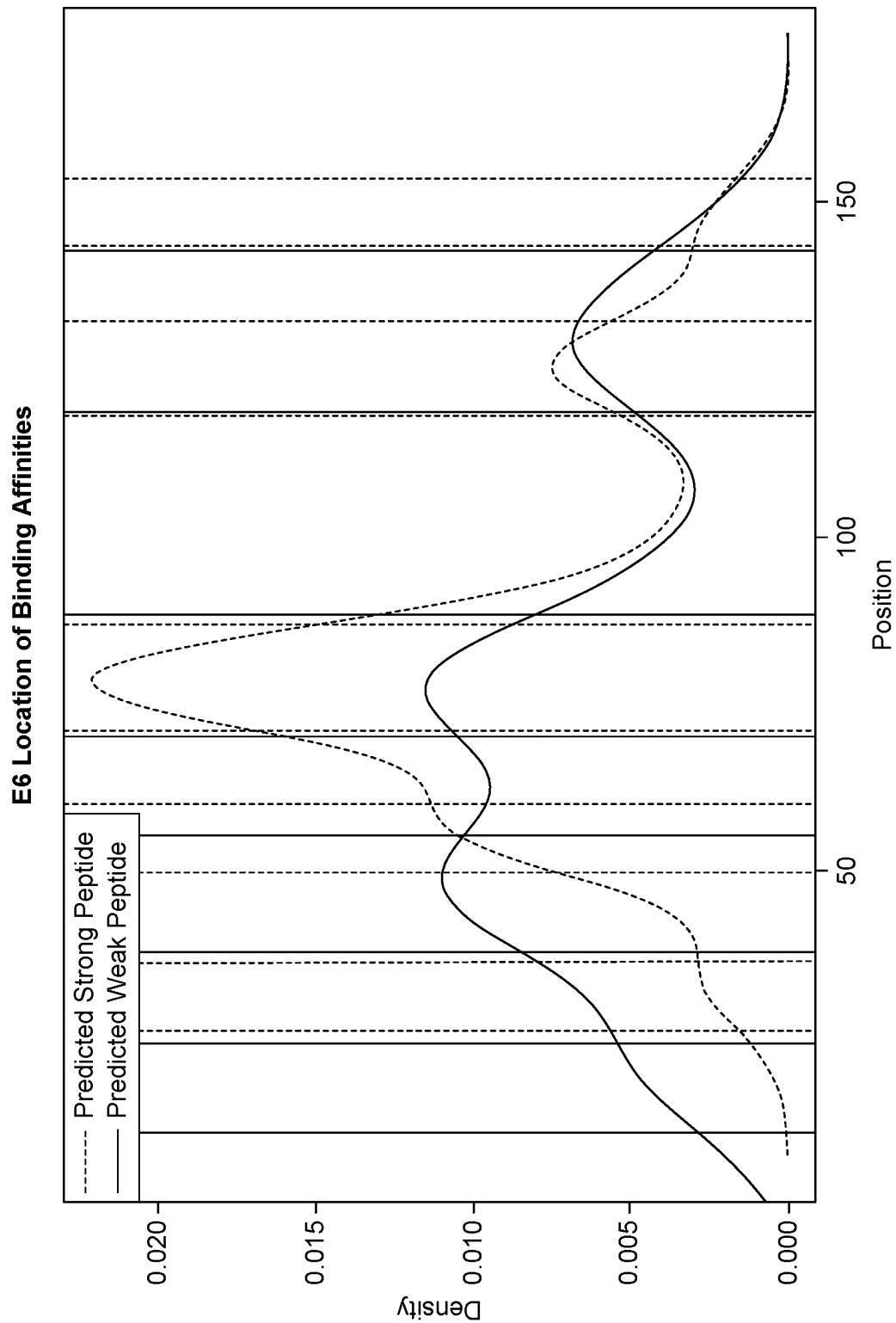
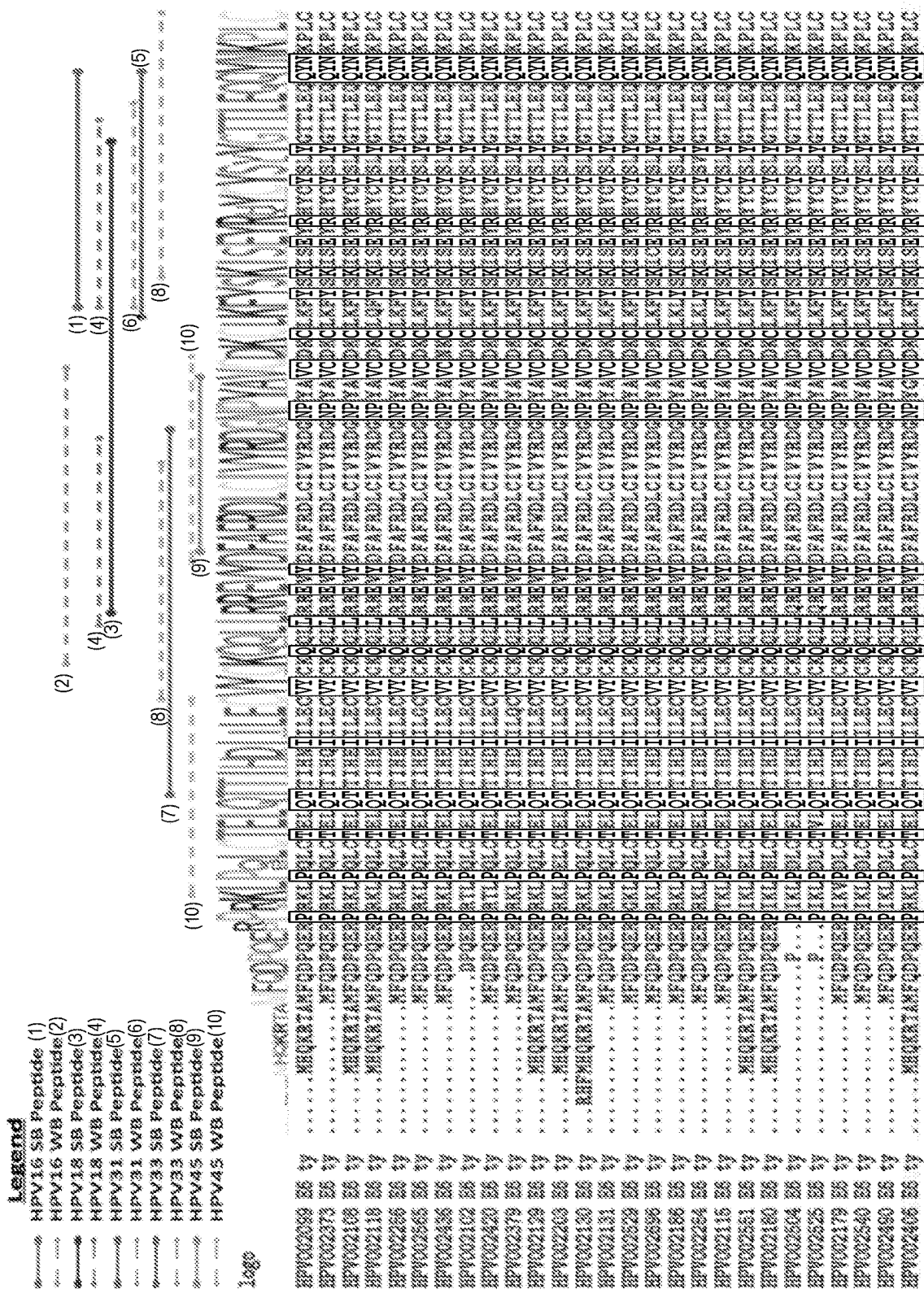


FIG. 7A



**FIG. 7B**



|               |                    |                                                                       |
|---------------|--------------------|-----------------------------------------------------------------------|
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | -----MFQDPQERPRKLPQCTALQTTIHDIILECVYCKQQLLRREVYDFAFRDGC               |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | <b>klpqlctev</b> MFQDPQERPRKLPQCTALQTTIHDIILECVYCKQQLLRREVYDFAFRDGC   |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | IVYRDGNPYAVCDKCLKFYISKISEYRHYCYSLYGTTLQEQYNKPLCDLLIRCINCQKPLC         |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | IVYRDGNPYAVCDKCLKFYISKISEYRHYCYSLYGTTLQEQYNKPLCDLLIRCINCQKPLC         |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | PEEKQRHLDKKQRFHNIRGRWTGRCMSCCRSSRTRRAAAA-----MPGDTPTLHE               |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | PEEKQRHLDKKQRFHNIRGRWTGRCMSCCRSSRTRRAAAA <b>qlynkplcdv</b> MPGDTPTLHE |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | YMLDLQPETTDLYGYEQNLNDSSEEEDEIDGPAGQAAPDRAHYNIVTFCKKCDSTLRRCVQ         |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | YMLDLQPETTDLYGYEQNLNDSSEEEDEIDGPAGQAAPDRAHYNIVTFCKKCDSTLRRCVQ         |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | STHVDIRTLEDLLMGTLGIVCPICSQKP-----MTNLDTASTTLLACFLLCFCVL               |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | STHVDIRTLEDLLMGTLGIVCPICSQKP <b>rtledllmgv</b> MTNLDTASTTLLACFLLCFCVL |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | LCVCLLIRPLLLSVSTYTRCFIVYIIFVYIPLFLIH-----MARFEDPTRRPYKLP              |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | LCVCLLIRPLLLSVSTYTRCFIVYIIFVYIPLFLIH <b>klpdlctev</b> MARFEDPTRRPYKLP |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | DLCTALNTSLQDIEITCVYCKTVLELTFEVFEFAFKDGFVVYRDSIPHAACHKCIDFYSRI         |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | DLCTALNTSLQDIEITCVYCKTVLELTFEVFEFAFKDGFVVYRDSIPHAACHKCIDFYSRI         |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | RELRHYSDSVYGDITLEKLTNTGLYNLLIRCLRCQKPLNPAEKLH-----EIDGV               |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | RELRHYSDSVYGDITLEKLTNTGLYNLLIRCLRCQKPLNPAEKLH <b>kltn</b> tglynveIDGV |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | NHQHLPARRAAPQRHTMLCMCKCEARIERVVLESSADDLRAFQQLFLNTLSFVCPWCASQ          |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | NHQHLPARRAAPQRHTMLCMCKCEARIERVVLESSADDLRAFQQLFLNTLSFVCPWCASQ          |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | Q-----MLSLIFLFCFCVCMYVCHVPLLPSPVVITSPATAFTVYLLPMLLLHH                 |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | <b>Qrafqqlflnv</b> MLSLIFLFCFCVCMYVCHVPLLPSPVVITSPATAFTVYLLPMLLLHH    |
| (MOD-1755822) | HPV 16 18 E6 E7 E5 | AILS                                                                  |
| (MOD-1755825) | HPV 16 18 E6 E7 E5 | AILS                                                                  |

FIG. 8

|                                                |                                                                                                                                                                        |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (MOD-1755804) HPV- 16 E6 (1-151)<br>native seq | -----MFQDPQERPRKLPQLCT <b>a</b> LQTTIHDIILECVYCKQQLLRREVYDFAFRD <b>g</b> CIV<br><b>mhqkrta</b> MFQDPQERPRKLPQLCT <b>e</b> LQTTIHDIILECVYCKQQLLRREVYDFAFRD <b>l</b> CIV |
| (MOD-1755804) HPV- 16 E6 (1-151)<br>native seq | YRDGNPYAVCDKCLKFYSKISEYRHYCYSLYGTTLEQQYNKPLCDLLIRCINCQKPLCPE<br>YRDGNPYAVCDKCLKFYSKISEYRHYCYSLYGTTLEQQYNKPLCDLLIRCINCQKPLCPE                                           |
| (MOD-1755804) HPV- 16 E6 (1-151)<br>native seq | EKQRHLDKKQRFHNIIRGRWTGRCMSCCRSSRTRR <b>aaaa</b><br>EKQRHLDKKQRFHNIIRGRWTGRCMSCCRSSRTRR <b>etq1</b>                                                                     |

FIG. 9A

|                                            |   |                                                                                                                      |
|--------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------|
| (MOD-1755805) HPV- 16 E7 (1-98) native seq | n | MpGDTPTLHEYMLDLQPETTDLYgYEQLNDSSEEEDEIDGPAGQApDRAHYNIVTFCKMhGDTPTLHEYMLDLQPETTDLYcYEQLNDSSEEEDEIDGPAGQaPDRAHYNIVTFCK |
| (MOD-1755805) HPV- 16 E7 (1-98) native seq | n | CDSTLRrCVQSTHVDIRTLEDLLMGTLGIVCPICSQKPCDSTLRlCVQSTHVDIRTLEDLLMGTLGIVCPICSQKP                                         |

FIG. 9B

|               |                     |                                                                   |
|---------------|---------------------|-------------------------------------------------------------------|
| (MOD-1755806) | HPV- 16 E5 (1-40:58 | MTNLDTASTTLLACFLLCFVLLCVCLLIRPLLLLSVSTYT-----RCF                  |
| native seq    |                     | MTNLDTASTTLLACFLLCFVLLCVCLLIRPLLLLSVSTYT <b>slilvlllwtasafRCF</b> |
| (MOD-1755806) | HPV- 16 E5 (1-40:58 | IVYIIFVYIPLFLIH-----                                              |
| native seq    |                     | IVYIIFVYIPLFLIH <b>tharflit</b>                                   |

FIG. 9C

|                                      |                                                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (MOD-1755807) HPV- 18 E6 (1-120) seq | MARFEDPTRRPYKLPDLCTaLNTSLQDIEITCVYCKTVLELTFEVEFAFKDD <b>g</b> FVVYRDSI<br>MARFEDPTRRPYKLPDLCT <b>e</b> LNTSLQDIEITCVYCKTVLELTFEVEFAFKDD <b>l</b> FVVYRDSI |
| (MOD-1755807) HPV- 18 E6 (1-120) seq | PHAACHKCIDFYSRIRELRHYSDSVYGDTLEKLTNTGLYNLLIRCLRCQKPLNPAEKLRLH<br>PHAACHKCIDFYSRIRELRHYSDSVYGDTLEKLTNTGLYNLLIRCLRCQKPLNPAEKLRLH                            |
| (MOD-1755807) HPV- 18 E6 (1-120) seq | -----<br>lnekrrfhniaghyrgqchscnrrarqerlqrrretqv                                                                                                           |

FIG. 10A



|                                                    |                                                                                                            |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| (MOD-1755808) HPV- 18 E7 (40-105)<br>native E7 seq | -----EIDGVNHQHLPARRAaPQRHT<br><b>mhgpkatlqdivlhlepqneipvdllcheqlsdseeend</b> EIDGVNHQHLPARRAePQRHT         |
| (MOD-1755808) HPV- 18 E7 (40-105)<br>native E7 seq | MLCMCKCEAR <b>E</b> VVESADDDLRAFQQLFLNTLSFVCPWCASQQ<br>MLCMCKCEAR <b>K</b> VVESADDDLRAFQQLFLNTLSFVCPWCASQQ |

FIG. 10B

|               |      |    |             |                              |                                                                                           |
|---------------|------|----|-------------|------------------------------|-------------------------------------------------------------------------------------------|
| (MOD-1755809) | HPV- | 18 | E5 (1-26:41 | MLSLIFLCFCVCMYVCCHVPLLP      | SV-----VVITSPATAFTVY-----LLP                                                              |
| native seq    |      |    |             | MLSLIFLCFCVCMYVCCHVPLLP      | SV <b>cmca</b> yaw <b>l</b> v <b>fv</b> y <b>i</b> VVITSPATAFTVY <b>vf</b> <b>cf</b> LLLP |
| (MOD-1755809) | HPV- | 18 | E5 (1-26:41 | MLLLHHAILS--                 |                                                                                           |
| native seq    |      |    |             | MLLLHHAILS <b>l</b> <b>q</b> |                                                                                           |

FIG. 10C

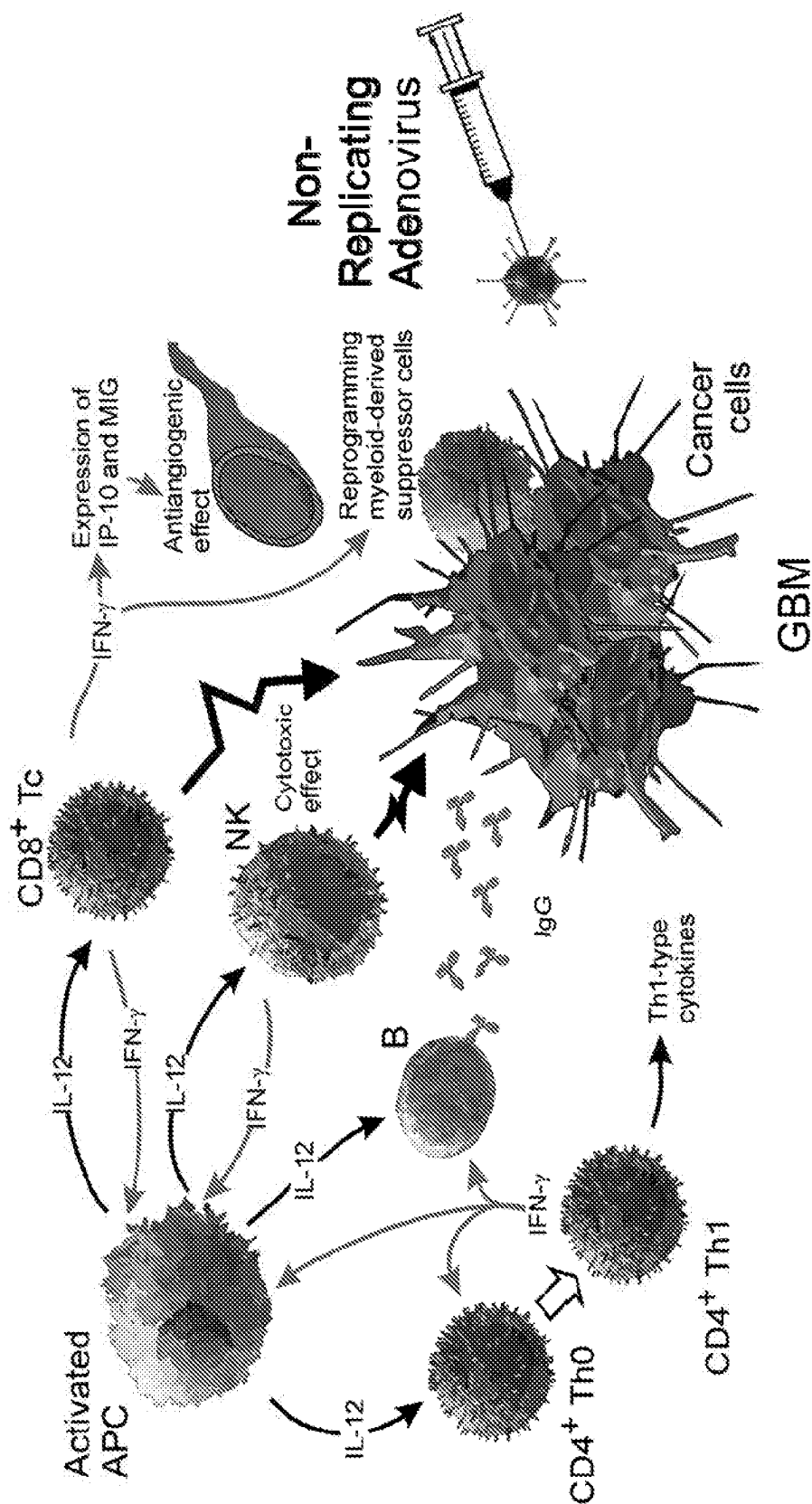


FIG. 11

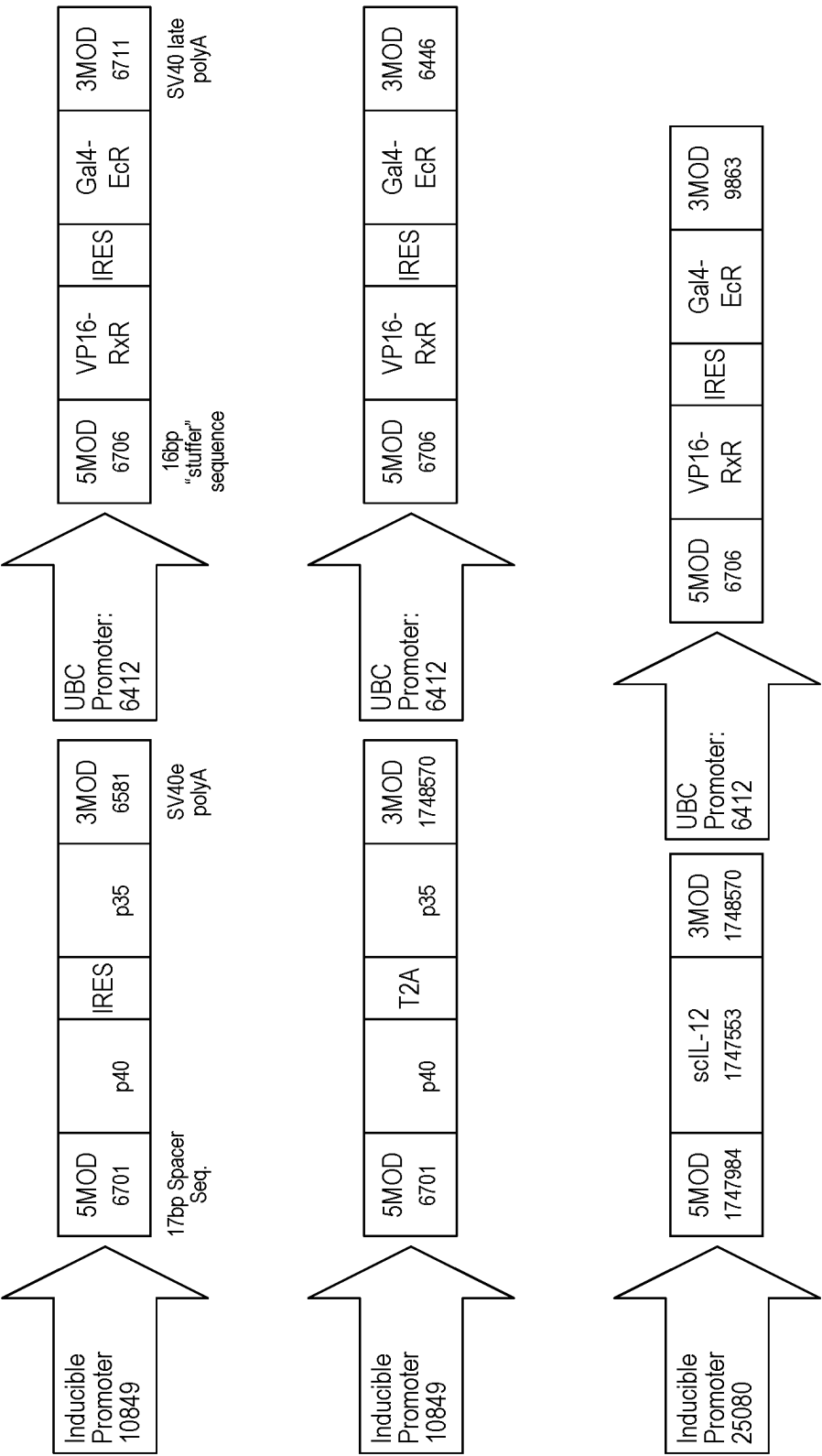


FIG. 12

1

## HUMAN PAPILLOMAVIRUS VACCINES AND USES OF THE SAME

The content of the electronically submitted sequence listing (Name: 2584\_156PC01\_SeqListing\_ST25; Size: 444, 182 bytes; Date of Creation: Mar. 5, 2019) filed with this application is incorporated herein by reference in its entirety.

### FIELD OF THE DISCLOSURE

The present disclosure relates to improved, broad-spectrum HPV molecular vaccines designed via use of advanced principles in bioinformatics and protein engineering.

### BACKGROUND OF THE DISCLOSURE

Globally, tens of millions of people are currently infected with human papillomavirus (HPV), and millions more become newly infected each year. HPV has been determined to be a precursor of cervical and other cancers, such as head and neck cancers. Tens of thousands of women get cervical cancer each year. Currently, vaccinations can protect against diseases caused by HPV when given to recommended age groups; however, there is a tremendous need for HPV vaccines with broad coverage (against multiple HPV strains) and functionality against HPV-associated cancers.

The present disclosure relates to improved, broad-spectrum HPV molecular vaccines designed via use of advanced principles in bioinformatics and protein engineering. These novel HPV vaccines can be used as a therapeutic vaccine against HPV related diseases.

### INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

### SUMMARY OF THE DISCLOSURE

Provided herein is a non-naturally occurring polynucleotide encoding a polypeptide comprising at least one of one or more immune response-inducing human papilloma virus (HPV) polypeptides.

In some embodiments, said non-naturally occurring polynucleotide encodes a polypeptide comprising two or more HPV polypeptides. In some embodiments, said two or more HPV polypeptides comprise one or more HPV-16 immune response-inducing polypeptide sequences. In some embodiments, said HPV-16 peptide comprises at least one of an E5 peptide, an E6 peptide or an E7 peptide. In some embodiments, said HPV-16 peptide comprises an E5 peptide, and said E5 peptide has a sequence as shown in SEQ ID NO: 47. In some embodiments, said HPV-16 peptide comprises an E6 peptide, and said E6 peptide has a sequence as shown in SEQ ID NO: 45. In some embodiments, said HPV-16 peptide comprises an E7 peptide, and said E7 peptide has a sequence as shown in SEQ ID NO: 46. In some embodiments, said one or more HPV peptides comprises an HPV-18 peptide. In some embodiments, said HPV-18 peptide comprises at least one of an E5 peptide, an E6 peptide or an E7 peptide. In some embodiments, said HPV-18 peptide comprises an E5 peptide, and said E5 peptide has a sequence as shown in SEQ ID NO: 50. In some embodiments, said

2

HPV-18 peptide comprises an E6 peptide, and said E6 peptide has a sequence as shown in SEQ ID NO: 48. In some embodiments, said HPV-18 peptide comprises an E7 peptide, and said E7 peptide has a sequence as shown in SEQ ID NO: 49. In some embodiments, said polypeptide has a sequence as shown in SEQ ID NO: 51. In some embodiments, at least one of said one or more HPV peptides is connected to an agonist peptide. In some embodiments, said agonist peptide has a sequence comprising an agonist peptide sequence as shown in Table 2. In some embodiments, said polypeptide has a sequence as shown in SEQ ID NO: 53.

Provided herein is a polynucleotide comprising any of the polynucleotides provided herein, further comprising one or more polynucleotides encoding a gene switch system for inducible control of heterologous gene expression, wherein heterologous gene expression is regulated by said gene switch system; and, wherein said heterologous gene comprises any of the polynucleotide described herein. In some embodiments, said gene switch system is an ecdysone receptor-based (EcR-based) gene switch system. In some embodiments, said one or more HPV polypeptides is for use in a vaccine.

Provided herein is a vector comprising any of the polynucleotides provided herein. In some embodiments, said vector is an adenoviral vector. In some embodiments, said adenoviral vector is a gorilla adenoviral vector.

Provided herein is a method of regulating the expression of a heterologous gene in a cell, the method comprising: introducing into said cell one or more polynucleotides that comprise (i) an repressible or inducible gene switch, and (ii) a heterologous immune response-inducing gene, wherein expression of said heterologous immune response-inducing gene is regulated by said gene switch, wherein said heterologous immune response-inducing gene encodes at least one of one or more HPV polypeptides; and exposing said cell to a compound in an amount sufficient to repress or induce expression of said heterologous immune response-inducing gene.

In some embodiments, said target cell is a mammalian cell in a method of regulating the expression of a heterologous gene in a cell described herein. In some embodiments, said gene switch comprises a ligand binding domain derived from at least one of an ecdysone receptor (EcR), a ubiquitously expressed receptor, an orphan receptor 1, an NER-1, a steroid hormone nuclear receptor 1, a retinoid X receptor interacting protein-15, a liver X receptor  $\beta$ , a steroid hormone receptor like protein, a liver X receptor, a liver X receptor  $\alpha$ , a farnesoid X receptor, a receptor interacting protein 14, and a farnesol receptor.

Provided herein is an E6 peptide, wherein said E6 peptide comprises an E18A amino acid substitution and at least one of an L50G, E148A, T149A, Q150A and L151A amino acid substitution as compared to a wildtype E6 peptide. In some embodiments, said E6 peptide comprises said E18A amino acid substitution and said L50G, E148A, T149A, Q150A and L151A amino acid substitutions. In some embodiments, said E6 peptide has a sequence as shown in SEQ ID NO: 45. In some embodiments, said E6 peptide is fused to an agonist peptide. In some embodiments, said agonist peptide is fused to at least one of a C-terminus and an N-terminus of said E6 peptide. In some embodiments, said wildtype E6 peptide is from HPV-16.

Provided herein is an E6 peptide, wherein said E6 peptide comprises a deletion as compared to a wildtype E6 peptide, wherein said deletion comprises a C-terminus of said wildtype E6 peptide. In some embodiments, said deletion com-

3

prises amino acids from amino acid 121 to a C-terminus of said wildtype E6 peptide. In some embodiments, said E6 peptide comprises at least one of an E18A and L50G substitution as compared to said wildtype E6 peptide. In some embodiments, said wildtype E6 peptide is from HPV-18. In some embodiments, said E6 peptide has a sequence as shown in SEQ ID NO: 48.

Provided herein is an E7 peptide, wherein said E7 peptide comprises a deletion as compared to a wildtype E7 peptide, wherein said deletion comprises an N-terminus of said wildtype E7 peptide. In some embodiments, said deletion comprises amino acids 1-39 of said wildtype E7 peptide. In some embodiments, said E7 peptide comprises at least one of an E55A and L74R substitution as compared to said wildtype E7 peptide. In some embodiments, said wildtype E7 peptide is from HPV-18. In some embodiments, said E7 peptide has a sequence as shown in SEQ ID NO: 49.

Provided herein is an E5 peptide, wherein said E5 peptide comprises a deletion as compared to a wildtype E5 peptide, wherein said deletion comprises amino acids 41-57 of said wildtype E5 peptide. In some embodiments, said E5 peptide has a sequence as shown in SEQ ID NO: 47. In some embodiments, said wildtype E5 peptide is from HPV-16.

Provided herein is an E5 peptide, wherein said E5 peptide comprises a deletion as compared to a wildtype E5 peptide, wherein said deletion comprises at least one of amino acids 27-40 or amino acids 54-57 of said wildtype E5 peptide. In some embodiments, said E5 peptide has a sequence as shown in SEQ ID NO: 50. In some embodiments, said wildtype E5 peptide is from HPV-18.

Provided herein is a polypeptide construct comprising any one of the presently described E5, E6, and E7.

Provided herein is a polypeptide construct, wherein said polypeptide construct comprises an HPV-16 E6 peptide, wherein said HPV-16 E6 peptide comprises an E18A amino acid substitution and at least one of an L50G, E148A, T149A, Q150A and L151A amino acid substitution as compared to a wildtype HPV-16 E6 peptide. In some embodiments, said HPV-16 E6 peptide comprises said E18A amino acid substitution and said L50G, E148A, T149A, Q150A and L151A amino acid substitutions. In some embodiments, said HPV-16 E6 peptide has a sequence as shown in SEQ ID NO: 45. In some embodiments, said polypeptide construct further comprises an HPV-16 E7 peptide, wherein said HPV-16 E7 peptide comprises at least one of an H2P, C24G, E46A and L67R amino acid substitution as compared to a wildtype HPV-16 E7 peptide. In some embodiments, said HPV-16 E7 peptide comprises said H2P, C24G, E46A and L67R amino acid substitutions. In some embodiments, said HPV-16 E7 peptide has a sequence as shown in SEQ ID NO: 46. In some embodiments, said polypeptide construct further comprises an HPV-16 E5 peptide. In some embodiments, said HPV-16 E5 peptide comprises a deletion of one or more amino acids as compared to a wildtype HPV-16 E5 peptide. In some embodiments, said deletion comprises amino acids 41-57 of said wildtype HPV-16 E5 peptide. In some embodiments, said HPV-16 E5 peptide has a sequence as shown in SEQ ID NO: 47.

In some embodiments, said polypeptide construct comprising an HPV-16 E6 peptide, wherein said HPV-16 E6 peptide comprises an E18A amino acid substitution and at least one of an L50G, E148A, T149A, Q150A and L151A amino acid substitution as compared to a wildtype HPV-16 E6 peptide, further comprises an HPV-18 E6 peptide. In some embodiments, said HPV-18 E6 peptide comprises an E18A and L50G substitution as compared to a wildtype

4

HPV-18 E6 peptide. In some embodiments, said HPV-18 E6 peptide comprises a deletion of at least one C-terminus amino acid relative to said wildtype HPV-18 E6 peptide. In some embodiments, said deletion comprises amino acids from amino acid 121 to said C-terminus of said wildtype HPV-18 E6 peptide. In some embodiments, said HPV-18 E6 peptide has a sequence as shown in SEQ ID NO: 48. In some embodiments, said polypeptide construct further comprises an HPV-18 E7 peptide. In some embodiments, said HPV-18 E7 peptide comprises an E55A and L74R substitution as compared to a wildtype HPV-18 E7 peptide. In some embodiments, said HPV-18 E7 peptide comprises a deletion of at least one amino acid from an N-terminus of said HPV-18 E7 peptide. In some embodiments, said deletion comprises amino acids 1-40 of said wildtype HPV-18 E7 peptide. In some embodiments, said HPV-18 E7 peptide has a sequence as shown in SEQ ID NO: 49. In some embodiments, said polypeptide construct further comprises an HPV-18 E5 peptide. In some embodiments, said HPV-18 E5 peptide comprises a deletion of at least one amino acid as compared to a wildtype HPV-18 E5 peptide. In some embodiments, said deletion comprises amino acids 27-40 or 54-57 of said wildtype HPV-18 E5 peptide. In some embodiments, said HPV-18 E5 peptide has a sequence as shown in SEQ ID NO: 50. In some embodiments, said polypeptide construct has a sequence as shown in SEQ ID NO: 51. In some embodiments, said polypeptide construct further comprises at least one agonist peptide. In some embodiments, said at least one agonist peptide has a sequence comprising an agonist peptide sequence as shown in Table 2. In some embodiments, said polypeptide construct has a sequence as shown in SEQ ID NO: 53.

Provided herein is a polypeptide construct comprising an ankyrin-like repeat domain and an HPV peptide. In some embodiments, said ankyrin-like repeat protein is a human ankyrin-like repeat protein. In some embodiments, said HPV peptide is linked to said ankyrin-like repeat protein by a linker. In some embodiments, said HPV peptide comprises at least one of an HPV-16 peptide or an HPV-18 peptide. In some embodiments, said HPV peptide comprises an HPV-16 peptide, and said HPV-16 peptide comprises at least one of an E5 peptide, an E6 peptide or an E7 peptide. In some embodiments, said HPV peptide comprises an HPV-18 peptide, and said HPV-18 peptide comprises at least one of an E6 peptide or an E7 peptide. In some embodiments, said HPV peptide comprises an HPV-16 E5 sequence, an HPV-16 E6 sequence, an HPV-16 E7 sequence, an HPV-18 E6 sequence or an HPV-18 E7 sequence as shown in Table 2. In some embodiments, said polypeptide construct has a sequence as shown in SEQ ID NO: 52. In some embodiments, said polypeptide construct further comprises at least one agonist peptide. In some embodiments, said polypeptide construct comprises three agonist peptides. In some embodiments, said polypeptide construct has a sequence as shown in SEQ ID NO: 54.

Provided herein is a polypeptide construct, wherein said polypeptide construct comprises at least two HPV amino acid sequences as shown in Table 2, wherein said at least two HPV amino acid sequences are connected by a peptide linker, wherein said peptide linker is a KK linker. In some embodiments, said at least two HPV amino acid sequences comprise at least one of an HPV-16 peptide or an HPV-18 peptide as shown in Table 2. In some embodiments, said at least two HPV amino acid sequences comprise an HPV-16 peptide, and said HPV-16 peptide comprises at least one of an HPV-16 E5 peptide, an HPV-16 E6 peptide or an HPV-16 E7 peptide as shown in Table 2. In some embodiments, said

5

at least two HPV amino acid sequences comprise an HPV-18 peptide, and said HPV-18 peptide comprises at least one of an HPV-18 E6 peptide or an HPV-18 E7 peptide as shown in Table 2. In some embodiments, said at least two HPV amino acid sequences comprise each of the amino acid sequences as shown in Table 2. In some embodiments, said each of the amino acid sequences is connected to another of said each of the amino acid sequences by said KK linker. In some embodiments, said polypeptide construct has a sequence as shown in SEQ ID NO: 55. In some embodiments, any of the polypeptide constructs described herein is for use in a vaccine.

Provided herein is a polynucleotide encoding any of the presently described polypeptide constructs. Also provided herein is a vector comprising said polynucleotide. In some embodiments, said vector is an adenoviral vector. In some embodiments, said adenoviral vector is a gorilla adenoviral vector.

Provided herein is a vector, wherein said vector comprises a polynucleotide that encodes at least one HPV peptide, wherein said vector is an adenoviral vector.

Provided herein is a vector, wherein said vector comprises a polynucleotide that encodes at least one HPV peptide, wherein said vector is an adenoviral vector, wherein said adenoviral vector is a gorilla adenoviral vector.

In some embodiments, any of the polypeptide constructs described herein is for use in a vaccine. Also provided herein is a polynucleotide encoding any of the polypeptide constructs presently described. Also provided herein is a vector comprising said polynucleotide. In some embodiments, said vector is an adenoviral vector. In some embodiments, said adenoviral vector is a gorilla adenoviral vector.

#### BRIEF DESCRIPTION OF THE DRAWINGS

The features of the present disclosure are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present disclosure will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosure are utilized, and the accompanying drawings of which:

FIG. 1 is a schematic diagram of HPV genome. The HPV genomes includes seven early genes (E1 to E7) and two late genes (L1 and L2), and each gene has specific functions. E5, E6, and E7 genes are associated with cancer development.

FIG. 2 is a schematic overall workflow implemented for designing HPV vaccine antigens.

FIG. 3 is schematic diagrams of HPV16 and HPV18 antigenic designs for HPV design 1 and HPV design 3. Consensus sequence information was utilized to select HPV16/HPV18 reference sequences for the design which included all major variants. The vaccine composition comprising different E6, E7 and E5 protein components with domain boundaries and mutation information is shown. These different domains contain most prevalent peptides deduced from IEDB predictions for MHC-I binding. HPV design 3 is similar to HPV design 1 with the addition of enhancer agonist peptides.

FIG. 4A and FIG. 4B show a homology model of HPV design 2 and HPV design 4, respectively. A homology model is used to assess the overall structural feature and to compare the HPV design against native ankyrin repeats. HPV designs are shown in the same orientation and suggest different structural conformations due to shuffled peptides although maintaining the same overall fold.

6

FIG. 5A shows HPV design 5 (subject) mapped onto HPV design 4 (Query) using protein blast. Strong and weak binders were identified using netMHC. FIG. 5B shows density plots of HPV designs 4 and 5 which were extracted based on the mapped positions. Similar patterns in predicted strong/weak binding peptides were observed. The binding affinity predictions on the matched regions of HPV designs 4 and 5 were similar.

FIG. 6 is a schematic illustration showing short and long primer and probe sets generated for RNA qPCR relative expression assay. Specific primers were designed for each HPV antigen design.

FIG. 7A shows NetMHC 4.0 antigenicity predictions. Predicted strong and weak binding peptide indices are plotted against peptide locations. FIG. 7B shows NetMHC 4.0 antigenicity prediction density plot. First/second order differentials are employed in order to identify peaks. FIG. 7C shows amino acid sequences aligned against consensus sequences in order to determine coverage across HPV subtypes.

FIG. 8 shows a comparison of HPV design1 (MOD-1755822) and HPV design 3 (MOD-1755825).

FIG. 9A shows HPV16 E6 sequence alignment to wild-type HPV16 E6 from UP P03126. FIG. 9B shows HPV16 E7 sequence alignment to wildtype HPV16 E7 from UP P03129. FIG. 9C shows HPV16 E5 sequence alignment to wildtype HPV16 E5 from UP P06927.

FIG. 10A shows HPV18 E6 sequence alignment to wild-type HPV18E6 from UP P06463. FIG. 10B shows HPV18 E7 sequence alignment to wildtype HPV18E7 from UP P06788. FIG. 10C shows HPV18 E5 sequence alignment to wildtype HPV18E5 from UP P06792.

FIG. 11 shows an overview of IL-12 promoting immune response by activating NK cells and T cells.

FIG. 12 shows various structural components of diverse IL-12 ligand-inducible gene switch vector systems.

#### DETAILED DESCRIPTION OF THE DISCLOSURE

The following description and examples illustrate embodiments of the present disclosure in detail.

It is to be understood that the present disclosure is not limited to the particular embodiments described herein and as such can vary. Those of skill in the art will recognize that there are variations and modifications of the present disclosure, which are encompassed within its scope.

All terms are intended to be understood as they would be understood by a person skilled in the art. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure pertains.

The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

Although various features of the disclosure can be described in the context of a single embodiment, the features can also be provided separately or in any suitable combination. Conversely, although the present disclosure can be described herein in the context of separate embodiments for clarity, the present disclosure can also be implemented in a single embodiment.

The following definitions supplement those in the art and are directed to the current application and are not to be imputed to any related or unrelated case, e.g., to any commonly owned patent or application. Although any methods and materials similar or equivalent to those described

herein can be used in the practice for testing of the present disclosure, the preferred materials and methods are described herein. Accordingly, the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

### Definitions

In this application, the use of the singular includes the plural unless specifically stated otherwise. It must be noted that, as used in the specification, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise.

In this application, the use of “or” means “and/or” unless stated otherwise. The terms “and/or” and “any combination thereof” and their grammatical equivalents as used herein, can be used interchangeably. These terms can convey that any combination is specifically contemplated. Solely for illustrative purposes, the following phrases “A, B, and/or C” or “A, B, C, or any combination thereof” can mean “A individually; B individually; C individually; A and B; B and C; A and C; and A, B, and C.” The term “or” can be used conjunctively or disjunctively, unless the context specifically refers to a disjunctive use.

Furthermore, use of the term “including” as well as other forms, such as “include,” “includes,” and “included,” is not limiting.

Reference in the specification to “some embodiments,” “an embodiment,” “one embodiment” or “other embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the present disclosures.

As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps. It is contemplated that any embodiment discussed in this specification can be implemented with respect to any method or composition of the disclosure, and vice versa. Furthermore, compositions of the present disclosure can be used to achieve methods of the present disclosure.

The term “about” in relation to a reference numerical value and its grammatical equivalents as used herein can include the numerical value itself and a range of values plus or minus 10% from that numerical value.

The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the art. Alternatively, “about” can mean a range of up to 20%, up to 10%, or up to 5%, or up to 1% of a given value. In another example, the amount “about 10” includes 10 and any amounts from 9 to 11. In yet another example, the term “about” in relation to a reference numerical value can also include a range of values plus or minus 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, or 1% from that value. Alternatively, particularly with respect to biological systems or processes, the term “about” can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value. Where particular values

are described in the application and claims, unless otherwise stated the term “about” meaning within an acceptable error range for the particular value should be assumed.

The term “isolated” and its grammatical equivalents as used herein refer to the removal of a nucleic acid from its natural environment. The term “purified” and its grammatical equivalents as used herein refer to a molecule or composition, whether removed from nature (including genomic DNA and mRNA) or synthesized (including cDNA) and/or amplified under laboratory conditions, that has been increased in purity, wherein “purity” is a relative term, not “absolute purity.” It is to be understood, however, that nucleic acids and proteins can be formulated with diluents or adjuvants and still for practical purposes be isolated. For example, nucleic acids typically are mixed with an acceptable carrier or diluent when used for introduction into cells. The term “substantially purified” and its grammatical equivalents as used herein refer to a nucleic acid sequence, polypeptide, protein or other compound which is essentially free, i.e., is more than about 50% free of, more than about 70% free of, more than about 90% free of, the polynucleotides, proteins, polypeptides and other molecules that the nucleic acid, polypeptide, protein or other compound is naturally associated with.

“Polynucleotide”, “oligonucleotide”, “polynucleotide construct”, “gene”, “gene construct”, “heterologous gene” and their grammatical equivalents as used herein refer to a polymeric form of nucleotides or nucleic acids of any length, either ribonucleotides or deoxyribonucleotides. This term refers only to the primary structure of the molecule. Thus, this term includes double and single stranded DNA, triplex DNA, as well as double and single stranded RNA. It also includes modified, for example, by methylation and/or by capping, and unmodified forms of the polynucleotide. The term is also meant to include molecules that include non-naturally occurring or synthetic nucleotides as well as nucleotide analogs. The nucleic acid sequences and vectors disclosed or contemplated herein can be introduced into a cell by, for example, transfection, transformation, or transduction.

“Transfection,” “transformation,” or “transduction” as used herein refer to the introduction of one or more exogenous polynucleotides into a host cell by using physical or chemical methods. Many transfection techniques are known in the art and include, for example, calcium phosphate DNA co-precipitation (see, e.g., Murray E. J. (ed.), *Methods in Molecular Biology*, Vol. 7, Gene Transfer and Expression Protocols, Humana Press (1991)); DEAE-dextran; electroporation; cationic liposome-mediated transfection; tungsten particle-facilitated microparticle bombardment (Johnston, *Nature*, 346: 776-777 (1990)); and strontium phosphate DNA co-precipitation (Brash et al., *Mol. Cell Biol.*, 7: 2031-2034 (1987)). Phage or viral vectors can be introduced into host cells, after growth of infectious particles in suitable packaging cells, many of which are commercially available.

“Polypeptide”, “peptide” “polypeptide construct” and “peptide construct” and their grammatical equivalents as used herein refer to a polymer of amino acid residues. A “mature protein” is a protein which is full-length and which, optionally, includes glycosylation or other modifications typical for the protein in a given cellular environment. Polypeptides and proteins disclosed herein (including functional portions and functional variants thereof) can comprise synthetic amino acids in place of one or more naturally-occurring amino acids. Such synthetic amino acids are known in the art, and include, for example, aminocyclo-



hexane carboxylic acid, norleucine,  $\alpha$ -amino n-decanoic acid, homoserine, S-acetylaminomethyl-cysteine, trans-3- and trans-4-hydroxyproline, 4-aminophenylalanine, 4-nitrophenylalanine, 4-chlorophenylalanine, 4-carboxyphenylalanine,  $\beta$ -phenyl serine  $\beta$ -hydroxyphenylalanine, phenylglycine,  $\alpha$ -naphthylalanine, cyclohexylalanine, cyclohexylglycine, indoline-2-carboxylic acid, 1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid, aminomalonic acid, aminomalonic acid monoamide, N'-benzyl-N'-methyl-lysine, N',N'-dibenzyl-lysine, 6-hydroxylysine, ornithine,  $\alpha$ -aminocyclopentane carboxylic acid,  $\alpha$ -aminocyclohexane carboxylic acid,  $\alpha$ -aminocycloheptane carboxylic acid,  $\alpha$ -(2-amino-2-norbornane)-carboxylic acid,  $\alpha,\gamma$ -diaminobutyric acid,  $\alpha,\beta$ -diaminopropionic acid, homophenylalanine, and  $\alpha$ -tert-butylglycine. The present disclosure further contemplates that expression of polypeptides described herein in an engineered cell can be associated with post-translational modifications of one or more amino acids of the polypeptide constructs. Non-limiting examples of post-translational modifications include phosphorylation, acylation including acetylation and formylation, glycosylation (including N-linked and O-linked), amidation, hydroxylation, alkylation including methylation and ethylation, ubiquitylation, addition of pyrrolidone carboxylic acid, formation of disulfide bridges, sulfation, myristoylation, palmitoylation, isoprenylation, farnesylation, geranylation, glypation, lipoylation and iodination.

Nucleic acids and/or nucleic acid sequences are "homologous" when they are derived, naturally or artificially, from a common ancestral nucleic acid or nucleic acid sequence. Proteins and/or protein sequences are "homologous" when their encoding DNAs are derived, naturally or artificially, from a common ancestral nucleic acid or nucleic acid sequence. The homologous molecules can be termed homologs. For example, any naturally occurring proteins, as described herein, can be modified by any available mutagenesis method. When expressed, this mutagenized nucleic acid encodes a polypeptide that is homologous to the protein encoded by the original nucleic acid. Homology is generally inferred from sequence identity between two or more nucleic acids or proteins (or sequences thereof). The precise percentage of identity between sequences that is useful in establishing homology varies with the nucleic acid and protein at issue, but as little as 25% sequence identity is routinely used to establish homology. Higher levels of sequence identity, e.g., 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 99% or more can also be used to establish homology. Methods for determining sequence identity percentages (e.g., BLASTP and BLASTN using default parameters) are described herein and are generally available.

The term "identical" and its grammatical equivalents as used herein or "sequence identity" in the context of two nucleic acid sequences or amino acid sequences of polypeptides refers to the residues in the two sequences which are the same when aligned for maximum correspondence over a specified comparison window. A "comparison window", as used herein, refers to a segment of at least about 20 contiguous positions, usually about 50 to about 200, more usually about 100 to about 150 in which a sequence can be compared to a reference sequence of the same number of contiguous positions after the two sequences are aligned optimally. Methods of alignment of sequences for comparison are well-known in the art. Optimal alignment of sequences for comparison can be conducted by the local homology algorithm of Smith and Waterman, *Adv. Appl. Math.*, 2:482 (1981); by the alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.*, 48:443 (1970); by the search

for similarity method of Pearson and Lipman, *Proc. Nat. Acad. Sci. U.S.A.*, 85:2444 (1988); by computerized implementations of these algorithms (including, but not limited to CLUSTAL in the PC/Gene program by Intelligentic, Mountain View Calif., GAP, BESTFIT, BLAST, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group (GCG), 575 Science Dr., Madison, Wis., U.S.A.); the CLUSTAL program is well described by Higgins and Sharp, *Gene*, 73:237-244 (1988) and Higgins and Sharp, *CABIOS*, 5:151-153 (1989); Corpet et al., *Nucleic Acids Res.*, 16:10881-10890 (1988); Huang et al., *Computer Applications in the Biosciences*, 8:155-165 (1992); and Pearson et al., *Methods in Molecular Biology*, 24:307-331 (1994). Alignment is also often performed by inspection and manual alignment. In one class of embodiments, the polypeptides herein are at least 80%, 85%, 90%, 98% 99% or 100% identical to a reference polypeptide, or a fragment thereof, e.g., as measured by BLASTP (or CLUSTAL, or any other available alignment software) using default parameters. Similarly, nucleic acids can also be described with reference to a starting nucleic acid, e.g., they can be 50%, 60%, 70%, 75%, 80%, 85%, 90%, 98%, 99% or 100% identical to a reference nucleic acid or a fragment thereof, e.g., as measured by BLASTN (or CLUSTAL, or any other available alignment software) using default parameters. When one molecule is said to have certain percentage of sequence identity with a larger molecule, it means that when the two molecules are optimally aligned, said percentage of residues in the smaller molecule finds a match residue in the larger molecule in accordance with the order by which the two molecules are optimally aligned.

The term "substantially identical" and its grammatical equivalents as applied to nucleic acid or amino acid sequences mean that a nucleic acid or amino acid sequence comprises a sequence that has at least 90% sequence identity or more, at least 95%, at least 98% and at least 99%, compared to a reference sequence using the programs described above, e.g., BLAST, using standard parameters. For example, the BLASTN program (for nucleotide sequences) uses as defaults a word length (W) of 11, an expectation (E) of 10, M=5, N=-4, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a word length (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff & Henikoff, *Proc. Natl. Acad. Sci. USA* 89:10915 (1992)). Percentage of sequence identity is determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window can comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. In embodiments, the substantial identity exists over a region of the sequences that is at least about 50 residues in length, over a region of at least about 100 residues, and in embodiments, the sequences are substantially identical over at least about 150 residues. In embodiments, the sequences are substantially identical over the entire length of the coding regions.

An "expression vector" or "vector" is any genetic element, e.g., a plasmid, chromosome, virus, transposon, behaving either as an autonomous unit of polynucleotide

replication within a cell. (i.e. capable of replication under its own control) or being rendered capable of replication by insertion into a host cell chromosome, having attached to it another polynucleotide segment, so as to bring about the replication and/or expression of the attached segment. Suitable vectors include, but are not limited to, plasmids, transposons, bacteriophages and cosmids. Vectors can contain polynucleotide sequences which are necessary to effect ligation or insertion of the vector into a desired host cell and to effect the expression of the attached segment. Such sequences differ depending on the host organism; they include promoter sequences to effect transcription, enhancer sequences to increase transcription, ribosomal binding site sequences and transcription and translation termination sequences. Alternatively, expression vectors can be capable of directly expressing nucleic acid sequence products encoded therein without ligation or integration of the vector into host cell DNA sequences. In some embodiments, the vector is an "episomal expression vector" or "episome," which is able to replicate in a host cell, and persists as an extrachromosomal segment of DNA within the host cell in the presence of appropriate selective pressure (see, e.g., Conese et al., *Gene Therapy*, 11:1735-1742 (2004)). Representative commercially available episomal expression vectors include, but are not limited to, episomal plasmids that utilize Epstein Barr Nuclear Antigen 1 (EBNA1) and the Epstein Barr Virus (EBV) origin of replication (oriP). The vectors pREP4, pCEP4, pREP7, and pcDNA3.1 from Invitrogen (Carlsbad, Calif.) and pBK-CMV from Stratagene (La Jolla, Calif.) represent non-limiting examples of an episomal vector that uses T-antigen and the SV40 origin of replication in lieu of EBNA1 and oriP. Vector also can comprise a selectable marker gene.

The term "adenovirus," as used herein, refers to an adenovirus that retains the ability to participate in the adenovirus life cycle and has not been physically inactivated by, for example, disruption (e.g., sonication), denaturing (e.g., using heat or solvents), or cross-linkage (e.g., via formalin cross-linking). The "adenovirus life cycle" includes (1) virus binding and entry into cells, (2) transcription of the adenoviral genome and translation of adenovirus proteins, (3) replication of the adenoviral genome, and (4) viral particle assembly (see, e.g., Fields Virology, 5<sup>th</sup> ed., Knipe et al. (eds.), Lippincott Williams & Wilkins, Philadelphia, PA (2006)). The term "adenoviral vector," as used herein, refers to an adenovirus in which the adenoviral genome has been manipulated to accommodate a nucleic acid sequence that is non-native with respect to the adenoviral genome. Typically, an adenoviral vector is generated by introducing one or more mutations (e.g., a deletion, insertion, or substitution) into the adenoviral genome of the adenovirus so as to accommodate the insertion of a non-native nucleic acid sequence, for example, for gene transfer, into the adenovirus.

The term "selectable marker gene" as used herein refers to a nucleic acid sequence that allows cells expressing the nucleic acid sequence to be specifically selected for or against, in the presence of a corresponding selective agent. Suitable selectable marker genes are known in the art and described in, e.g., International Patent Application Publications WO 1992/08796 and WO 1994/28143; Wigler et al., *Proc. Natl. Acad. Sci. USA*, 77: 3567 (1980); O'Hare et al., *Proc. Natl. Acad. Sci. USA*, 78: 1527 (1981); Mulligan & Berg, *Proc. Natl. Acad. Sci. USA*, 78: 2072 (1981); Colberre-Garapin et al., *J. Mol. Biol.*, 150:1 (1981); Santerre et al., *Gene*, 30: 147 (1984); Kent et al., *Science*, 237: 901-903 (1987); Wigler et al., *Cell*, 11: 223 (1977); Szybalska &

Szybalski, *Proc. Natl. Acad. Sci. USA*, 48: 2026 (1962); Lowy et al., *Cell*, 22: 817 (1980); and U.S. Pat. Nos. 5,122,464 and 5,770,359.

The term "coding sequence" as used herein refers to a segment of a polynucleotide that codes for protein. The region or sequence is bounded nearer the 5' end by a start codon and nearer the 3' end with a stop codon. Coding sequences can also be referred to as open reading frames.

The term "operably linked" as used herein refers to refers to the physical and/or functional linkage of a DNA segment to another DNA segment in such a way as to allow the segments to function in their intended manners. A DNA sequence encoding a gene product is operably linked to a regulatory sequence when it is linked to the regulatory sequence, such as, for example, promoters, enhancers and/or silencers, in a manner which allows modulation of transcription of the DNA sequence, directly or indirectly. For example, a DNA sequence is operably linked to a promoter when it is ligated to the promoter downstream with respect to the transcription initiation site of the promoter, in the correct reading frame with respect to the transcription initiation site and allows transcription elongation to proceed through the DNA sequence. An enhancer or silencer is operably linked to a DNA sequence coding for a gene product when it is ligated to the DNA sequence in such a manner as to increase or decrease, respectively, the transcription of the DNA sequence. Enhancers and silencers can be located upstream, downstream or embedded within the coding regions of the DNA sequence. A DNA for a signal sequence is operably linked to DNA coding for a polypeptide if the signal sequence is expressed as a pre-protein that participates in the secretion of the polypeptide. Linkage of DNA sequences to regulatory sequences is typically accomplished by ligation at suitable restriction sites or via adapters or linkers inserted in the sequence using restriction endonucleases known to one of skill in the art.

The terms "induce", "induction" and its grammatical equivalents as used herein refer to an increase in nucleic acid sequence transcription, promoter activity and/or expression brought about by a transcriptional regulator, relative to some basal level of transcription.

The term "transcriptional regulator" refers to a biochemical element that acts to prevent or inhibit the transcription of a promoter-driven DNA sequence under certain environmental conditions (e.g., a repressor or nuclear inhibitory protein), or to permit or stimulate the transcription of the promoter-driven DNA sequence under certain environmental conditions (e.g., an inducer or an enhancer).

The term "enhancer" as used herein, refers to a DNA sequence that increases transcription of, for example, a nucleic acid sequence to which it is operably linked. Enhancers can be located many kilobases away from the coding region of the nucleic acid sequence and can mediate the binding of regulatory factors, patterns of DNA methylation, or changes in DNA structure. A large number of enhancers from a variety of different sources are well known in the art and are available as or within cloned polynucleotides (from, e.g., depositories such as the ATCC as well as other commercial or individual sources). A number of polynucleotides comprising promoters (such as the commonly-used CMV promoter) also comprise enhancer sequences. Enhancers can be located upstream, within, or downstream of coding sequences. The term "Ig enhancers" refers to enhancer elements derived from enhancer regions mapped within the immunoglobulin (Ig) locus (such enhancers include for example, the heavy chain (mu) 5' enhancers, light chain (kappa) 5' enhancers, kappa and mu intronic

enhancers, and 3' enhancers (see generally Paul W. E. (ed), *Fundamental Immunology*, 3rd Edition, Raven Press, New York (1993), pages 353-363; and U.S. Pat. No. 5,885,827).

The term "promoter" refers to a region of a polynucleotide that initiates transcription of a coding sequence. Promoters are located near the transcription start sites of genes, on the same strand and upstream on the DNA (towards the 5' region of the sense strand). Some promoters are constitutive as they are active in all circumstances in the cell, while others are regulated becoming active in response to specific stimuli, e.g., an inducible promoter. The term "promoter activity" and its grammatical equivalents as used herein refer to the extent of expression of nucleotide sequence that is operably linked to the promoter whose activity is being measured. Promoter activity can be measured directly by determining the amount of RNA transcript produced, for example by Northern blot analysis or indirectly by determining the amount of product coded for by the linked nucleic acid sequence, such as a reporter nucleic acid sequence linked to the promoter.

"Inducible promoter" as used herein refers to a promoter which is induced into activity by the presence or absence of transcriptional regulators, e.g., biotic or abiotic factors. Inducible promoters are useful because the expression of genes operably linked to them can be turned on or off at certain stages of development of an organism or in a particular tissue. Non-limiting examples of inducible promoters include alcohol-regulated promoters, tetracycline-regulated promoters, steroid-regulated promoters, metal-regulated promoters, pathogenesis-regulated promoters, temperature-regulated promoters and light-regulated promoters. The inducible promoter can be part of a gene switch or genetic switch. The inducible promoter can be a gene switch ligand inducible promoter. In some cases, an inducible promoter can be a small molecule ligand-inducible two polypeptide ecdysone receptor-based gene switch. In some cases, a gene switch can be selected from ecdysone-based receptor components as described in, but without limitation to, any of the systems described in: International Patent Applications WO 2001/070816; WO 2002/029075; WO 2002/066613; WO 2002/066614; WO 2002/066612; WO 2002/066615; WO 2003/027266; WO 2003/027289; WO 2005/108617; WO 2009/045370; WO 2009/048560; WO 2010/042189; WO 2010/042189; WO 2011/119773; and WO 2012/122025; and U.S. Pat. Nos. 7,091,038; 7,776,587; 7,807,417; 8,202,718; 8,105,825; 8,168,426; 7,531,326; 8,236,556; 8,598,409; 8,715,959; 7,601,508; 7,829,676; 7,919,269; 8,030,067; 7,563,879; 8,021,878; 8,497,093; 7,935,510; 8,076,454; 9,402,919; 9,493,540; 9,249,207; and 9,492,482, each of which is incorporated by reference in its entirety).

The term "gene switch" or "genetic switch" refers to the combination of a response element associated with a promoter, and for instance, an EcR based system, which, in the presence of one or more ligands, modulates the expression of a gene into which the response element and promoter are incorporated. Tightly regulated inducible gene expression systems or gene switches are useful for various applications such as gene therapy, large scale production of proteins in cells, cell based high throughput screening assays, functional genomics and regulation of traits in transgenic plants and animals. Such inducible gene expression systems can include ligand inducible heterologous gene expression systems.

"Sleeping Beauty (SB) Transposon System" refers a synthetic DNA transposon system for to introducing DNA sequences into the chromosomes of vertebrates. Some

exemplary embodiments of the system are described, for example, in U.S. Pat. Nos. 6,489,458, 8,227,432, 9,228,180 and WO/2016/145146. The Sleeping Beauty transposon system is composed of a Sleeping Beauty (SB) transposase and a SB transposon. In embodiments, the Sleeping Beauty transposon system can include the SB11 transposon system, the SB100X transposon system, or the SB110 transposon system.

"Transposon" or "transposable element" (TE) is a vector DNA sequence that can change its position within the genome, sometimes creating or reversing mutations and altering the cell's genome size. Transposition often results in duplication of the TE. Class I TEs are copied in two stages: first they are transcribed from DNA to RNA, and the RNA produced is then reverse transcribed to DNA. This copied DNA is then inserted at a new position into the genome. The reverse transcription step is catalyzed by a reverse transcriptase, which can be encoded by the TE itself. The characteristics of retrotransposons are similar to retroviruses, such as HIV. The cut-and-paste transposition mechanism of class II TEs does not involve an RNA intermediate. The transpositions are catalyzed by several transposase enzymes. Some transposases non-specifically bind to any target site in DNA, whereas others bind to specific DNA sequence targets. The transposase makes a staggered cut at the target site resulting in single-strand 5' or 3' DNA overhangs (sticky ends). This step cuts out the DNA transposon, which is then ligated into a new target site; this process involves activity of a DNA polymerase that fills in gaps and of a DNA ligase that closes the sugar-phosphate backbone. This results in duplication of the target site. The insertion sites of DNA transposons can be identified by short direct repeats which can be created by the staggered cut in the target DNA and filling in by DNA polymerase, followed by a series of inverted repeats important for the TE excision by transposase. Cut-and-paste TEs can be duplicated if their transposition takes place during S phase of the cell cycle when a donor site has already been replicated, but a target site has not yet been replicated. Transposition can be classified as either autonomous or non-autonomous in both Class I and Class II TEs. Autonomous TEs can move by themselves while non-autonomous TEs require the presence of another TE to move. This is often because non-autonomous TEs lack transposase (for class II) or reverse transcriptase (for class I).

"Transposase" refers an enzyme that binds to the end of a transposon and catalyzes the movement of the transposon to another part of the genome by a cut and paste mechanism or a replicative transposition mechanism.

"T cell" or "T lymphocyte" as used herein is a type of lymphocyte that plays a central role in cell-mediated immunity. They can be distinguished from other lymphocytes, such as B cells and natural killer cells (NK cells), by the presence of a T-cell receptor (TCR) on the cell surface.

"T helper cells" ( $T_H$  cells) assist other white blood cells in immunologic processes, including maturation of B cells into plasma cells and memory B cells, and activation of cytotoxic T cells and macrophages. These cells are also known as CD4+ T cells because they express the CD4 glycoprotein on their surfaces. Helper T cells become activated when they are presented with peptide antigens by MHC class II molecules, which are expressed on the surface of antigen-presenting cells (APCs). Once activated, they divide rapidly and secrete small proteins called cytokines that regulate or assist in the active immune response. These cells can differentiate into one of several subtypes, including  $T_H1$ ,  $T_H2$ ,  $T_H3$ ,  $T_H9$ ,  $T_H17$ ,  $T_H22$  or  $T_{FH}$  (T follicular helper cells),

which secrete different cytokines to facilitate different types of immune responses. Signaling from the APCs directs T cells into particular subtypes.

“Cytotoxic T cells” (TC cells, or CTLs) or “cytotoxic T lymphocytes” destroy virus-infected cells and tumor cells, and are also implicated in transplant rejection. These cells are also known as CD8+ T cells since they express the CD8 glycoprotein at their surfaces. These cells recognize their targets by binding to antigen associated with MEW class I molecules, which are present on the surface of all nucleated cells. Through IL-10, adenosine, and other molecules secreted by regulatory T cells, the CD8+ cells can be inactivated to an anergic state, which prevents autoimmune diseases.

“Memory T cells” are a subset of antigen-specific T cells that persist long-term after an infection has resolved. They quickly expand to large numbers of effector T cells upon re-exposure to their cognate antigen, thus providing the immune system with memory against past infections. Memory T cells comprise three subtypes: central memory T cells ( $T_{CM}$  cells) and two types of effector memory T cells ( $T_{EM}$  cells and  $T_{EMRA}$  cells). Memory cells can be either CD4+ or CD8+. Memory T cells typically express the cell surface proteins CD45RO, CD45RA and/or CCR7.

“Regulatory T cells” (Treg cells), formerly known as suppressor T cells, play a role in the maintenance of immunological tolerance. Their major role is to shut down T cell-mediated immunity toward the end of an immune reaction and to suppress autoreactive T cells that escaped the process of negative selection in the thymus.

“Natural killer T cells” (NKT cells—not to be confused with natural killer cells of the innate immune system) bridge the adaptive immune system with the innate immune system. Unlike conventional T cells that recognize peptide antigens presented by major histocompatibility complex (MHC) molecules, NKT cells recognize glycolipid antigen presented by a molecule called CD1d. Once activated, these cells can perform functions ascribed to both T helper ( $T_H$ ) and cytotoxic T (TC) cells (i.e., cytokine production and release of cytolytic/cell killing molecules). They are also able to recognize and eliminate some tumor cells and cells infected with herpes viruses.

“Adoptive T cell transfer” refers to the isolation and ex vivo expansion of tumor specific T cells to achieve greater number of T cells than what could be obtained by vaccination alone or the patient’s natural tumor response. The tumor specific T cells are then infused into patients with cancer in an attempt to give their immune system the ability to overwhelm remaining tumor via T cells which can attack and kill cancer. There are many forms of adoptive T cell therapy being used for cancer treatment; culturing tumor infiltrating lymphocytes or TIL, isolating and expanding one particular T cell or clone, and even using T cells that have been engineered to potentially recognize and attack tumors.

“Antibody” as used herein refers to monoclonal or polyclonal antibodies. The term “monoclonal antibodies,” as used herein, refers to antibodies that are produced by a single clone of B-cells and bind to the same epitope. In contrast, “polyclonal antibodies” refer to a population of antibodies that are produced by different B-cells and bind to different epitopes of the same antigen. A whole antibody typically consists of four polypeptides: two identical copies of a heavy (H) chain polypeptide and two identical copies of a light (L) chain polypeptide. Each of the heavy chains contains one N-terminal variable (VH) region and three C-terminal constant (CH1, CH2 and CH3) regions, and each light chain contains one N-terminal variable (VL) region and

one C-terminal constant (CL) region. The variable regions of each pair of light and heavy chains form the antigen binding site of an antibody. The VH and VL regions have a similar general structure, with each region comprising four framework regions, whose sequences are relatively conserved. The framework regions are connected by three complementarity determining regions (CDRs). The three CDRs, known as CDR1, CDR2, and CDR3, form the “hypervariable region” of an antibody, which is responsible for antigen binding.

“Antibody like molecules” can be for example proteins that are members of the Ig-superfamily which are able to selectively bind a partner. MEW molecules and T cell receptors are such molecules. In one embodiment, the antibody-like molecule is an TCR. In one embodiment, the TCR has been modified to increase its MEW binding affinity.

The terms “fragment of an antibody,” “antibody fragment,” “functional fragment of an antibody,” “antigen-binding portion” or its grammatical equivalents are used interchangeably herein to mean one or more fragments or portions of an antibody that retain the ability to specifically bind to an antigen (see, generally, Holliger et al., *Nat. Biotech.*, 23(9):1126-1129 (2005)). The antibody fragment desirably comprises, for example, one or more CDRs, the variable region (or portions thereof), the constant region (or portions thereof), or combinations thereof. Non-limiting examples of antibody fragments include (i) a Fab fragment, which is a monovalent fragment consisting of the VL, VH, CL, and CH1 domains; (ii) a F(ab')<sub>2</sub> fragment, which is a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the stalk region; (iii) a Fv fragment consisting of the VL and VH domains of a single arm of an antibody; (iv) a single chain Fv (scFv), which is a monovalent molecule consisting of the two domains of the Fv fragment (i.e., VL and VH) joined by a synthetic linker which enables the two domains to be synthesized as a single polypeptide chain (see, e.g., Bird et al., *Science*, 242: 423-426 (1988); Huston et al., *Proc. Natl. Acad. Sci. USA*, 85: 5879-5883 (1988); and Osbourn et al., *Nat. Biotechnol.*, 16: 778 (1998)) and (v) a diabody, which is a dimer of polypeptide chains, wherein each polypeptide chain comprises a VH connected to a VL by a peptide linker that is too short to allow pairing between the VH and VL on the same polypeptide chain, thereby driving the pairing between the complementary domains on different VH-VL polypeptide chains to generate a dimeric molecule having two functional antigen binding sites. Antibody fragments are known in the art and are described in more detail in, e.g., U.S. Pat. No. 8,603,950.

“Antigen recognition moiety” or “antigen recognition domain” refers to a molecule or portion of a molecule that specifically binds to an antigen. In one embodiment, the antigen recognition moiety is an antibody, antibody like molecule or fragment thereof and the antigen is a tumor antigen.

The term “conservative amino acid substitution” or “conservative mutation” refers to the replacement of one amino acid by another amino acid with a common property. A functional way to define common properties between individual amino acids is to analyze the normalized frequencies of amino acid changes between corresponding proteins of homologous organisms (Schulz, G. E. and Schirmer, R. H., *Principles of Protein Structure*, Springer-Verlag, New York (1979)). According to such analyses, groups of amino acids can be defined where amino acids within a group exchange preferentially with each other, and therefore resemble each other most in their impact on the overall protein structure

(Schulz, G. E. and Schirmer, R. H., supra). Examples of conservative mutations include amino acid substitutions of amino acids within the sub-groups above, for example, lysine for arginine and vice versa such that a positive charge can be maintained; glutamic acid for aspartic acid and vice versa such that a negative charge can be maintained; serine for threonine such that a free —OH can be maintained; and glutamine for asparagine such that a free —NH<sub>2</sub> can be maintained. Alternatively or additionally, the functional variants can comprise the amino acid sequence of the reference protein with at least one non-conservative amino acid substitution.

The term “non-conservative mutations” involve amino acid substitutions between different groups, for example, lysine for tryptophan, or phenylalanine for serine, etc. In this case, it is preferable for the non-conservative amino acid substitution to not interfere with, or inhibit the biological activity of, the functional variant. The non-conservative amino acid substitution can enhance the biological activity of the functional variant, such that the biological activity of the functional variant is increased as compared to the homologous parent protein.

The term “ankyrin” refers to a family of adaptor proteins that mediate the attachment of integral membrane proteins to the spectrin-actin based membrane cytoskeleton. Ankyrins have binding sites for the beta subunit of spectrin and at least 12 families of integral membrane proteins. This linkage is required to maintain the integrity of the plasma membranes and to anchor specific ion channels, ion exchangers and ion transporters in the plasma membrane. Ankyrins contain four functional domains: an N-terminal domain that contains 24 tandem ankyrin repeats, a central domain that binds to spectrin, a death domain that binds to proteins involved in apoptosis, and a C-terminal regulatory domain that is highly variable between different ankyrin proteins. The 24 tandem ankyrin repeats are responsible for the recognition of a wide range of membrane proteins. These 24 repeats contain 3 structurally distinct binding sites ranging from repeat 1-14. These binding sites are quasi-independent of each other and can be used in combination. The interactions the sites use to bind to membrane proteins are non-specific and consist of: hydrogen bonding, hydrophobic interactions and electrostatic interactions. These non-specific interactions gives ankyrin the property to recognize a large range of proteins as the sequence doesn't have to be conserved just the properties of the amino acids. The quasi-independence means that if a binding site is not used, it won't have a large effect on the overall binding. These two properties in combination give rise to large repertoire of proteins ankyrin can recognize. Ankyrins are encoded by three genes (ANK1, ANK2 and ANK3) in mammals. Each gene in turn produces multiple proteins through alternative splicing.

The term “proliferative disease” as referred to herein refers to a unifying concept in which excessive proliferation of cells and/or turnover of cellular matrix contributes significantly to the pathogenesis of the disease, including cancer.

“Patient” or “subject” as used herein refers to a mammalian subject diagnosed with or suspected of having or developing a proliferative disorder such as cancer. In some embodiments, the term “patient” refers to a mammalian subject with a higher than average likelihood of developing a proliferative disorder such as cancer. Exemplary patients can be humans, apes, dogs, pigs, cattle, cats, horses, goats, sheep, rodents and other mammals that can benefit from the therapies disclosed herein. Exemplary human patients can be male and/or female. “Patient in need thereof” or

“subject in need thereof” is referred to herein as a patient diagnosed with or suspected of having a disease or disorder, for instance, but not restricted to human papilloma virus (HPV) infection.

“Administering” is referred to herein as providing one or more compositions described herein to a patient or a subject. By way of example and not limitation, composition administration, e.g., injection, can be performed by intravenous (i.v.) injection, sub-cutaneous (s.c.) injection, intradermal (i.d.) injection, intraperitoneal (i.p.) injection, or intramuscular (i.m.) injection. One or more such routes can be employed. Parenteral administration can be, for example, by bolus injection or by gradual perfusion over time. Alternatively, or concurrently, administration can be by the oral route. Additionally, administration can also be by surgical deposition of a bolus or pellet of cells, or positioning of a medical device. In an embodiment, a composition of the present disclosure can comprise engineered cells or host cells expressing nucleic acid sequences described herein, or a vector comprising at least one nucleic acid sequence described herein, in an amount that is effective to treat or prevent proliferative disorders. A pharmaceutical composition can comprise a target cell population as described herein, in combination with one or more pharmaceutically or physiologically acceptable carriers, diluents or excipients. Such compositions can comprise buffers such as neutral buffered saline, phosphate buffered saline and the like; carbohydrates such as glucose, mannose, sucrose or dextrans, mannitol; proteins; polypeptides or amino acids such as glycine; antioxidants; chelating agents such as EDTA or glutathione; adjuvants (e.g., aluminum hydroxide); and preservatives.

As used herein, the term “treatment”, “treating”, or its grammatical equivalents refers to obtaining a desired pharmacologic and/or physiologic effect. In embodiments, the effect is therapeutic, i.e., the effect partially or completely cures a disease and/or adverse symptom attributable to the disease. To this end, the inventive method comprises administering a therapeutically effective amount of the composition comprising the host cells expressing the inventive nucleic acid sequence, or a vector comprising the inventive nucleic acid sequences.

The term “therapeutically effective amount”, therapeutic amount”, “immunologically effective amount”, “anti-tumor effective amount”, “tumor-inhibiting effective amount” or its grammatical equivalents refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired therapeutic result. The therapeutically effective amount can vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of a composition described herein to elicit a desired response in one or more subjects. The precise amount of the compositions of the present disclosure to be administered can be determined by a physician with consideration of individual differences in age, weight, tumor size, extent of infection or metastasis, and condition of the patient (subject).

Alternatively, the pharmacologic and/or physiologic effect of administration of one or more compositions described herein to a patient or a subject of can be “prophylactic,” i.e., the effect completely or partially prevents a disease or symptom thereof. A “prophylactically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired prophylactic result (e.g., prevention of disease onset).

HPV Molecular Vaccine

Human papillomavirus (HPV) is a group of more than 200 related viruses. Each HPV virus in this large group is given

a number which is called its HPV type (or serotype). HPV is a small, non-enveloped deoxyribonucleic acid (DNA) virus that infects skin or mucosal cells. The circular, double-stranded viral genome is approximately 8-kb in length. The genome encodes for seven early proteins (E1 to E7) responsible for virus replication and two late proteins (L1 and L2), which are the viral structural proteins. As depicted in FIG. 1, each gene has specific functions. At least 13 of more than 200 known HPV types can cause cancer of the cervix and are associated with other anogenital cancers and cancers of the head and neck. The two most common "high-risk" serotypes (HPV-16 and HPV-18) cause approximately 70% of all cervical cancers. In HPV-16 and HPV-18, two primary oncoproteins, E6 and E7, are constitutively expressed by HPV-associated tumors and are critical for the induction and maintenance of cellular transformation in HPV-infected cells. Recent evidence also suggests the E5 protein also impacts viral transformation. HPV types 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 68, 73, and 82 are considered carcinogenic. Two "low-risk" HPV 6 and 11 are known to cause genital warts, a common benign condition of the external genitalia that causes significant morbidity. HPV is highly transmissible, with peak incidence soon after the onset of sexual activity, and most persons acquire infection at some time in their lives.

Prophylactic vaccines composed of virus-like particles, only induce immunity to the capsid structure and not to the non-structural proteins responsible for cell transformation. Initial HPV studies in animal models showed that inoculation with species-specific papillomaviruses induced an immune response that conferred protection against homologous virus challenge. However, native papillomaviruses are not good substrates for vaccine development as they cannot be grown easily in tissue culture. Subsequent studies were initiated on the production of viral particles from expression of the structural proteins in heterologous expression systems, such as yeast or baculovirus vectors. Results showed that expression of L1 alone led to the production of virus-like particles (VLPs) which morphologically resemble the authentic HPV virions but contain no viral DNA. These VLPs are produced by self-assembly of the L1 protein when expressed in a heterologous cell substrate. In animal studies, VLPs were shown to protect against high dose experimental infection by homologous virus. HPV VLPs are highly immunogenic in mice or rabbits, and the resulting antibodies have been shown to be neutralizing and type restricted when tested in a pseudovirion neutralization assay. Immunization with denatured particles does not result in the production of neutralizing antibodies, or protect from experimental virus challenge, indicating that neutralizing epitopes are conformation dependent.

Provided herein are compositions, kits, and systems comprising methods of making HPV recombinant vaccines. The HPV recombinant vaccines (e.g., HPV designs 1-5) in the present disclosure are engineered through protein engineering of E5, E6, and E7. These vaccines include a greater protein level sequence consensus amongst different HPV-16 and HPV-18 isolates. Moreover, they include genetically modified mutations to avoid oncogenic activities, improve their expression, and trigger broader immune response. Also provided herein are multi-epitope recombinant antigens containing immunogenic epitopes of the E5, E6, and E7 proteins. Along with the present HPV vaccine designs, certain enhancer agonist peptides previously shown to activate HPV-specific T cells are included. In some embodiments, the agonist peptides comprise one or more peptide sequences as shown in Table 2.

Each HPV vaccine antigen design was inspired via inventor-selected combinatorial use of bioinformatics analysis and in silico protein engineering methods (e.g., selection of antigenic sequences based on consensus sequences, antigenicity predictions, and T cell epitope mapping, targeted to induce MHC-I binding and cytokine production following T cell activation). The overall workflow of the presently disclosed HPV vaccine designs is shown in FIG. 2 and is further detailed in Example 2.

The naturally existing sequence variations on numerous HPV strains present a significant hurdle to the development of effective, broad spectrum HPV vaccines. As a solution to this problem, the present vaccine design approach utilizes advanced bioinformatics and protein engineering approaches to select and design antigen sequences with broad coverage of T cell epitopes, novel mutations, and enhancer agonist peptides. Drawing on information available of extended coverage of antigen regions with CTL-specific epitopes and in silico prediction results, the designed HPV vaccine antigens are targeted to induce robust HPV-16 and HPV-18 specific responses and provide therapeutic benefit to persons at risk of HPV-derived cancers.

The present disclosure provides five HPV antigen designs (HPV designs 1-5) constructed in a multi-deleted gorilla adenovector (GC46). Detailed methods for each of the HPV antigen designs are elaborated in Examples 1-3. For RNA qPCR relative expression assay, 5'-TGC-CAAGAGTGACGTGTCCA-3' (SEQ ID NO: 110) was used as a splice primer, and 5'-CCCAGGTC-CAACTGCAGCCGG-3' (SEQ ID NO: 111) was used as a splice probe. Specific primers designed for each antigen were used as reverse primers (FIG. 6).

#### Delivery System

##### Gorilla Adenovirus Shuttle Vector

Certain aspects of the present disclosure are directed to a vector comprising a polynucleotide encoding a polypeptide comprising one or more immune response-inducing HPV polypeptides described herein. In certain embodiments, the vector is a viral vector. In particular embodiments, the vector is an adenoviral vector. Adenoviruses are generally associated with benign pathologies in humans, and the genomes of adenoviruses isolated from a variety of species, including humans, have been extensively studied. Adenovirus is a medium-sized (90-100 nm), non-enveloped icosahedral virus containing approximately 36 kb of double-stranded DNA. The adenovirus capsid mediates the key interactions of the early stages of the infection of a cell by the virus, and is required for packaging adenovirus genomes at the end of the adenovirus life cycle. The capsid comprises 252 capsomeres, which includes 240 hexons, 12 penton base proteins, and 12 fibers (Ginsberg et al., *Virology*, 28: 782-83 (1966)). The hexon comprises three identical proteins, namely polypeptide II (Roberts et al., *Science*, 232: 1148-51 (1986)). The penton base comprises five identical proteins and the fiber comprises three identical proteins. Proteins IIIa, VI, and IX are present in the adenoviral coat and are believed to stabilize the viral capsid (Stewart et al., *Cell*, 67: 145-54 (1991), and Stewart et al., *EMBO J.*, 12(7): 2589-99 (1993)). The expression of the capsid proteins, with the exception of pIX, is dependent on the adenovirus polymerase protein. Therefore, major components of an adenovirus particle are expressed from the genome only when the polymerase protein gene is present and expressed.

Several features of adenoviruses make them ideal vehicles for transferring genetic material to cells for therapeutic applications (i.e., "gene therapy"), or for use as antigen delivery systems for vaccine applications. For example,

adenoviruses can be produced in high titers (e.g., about  $10^{13}$  particle units (pu)), and can transfer genetic material to non-replicating and replicating cells. The adenoviral genome can be manipulated to carry a large amount of exogenous DNA (up to about 8 kb), and the adenoviral capsid can potentiate the transfer of even longer sequences (Curiel et al., *Hum. Gene Ther.*, 3: 147-154 (1992)). Additionally, adenoviruses generally do not integrate into the host cell chromosome, but rather are maintained as a linear episome, thereby minimizing the likelihood that a recombinant adenovirus will interfere with normal cell function.

In some embodiments, the adenovirus described herein is isolated from a gorilla. There are four widely recognized gorilla subspecies within the two species of Eastern Gorilla (*Gorilla beringei*) and Western Gorilla (*Gorilla gorilla*). The Western Gorilla species includes the subspecies Western Lowland Gorilla (*Gorilla gorilla gorilla*) and Cross River Gorilla (*Gorilla gorilla diehli*). The Eastern Gorilla species includes the subspecies Mountain Gorilla (*Gorilla beringei beringei*) and Eastern Lowland Gorilla (*Gorilla beringei graueri*) (see, e.g., Wilson and Reeder, eds., *Mammalian Species of the World*, 3<sup>rd</sup> ed., Johns Hopkins University Press, Baltimore, Maryland (2005)). In some embodiments, the adenovirus of the present disclosure is isolated from Mountain Gorilla (*Gorilla beringei beringei*).

Various Gorilla adenoviruses or adenoviral vectors are described in International Patent Application Publications WO 2013/052832; WO 2013/052811; and WO 2013/052799, each of which is herein incorporated by reference in its entirety.

The genomes of several such adenoviruses have been analyzed, and it has been determined that the adenovirus can have the nucleic acid sequence of, for example, SEQ ID NO: 21, SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, or SEQ ID NO: 25, each of which includes a number of sub-sequences that serve to uniquely define the adenovirus, namely the nucleic acid sequences SEQ ID NOs: 1-10, and amino acid sequences SEQ ID NOs: 11-20. SEQ ID NOs: 6-10 encode the amino acid sequences of SEQ ID NOs: 16-20, respectively. SEQ ID NOs: 1-5 are a subset of the nucleic acid sequences of SEQ ID NOs: 6-10, respectively. SEQ ID NOs: 11-15 are a subset of the amino acid sequences of SEQ ID NOs: 16-20, respectively.

The adenovirus can be modified in the same manner as previously known adenoviruses to be used as an adenoviral vector, e.g., a gene delivery vehicle. The adenovirus and adenoviral vector can be replication-competent, conditionally replication-competent, or replication-deficient.

A replication-competent adenovirus or adenoviral vector can replicate in typical host cells, i.e., cells typically capable of being infected by an adenovirus. A replication-competent adenovirus or adenoviral vector can have one or more mutations as compared to the wild-type adenovirus (e.g., one or more deletions, insertions, and/or substitutions) in the adenoviral genome that do not inhibit viral replication in host cells. For example, the adenovirus or adenoviral vector can have a partial or entire deletion of the adenoviral early region known as the E3 region, which is not essential for propagation of the adenovirus or adenoviral genome.

A conditionally-replicating adenovirus or adenoviral vector is an adenovirus or adenoviral vector that has been engineered to replicate under pre-determined conditions. For example, replication-essential gene functions, e.g., gene functions encoded by the adenoviral early regions, can be operably linked to an inducible, repressible, or tissue-specific transcription control sequence, e.g., promoter. In such an embodiment, replication requires the presence or absence

of specific factors that interact with the transcription control sequence. Conditionally-replicating adenoviral vectors are further described in U.S. Pat. No. 5,998,205.

A replication-deficient adenovirus or adenoviral vector is an adenovirus or adenoviral vector that requires complementation of one or more gene functions or regions of the adenoviral genome that are required for replication, as a result of, for example, a deficiency in one or more replication-essential gene function or regions, such that the adenovirus or adenoviral vector does not replicate in typical host cells, especially those in a human to be infected by the adenovirus or adenoviral vector.

A deficiency in a gene function or genomic region, as used herein, is defined as a disruption (e.g., deletion) of sufficient genetic material of the adenoviral genome to obliterate or impair the function of the gene (e.g., such that the function of the gene product is reduced by at least about 2-fold, 5-fold, 10-fold, 20-fold, 30-fold, or 50-fold) whose nucleic acid sequence was disrupted (e.g., deleted) in whole or in part. Deletion of an entire gene region often is not required for disruption of a replication-essential gene function. However, for the purpose of providing sufficient space in the adenoviral genome for one or more transgenes, removal of a majority of one or more gene regions can be desirable. While deletion of genetic material is preferred, mutation of genetic material by addition or substitution also is appropriate for disrupting gene function. Replication-essential gene functions are those gene functions that are required for adenovirus replication (e.g., propagation) and are encoded by, for example, the adenoviral early regions (e.g., the E1, E2, and E4 regions), late regions (e.g., the L1, L2, L3, L4, and L5 regions), genes involved in viral packaging (e.g., the IVa2 gene), and virus-associated RNAs (e.g., VA-RNA-1 and/or VA-RNA-2).

Whether the adenovirus or adenoviral vector is replication-competent or replication-deficient, the adenovirus or adenoviral vector retains at least a portion of the adenoviral genome. The adenovirus or adenoviral vector can comprise any portion of the adenoviral genome, including protein coding and non-protein coding regions. Desirably, the adenovirus or adenoviral vector comprises at least one nucleic acid sequence that encodes an adenovirus protein. The adenovirus or adenoviral vector can comprise a nucleic acid sequence that encodes any suitable adenovirus protein, such as, for example, a protein encoded by any one of the early region genes (i.e., E1A, E1B, E2A, E2B, E3, and/or E4 regions), or a protein encoded by any one of the late region genes, which encode the virus structural proteins (i.e., L1, L2, L3, L4, and L5 regions).

The adenovirus or adenoviral vector desirably comprises one or more nucleic acid sequences that encode the pIX protein, the DNA polymerase protein, the penton protein, the hexon protein, and/or the fiber protein. The adenovirus or adenoviral vector can comprise a full-length nucleic acid sequence that encodes a full-length amino acid sequence of an adenovirus protein. Alternatively, the adenovirus or adenoviral vector can comprise a portion of a full-length nucleic acid sequence that encodes a portion of a full-length amino acid sequence of an adenovirus protein.

A "portion" of a nucleic acid sequence comprises at least ten nucleotides (e.g., about 10 to about 5000 nucleotides). Preferably, a "portion" of a nucleic acid sequence comprises 10 or more (e.g., 15 or more, 20 or more, 25 or more, 30 or more, 35 or more, 40 or more, 45 or more, 50 or more, or 100 or more) nucleotides, but less than 5,000 (e.g., 4900 or less, 4000 or less, 3000 or less, 2000 or less, 1000 or less, 800 or less, 500 or less, 300 or less, or 100 or less)



nucleotides. Preferably, a portion of a nucleic acid sequence is about 10 to about 3500 nucleotides (e.g., about 10, 20, 30, 50, 100, 300, 500, 700, 1000, 1500, 2000, 2500, or 3000 nucleotides), about 10 to about 1000 nucleotides (e.g., about 25, 55, 125, 325, 525, 725, or 925 nucleotides), or about 10 to about 500 nucleotides (e.g., about 15, 30, 40, 50, 60, 70, 80, 90, 150, 175, 250, 275, 350, 375, 450, 475, 480, 490, 495, or 499 nucleotides), or a range defined by any two of the foregoing values. More preferably, a "portion" of a nucleic acid sequence comprises no more than about 3200 nucleotides (e.g., about 10 to about 3200 nucleotides, about 10 to about 3000 nucleotides, or about 30 to about 500 nucleotides, or a range defined by any two of the foregoing values).

A "portion" of an amino acid sequence comprises at least three amino acids (e.g., about 3 to about 1,200 amino acids). Preferably, a "portion" of an amino acid sequence comprises 3 or more (e.g., 5 or more, 10 or more, 15 or more, 20 or more, 25 or more, 30 or more, 40 or more, or 50 or more) amino acids, but less than 1,200 (e.g., 1,000 or less, 800 or less, 700 or less, 600 or less, 500 or less, 400 or less, 300 or less, 200 or less, or 100 or less) amino acids. Preferably, a portion of an amino acid sequence is about 3 to about 500 amino acids (e.g., about 10, 100, 200, 300, 400, or 500 amino acids), about 3 to about 300 amino acids (e.g., about 20, 50, 75, 95, 150, 175, or 200 amino acids), or about 3 to about 100 amino acids (e.g., about 15, 25, 35, 40, 45, 60, 65, 70, 80, 85, 90, 95, or 99 amino acids), or a range defined by any two of the foregoing values. More preferably, a "portion" of an amino acid sequence comprises no more than about 500 amino acids (e.g., about 3 to about 400 amino acids, about 10 to about 250 amino acids, or about 50 to about 100 amino acids, or a range defined by any two of the foregoing values).

The adenovirus pIX protein is present in the adenovirus capsid, has been shown to strengthen hexon nonamer interactions, and is essential for the packaging of full-length genomes (see, e.g., Boulanger et al., *J. Gen. Virol.*, 44: 783-800 (1979); Horwitz M. S., "Adenoviridae and their replication" in *Virology*, 2<sup>nd</sup> ed., B. N. Fields et al. (eds.), Raven Press, Ltd., New York, pp. 1679-1721 (1990), Ghosh-Choudhury et al., *EMBO J.*, 6: 1733-1739 (1987), and van Oostrum et al., *J. Virol.*, 56: 439-448 (1985)). In addition to its contribution to adenovirus structure, pIX also has been shown to exhibit transcriptional properties, such as stimulation of adenovirus major late promoter (MLP) activity (see, e.g., Lutz et al., *J. Virol.*, 71(7): 5102-5109 (1997)). Nucleic acid sequences that encode all or a portion of an adenovirus pIX protein include, for example, SEQ ID NO: 6 and SEQ ID NO: 1. Amino acid sequences that comprise a full-length pIX protein, or a portion thereof, include, for example, SEQ ID NO: 16 and SEQ ID NO: 11.

The adenovirus DNA polymerase protein is essential for viral DNA replication both in vitro and in vivo. The polymerase co-purifies in a complex with the precursor (pTP) of the terminal protein (TP), which is covalently attached to the 5' ends of adenovirus DNA (Field et al., *J. Biol. Chem.*, 259: 9487-9495 (1984)). Both the adenovirus DNA polymerase and pTP are encoded by the E2 region. The polymerase protein is required for the expression of all the structural proteins except for pIX. Without the gene sequence for polymerase protein, polymerase protein is not produced. As a result, the viral genome is not replicated, the Major Late Promoter is not activated, and the capsid proteins are not expressed. Nucleic acid sequences that encode all or a portion of an adenovirus DNA polymerase protein include, for example, SEQ ID NO: 7 and SEQ ID NO: 2. Amino acid

sequences that comprise a full-length adenovirus DNA polymerase, or a portion thereof, include, for example, SEQ ID NO: 17 and SEQ ID NO: 12.

The adenovirus hexon protein is the largest and most abundant protein in the adenovirus capsid. The hexon protein is essential for virus capsid assembly, determination of the icosahedral symmetry of the capsid (which in turn defines the limits on capsid volume and DNA packaging size), and integrity of the capsid. In addition, hexon is a primary target for modification in order to reduce neutralization of adenoviral vectors (see, e.g., Gall et al., *J. Virol.*, 72: 10260-264 (1998), and Rux et al., *J. Virol.*, 77(17): 9553-9566 (2003)). The major structural features of the hexon protein are shared by adenoviruses across serotypes, but the hexon protein differs in size and immunological properties between serotypes (Jornvall et al., *J. Biol. Chem.*, 256(12): 6181-6186 (1981)). A comparison of 15 adenovirus hexon proteins revealed that the predominant antigenic and serotype-specific regions of the hexon appear to be in loops 1 and 2 (i.e., LI or LI, and LII or LI2, respectively), within which are seven discrete hypervariable regions (HVR1 to HVR7) varying in length and sequence between adenoviral serotypes (Crawford-Miksza et al., *J. Virol.*, 70(3): 1836-1844 (1996)). Nucleic acid sequences that encode all or a portion of an adenovirus hexon protein include, for example, SEQ ID NO: 9 and SEQ ID NO: 4. Amino acid sequences that comprise a full-length adenovirus hexon protein, or a portion thereof, include, for example, SEQ ID NO: 19 and SEQ ID NO: 14.

The adenovirus fiber protein is a homotrimer of the adenoviral polypeptide IV that has three domains: the tail, shaft, and knob. (Devaux et al., *J. Molec. Biol.*, 215: 567-88 (1990), Yeh et al., *Virus Res.*, 33: 179-98 (1991)). The fiber protein mediates primary viral binding to receptors on the cell surface via the knob and the shaft domains (Henry et al., *J. Virol.*, 68(8): 5239-46 (1994)). The amino acid sequences for trimerization are located in the knob, which appears necessary for the amino terminus of the fiber (the tail) to properly associate with the penton base (Novelli et al., *Virology*, 185: 365-76 (1991)). In addition to recognizing cell receptors and binding the penton base, the fiber contributes to serotype identity. Fiber proteins from different adenoviral serotypes differ considerably (see, e.g., Green et al., *EMBO J.*, 2: 1357-65 (1983), Chroboczek et al., *Virology*, 186: 280-85 (1992), and Signas et al., *J. Virol.*, 53: 672-78 (1985)). Thus, the fiber protein has multiple functions key to the life cycle of adenovirus. Nucleic acid sequences that encode all or a portion of an adenovirus fiber protein include, for example, SEQ ID NO: 10 and SEQ ID NO: 5. Amino acid sequences that comprise a full-length adenovirus fiber protein, or a portion thereof, include, for example, SEQ ID NO: 20 and SEQ ID NO: 15.

The adenovirus penton base protein is located at the vertices of the icosahedral capsid and comprises five identical monomers. The penton base protein provides a structure for bridging the hexon proteins on multiple facets of the icosahedral capsid, and provides the essential interface for the fiber protein to be incorporated in the capsid. Each monomer of the penton base contains an RGD tripeptide motif (Neumann et al., *Gene*, 69: 153-157 (1988)). The RGD tripeptide mediates binding to  $\alpha v$  integrins and adenoviruses that have point mutations in the RGD sequence of the penton base are restricted in their ability to infect cells (Bai et al., *J. Virol.*, 67: 5198-5205 (1993)). Thus, the penton base protein is essential for the architecture of the capsid and for maximum efficiency of virus-cell interaction. Nucleic acid sequences that encode all or a portion of an adenovirus



penton base protein include, for example, SEQ ID NO: 8 and SEQ ID NO: 3. Amino acid sequences that comprise a full-length adenovirus penton base protein, or a portion thereof, include, for example, SEQ ID NO: 18 and SEQ ID NO: 13.

Nucleic acid or amino acid sequence "identity," as described herein, can be determined by comparing a nucleic acid or amino acid sequence of interest to a reference nucleic acid or amino acid sequence. The numbers of nucleotides or amino acid residues that have been changed and/or modified (such as, e.g., by point mutations, insertions, or deletions) in the reference sequence so as to result in the sequence of interest are counted. The total number of such changes is subtracted from the total length of the sequence of interest, and the difference is divided by the length of the sequence of interest and expressed as a percentage. A number of mathematical algorithms for obtaining the optimal alignment and calculating identity between two or more sequences are known and incorporated into a number of available software programs. Examples of such programs include CLUSTAL-W, T-Coffee, and ALIGN (for alignment of nucleic acid and amino acid sequences), BLAST programs (e.g., BLAST 2.1, BL2SEQ, and later versions thereof) and FASTA programs (e.g., FASTA3x, FASTM, and SSEARCH) (for sequence alignment and sequence similarity searches). Sequence alignment algorithms also are disclosed in, for example, Altschul et al., *J. Molecular Biol.*, 215(3): 403-410 (1990), Beigert et al., *Proc. Natl. Acad. Sci. USA*, 106(10): 3770-3775 (2009), Durbin et al., eds., *Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids*, Cambridge University Press, Cambridge, UK (2009), Soding, *Bioinformatics*, 21(7): 951-960 (2005), Altschul et al., *Nucleic Acids Res.*, 25(17): 3389-3402 (1997), and Gusfield, *Algorithms on Strings, Trees and Sequences*, Cambridge University Press, Cambridge UK (1997).

The adenovirus or adenoviral vector can comprise one, two, three, four, or all five of the aforementioned sequences alone or in any combination. In this respect, the adenovirus or adenoviral vector can comprise any combination of any two of the aforementioned sequences, any combination of any three of the aforementioned sequences, any combination of any four of the aforementioned sequences, or all five of the aforementioned sequences.

As discussed herein, the adenovirus or adenoviral vector can be replication-competent, conditionally-replicating, or replication-deficient. Preferably, the adenovirus or adenoviral vector is replication-deficient, such that the replication-deficient adenovirus or adenoviral vector requires complementation of at least one replication-essential gene function of one or more regions of the adenoviral genome for propagation (e.g., to form adenoviral vector particles).

The replication-deficient adenovirus or adenoviral vector can be modified in any suitable manner to cause the deficiencies in the one or more replication-essential gene functions in one or more regions of the adenoviral genome for propagation. The complementation of the deficiencies in the one or more replication-essential gene functions of one or more regions of the adenoviral genome refers to the use of exogenous means to provide the deficient replication-essential gene functions. Such complementation can be effected in any suitable manner, for example, by using complementing cells and/or exogenous DNA (e.g., helper adenovirus) encoding the disrupted replication-essential gene functions.

The adenovirus or adenoviral vector can be deficient in one or more replication-essential gene functions of only the early regions (i.e., E1-E4 regions) of the adenoviral genome,

only the late regions (i.e., L1-L5 regions) of the adenoviral genome, both the early and late regions of the adenoviral genome, or all adenoviral genes (i.e., a high capacity adenovector (HC-Ad). See Morsy et al., *Proc. Natl. Acad. Sci. USA*, 95: 965-976 (1998); Chen et al., *Proc. Natl. Acad. Sci. USA*, 94: 1645-1650 (1997); and Kochanek et al., *Hum. Gene Ther.*, 10: 2451-2459 (1999). Examples of replication-deficient adenoviral vectors are disclosed in U.S. Pat. Nos. 5,837,511; 5,851,806; 5,994,106; 6,127,175; 6,482,616; and 7,195,896, and International Patent Application Publications WO 1994/028152, WO 1995/002697, WO 1995/016772, WO 1995/034671, WO 1996/022378, WO 1997/012986, WO 1997/021826, and WO 2003/022311.

The early regions of the adenoviral genome include the E1, E2, E3, and E4 regions. The E1 region comprises the E1A and E1B subregions, and one or more deficiencies in replication-essential gene functions in the E1 region can include one or more deficiencies in replication-essential gene functions in either or both of the E1A and E1B subregions, thereby requiring complementation of the E1A subregion and/or the E1B subregion of the adenoviral genome for the adenovirus or adenoviral vector to propagate (e.g., to form adenoviral vector particles). The E2 region comprises the E2A and E2B subregions, and one or more deficiencies in replication-essential gene functions in the E2 region can include one or more deficiencies in replication-essential gene functions in either or both of the E2A and E2B subregions, thereby requiring complementation of the E2A subregion and/or the E2B subregion of the adenoviral genome for the adenovirus or adenoviral vector to propagate (e.g., to form adenoviral vector particles).

The E3 region does not include any replication-essential gene functions, such that a deletion of the E3 region in part or in whole does not require complementation of any gene functions in the E3 region for the adenovirus or adenoviral vector to propagate (e.g., to form adenoviral vector particles). In the context of the present disclosure, the E3 region is defined as the region that initiates with the open reading frame that encodes a protein with high homology to the 12.5K protein from the E3 region of human adenovirus 5 (NCBI reference sequence AP\_000218) and ends with the open reading frame that encodes a protein with high homology to the 14.7K protein from the E3 region of human adenovirus 5 (NCBI reference sequence AP\_000224.1). The E3 region can be deleted in whole or in part, or retained in whole or in part. The size of the deletion can be tailored so as to retain an adenovirus or adenoviral vector whose genome closely matches the optimum genome packaging size. A larger deletion will accommodate the insertion of larger heterologous nucleic acid sequences in the adenovirus or adenoviral genome. In one embodiment of the present disclosure, the L4 polyadenylation signal sequences, which reside in the E3 region, are retained.

The E4 region comprises multiple open reading frames (ORFs). An adenovirus or adenoviral vector with a deletion of all of the open reading frames of the E4 region except ORF6, and in some cases ORF3, does not require complementation of any gene functions in the E4 region for the adenovirus or adenoviral vector to propagate (e.g., to form adenoviral vector particles). Conversely, an adenovirus or adenoviral vector with a disruption or deletion of ORF6, and in some cases ORF3, of the E4 region (e.g., with a deficiency in a replication-essential gene function based in ORF6 and/or ORF3 of the E4 region), with or without a disruption or deletion of any of the other open reading frames of the E4 region or the native E4 promoter, polyadenylation sequence, and/or the right-side inverted terminal repeat (ITR), requires

complementation of the E4 region (specifically, of ORF6 and/or ORF3 of the E4 region) for the adenovirus or adenoviral vector to propagate (e.g., to form adenoviral vector particles). The late regions of the adenoviral genome include the L1, L2, L3, L4, and L5 regions. The adenovirus or adenoviral vector also can have a mutation in the major late promoter (MLP), as discussed in International Patent Application Publication WO 2000/000628, which can render the adenovirus or adenoviral vector replication-deficient if desired.

The one or more regions of the adenoviral genome that contain one or more deficiencies in replication-essential gene functions desirably are one or more early regions of the adenoviral genome, i.e., the E1, E2, and/or E4 regions, optionally with the deletion in part or in whole of the E3 region.

The replication-deficient adenovirus or adenoviral vector also can have one or more mutations as compared to the wild-type adenovirus (e.g., one or more deletions, insertions, and/or substitutions) in the adenoviral genome that do not inhibit viral replication in host cells. Thus, in addition to one or more deficiencies in replication-essential gene functions, the adenovirus or adenoviral vector can be deficient in other respects that are not replication-essential. For example, the adenovirus or adenoviral vector can have a partial or entire deletion of the adenoviral early region known as the E3 region, which is not essential for propagation of the adenovirus or adenoviral genome.

In one embodiment, the adenovirus or adenoviral vector is replication-deficient and requires, at most, complementation of the E1 region or the E4 region of the adenoviral genome, for propagation (e.g., to form adenoviral vector particles). Thus, the replication-deficient adenovirus or adenoviral vector requires complementation of at least one replication-essential gene function of the E1A subregion and/or the E1B region of the adenoviral genome (denoted an E1-deficient adenoviral vector) or the E4 region of the adenoviral genome (denoted an E4-deficient adenoviral vector) for propagation (e.g., to form adenoviral vector particles). The adenovirus or adenoviral vector can be deficient in at least one replication-essential gene function (desirably all replication-essential gene functions) of the E1 region of the adenoviral genome and at least one gene function of the nonessential E3 region of the adenoviral genome (denoted an E1/E3-deficient adenoviral vector). The adenovirus or adenoviral vector can be deficient in at least one replication-essential gene function (desirably all replication-essential gene functions) of the E4 region of the adenoviral genome and at least one gene function of the nonessential E3 region of the adenoviral genome (denoted an E3/E4-deficient adenoviral vector).

In one embodiment, the adenovirus or adenoviral vector is replication-deficient and requires, at most, complementation of the E2 region, preferably the E2A subregion, of the adenoviral genome, for propagation (e.g., to form adenoviral vector particles). Thus, the replication-deficient adenovirus or adenoviral vector requires complementation of at least one replication-essential gene function of the E2A subregion of the adenoviral genome (denoted an E2A-deficient adenoviral vector) for propagation (e.g., to form adenoviral vector particles). The adenovirus or adenoviral vector can be deficient in at least one replication-essential gene function (desirably all replication-essential gene functions) of the E2A region of the adenoviral genome and at least one gene function of the nonessential E3 region of the adenoviral genome (denoted an E2A/E3-deficient adenoviral vector).

In one embodiment, the adenovirus or adenoviral vector is replication-deficient and requires, at most, complementation of the E1 and E4 regions of the adenoviral genome for propagation (e.g., to form adenoviral vector particles). Thus, the replication-deficient adenovirus or adenoviral vector requires complementation of at least one replication-essential gene function of both the E1 and E4 regions of the adenoviral genome (denoted an E1/E4-deficient adenoviral vector) for propagation (e.g., to form adenoviral vector particles). The adenovirus or adenoviral vector can be deficient in at least one replication-essential gene function (desirably all replication-essential gene functions) of the E1 region of the adenoviral genome, at least one replication-essential gene function of the E4 region of the adenoviral genome, and at least one gene function of the nonessential E3 region of the adenoviral genome (denoted an E1/E3/E4-deficient adenoviral vector). The adenovirus or adenoviral vector preferably requires, at most, complementation of the E1 region of the adenoviral genome for propagation, and does not require complementation of any other deficiency of the adenoviral genome for propagation. More preferably, the adenovirus or adenoviral vector requires, at most, complementation of the E1 and E4 regions of the adenoviral genome for propagation, and does not require complementation of any other deficiency of the adenoviral genome for propagation.

The adenovirus or adenoviral vector, when deficient in multiple replication-essential gene functions of the adenoviral genome (e.g., an E1/E4-deficient adenoviral vector), can include a spacer sequence to provide viral growth in a complementing cell line similar to that achieved by adenoviruses or adenoviral vectors deficient in a single replication-essential gene function (e.g., an E1-deficient adenoviral vector). The spacer sequence can contain any nucleotide sequence or sequences which are of a desired length, such as sequences at least about 15 base pairs (e.g., between about 15 nucleotides and about 12,000 nucleotides), preferably about 100 nucleotides to about 10,000 nucleotides, more preferably about 500 nucleotides to about 8,000 nucleotides, even more preferably about 1,500 nucleotides to about 6,000 nucleotides, and most preferably about 2,000 to about 3,000 nucleotides in length, or a range defined by any two of the foregoing values. The spacer sequence can be coding or non-coding and native or non-native with respect to the adenoviral genome, but does not restore the replication-essential function to the deficient region. The spacer also can contain an expression cassette. More preferably, the spacer comprises a polyadenylation sequence and/or a gene that is non-native with respect to the adenovirus or adenoviral vector. The use of a spacer in an adenoviral vector is further described in, for example, U.S. Pat. No. 5,851,806 and International Patent Application Publication WO 1997/021826.

By removing all or part of the adenoviral genome, for example, the E1, E3, and E4 regions of the adenoviral genome, the resulting adenovirus or adenoviral vector is able to accept inserts of exogenous nucleic acid sequences while retaining the ability to be packaged into adenoviral capsids. An exogenous nucleic acid sequence can be inserted at any position in the adenoviral genome so long as insertion in the position allows for the formation of adenovirus or the adenoviral vector particle. The exogenous nucleic acid sequence preferably is positioned in the E1 region, the E3 region, or the E4 region of the adenoviral genome.

The replication-deficient adenovirus or adenoviral vector of the present disclosure can be produced in complementing cell lines that provide gene functions not present in the

replication-deficient adenovirus or adenoviral vector, but required for viral propagation, at appropriate levels in order to generate high titers of viral vector stock. Such complementing cell lines are known and include, but are not limited to, 293 cells (described in, e.g., Graham et al., *J. Gen. Virol.*, 36: 59-72 (1977)), PER.C6 cells (described in, e.g., International Patent Application Publication WO 1997/000326, and U.S. Pat. Nos. 5,994,128 and 6,033,908), and 293-ORF6 cells (described in, e.g., International Patent Application Publication WO 95/34671 and Brough et al., *J. Virol.*, 71: 9206-9213 (1997)). Other suitable complementing cell lines to produce the replication-deficient adenovirus or adenoviral vector of the present disclosure include complementing cells that have been generated to propagate adenoviral vectors encoding transgenes whose expression inhibits viral growth in host cells (see, e.g., U.S. Patent Application Publication No. 2008/0233650). Additional suitable complementing cells are described in, for example, U.S. Pat. Nos. 6,677,156 and 6,682,929, and International Patent Application Publication WO 2003/020879. In some instances, the cellular genome need not comprise nucleic acid sequences, the gene products of which complement for all of the deficiencies of a replication-deficient adenoviral vector. One or more replication-essential gene functions lacking in a replication-deficient adenoviral vector can be supplied by a helper virus, e.g., an adenoviral vector that supplies in trans one or more essential gene functions required for replication of the replication-deficient adenovirus or adenoviral vector. Alternatively, the inventive adenovirus or adenoviral vector can comprise a non-native replication-essential gene that complements for the one or more replication-essential gene functions lacking in the inventive replication-deficient adenovirus or adenoviral vector. For example, an E1/E4-deficient adenoviral vector can be engineered to contain a nucleic acid sequence encoding E4 ORF 6 that is obtained or derived from a different adenovirus (e.g., an adenovirus of a different serotype than the inventive adenovirus or adenoviral vector, or an adenovirus of a different species than the inventive adenovirus or adenoviral vector).

The adenovirus or adenoviral vector can further comprise a transgene. The term "transgene" is defined herein as a non-native nucleic acid sequence that is operably linked to appropriate regulatory elements (e.g., a promoter), such that the non-native nucleic acid sequence can be expressed to produce a protein (e.g., peptide or polypeptide). The regulatory elements (e.g., promoter) can be native or non-native to the adenovirus or adenoviral vector.

A "non-native" nucleic acid sequence is any nucleic acid sequence (e.g., DNA, RNA, or cDNA sequence) that is not a naturally occurring nucleic acid sequence of an adenovirus in a naturally occurring position. Thus, the non-native nucleic acid sequence can be naturally found in an adenovirus, but located at a non-native position within the adenoviral genome and/or operably linked to a non-native promoter. The terms "non-native nucleic acid sequence," "heterologous nucleic acid sequence," and "exogenous nucleic acid sequence" are synonymous and can be used interchangeably in the context of the present disclosure. The non-native nucleic acid sequence preferably is DNA and preferably encodes a protein (i.e., one or more nucleic acid sequences encoding one or more proteins).

The non-native nucleic acid sequence can encode a therapeutic protein that can be used to prophylactically or therapeutically treat a mammal for a disease. Examples of suitable therapeutic proteins include cytokines, toxins, tumor suppressor proteins, growth factors, hormones, receptors, mitogens, immunoglobulins, neuropeptides, neurotransmit-

ters, and enzymes. Alternatively, the non-native nucleic acid sequence can encode an antigen of a pathogen (e.g., a bacterium or a virus), and the adenovirus or adenoviral vector can be used as a vaccine.

#### 5 Viral Based Delivery System

The present disclosure also provides delivery systems, such as viral-based systems, in which a nucleic acid described herein is inserted. Representative viral expression vectors include, but are not limited to, adeno-associated viral vectors, adenovirus-based vectors, lentivirus-based vectors, retroviral vectors, and herpes virus-based vectors. In an embodiment, the viral vector is a lentivirus vector. Vectors derived from retroviruses such as the lentivirus are suitable tools to achieve long-term gene transfer since they allow long-term, stable integration of a transgene and its propagation in daughter cells. Lentiviral vectors have the added advantage over vectors derived from onco-retroviruses such as murine leukemia viruses in that they can transduce non-proliferating cells, such as hepatocytes. They also have the added advantage of low immunogenicity. In an additional embodiment, the viral vector is an adeno-associated viral vector. In a further embodiment, the viral vector is a retroviral vector. In general, and in embodiments, a suitable vector contains an origin of replication functional in at least one organism, a promoter sequence, convenient restriction endonuclease sites, and one or more selectable markers.

Additional suitable vectors include integrating expression vectors, which can randomly integrate into the host cell's DNA, or can include a recombination site to enable the specific recombination between the expression vector and the host cell's chromosome. Such integrating expression vectors can utilize the endogenous expression control sequences of the host cell's chromosomes to effect expression of the desired protein. Examples of vectors that integrate in a site specific manner include, for example, components of the flp-in system from Invitrogen (Carlsbad, Calif.) (e.g., pcDNATM5/FRT), or the cre-lox system, such as can be found in the pExchange-6 Core Vectors from Stratagene (La Jolla, Calif.). Examples of vectors that randomly integrate into host cell chromosomes include, for example, pcDNA3.1 (when introduced in the absence of T-antigen) from Invitrogen (Carlsbad, Calif.), and pCI or pFN10A (ACT) FLEXI™ from Promega (Madison, Wis.). Additional promoter elements, e.g., enhancers, regulate the frequency of transcriptional initiation. Typically, these are located in the region 30-110 bp upstream of the start site, although a number of promoters have recently been shown to contain functional elements downstream of the start site as well. The spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. In the thymidine kinase (tk) promoter, the spacing between promoter elements can be increased to 50 bp apart before activity begins to decline. Depending on the promoter, it appears that individual elements can function either cooperatively or independently to activate transcription.

One example of a suitable promoter is the immediate early cytomegalovirus (CMV) promoter sequence. This promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto.

However, other constitutive promoter sequences can also be used, including, but not limited to the simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate

early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter. Further, the present disclosure should not be limited to the use of constitutive promoters. Inducible promoters are also contemplated as part of the present disclosure. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. Examples of inducible promoters include, but are not limited to a metallothionein promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.

Reporter genes can be used for identifying potentially transfected cells and for evaluating the functionality of regulatory sequences. In general, a reporter gene is a gene that is not present in or expressed by the recipient organism or tissue and that encodes a polypeptide whose expression is manifested by some easily detectable property, e.g., enzymatic activity. Expression of the reporter gene is assayed at a suitable time after the DNA has been introduced into the recipient cells. Suitable reporter genes can include genes encoding luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase, or the green fluorescent protein gene (e.g., Ui-Tei et al., *FEBS Letters* 479: 79-82 (2000)). Suitable expression systems are well known and can be prepared using known techniques or obtained commercially. In general, the construct with the minimal 5' flanking region showing the highest level of expression of reporter gene is identified as the promoter. Such promoter regions can be linked to a reporter gene and used to evaluate agents for the ability to modulate promoter-driven transcription.

Methods of introducing and expressing genes into a cell are known in the art. In the context of an expression vector, the vector can be readily introduced into a host cell, e.g., mammalian, bacterial, yeast, or insect cell by any method in the art. For example, the expression vector can be transferred into a host cell by physical, chemical, or biological means.

Physical methods for introducing a polynucleotide into a host cell include calcium phosphate precipitation, lipofection, particle bombardment, microinjection, electroporation, and the like. Methods for producing cells comprising vectors and/or exogenous nucleic acids are well-known in the art. See, for example, Sambrook et al. (*Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Laboratory, New York (2001)). In embodiments, a method for the introduction of a polynucleotide into a host cell is calcium phosphate transfection or polyethylenimine (PEI) transfection.

Biological methods for introducing a polynucleotide of interest into a host cell include the use of DNA and RNA vectors. Viral vectors, and especially retroviral vectors, have become the most widely used method for inserting genes into mammalian, e.g., human cells. Other viral vectors can be derived from lentivirus, poxviruses, herpes simplex virus I, adenoviruses and adeno-associated viruses, and the like. See, for example, U.S. Pat. Nos. 5,350,674 and 5,585,362. Non-Viral Based Delivery System

Chemical means for introducing a polynucleotide into a host cell include colloidal dispersion systems, such as macromolecule complexes, nanocapsules, microspheres, beads, and lipid-based systems including oil-in-water emulsions, micelles, mixed micelles, and liposomes. An exemplary colloidal system for use as a delivery vehicle in vitro and in vivo is a liposome (e.g., an artificial membrane vesicle).

The use of lipid formulations is contemplated for the introduction of the nucleic acids into a host cell (in vitro, ex vivo or in vivo). In another aspect, the nucleic acid can be associated with a lipid. The nucleic acid associated with a lipid can be encapsulated in the aqueous interior of a liposome, interspersed within the lipid bilayer of a liposome, attached to a liposome via a linking molecule that is associated with both the liposome and the oligonucleotide, entrapped in a liposome, complexed with a liposome, dispersed in a solution containing a lipid, mixed with a lipid, combined with a lipid, contained as a suspension in a lipid, contained or complexed with a micelle, or otherwise associated with a lipid. Lipid, lipid/DNA or lipid/expression vector associated compositions are not limited to any particular structure in solution. For example, they can be present in a bilayer structure, as micelles, or with a "collapsed" structure. They can also simply be interspersed in a solution, possibly forming aggregates that are not uniform in size or shape. Lipids are fatty substances which can be naturally occurring or synthetic lipids. For example, lipids include the fatty droplets that naturally occur in the cytoplasm as well as the class of compounds which contain long-chain aliphatic hydrocarbons and their derivatives, such as fatty acids, alcohols, amines, amino alcohols, and aldehydes.

Lipids suitable for use can be obtained from commercial sources. For example, dimyristyl phosphatidylcholine ("DMPC") can be obtained from Sigma, St. Louis, Mo.; dicetyl phosphate ("DCP") can be obtained from K & K Laboratories (Plainview, N.Y.); cholesterol ("Choi") can be obtained from Calbiochem-Behring; dimyristyl phosphatidylglycerol ("DMPG") and other lipids can be obtained from Avanti Polar Lipids, Inc. (Birmingham, Ala.). Stock solutions of lipids in chloroform or chloroform/methanol can be stored at about -20°C. Chloroform is used as the only solvent since it is more readily evaporated than methanol. "Liposome" is a generic term encompassing a variety of single and multilamellar lipid vehicles formed by the generation of enclosed lipid bilayers or aggregates. Liposomes can be characterized as having vesicular structures with a phospholipid bilayer membrane and an inner aqueous medium. Multilamellar liposomes have multiple lipid layers separated by aqueous medium. They form spontaneously when phospholipids are suspended in an excess of aqueous solution. The lipid components undergo self-rearrangement before the formation of closed structures and entrap water and dissolved solutes between the lipid bilayers (Ghosh et al., *Glycobiology* 5: 505-10 (1991)). However, compositions that have different structures in solution than the normal vesicular structure are also encompassed. For example, the lipids can assume a micellar structure or merely exist as nonuniform aggregates of lipid molecules. Also contemplated are lipofectamine-nucleic acid complexes.

In some instances, polynucleotides encoding polypeptides can also be introduced into cells using non-viral based delivery systems, such as the "Sleeping Beauty (SB) Transposon System," which refers to a synthetic DNA transposon system for introducing DNA sequences into the chromosomes of vertebrates. Some exemplary embodiments of the system are described, for example, in U.S. Pat. Nos. 6,489,458 and 8,227,432. The Sleeping Beauty transposon system is composed of a Sleeping Beauty (SB) transposase and a SB transposon. In embodiments, the Sleeping Beauty transposon system can include the SB11 transposon system, the SB100X transposon system, or the SB110 transposon system.

DNA transposons translocate from one DNA site to another in a simple, cut-and-paste manner. Transposition is a precise process in which a defined DNA segment is excised from one DNA molecule and moved to another site in the same or different DNA molecule or genome. As do other Tc1/mariner-type transposases, SB transposase inserts a transposon into a TA dinucleotide base pair in a recipient DNA sequence. The insertion site can be elsewhere in the same DNA molecule, or in another DNA molecule (or chromosome). In mammalian genomes, including humans, there are approximately 200 million TA sites. The TA insertion site is duplicated in the process of transposon integration. This duplication of the TA sequence is a hallmark of transposition and used to ascertain the mechanism in some experiments. The transposase can be encoded either within the transposon or the transposase can be supplied by another source, for instance a DNA or mRNA source, in which case the transposon becomes a non-autonomous element. Non-autonomous transposons are most useful as genetic tools because after insertion they cannot independently continue to excise and re-insert. SB transposons envisaged to be used as non-viral vectors for introduction of genes into genomes of vertebrate animals and for gene therapy.

Regardless of the method used to introduce exogenous nucleic acids into a host cell or otherwise expose a cell to the inhibitor of the present disclosure, in order to confirm the presence of the recombinant DNA sequence in the host cell, a variety of assays can be performed. Such assays include, for example, molecular assays well known to those of skill in the art, such as Southern and Northern blotting, RT-PCR and PCR; "biochemical" assays, such as detecting the presence or absence of a particular peptide, e.g., by immunological means (ELISAs and Western blots) or by assays described herein to identify agents falling within the scope of the present disclosure.

In embodiments, a modified effector cell described herein and other genetic elements are delivered to a cell using the SB11 transposon system, the SB100X transposon system, the SB110 transposon system, the piggyBac transposon system (see, e.g., Wilson et al., "PiggyBac Transposon-mediated Gene Transfer in Human Cells," *Molecular Therapy* 15:139-145 (2007), incorporated herein by reference in its entirety) and/or the piggyBac transposon system (see, e.g., Mitra et al., "Functional characterization of piggyBac from the bat *Myotis lucifugus* unveils an active mammalian DNA transposon," *Proc. Natl. Acad. Sci USA* 110:234-239 (2013). Additional transposases and transposon systems are provided in U.S. Pat. Nos. 6,489,458; 6,613,752; 7,148,203; 7,985,739; 8,227,432; 9,228,180; U.S. Patent Pub. No. 2011/0117072; Mates et al., *Nat Genet*, 41(6): 753-61 (2009). doi: 10.1038/ng.343. Epub 2009 May 3, *Gene Ther.*, 18(9):849-56 (2011). doi: 10.1038/gt.2011.40. Epub 2011 Mar. 31 and in Ivics et al., *Cell*, 91(4):501-10, (1997), each of which is incorporated herein by reference in its entirety.

Additional suitable non-viral systems can include integrating expression vectors, which can randomly integrate into the host cell's DNA, or can include a recombination site to enable the specific recombination between the expression vector and the host cell's chromosome. Targeted integration of transgenes into predefined genetic loci is a desirable goal for many applications. First, a first recombination site for a site-specific recombinase is inserted at a genomic site, either at a random or at a predetermined location. Subsequently, the cells are transfected with a plasmid carrying the gene or DNA of interest and the second recombination site and a

source for recombinase (expression plasmid, RNA, protein, or virus-expressing recombinase). Recombination between the first and second recombination sites leads to integration of plasmid DNA.

Such integrating expression vectors can utilize the endogenous expression control sequences of the host cell's chromosomes to effect expression of the desired protein. In some embodiments, targeted integration is promoted by the presence of sequences on the donor polynucleotide that are homologous to sequences flanking the integration site. For example, targeted integration using the donor polynucleotides described herein can be achieved following conventional transfection techniques, e.g. techniques used to create gene knockouts or knockins by homologous recombination. In other embodiments, targeted integration is promoted both by the presence of sequences on the donor polynucleotide that are homologous to sequences flanking the integration site, and by contacting the cells with donor polynucleotide in the presence of a site-specific recombinase. By a site-specific recombinase, or simply a recombinase, it is meant is a polypeptide that catalyzes conservative site-specific recombination between its compatible recombination sites. As used herein, a site-specific recombinase includes native polypeptides as well as derivatives, variants and/or fragments that retain activity, and native polynucleotides, derivatives, variants, and/or fragments that encode a recombinase that retains activity.

Also provided herein is a system for integrating heterologous genes in a host cell, said system comprising one or more gene expression cassettes. In some instances, the system includes a first gene expression cassette comprising a first polynucleotide encoding a first polypeptide construct. In other instances, the system can include a second gene expression cassette comprising a second polynucleotide encoding a second polypeptide construct. In yet other instances, the system can include a third gene expression cassette. In one embodiment, one of the gene expression cassettes can comprise a gene switch polynucleotide encoding one or more of: (i) a transactivation domain; (ii) nuclear receptor ligand binding domain; (iii) a DNA-binding domain; and (iv) ecdysone receptor binding domain. In another embodiment, the system further includes recombinant attachment sites; and a serine recombinase; such that upon contacting said host cell with at least said first gene expression cassette, in the presence of said serine recombinase, said heterologous genes are integrated in said host cell.

In some instances, the system further comprises a ligand; such that upon contacting said host cell, in the presence of said ligand, said heterologous gene are expressed in said host cell. In one instance, the system also includes recombinant attachment sites. In some instances, one recombination attachment site is a phage genomic recombination attachment site (attP) or a bacterial genomic recombination attachment site (attB). In one instance, the host cell is an eukaryotic cell. In another instance, the host cell is a human cell. In further instances, the host cell is a T cell or NK cell. Promoters

"Promoter" refers to a region of a polynucleotide that initiates transcription of a coding sequence. Promoters are located near the transcription start sites of genes, on the same strand and upstream on the DNA (towards the 5' region of the sense strand). Some promoters are constitutive as they are active in all circumstances in the cell, while others are regulated becoming active in response to specific stimuli, e.g., an inducible promoter. Yet other promoters are tissue specific or activated promoters, including but not limited to T-cell specific promoters.

The term “promoter activity” and its grammatical equivalents as used herein refer to the extent of expression of nucleotide sequence that is operably linked to the promoter whose activity is being measured. Promoter activity can be measured directly by determining the amount of RNA transcript produced, for example by Northern blot analysis or indirectly by determining the amount of product coded for by the linked nucleic acid sequence, such as a reporter nucleic acid sequence linked to the promoter.

“Inducible promoter” as used herein refers to a promoter which is induced into activity by the presence or absence of transcriptional regulators, e.g., biotic or abiotic factors. Inducible promoters are useful because the expression of genes operably linked to them can be turned on or off at certain stages of development of an organism or in a particular tissue. Examples of inducible promoters are alcohol-regulated promoters, tetracycline-regulated promoters, steroid-regulated promoters, metal-regulated promoters, pathogenesis-regulated promoters, temperature-regulated promoters and light-regulated promoters. In one embodiment, the inducible promoter is part of a genetic switch. The inducible promoter can be a gene switch ligand inducible promoter. In some cases, an inducible promoter can be a small molecule ligand-inducible two polypeptide ecdysone receptor-based gene switch, such as RHEOSWITCH® gene switch. In some cases, a gene switch can be selected from ecdysone-based receptor components as described in, but without limitation to, any of the systems described in: PCT/US2001/009050 (WO 2001/070816); U.S. Pat. Nos. 7,091,038; 7,776,587; 7,807,417; 8,202,718; PCT/US2001/030608 (WO 2002/029075); U.S. Pat. Nos. 8,105,825; 8,168,426; PCT/US2002/005235 (WO 2002/066613); U.S. application Ser. No. 10/468,200 (U.S. Pub. No. 20120167239); PCT/US2002/005706 (WO 2002/066614); U.S. Pat. Nos. 7,531,326; 8,236,556; 8,598,409; PCT/US2002/005090 (WO 2002/066612); U.S. Pat. No. 8,715,959 (U.S. Pub. No. 20060100416); PCT/US2002/005234 (WO 2003/027266); U.S. Pat. Nos. 7,601,508; 7,829,676; 7,919,269; 8,030,067; PCT/US2002/005708 (WO 2002/066615); U.S. application Ser. No. 10/468,192 (U.S. Pub. No. 20110212528); PCT/US2002/005026 (WO 2003/027289); U.S. Pat. Nos. 7,563,879; 8,021,878; 8,497,093; PCT/US2005/015089 (WO 2005/108617); U.S. Pat. Nos. 7,935,510; 8,076,454; PCT/US2008/011270 (WO 2009/045370); U.S. application Ser. No. 12/241,018 (U.S. Pub. No. 20090136465); PCT/US2008/011563 (WO 2009/048560); U.S. application Ser. No. 12/247,738 (U.S. Pub. No. 20090123441); PCT/US2009/005510 (WO 2010/042189); U.S. application Ser. No. 13/123,129 (U.S. Pub. No. 20110268766); PCT/US2011/029682 (WO 2011/119773); U.S. application Ser. No. 13/636,473 (U.S. Pub. No. 20130195800); PCT/US2012/027515 (WO 2012/122025); and, U.S. Pat. No. 9,402,919 each of which is incorporated by reference in its entirety).

Provided herein are methods comprising administering to a subject at least one non-viral vector comprising a polynucleotide encoding a polypeptide sequence described herein comprising at least two functional proteins or portions thereof; at least one promoter; and at least one engineered recombination site; wherein said at least one promoter drives expression of said at least two functional proteins. In some cases, at least one promoter can be constitutive. In some cases, at least one promoter can be tissue-specific. In some cases, at least one promoter can be inducible. In some cases, an inducible promoter is a small molecule ligand-inducible two polypeptide ecdysone receptor-based gene switch. In other cases, a combination of

promoters wherein at least one promoter can be inducible and at least one promoter can be activation specific can be utilized.

An inducible promoter utilizes a ligand for dose-regulated control of expression of said at least two genes. In some cases, a ligand can be selected from a group consisting of ecdysteroid, 9-cis-retinoic acid, synthetic analogs of retinoic acid, N,N'-diacylhydrazines, oxadiazolines, dibenzoylalkyl cyanohydrazines, N-alkyl-N,N'-diaroylhydrazines, N-acyl-N-alkylcarbonylhydrazines, N-aroyl-N-alkyl-N'-aroylhydrazines, amidoketones, 3,5-di-tert-butyl-4-hydroxy-N-isobutyl-benzamide, 8-O-acetylharpagide, oxysterols, 22(R) hydroxycholesterol, 24(S) hydroxycholesterol, 25-epoxycholesterol, T0901317, 5-alpha-6-alpha-epoxycholesterol-3-sulfate (ECHS), 7-ketocholesterol-3-sulfate, framesol, bile acids, 1,1-biphosphonate esters, juvenile hormone III, RG-115819 (3,5-Dimethyl-benzoic acid N-(1-ethyl-2,2-dimethyl-propyl)-N'-(2-methyl-3-methoxy-benzoyl)-hydrazide-), RG-115932 ((R)-3,5-Dimethyl-benzoic acid N-(1-tert-butyl-butyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide), and RG-115830 (3,5-Dimethyl-benzoic acid N-(1-tert-butyl-butyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide), and any combination thereof.

In some embodiments, a promoter is an inducible promoter. In some embodiments, a promoter is a non-inducible promoter. In some cases, a promoter can be a tissue-specific promoter. Herein “tissue-specific” refers to regulated expression of a gene in a subset of tissues or cell types. In some cases, a tissue-specific promoter can be regulated spatially such that the promoter drives expression only in certain tissues or cell types of an organism. In other cases, a tissue-specific promoter can be regulated temporally such that the promoter drives expression in a cell type or tissue differently across time, including during development of an organism. In some cases, a tissue-specific promoter is regulated both spatially and temporally. In certain embodiments, a tissue-specific promoter is activated in certain cell types either constitutively or intermittently at particular times or stages of the cell type. For example, a tissue-specific promoter can be a promoter that is activated when a specific cell such as a T cell or a NK cell is activated. T cells can be activated in a variety of ways, for example, when presented with peptide antigens by MHC class II molecules.

In one case, at least one promoter is an engineered promoter or variants thereof. As described herein, the promoter can incorporate minimal promoter sequences from IL-2 and one or more of the following: nuclear factor of activated T-cells (NFAT) response element(s); NFIL2D response element, NFkB/TCF response element, NFAT/NFIL2B response element or NFIL2A/OCT response element. Examples of response elements are described in Mattila et al., i EMBO J. 9(13):4425-33 (1990); incorporated herein in its entirety.

In some embodiments, at least one promoter comprises IL-2 core promoter (SEQ ID NO: 26). In one embodiment, at least one promoter comprises IL-2 minimal promoter (SEQ ID NO: 27). In another embodiment, at least one promoter comprises IL-2 enhancer and promoter variant (SEQ ID NOS: 26-28). In yet another embodiment, at least one promoter comprises NF-kB binding site (SEQ ID NOS: 30-32). In some embodiments, at least one promoter comprises (NF-kB)<sub>1</sub>-IL2 promoter variant (SEQ ID NO: 30). In some embodiments, at least one promoter comprises (NF-kB)<sub>3</sub>-IL2 promoter variant (SEQ ID NO: 31). In some embodiments, at least one promoter comprises (NF-kB)<sub>6</sub>-IL2 promoter variant (SEQ ID NO: 32). In one embodiment, at least one promoter comprises 1x nuclear factor of acti-

vated T-cells (NFAT) response elements-IL2 promoter variant (SEQ ID NO: 33). In another embodiment, at least one promoter comprises 3× NFAT response element (SEQ ID NOS: 34-35). In yet another embodiment, at least one promoter comprises 6×NFAT response elements-IL2 promoter variant (SEQ ID NOS: 36-39). In some embodiments, at least one promoter comprises human EF1A1 promoter variant (SEQ ID NOS: 40-41). In some embodiment, at least one promoter comprises human EF1A1 promoter and enhancer (SEQ ID NO: 42). In some embodiments, at least one promoter comprises human UBC promoter (SEQ ID NO: 43). In some embodiments, at least one promoter comprises 6 site GAL4-inducible proximal factor binding element (PFB). In some embodiment, at least one promoter comprises synthetic minimal promoter 1 (inducible promoter) (SEQ ID NO: 44).

Use of gene switch for ligand inducible control of IL-12 expression described herein can improve the safety profile of IL-12 by for example allowing for regulated expression and improving therapeutic index. However, a condition for ligand dose dependent expression of IL-12 using gene switch(es) is the presence or absence of activator ligand (e.g. veledimex). In certain embodiments, an additional conditional control for induction of IL-12 expression is contemplated. Gene switch components under the control of T cell activated specific promoters are provided. This results in conditional expression (e.g., T cell activation) of gene switch components necessary for veledimex controlled expression of transgene(s) under control of a gene switch. In some embodiments, this results in preferential expression of cytokines such as IL-12 or IL-15 by tumor specific T cells when veledimex is present and T cells are activated. This can lead to increased localized levels of gene switch controlled transgene expression.

For example, T cell activation specific expression of gene switch components can be controlled by promoter comprising Nuclear Factor of Activated T-cells (NFAT) response element(s). NFAT transcription factors are key modulators of effector T-cell states. NFATs are early transcriptional checkpoint progressively driving exhaustion. NFATs are quickly activated in T cells following TCR stimulation and form a protein complex with AP-1 induced by appropriate co-stimulation signaling and regulate effector genes and T-cell functions. NFAT response element(s) can be fused with other minimal promoter sequences (e.g. IL2 minimal promoter) to drive expression of transgenes in response to T cell activation.

Other examples of activation specific promoters include but are not limited to interleukin-2 (IL2) promoter and Programmed Death (PD)-1 (CD279) promoter. Gene switch components can also be conditionally expressed upon immune cell activation by fusing binding sites for other nuclear factors like NF-κB of proinflammatory signaling pathway to minimal promoter sequence (e.g. IL2).

In certain embodiments, the promoter can be any one or more of: IL-2 core promoter, IL-2 minimal promoter, IL-2 enhancer and promoter variant, (NF-κB)<sub>1</sub>-IL2 promoter variant, (NF-κB)<sub>3</sub>-IL2 promoter variant, (NF-κB)<sub>6</sub>-IL2 promoter variant, 1× NFAT response elements-IL2 promoter variant, 3×NFAT response elements-IL2 promoter variant, 6×NFAT response elements-IL2 promoter variant, human EE1A1 promoter variant, human EE1A1 promoter and enhancer, human UBC promoter and synthetic minimal promoter 1. In certain embodiments, the promoter nucleotides can comprise SEQ ID NOS: 26-44.

#### Gene Switch

Provided herein are gene switch polypeptides, polynucleotides encoding ligand-inducible gene switch polypeptides, and methods and systems incorporating these polypeptides and/or polynucleotides. In certain aspects, the present disclosure is directed to a polynucleotide comprising one or more polynucleotides encoding a gene switch system for inducible control of heterologous gene expression, wherein the heterologous gene expression is regulated by said gene switch system; and, wherein said heterologous gene comprises a polynucleotide encoding a polypeptide comprising one or more immune response-inducing human papilloma virus (HPV) polypeptides, disclosed herein.

The term "gene switch" refers to the combination of a response element associated with a promoter, and for instance, an EcR based system which, in the presence of one or more ligands, modulates the expression of a gene into which the response element and promoter are incorporated. Tightly regulated inducible gene expression systems or gene switches are useful for various applications such as gene therapy, large scale production of proteins in cells, cell based high throughput screening assays, functional genomics and regulation of traits in transgenic plants and animals. Such inducible gene expression systems can include ligand inducible heterologous gene expression systems.

An early version of EcR-based gene switch used *Drosophila melanogaster* EcR (DmEcR) and *Mus musculus* RXR (MmRXR) polypeptides and showed that these receptors in the presence of steroid, ponasteroneA, transactivate reporter genes in mammalian cell lines and transgenic mice (Christopherson et al., *Proc. Natl. Acad. Sci. USA* 89(14):6314-18 (1992); No et al., *Proc. Natl. Acad. Sci. USA* 93(8):3346-51 (1996)). Later, Suhr et al. (*Proc. Natl. Acad. Sci. USA* 95(14):7999-8004 (1998)) showed that non-steroidal ecdysone agonist, tebufenozide, induced high level of transactivation of reporter genes in mammalian cells through *Bombyx mori* EcR (BmEcR) in the absence of exogenous heterodimer partner.

International Patent Applications No. PCT/US97/05330 (WO 97/38117) and PCT/US99/08381 (WO99/58155) disclose methods for modulating the expression of an exogenous gene in which a DNA construct comprising the exogenous gene and an ecdysone response element is activated by a second DNA construct comprising an ecdysone receptor that, in the presence of a ligand therefor, and optionally in the presence of a receptor capable of acting as a silent partner, binds to the ecdysone response element to induce gene expression. In this example, the ecdysone receptor was isolated from *Drosophila melanogaster*. Typically, such systems require the presence of the silent partner, preferably retinoid X receptor (RXR), in order to provide optimum activation. In mammalian cells, insect ecdysone receptor (EcR) is capable of heterodimerizing with mammalian retinoid X receptor (RXR) and, thereby, be used to regulate expression of target genes or heterologous genes in a ligand dependent manner. International Patent Application No. PCT/US98/14215 (WO 99/02683) discloses that the ecdysone receptor isolated from the silk moth *Bombyx mori* is functional in mammalian systems without the need for an exogenous dimer partner.

U.S. Pat. No. 6,265,173 discloses that various members of the steroid/thyroid superfamily of receptors can combine with *Drosophila melanogaster* ultraspiracle receptor (USP) or fragments thereof comprising at least the dimerization domain of USP for use in a gene expression system. U.S. Pat. No. 5,880,333 discloses a *Drosophila melanogaster* EcR and ultraspiracle (USP) heterodimer system used in plants in which the transactivation domain and the DNA



binding domain are positioned on two different hybrid proteins. In each of these cases, the transactivation domain and the DNA binding domain (either as native EcR as in International Patent Application No. PCT/US98/14215 or as modified EcR as in International Patent Application No. PCT/US97/05330) were incorporated into a single molecule and the other heterodimeric partners, either USP or RXR, were used in their native state.

International Patent Application No. PCT/US01/0905 discloses an ecdysone receptor-based inducible gene expression system in which the transactivation and DNA binding domains are separated from each other by placing them on two different proteins results in greatly reduced background activity in the absence of a ligand and significantly increased activity over background in the presence of a ligand. This two-hybrid system is a significantly improved inducible gene expression modulation system compared to the two systems disclosed in applications PCT/US97/05330 and PCT/US98/14215. The two-hybrid system is believed to exploit the ability of a pair of interacting proteins to bring the transcription activation domain into a more favorable position relative to the DNA binding domain such that when the DNA binding domain binds to the DNA binding site on the gene, the transactivation domain more effectively activates the promoter (see, for example, U.S. Pat. No. 5,283,173). The two-hybrid gene expression system comprises two gene expression cassettes; the first encoding a DNA binding domain fused to a nuclear receptor polypeptide, and the second encoding a transactivation domain fused to a different nuclear receptor polypeptide. In the presence of ligand, it is believed that a conformational change is induced which promotes interaction of the first polypeptide with the second polypeptide thereby resulting in dimerization of the DNA binding domain and the transactivation domain. Since the DNA binding and transactivation domains reside on two different molecules, the background activity in the absence of ligand is greatly reduced.

Another surprising discovery was that certain modifications of the two-hybrid system could also provide improved sensitivity to non-steroidal ligands for example, diacylhydrazines, when compared to steroidal ligands for example, ponasterone A ("PonA") or muristerone A ("MurA"). That is, when compared to steroids, the non-steroidal ligands provided higher gene transcription activity at a lower ligand concentration. Furthermore, the two-hybrid system avoids some side effects due to overexpression of RXR that can occur when unmodified RXR is used as a switching partner. In a preferred two-hybrid system, native DNA binding and transactivation domains of EcR or RXR are eliminated and as a result, these hybrid molecules have less chance of interacting with other steroid hormone receptors present in the cell, thereby resulting in reduced side effects.

The ecdysone receptor (EcR) is a member of the nuclear receptor superfamily and is classified into subfamily 1, group H (referred to herein as "Group H nuclear receptors"). The members of each group share 40-60% amino acid identity in the E (ligand binding) domain (Laudet et al., A Unified Nomenclature System for the Nuclear Receptor Subfamily, 1999; *Cell* 97: 161-163). In addition to the ecdysone receptor, other members of this nuclear receptor subfamily 1, group H include: ubiquitous receptor (UR), Orphan receptor 1 (OR-1), steroid hormone nuclear receptor 1 (NER-1), RXR interacting protein-15 (RIP-15), liver x receptor  $\beta$  (LXR $\beta$ ), steroid hormone receptor like protein (RLD-1), liver x receptor (LXR), liver x receptor  $\alpha$  (LXR $\alpha$ ), farnesoid x receptor (FXR), receptor interacting protein 14 (RIP-14), and farnesol receptor (HRR-1).

In some cases, an inducible promoter can be a small molecule ligand-inducible two polypeptide ecdysone receptor-based gene switch, such as Intrexon Corporation's RHEOSWITCH® gene switch. In some cases, a gene switch can be selected from ecdysone-based receptor components as described in, but without limitation to, any of the systems described in: PCT/US2001/009050 (WO 2001/070816); U.S. Pat. Nos. 7,091,038; 7,776,587; 7,807,417; 8,202,718; PCT/US2001/030608 (WO 2002/029075); U.S. Pat. Nos. 8,105,825; 8,168,426; PCT/1J52002/005235 (WO 2002/066613); U.S. application Ser. No. 10/468,200 (U.S. Pub. No. 20120167239); PCT/US2002/005706 (WO 2002/066614); U.S. Pat. Nos. 7,531,326; 8,236,556; 8,598,409; PCT/US2002/005090 (WO 2002/066612); U.S. Pat. No. 8,715,959 (U.S. Pub. No. 20060100416); PCT/US2002/005234 (WO 2003/027266); U.S. Pat. Nos. 7,601,508; 7,829,676; 7,919,269; 8,030,067; PCT/US2002/005708 (WO 2002/066615); U.S. application Ser. No. 10/468,192 (U.S. Pub. No. 20110212528); PCT/US2002/005026 (WO 2003/027289); U.S. Pat. Nos. 7,563,879; 8,021,878; 8,497,093; PCT/US2005/015089 (WO 2005/108617); U.S. Pat. Nos. 7,935,510; 8,076,454; PCT/US2008/011270 (WO 2009/045370); U.S. application Ser. No. 12/241,018 (U.S. Pub. No. 20090136465); PCT/US2008/011563 (WO 2009/048560); U.S. application Ser. No. 12/247,738 (U.S. Pub. No. 20090123441); PCT/US2009/005510 (WO 2010/042189); U.S. application Ser. No. 13/123,129 (U.S. Pub. No. 20110268766); PCT/US2011/029682 (WO 2011/119773); U.S. application Ser. No. 13/636,473 (U.S. Pub. No. 20130195800); PCT/US2012/027515 (WO 2012/122025); and, U.S. Pat. No. 9,402,919 each of which is incorporated by reference in its entirety.

Provided are systems for modulating the expression of a heterologous gene and an interleukin in a host cell, comprising polynucleotides expressing gene-switch polypeptides disclosed herein.

In some embodiments are systems for modulating the expression of a heterologous gene and a cytokine in a host cell, comprising a first gene expression cassette comprising a first polynucleotide encoding a first polypeptide; a second gene expression cassette comprising a second polynucleotide encoding a second polypeptide; and a ligand; wherein said first and second polypeptides comprise one or more of: (i) a transactivation domain; (ii) a DNA-binding domain; and (iii) a ligand binding domain; (iv) said heterologous gene; and (vi) said cytokine such that upon contacting said host cell with said first gene expression cassette and said second gene expression cassette in the presence of said ligand, said heterologous gene and said cytokine are expressed in said host cell. In some cases, the heterologous gene comprises an antigen binding polypeptide described herein. In some cases, the cytokine comprises at least one chemokine, interferon, interleukin, lymphokine, tumor necrosis factor, or variant or combination thereof. In some cases, the cytokine is an interleukin. In some cases the interleukin is at least one of IL12, IL2, IL15, IL21, and functional variants and fragments thereof. In some embodiments, the cytokines can be membrane bound or secreted. In other embodiments, the cytokines can be intracellular. The interleukin can comprise membrane bound IL-15 (mbIL-15) or a fusion of IL-15 and IL-15R $\alpha$ . In some embodiments, a mbIL-15 is a membrane-bound chimeric IL-15 which can be co-expressed with a modified effector cell described herein. In some embodiments, the mbIL-15 comprises a full-length IL-15 (e.g., a native IL-15 polypeptide) or fragment or variant thereof, fused in frame with a full length IL-15R $\alpha$ , functional fragment or variant thereof. In some cases, the



IL-15 is indirectly linked to the IL-15R $\alpha$  through a linker. In some instances, the mbIL-15 is as described in Hurton et al., "Tethered IL-15 augments antitumor activity and promotes a stem-cell memory subset in tumor-specific T cells." *Proc. Natl. Acad. Sci. USA* 113(48):E7788-E7797 (2016). In another aspect, the interleukin can comprise IL-12. In some embodiments, the IL-12 is a single chain IL-12 (scIL-12), protease sensitive IL-12, destabilized IL-12, membrane bound IL-12, intercalated IL-12. In some instances, the IL-12 variants are as described in WO2015/095249, WO2016/048903, WO2017/062953, all of which is incorporated by reference in their entireties.

Provided herein are polynucleotides encoding gene switch polypeptides, wherein said gene switch polypeptides comprise: a) a first gene switch polypeptide comprising a DNA-binding domain fused to a nuclear receptor ligand binding domain, and b) a second gene switch polypeptide comprising a transactivation domain fused to a nuclear receptor ligand binding domain, wherein the first gene switch polypeptide and the second gene switch polypeptide are connected by a linker. In some cases, the linker can be a linker described herein, for instance GSG linker, furinlink, a 2A linker such as F/T2A, T2A, p2A, GSG-p2A, variants and derivatives thereof. In other instances, the linker can be an IRES.

In some cases, the DNA binding domain (DBD) comprises a DBD described herein, for instance at least one of GAL4 (GAL4 DBD), a LexA DBD, a transcription factor DBD, a steroid/thyroid hormone nuclear receptor superfamily member DBD, a bacterial LacZ DBD, and a yeast DBD. The transactivation domain can comprise a transactivation domain described herein, for instance one of a VP16 transactivation domain, a p53 transactivation domain and a B42 acidic activator transactivation domain. The Nuclear receptor ligand binding domain can comprise at least one of an ecdysone receptor (EcR), a ubiquitous receptor, an orphan receptor 1, a NER-1, a steroid hormone nuclear receptor 1, a retinoid X receptor interacting protein-15, a liver X receptor  $\beta$ , a steroid hormone receptor like protein, a liver X receptor, a liver X receptor  $\alpha$ , a farnesoid X receptor, a receptor interacting protein 14, and a farnesol receptor.

In some cases, the gene switch polypeptides connected by a polypeptide linker or ribosome-skipping sequence exhibit improved dose-dependent ligand-inducible control of gene expression compared to a ligand-inducible gene switch wherein the gene switch polypeptides are connected by non-coding sequences, such as an IRES. In some cases, the gene switch polypeptides connected by a 2A linker can exhibit improved dose-dependent ligand-inducible control of heterologous gene expression compared to a gene switch wherein said gene switch polypeptides are separated by an IRES.

In some embodiments, the gene switch comprises a VP16 transactivation domain. In one embodiment, the gene switch comprises at least one of an ecdysone receptor (EcR), a ubiquitous receptor, an orphan receptor 1, a NER-1, a steroid hormone nuclear receptor 1, a retinoid X receptor interacting protein-15, a liver X receptor  $\beta$ , a steroid hormone receptor like protein, a liver X receptor, a liver X receptor  $\alpha$ , a farnesoid X receptor, a receptor interacting protein 14, and a farnesol receptor. In another embodiment, a DNA-binding domain (DBD) of the gene switch comprises at least one of GAL4 (GAL4 DBD), a LexA DBD, a transcription factor DBD, a steroid/thyroid hormone nuclear receptor superfamily member DBD, a bacterial LacZ DBD, and a yeast DBD. In yet another case, the gene switch further comprises at least one of ultraspiracle protein (USP), retinoid receptor X

(RXR), functional fragments and variants thereof wherein said functional fragments and variants are capable of binding to an EcR.

The polypeptides and polynucleotides as described herein can be expressed in an engineered cell. Herein an engineered cell is a cell which has been modified from its natural or endogenous state. An example of an engineered cell is a cell described herein which has been modified (e.g., by transfection of a polynucleotide into the cell) to encode for example, gene switch polypeptides, gene of interest (GOI), cell tags, heterologous genes and any other polypeptides and polynucleotides described herein.

#### Ligands

In some embodiments, a ligand used for inducible gene switch regulation can be selected from any of, but without limitation to, following: N-[(1R)-1-(1,1-dimethylethyl)butyl]-N-(2-ethyl-3-methoxybenzoyl)-3,5-dimethylbenzohydrazide (also referred to as veledimex), (2S,3R,5R,9R,10R,13R,14S,17R)-17-[(2S,3R)-3,6-dihydroxy-6-methylheptan-2-yl]-2,3,14-trihydroxy-10,13-dimethyl-2,3,4,5,9,11,12,15,16,17-decahydro-1H-cyclopenta[a]phenanthren-6-one; N'-(3,5-Dimethylbenzoyl)-N'-[(3R)-2,2-dimethyl-3-hexanyl]-2-ethyl-3-methoxybenzohydrazide; 5-Methyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(3,5-dimethyl-benzoyl)-N'-(1-ethyl-2,2-dimethyl-propyl)-hydrazide; 5-Methyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(3,5-dimethoxy-4-methyl-benzoyl)-N'-(1-ethyl-2,2-dimethyl-propyl)-hydrazide; 5-Methyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(1-tert-butyl-butyl)-N'-(3,5-dimethyl-benzoyl)-hydrazide; 5-Methyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(1-tert-butyl-butyl)-N'-(3,5-dimethoxy-4-methyl-benzoyl)-hydrazide; 5-Ethyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(3,5-dimethyl-benzoyl)-N'-(1-ethyl-2,2-dimethyl-propyl)-hydrazide; 5-Ethyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(3,5-dimethoxy-4-methyl-benzoyl)-N'-(1-ethyl-2,2-dimethyl-propyl)-hydrazide; 5-Ethyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(1-tert-butyl-butyl)-N'-(3,5-dimethyl-benzoyl)-hydrazide; 5-Ethyl-2,3-dihydro-benzo[1,4]dioxine-6-carboxylic acid N'-(1-tert-butyl-butyl)-N'-(3,5-dimethoxy-4-methyl-benzoyl)-hydrazide; 3,5-Dimethylbenzoic acid N-(1-ethyl-2,2-dimethyl-propyl)-N'-(3-methoxy-2-methyl-benzoyl)-hydrazide; 3,5-Dimethoxy-4-methyl-benzoic acid N-(1-ethyl-2,2-dimethyl-propyl)-N'-(3-methoxy-2-methyl-benzoyl)-hydrazide; 3,5-Dimethoxy-4-methyl-benzoic acid N-(1-tert-butyl-butyl)-N'-(3-methoxy-2-methyl-benzoyl)-hydrazide; 3,5-Dimethoxy-4-methyl-benzoic acid N-(1-tert-butyl-butyl)-N'-(3-methoxy-2-methyl-benzoyl)-hydrazide; 3,5-Dimethylbenzoic acid N-(1-ethyl-2,2-dimethyl-propyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide; 3,5-Dimethoxy-4-methyl-benzoic acid N-(1-ethyl-2,2-dimethyl-propyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide; 3,5-Dimethylbenzoic acid N-(1-tert-butyl-butyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide; 3,5-Dimethoxy-4-methyl-benzoic acid N-(1-tert-butyl-butyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide; 2-Methoxy-nicotinic acid N-(1-tert-butyl-pentyl)-N'-(4-ethyl-benzoyl)-hydrazide; 3,5-Dimethylbenzoic acid N-(2,2-dimethyl-1-phenyl-propyl)-N'-(4-ethyl-benzoyl)-hydrazide; 3,5-Dimethylbenzoic acid N-(1-tert-butyl-pentyl)-N'-(3-methoxy-2-methyl-benzoyl)-hydrazide; and 3,5-Dimethoxy-4-methyl-benzoic acid N-(1-tert-butyl-pentyl)-N'-(3-methoxy-2-methyl-benzoyl)-hydrazide.

In some cases, a ligand used for dose-regulated control of ecdysone receptor-based inducible gene switch can be selected from any of, but without limitation to, an ecdysteroid, such as ecdysone, 20-hydroxyecdysone, ponasterone

A, muristerone A, and the like, 9-cis-retinoic acid, synthetic analogs of retinoic acid, N,N'-diacylhydrazines such as those disclosed in U.S. Pat. Nos. 6,013,836; 5,117,057; 5,530,028; and 5,378,726 and U.S. Published Application Nos. 2005/0209283 and 2006/0020146; oxadiazolines as described in U.S. Published Application No. 2004/0171651; dibenzoylalkyl cyanohydrazines such as those disclosed in European Application No. 461,809; N-alkyl-N,N'-diaroylhydrazines such as those disclosed in U.S. Pat. No. 5,225,443; N-acyl-N-alkylcarbonylhydrazines such as those disclosed in European Application No. 234,994; N-aryl-N-alkyl-N'-aroylhydrazines such as those described in U.S. Pat. No. 4,985,461; amidoketones such as those described in U.S. Published Application No. 2004/0049037; each of which is incorporated herein by reference and other similar materials including 3,5-di-tert-butyl-4-hydroxy-N-isobutylbenzamide, 8-O-acetylharpagide, oxysterols, 22(R) hydroxycholesterol, 24(S) hydroxycholesterol, 25-epoxycholesterol, T0901317, 5-alpha-6-alpha-epoxycholesterol-3-sulfate (ECHS), 7-ketocholesterol-3-sulfate, framesol, bile acids, 1,1-biphosphonate esters, juvenile hormone III, and the like. Examples of diacylhydrazine ligands useful in the present disclosure include RG-115819 (3,5-Dimethylbenzoic acid N-(1-ethyl-2,2-dimethyl-propyl)-N'-(2-methyl-3-methoxy-benzoyl)-hydrazide-), RG-115932 ((R)-3,5-Dimethylbenzoic acid N-(1-tert-butyl-butyl)-N'-(2-ethyl-1,3-methoxy-benzoyl)-hydrazide), and RG-115830 (3,5-Dimethylbenzoic acid N-(1-tert-butyl-butyl)-N'-(2-ethyl-3-methoxy-benzoyl)-hydrazide). See, e.g., U.S. patent application Ser. No. 12/155,111, and PCT Appl. No. PCT/US2008/006757, both of which are incorporated herein by reference in their entireties.

#### Cytokines

In certain embodiments, HPV vaccine antigens provided herein may be co-delivered and/or co-expressed (e.g., as part of the same HPV antigen delivery vector or via a separate vector) along with other cytokines. Provided herein are polynucleotides encoding gene-switch polypeptides and a cytokine, or variant or derivative thereof, and methods and systems incorporating the same. Cytokine is a category of small proteins between about 5-20 kDa that are involved in cell signaling. In some instances, cytokines include chemokines, interferons, interleukins, colony-stimulating factors or tumor necrosis factors. In some embodiments, chemokines play a role as a chemoattractant to guide the migration of cells, and is classified into four subfamilies: CXC, CC, CX3C, and XC. Exemplary chemokines include chemokines from the CC subfamily: CCL1, CCL2 (MCP-1), CCL3, CCL4, CCL5 (RANTES), CCL6, CCL7, CCL8, CCL9 (or CCL10), CCL11, CCL12, CCL13, CCL14, CCL15, CCL16, CCL17, CCL18, CCL19, CCL20, CCL21, CCL22, CCL23, CCL24, CCL25, CCL26, CCL27, and CCL28; the CXC subfamily: CXCL1, CXCL2, CXCL3, CXCL4, CXCL5, CXCL6, CXCL7, CXCL8, CXCL9, CXCL10, CXCL11, CXCL12, CXCL13, CXCL14, CXCL15, CXCL16, and CXCL17; the XC subfamily: XCL1 and XCL2; and the CX3C subfamily CX3CL1.

In certain embodiments, HPV vaccine antigens provided herein may be co-delivered and/or co-expressed (e.g., as part of the same HPV antigen delivery vector or via a separate vector) along with interferons. Interferons (IFNs) comprise interferon type I (e.g. IFN- $\alpha$ , IFN- $\beta$ , IFN- $\epsilon$ , IFN- $\kappa$ , and IFN- $\omega$ ), interferon type II (e.g. IFN- $\gamma$ ), and interferon type III. In some embodiments, IFN- $\alpha$  is further classified into about 13 subtypes including IFNA1, IFNA2, IFNA4, IFNA5, IFNA6, IFNA7, IFNA8, IFNA10, IFNA13, IFNA14, IFNA16, IFNA17, and IFNA21.

In certain embodiments, HPV vaccine antigens provided herein may be co-delivered and/or co-expressed (e.g., as part of the same HPV antigen delivery vector or via a separate vector) along with other interleukins. Interleukins are expressed by leukocytes or white blood cells and they promote the development and differentiation of T and B lymphocytes and hematopoietic cells. Exemplary interleukins include IL-1, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8 (CXCL8), IL-9, IL-10, IL-11, IL-12, IL-13, IL-14, IL-15, IL-16, IL-17, IL-18, IL-19, IL-20, IL-21, IL-22, IL-23, IL-24, IL-25, IL-26, IL-27, IL-28, IL-29, IL-30, IL-31, IL-32, IL-33, IL-35, and IL-36. In some embodiments, interleukins are IL-2, IL-12, IL-15, IL-21 or a fusion of IL-15 and IL-15 $\alpha$ .

In some aspects, the interleukin can comprise IL-12. In some embodiments, the IL-12 is a single chain IL-12 (scIL-12), protease sensitive IL-12, destabilized IL-12, membrane bound IL-12, intercalated IL-12. In some instances, the IL-12 variants are as described in WO2015/095249, WO2016/048903, WO2017/062953, all of which is incorporated by reference in their entireties.

In some embodiments, an interleukin comprises mbIL-15. In some embodiments, a mbIL-15 is a membrane-bound chimeric IL-15 which can be co-expressed with a modified effector cell described herein. In some embodiments, the mbIL-15 comprises a full-length IL-15 (e.g., a native IL-15 polypeptide) or fragment or variant thereof, fused in frame with a full length IL-15R $\alpha$ , functional fragment or variant thereof. In some cases, the IL-15 is indirectly linked to the IL-15R $\alpha$  through a linker. In some instances, the mbIL-15 is as described in Hurton et al., "Tethered IL-15 augments antitumor activity and promotes a stem-cell memory subset in tumor-specific T cells," PNAS 2016.

In certain embodiments, HPV vaccine antigens provided herein may be co-delivered and/or co-expressed (e.g., as part of the same HPV antigen delivery vector or via a separate vector) along with tumor necrosis factors. Tumor necrosis factors (TNFs) are a group of cytokines that modulate apoptosis. In some instances, there are about 19 members within the TNF family, including, not limited to, TNF $\alpha$ , lymphotoxin-alpha (LT-alpha), lymphotoxin-beta (LT-beta), T cell antigen gp39 (CD40L), CD27L, CD30L, FASL, 4-1BBL, OX40L, and TNF-related apoptosis inducing ligand (TRAIL).

In certain embodiments, HPV vaccine antigens provided herein may be co-delivered and/or co-expressed (e.g., as part of the same HPV antigen delivery vector or via a separate vector) along with colony stimulating factors. Colony-stimulating factors (CSFs) are secreted glycoproteins that interact with receptor proteins on the surface of hemopoietic stem cells, which subsequently modulates cell proliferation and differentiation into specific kind of blood cells. In some instances, a CSF comprises macrophage colony-stimulating factor, granulocyte macrophage colony-stimulating factor (GM-CSF), granulocyte colony-stimulating factor (G-CSF) or promegapoietin.

In some embodiments, the cytokine is a membrane-bound cytokine, which is co-expressed with a chimeric antigen receptor described herein. In some embodiments, one or more methods described herein further comprise administration of a cytokine. In some instances, the cytokine comprises a chemokine, an interferon, an interleukin, a colony-stimulating factor or a tumor necrosis factor. In some instances, one or more methods described herein further comprise administration of a cytokine selected from a chemokine, an interferon, an interleukin, a colony-stimulating factor or a tumor necrosis factor. In some instances, one

or more methods described herein further comprise administration of a cytokine selected from IL2, IL7, IL12, IL15, a fusion of IL-15 and IL-15R $\alpha$ , IL21, IFN $\gamma$  or TNF- $\alpha$ . Interleukin-12

In particular embodiments, HPV vaccine antigens provided herein may be co-delivered and/or co-expressed (e.g., as part of the same HPV antigen delivery vector or via a separate vector) along with Interleukin-12. Interleukin 12 (IL-12) is an interleukin that is naturally produced by dendritic cells, macrophages, neutrophils, and human B-lymphoblastoid cells (NC-37) in response to antigenic stimulation. IL-12 is composed of a bundle of four alpha helices. It is a heterodimeric cytokine encoded by two separate genes, IL-12A (p35) and IL-12B (p40). The active heterodimer (referred to as p'70), and a homodimer of p40 are formed following protein synthesis. IL-12 is the master regulator of the immune system. IL-12 promotes immune response by activating NK cells and T cells (FIG. 11).

Provided herein are compositions, kits, and system comprising and methods of making HPV recombinant vaccines. The present disclosure provides HPV antigen designs (HPV designs 1-5) constructed in a multi-deleted gorilla adeno-vector (GC46) (SEQ ID NOS: 61-63). Also provided herein are polynucleotides encoding gene-switch polypeptides and IL-12 or variant or derivative thereof, and methods and systems incorporating the same (FIG. 12).

#### Linkers

Also disclosed are constructs comprising a linker to facilitate the expression and functionality of the polynucleotides and polypeptides described herein. In some embodiments, a polynucleotide linker can be utilized in a polynucleotide described herein. A polynucleotide linker can be a double-stranded segment of DNA containing desired restriction sites that can be added to create end structures that are compatible with a vector comprising a polynucleotide described herein. In some cases, a polynucleotide linker can be useful for modifying vectors comprising polynucleotides described herein. For example, a vector modification comprising a polynucleotide linker can be a change in a multiple cloning site, or the addition of a poly-histidine tail. Polynucleotide linkers can also be used to adapt the ends of blunt insert DNA for cloning into a vector cleaved with a restriction enzyme with cohesive end termini. The use of polynucleotide linkers can be more efficient than a blunt ligation into a vector and can provide a method of

releasing an insert from a vector in downstream applications. In some cases an insert can be a polynucleotide sequence encoding polypeptides useful for therapeutic applications. In some cases, a linker can be a cleavable linker.

A polynucleotide linker can be an oligomer. A polynucleotide linker can be a DNA double strand, single strand, or a combination thereof. In some cases, a linker can be RNA. A polynucleotide linker can be ligated into a vector comprising a polynucleotide described herein by a T4 ligase in some cases. To facilitate a ligation an excess of polynucleotide linkers can be added to a composition comprising an insert and a vector. In some cases, an insert and vector are pre-treated before a linker is introduced. For example, pre-treatment with a methylase can prevent unwanted cleavage of insert DNA.

In certain embodiments, two or more polypeptides encoded by a polynucleotide described herein can be separated by an intervening sequence encoding an intervening linker polypeptide. Herein the term "intervening linker polypeptide" referring to an amino acid sequence separating two or more polypeptides encoded by a polynucleotide is distinguished from the term "peptide linker" which refers to the sequence of amino acids which is optionally included in a polypeptide construct disclosed herein to connect the transmembrane domain to the cell surface polypeptide (e.g., comprising a truncated variant of a natural polypeptide). In certain cases, the intervening linker is a cleavage-susceptible intervening linker polypeptide. In some embodiments, the linker is a cleavable or ribosome skipping linker. In some embodiments, the cleavable linker or ribosome skipping linker sequence is selected from the group consisting of 2A, GSG-2A, GSG linker, SGSG linker, furinlink variants and derivatives thereof. In some embodiments, the 2A linker is a p2A linker, a T2A linker, F2A linker or E2A linker. In some embodiments, polypeptides of interest are expressed as fusion proteins linked by a cleavage-susceptible intervening linker polypeptide. In certain embodiments, cleavage-susceptible intervening linker polypeptide(s) can be any one or more of: F/T2A, T2A, p2A, 2A, GSG-p2A, GSG linker, and furinlink variants. Linkers (polynucleotide and polypeptide sequences) as disclosed in PCT/US2016/061668 (WO2017083750) published 18 May 2017 are incorporated by reference herein. In certain embodiments, the linker polypeptide comprises disclosed in the table below:

TABLE 1

| Linker amino acid sequences and polynucleotide sequences |            |                                                                                               |            |                                                  |
|----------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------|------------|--------------------------------------------------|
| Linker Name                                              | SEQ ID NO: | Polynucleotide Sequence (5' to 3' where applicable)                                           | SEQ ID NO: | Amino Acids Sequence (5' to 3' where applicable) |
| Whitlow Linker                                           | 64         | GGCAGCACCTCCGGCAGCG<br>GCAAGCCTGGCAGCGGCGA<br>GGGCAGCACCAAGGGC                                | 81         | GSTSGSGKPGSGEGSTKG                               |
| Linker                                                   | 65         | TCTGGCGGAGGATCTGGAG<br>GAGGCGGATCTGGAGGAGG<br>AGGCAGTGGAGGCGGAGGA<br>TCTGGCGGAGGATCTTGCA<br>G | 82         | SGGSGGGGSGGGSGGG<br>SGGGSLQ                      |
| GSG linker                                               | 66         | GGAAGCGGA                                                                                     | 83         | GSG                                              |
| SGSG linker                                              | 67         | AGTGGCAGCGGC                                                                                  | 84         | SGSG                                             |
| (G4S) <sub>3</sub> linker                                | 68         | GGTGGCGGTGGCTCGGGCG<br>GTGGTGGGTGGGTGGCGG<br>CGGATCT                                          | 85         | GGGSGGGSGGGGS                                    |

TABLE 1-continued

| Linker amino acid sequences and polynucleotide sequences |            |                                                                                                                |            |                                                  |
|----------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------|
| Linker Name                                              | SEQ ID NO: | Polynucleotide Sequence (5' to 3' where applicable)                                                            | SEQ ID NO: | Amino Acids Sequence (5' to 3' where applicable) |
| Furin cleavage site/<br>Furinlink1                       | 69         | CGTGCAAAGCGT                                                                                                   | 86         | RAKR                                             |
| Fmdv                                                     | 70         | AGAGCCAAGAGGGCACC GG<br>TGAACACAGACTTTGAATTTT<br>GACCTTCTGAAGTTGGCAGG<br>AGACGTTGAGTCCAACCTG<br>GGCC           | 87         | RAKRAPVKQTLNFDLLKL<br>AGDVESNPGP                 |
| Thosea asigna<br>virus 2A<br>region (T2A)                | 71         | GAGGGCAGAGGAAGTCTGC<br>TAACATGCGGTGACGTCGA<br>GGAGAATCCTGGACCT                                                 | 88         | EGRGSLTTCGDVEENPGP                               |
| Furin-GSG-<br>T2A                                        | 72         | AGAGCTAAGAGGGGAAGCG<br>GAGAGGGCAGAGGAAGTCT<br>GCTAACATGCGGTGACGTCG<br>AGGAGAATCCTGGACCT                        | 89         | RAKRSGEGRGSLTTCGD<br>VEENPGP                     |
| Furin-SGSG-<br>T2A                                       | 73         | AGGGCCAAGAGGAGTGGCA<br>GCGGCGAGGGCAGAGGAAG<br>TCTTCTAACATGCGGTGACG<br>TGGAGGAGAATCCCGGCCC<br>T                 | 90         | RAKRSGSGEGRGSLTTCGD<br>VEENPGP                   |
| Porcine<br>teschovirus-1<br>2A region<br>(P2A)           | 74         | GCAACGAACCTTCTCTCCT<br>AAAACAGGCTGGTGATGTG<br>GAGGAGAATCCTGGTCCA                                               | 91         | ATNFSLLKQAGDVEENPGP                              |
| GSG-P2A                                                  | 75         | GGAAGCGGAGCTACTAACT<br>TCAGCCTGCTGAAGCAGGCT<br>GGAGACGTGGAGGAGAACC<br>CTGGACCT                                 | 92         | GSGATNFSLLKQAGDVEE<br>NPGP                       |
| Equine rhinitis<br>A virus 2A<br>region (E2A)            | 76         | CAGTGTAATAATTATGCTCT<br>CTTGAAATTGGCTGGAGATG<br>TTGAGAGCAACCTGGACCT                                            | 93         | QCTNYALLKLAGDVESNP<br>GP                         |
| Foot-and-<br>mouth disease<br>virus 2A<br>region (F2A)   | 77         | GTCAAACAGACCCTAAACTT<br>TGATCTGCTAAAACGGCCG<br>GGGATGTGGAAAGTAATCC<br>CGGCCCC                                  | 94         | VKQTLNFDLLKLAGDVES<br>NPGP                       |
| FP2A                                                     | 78         | CGTGCAAAGCGTGCACCGG<br>TGAACACAGGGAAGCGGAGC<br>TACTAACTTCAGCCTGCTGA<br>AGCAGGCTGGAGACGTGGA<br>GGAGAACCCTGGACCT | 95         | RAKRAPVKQSGGATNFSLL<br>KQAGDVEENPGP              |
| Linker-GSG                                               | 79         | GCACCGGTGAAACAGGGAA<br>GCGGA                                                                                   | 96         | APVKQSGG                                         |
| Linker                                                   | 80         | GCACCGGTGAAACAG                                                                                                | 97         | APVKQ                                            |

In some embodiments, an intervening linker polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 99%, 99.5% or 100% identity with the amino acid sequence of Whitlow linker (SEQ ID NO: 64), GSG linker (SEQ ID NO: 66), SGSG linker (SEQ ID NO: 67), (G4S)<sub>3</sub> linker (SEQ ID NO: 68), Furin cleavage site/Furinlink1 (SEQ ID NO: 69), Fmdv linker (SEQ ID NO: 70), *Thosea asigna* virus 2A region (T2A) (SEQ ID NO: 71), Furin-GSG-T2A (SEQ ID NO: 72), Furin-SGSG-T2A (SEQ ID NO: 73), porcine teschovirus-1 2A region (P2A) (SEQ ID NO: 74), GSG-P2A (SEQ ID NO: 75), equine rhinitis A virus 2A region (E2A) (SEQ ID NO: 76), or foot-and-mouth disease virus 2A region (F2A) (SEQ ID NO: 78) (Table 1). In some cases, an intervening linker polypeptide comprises

an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 99%, 99.5% or 100% identity with the amino acid sequence of linkers (SEQ ID NOS: 65, 79 80) In some cases, a viral 2A sequence can be used. 2A elements can be shorter than IRES, having from 5 to 100 base pairs. In some cases, a 2A sequence can have 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, or 100 nucleotides in length. 2A linked genes can be expressed in one single open reading frame and "self-cleavage" can occur co-translationally between the last two amino acids, GP, at the C-terminus of the 2A polypeptide, giving rise to equal amounts of co-expressed proteins.

A viral 2A sequence can be about 20 amino acids. In some cases, a viral 2A sequence can contain a consensus motif

Asp-Val/Ile-Glu-X-Asn-Pro-Gly-Pro. A consensus motif sequence can act co-translationally. For example, formation of a normal peptide bond between a glycine and proline residue can be prevented, which can result in ribosomal skipping and cleavage of a nascent polypeptide. This effect can produce multiple genes at equimolar levels.

A 2A peptide can allow translation of multiple proteins in a single open reading frame into a polypeptide that can be subsequently cleaved into individual polypeptide through a ribosome-skipping mechanism (Funston et al., *J. Gen. Virol.* 89(Pt 2):389-96 (2008)). In some embodiments, a 2A sequence can include: F/T2A, T2A, p2A, 2A, T2A, E2A, F2A, and BmCPV2A, BmIFV2A, and any combination thereof.

In some cases, a vector can comprise an IRES sequence and a 2A linker sequence. In other cases, expression of multiple genes linked with 2A peptides can be facilitated by a spacer sequence (GSG) ahead of the 2A peptides. In some cases, constructs can combine a spacers, linkers, adaptors, promoters, or combinations thereof. For example, a linker can have a spacer (SGSG or GSG or Whitlow linker) and furin linker (R-A-K-R) cleavage site with different 2A peptides. A spacer can be an I-Ceui. In some cases, a linker can be engineered. For example, a linker can be designed to comprise chemical characteristics such as hydrophobicity. In some cases, at least two linker sequences can produce the same protein. In other cases, multiple linkers can be used in a vector. For example, genes of interest can be separated by at least two linkers.

In certain embodiments, two or more polypeptides encoded by a polynucleotide described herein can be separated by an intervening sequence encoding a linker polypeptide. In certain cases, the linker is a cleavage-susceptible linker. In some embodiments, polypeptides of interest are expressed as fusion proteins linked by a cleavage-susceptible linker polypeptide. In certain embodiments, cleavage-susceptible linker polypeptide(s) can be any one or two of: Furinlink, fmdv, p2a, GSG-p2a, and/or fp2a described below. In some cases, a linker is APVKQGS (SEQ ID NO: 96).

In certain cases, a linker polypeptide can comprise an amino acid sequence "RAKR" (SEQ ID NO: 86). In certain cases, a Furin linker polypeptide can be encoded by a polynucleotide sequence polynucleotide sequence comprising "CGTGCAAAGCGT" (SEQ ID NO: 69) or "AGAGCTAAGAGG" (SEQ ID NO: 112).

In some embodiments, a linker can be utilized in a polynucleotide described herein. A linker can be a flexible linker, a rigid linker, an in vivo cleavable linker, or any combination thereof. In some cases, a linker can link functional domains together (as in flexible and rigid linkers) or releasing free functional domain in vivo as in in vivo cleavable linkers.

Linkers can improve biological activity, increase expression yield, and achieving desirable pharmacokinetic profiles. A linker can also comprise hydrazone, peptide, disulfide, or thioether.

In some cases, a linker sequence described herein can include a flexible linker. Flexible linkers can be applied when a joined domain requires a certain degree of movement or interaction. Flexible linkers can be composed of small, non-polar (e.g., Gly) or polar (e.g., Ser or Thr) amino acids. A flexible linker can have sequences consisting primarily of stretches of Gly and Ser residues ("GS" linker). An example of a flexible linker can have the sequence of (Gly-Gly-Gly-Gly-Ser)<sub>n</sub> (SEQ ID NO: 85). By adjusting the copy number "n", the length of this exemplary GS linker can

be optimized to achieve appropriate separation of functional domains, or to maintain necessary inter-domain interactions. Besides GS linkers, other flexible linkers can be utilized for recombinant fusion proteins. In some cases, flexible linkers can also be rich in small or polar amino acids such as Gly and Ser, but can contain additional amino acids such as Thr and Ala to maintain flexibility. In other cases, polar amino acids such as Lys and Glu can be used to improve solubility.

Flexible linkers included in linker sequences described herein, can be rich in small or polar amino acids such as Gly and Ser to provide good flexibility and solubility. Flexible linkers can be suitable choices when certain movements or interactions are desired for fusion protein domains. In addition, although flexible linkers cannot have rigid structures, they can serve as a passive linker to keep a distance between functional domains. The length of flexible linkers can be adjusted to allow for proper folding or to achieve optimal biological activity of the fusion proteins.

A linker described herein can further include a rigid linker in some cases. A rigid linker can be utilized to maintain a fixed distance between domains of a polypeptide. Examples of rigid linkers can be: Alpha helix-forming linkers, Pro-rich sequence, (XP)<sub>n</sub>, X-Pro backbone, A(EAAAK)<sub>n</sub> (n=2-5), to name a few. Rigid linkers can exhibit relatively stiff structures by adopting  $\alpha$ -helical structures or by containing multiple Pro residues in some cases.

A linker described herein can be cleavable in some cases. In other cases a linker is not cleavable. Linkers that are not cleavable can covalently join functional domains together to act as one molecule throughout an in vivo processes or an ex vivo process. A linker can also be cleavable in vivo. A cleavable linker can be introduced to release free functional domains in vivo. A cleavable linker can be cleaved by the presence of reducing reagents, proteases, to name a few. For example, a reduction of a disulfide bond can be utilized to produce a cleavable linker. In the case of a disulfide linker, a cleavage event through disulfide exchange with a thiol, such as glutathione, could produce a cleavage. In other cases, an in vivo cleavage of a linker in a recombinant fusion protein can also be carried out by proteases that can be expressed in vivo under pathological conditions (e.g. cancer or inflammation), in specific cells or tissues, or constrained within certain cellular compartments. In some cases, a cleavable linker can allow for targeted cleavage. For example, the specificity of many proteases can offer slower cleavage of a linker in constrained compartments. A cleavable linker can also comprise hydrazone, peptides, disulfide, or thioether. For example, a hydrazone can confer serum stability. In other cases, a hydrazone can allow for cleavage in an acidic compartment. An acidic compartment can have a pH up to 7. A linker can also include a thioether. A thioether can be nonreducible A thioether can be designed for intracellular proteolytic degradation.

In certain embodiments, an fmdv linker polypeptide comprises a sequence that can be at least about 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 87. In certain embodiments, an fmdv linker polypeptide is one or more of the linkers encoded in a single vector linking two or more fusion proteins. In certain cases, an fmdv linker polypeptide can be encoded by a polynucleotide open reading frame (ORF) nucleic acid sequence. In some cases, an ORF encoding fmdv comprises or consists of a sequence of SEQ ID NO: 70). In certain embodiments, a polynucleotide encoding fmdv is at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 70).

In certain cases, a linker polypeptide can be a “p2a” linker. In certain embodiments, a p2a polypeptide can comprise a sequence that can be about at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 91). In certain embodiments, the p2a linker polypeptide can be one or more of the linkers encoded in a single vector linking two or more fusion proteins. In some cases, a p2a linker polypeptide can be encoded by a polynucleotide open reading frame (ORF) nucleic acid sequence. In certain embodiments, an ORF encoding p2a comprises or consists of the sequence of SEQ ID NO: 74). In certain cases, a polynucleotide encoding p2a can be or can be about at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 74).

In some cases, a linker polypeptide can be a “GSG-p2a” linker. In certain embodiments, a GSG-p2a linker polypeptide can comprise a sequence that can be about at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 92). In certain embodiments, a GSG-p2a linker polypeptide can be one or more of the linkers encoded in a single vector linking two or more fusion proteins. In some cases, a GSG-p2a linker polypeptide can be encoded by a polynucleotide open-reading frame (ORF) nucleic acid sequence. An ORF encoding GSG p2a can comprise the sequence of SEQ ID NO: 75). In some cases, a polynucleotide encoding GSG-p2a can be or can be about at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 75).

A linker polypeptide can be an “fp2a” linker as provided herein. In certain embodiments, a fp2a linker polypeptide can comprise a sequence that can be about at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 95). In certain cases, an fp2a linker polypeptide can be one or more of the linkers encoded in a single vector linking two or more fusion proteins. In some cases, a fp2a linker polypeptide can be encoded by a polynucleotide open reading frame (ORF) nucleic acid sequence. In certain embodiments, a polynucleotide encoding an fp2a linker can be or can be about at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99% or 100% identical to SEQ ID NO: 78).

In some cases, a linker can be engineered. For example, a linker can be designed to comprise chemical characteristics such as hydrophobicity. In some cases, at least two linker sequences can produce the same protein. A sequence can be or can be about 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to a polypeptide sequence of SEQ ID NO 82, 96 or 97. In other cases, multiple linkers can be used in a vector. For example, genes of interest, and one or more gene switch polypeptide sequences described herein can be separated by at least two linkers. In some cases, genes can be separated by 2, 3, 4, 5, 6, 7, 8, 9, or up to 10 linkers.

A linker can be an engineered linker. Methods of designing linkers can be computational. In some cases, computational methods can include graphic techniques. Computation methods can be used to search for suitable peptides from libraries of three-dimensional peptide structures derived from databases. For example, a Brookhaven Protein Data Bank (PDB) can be used to span the distance in space between selected amino acids of a linker.

In some embodiments are polynucleotides encoding a polypeptide construct comprising a furin polypeptide and a 2A polypeptide, wherein the furin polypeptide and the 2A polypeptide are connected by a polypeptide linker compris-

ing at least three hydrophobic amino acids. In some cases, at least three hydrophobic amino acids are selected from the list consisting of glycine (Gly)(G), alanine (Ala)(A), valine (Val)(V), leucine (Leu)(L), isoleucine (Ile)(I), proline (Pro)(P), phenylalanine (Phe)(F), methionine (Met)(M), tryptophan (Trp)(W). In some cases, a polypeptide linker can also include one or more GS linker sequences, for instance (GS)<sub>n</sub>, (SG)<sub>n</sub>, (GSG)<sub>n</sub>, and (SGSG)<sub>n</sub>, wherein n can be any number from zero to fifteen.

Provided are methods of obtaining an improved expression of a polypeptide construct comprising: providing a polynucleotide encoding said polypeptide construct comprising a first functional polypeptide and a second functional polypeptide, wherein said first functional polypeptide and second functional polypeptide are connected by a linker polypeptide comprising a sequence with at least 60% identity to the sequence APVKQ; and expressing said polynucleotide in a host cell, wherein said expressing results in an improved expression of the polypeptide construct as compared to a corresponding polypeptide construct that does not have a linker polypeptide comprising a sequence with at least 60% identity to the sequence APVKQ.

#### IRES Elements

Also disclosed herein are constructs comprising an IRES element to facilitate the expression and functionality of the polynucleotides and polypeptides described herein. The term “internal ribosome entry site (IRES)” as used herein can be intended to mean internal ribosomal entry site. In a vector comprising an IRES sequence, a first gene can be translated by a cap-dependent, ribosome scanning, mechanism with its own 5'-UTR, whereas translation of a subsequent gene can be accomplished by direct recruitment of a ribosome to an IRES in a cap-independent manner. An IRES sequence can allow eukaryotic ribosomes to bind and begin translation without binding to a 5' capped end. An IRES sequence can allow expression of multiple genes from one transcript (Mountford and Smith, *Trends Genet.* 11(5):179-84 (1995)).

The term “CAP” or “cap” as used herein refers to a modified nucleotide, generally a 7-methyl guanosine, linked 3' to 5' (7meG-ppp-G), to the 5' end of a eukaryotic mRNA, that serves as a required element in the normal translation initiation pathway during expression of protein from that mRNA.

In certain cases, an IRES region can be derived from a virus, such as picornavirus, encephalomyocarditis virus, hepatitis C virus IRES sequence. In other cases, an IRES sequence can be derived from an encephalomyocarditis virus. The term “EMCV” or “encephalomyocarditis virus” as used herein refers to any member isolate or strain of the encephalomyocarditis virus species of the genus of the family Picornaviridae. Examples are: EMCV-R (Rueckert) strain virus, Columbia-SK virus. In some cases, a cellular IRES element, such as eukaryotic initiation factor 4G, immunoglobulin heavy chain binding protein, c-myc proto-oncogene, vascular endothelial growth factor, fibroblast growth factor-1 IRES, or any combination or modification thereof can be used. In some cases, a cellular IRES can have increased gene expression when compared to a viral IRES.

An IRES sequence of viral, cellular or a combination thereof can be utilized in a vector. An IRES can be from encephalomyocarditis (EMCV) or poliovirus (PV). In some cases, an IRES element is selected from a group consisting of Poliovirus (PV), Encephalomyelitis virus (EMCV), Foot-and-mouth disease virus (FMDV), Porcine teschovirus-1 (PTV-1), Aichivirus (AiV), Seneca Valley virus (SVV), Hepatitis C virus (HCV), Classical swine fever virus

(CSFV), Human immunodeficiency virus-2 (HIV-2), Human immunodeficiency virus-1 (HIV-1), Moloney murine leukemia virus (MoMLV), Feline immunodeficiency virus (FIV), Mouse mammary tumor virus (MMTV), Human cytomegalovirus latency (pUL138), Epstein-Barr virus (EBNA-1), Herpes virus Marek's disease (MDV RLORF9), SV40 polycistronic 19S (SV40 19S), Rhopalosiphum padi virus (RhPV), Cricket paralysis virus (CrPV), Ectropis obliqua picorna-like virus (EoPV), Plautia stali intestine virus (PSIV), Triatoma virus (TrV), Bee paralysis dicistrovirus (IAPV, KBV), Black currant reversion virus (BRV), Pelargonium flower break virus (PFBV), Hibiscus chlorotic ringspot virus (HCRSV), Crucifer-infecting tobamovirus (CrTMV), Potato leaf roll polerovirus (PLRV), Tobacco etch virus (TEV), Giardiavirus (GLV), Leishmania RNA virus-1 (LRV-1), and combinations or modifications thereof. In some cases, an IRES is selected from a group consisting of Apaf-1, XIAP, HIAP2/c-IAP1, DAP5, Bcl-2, c-myc, CAT-1, INR, Differentiation LEF-1, PDGF2, HIF-1a, VEGF, FGF2, BiP, BAG-1, CIRP, p53, SHMT1, PITSL-REp58, CDK1, Rpr, hid, hsp70, grim, skl, Antennapedia, dFoxO, dInR, Adh-Adhr, HSP101, ADH, URE-2, GPR1, NCE102, YMR181a, MSN1, BOI1, FLO8, GIC1, and any combination or modification thereof. When an IRES element is included between two open reading frames (ORFs), initiation of translation can occur by a canonical 5'-m7GpppN cap-dependent mechanism in a first ORF and a cap-independent mechanism in a second ORF downstream of the IRES element.

In some cases, genes can be linked by an internal ribosomal entry site (IRES). An IRES can allow simultaneous expression of multiple genes. For example, an IRES sequence can permit production of multiple proteins from a single mRNA transcript. A ribosome can bind to an IRES in a 5'-cap independent manner and initiate translation.

In some cases, an IRES sequence can be or can be about 500 base pairs. An IRES sequence can be from 300 base pairs to 1000 base pairs. For example, an IRES can be 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, or 1000 base pairs long.

In some cases, expression of a downstream gene within a vector comprising an IRES sequence can be reduced. For example, a gene following an IRES sequence can have reduced expression over a gene preceding an IRES sequence. Reduced expression can be from 1% to 99% reduction over a preceding gene.

#### Methods of Regulating Expression

In one embodiment, a method of regulating the expression of a heterologous gene in an engineered cell is provided. Polynucleotides encoding for gene switch polypeptides for ligand inducible control of a heterologous gene expression, an antigen binding polypeptide and a heterologous gene is provided. In some instances, the polynucleotides are in one or more gene expression cassettes as depicted in any one of FIGS. 1 through 16. In another instance, the polynucleotides are incorporated into an engineered cell via viral or non-viral vectors. Viral vectors can include lentiviral vectors, retroviral vectors or adenoviral vectors. Non-viral vectors can include Sleeping Beauty transposons. In other instances, the polynucleotides are incorporated into an engineered cell via recombinases or gene editing techniques. Examples of recombinases are serine recombinases as described herein. Examples of gene editing techniques can include CRISPR or Argonaute systems. Herein a "CRISPR gene editing system" or "CRISPR system" refers to any RNA-guided Cas protein-mediated process for targeting a change in DNA sequence to a specific region of a genome. Herein "Argonaute gene

editing system" refers to any single-stranded DNA guided Argonaute endonuclease-mediated process for targeting a change in DNA sequence to a specific region of a genome. Pharmaceutical Compositions and Dosage

The present disclosure provides a composition comprising the adenovirus or adenoviral vector described herein and a carrier therefor (e.g., a pharmaceutically acceptable carrier). The composition desirably is a physiologically acceptable (e.g., pharmaceutically acceptable) composition, which comprises a carrier, preferably a physiologically (e.g., pharmaceutically) acceptable carrier, and the adenovirus or adenoviral vector. Any suitable carrier can be used within the context of the present disclosure, and such carriers are well known in the art. The choice of carrier will be determined, in part, by the particular use of the composition (e.g., administration to an animal) and the particular method used to administer the composition. Ideally, in the context of replication-deficient adenoviral vectors, the pharmaceutical composition preferably is free of replication-competent adenovirus. The pharmaceutical composition optionally can be sterile.

Suitable compositions include aqueous and non-aqueous isotonic sterile solutions, which can contain anti-oxidants, buffers, and bacteriostats, and aqueous and non-aqueous sterile suspensions that can include suspending agents, solubilizers, thickening agents, stabilizers, and preservatives. The composition can be presented in unit-dose or multi-dose sealed containers, such as ampules and vials, and can be stored in a freeze-dried (lyophilized) condition requiring only the addition of the sterile liquid carrier, for example, water, immediately prior to use. Extemporaneous solutions and suspensions can be prepared from sterile powders, granules, and tablets. Preferably, the carrier is a buffered saline solution. More preferably, the adenovirus or adenoviral vector is part of a composition formulated to protect the adenovirus or adenoviral vector from damage prior to administration. For example, the composition can be formulated to reduce loss of the adenovirus or adenoviral vector on devices used to prepare, store, or administer the adenovirus or adenoviral vector, such as glassware, syringes, or needles. The composition can be formulated to decrease the light sensitivity and/or temperature sensitivity of the adenovirus or adenoviral vector. To this end, the composition preferably comprises a pharmaceutically acceptable liquid carrier, such as, for example, those described above, and a stabilizing agent selected from the group consisting of polysorbate 80, L-arginine, polyvinylpyrrolidone, trehalose, and combinations thereof. Use of such a composition will extend the shelf life of the adenovirus or adenoviral vector, and facilitate its administration. Formulations for adenovirus or adenoviral vector-containing compositions are further described in, for example, U.S. Pat. Nos. 6,225,289, 6,514, 943, and International Patent Application Publication WO 2000/034444.

The composition also can be formulated to enhance transduction efficiency. In addition, one of ordinary skill in the art will appreciate that the adenovirus or adenoviral vector can be present in a composition with other therapeutic or biologically-active agents. For example, factors that control inflammation, such as ibuprofen or steroids, can be part of the composition to reduce swelling and inflammation associated with in vivo administration of the adenovirus or adenoviral vector. If the adenovirus or adenoviral vector is used to deliver an antigen-encoding nucleic acid sequence to a host, immune system stimulators or adjuvants, e.g., interleukins, lipopolysaccharide, or double-stranded RNA, can be administered to enhance or modify any immune response

to the antigen. Antibiotics, i.e., microbicides and fungicides, can be present to treat existing infection and/or reduce the risk of future infection, such as infection associated with gene transfer procedures.

In some embodiments, disclosed herein are compositions comprising a polynucleotide or polypeptide disclosed herein for administration in a subject. In some instances, are modified effector cell compositions encoding a polynucleotide or polypeptide disclosed herein, and optionally containing a cytokine and/or an additional therapeutic agent. In some instances, also included herein are vectors encoding gene-switch polypeptides for regulating expression of a chimeric antigen receptor for modification of an effector cell.

In some instances, pharmaceutical compositions of a modified effector cell or a vector encoding gene-switch polypeptides and a chimeric antigen receptor are formulated in a conventional manner using one or more physiologically acceptable carriers including excipients and auxiliaries which facilitate processing of the active compounds into preparations which can be used pharmaceutically. Proper formulation is dependent upon the route of administration chosen. A summary of pharmaceutical compositions described herein is found, for example, in Remington: The Science and Practice of Pharmacy, Nineteenth Ed (Easton, Pa.: Mack Publishing Company, 1995); Hoover, John E., Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pennsylvania 1975; Liberman, H. A. and Lachman, L., Eds., Pharmaceutical Dosage Forms, Marcel Dekker, New York, N.Y., 1980; and Pharmaceutical Dosage Forms and Drug Delivery Systems, Seventh Ed. (Lippincott Williams & Wilkins 1999).

Pharmaceutical compositions are optionally manufactured in a conventional manner, such as, by way of example only, by means of conventional mixing, dissolving, granulating, dragee-making, levigating, emulsifying, encapsulating, entrapping or compression processes.

In certain embodiments, compositions can also include one or more pH adjusting agents or buffering agents, including acids such as acetic, boric, citric, lactic, phosphoric and hydrochloric acids; bases such as sodium hydroxide, sodium phosphate, sodium borate, sodium citrate, sodium acetate, sodium lactate and tris-hydroxymethylaminomethane; and buffers such as citrate/dextrose, sodium bicarbonate and ammonium chloride. Such acids, bases and buffers are included in an amount required to maintain pH of the composition in an acceptable range.

In other embodiments, compositions can also include one or more salts in an amount required to bring osmolality of the composition into an acceptable range. Such salts include those having sodium, potassium or ammonium cations and chloride, citrate, ascorbate, borate, phosphate, bicarbonate, sulfate, thiosulfate or bisulfite anions; suitable salts include sodium chloride, potassium chloride, sodium thiosulfate, sodium bisulfite and ammonium sulfate.

The pharmaceutical compositions described herein are administered by any suitable administration route, including but not limited to, oral, parenteral (e.g., intravenous, subcutaneous, intramuscular, intracerebral, intracerebroventricular, intra-articular, intraperitoneal, or intracranial), intranasal, buccal, sublingual, or rectal administration routes. In some instances, the pharmaceutical composition is formulated for parenteral (e.g., intravenous, subcutaneous, intramuscular, intracerebral, intracerebroventricular, intra-articular, intraperitoneal, or intracranial) administration.

The pharmaceutical compositions described herein are formulated into any suitable dosage form, including but not

limited to, aqueous oral dispersions, liquids, gels, syrups, elixirs, slurries, suspensions and the like, for oral ingestion by an individual to be treated, solid oral dosage forms, aerosols, controlled release formulations, fast melt formulations, effervescent formulations, lyophilized formulations, tablets, powders, pills, dragees, capsules, delayed release formulations, extended release formulations, pulsatile release formulations, multiparticulate formulations, and mixed immediate release and controlled release formulations. In some embodiments, the pharmaceutical compositions are formulated into capsules. In some embodiments, the pharmaceutical compositions are formulated into solutions (for example, for IV administration). In some cases, the pharmaceutical composition is formulated as an infusion. In some cases, the pharmaceutical composition is formulated as an injection.

The pharmaceutical solid dosage forms described herein optionally include a compound described herein and one or more pharmaceutically acceptable additives such as a compatible carrier, binder, filling agent, suspending agent, flavoring agent, sweetening agent, disintegrating agent, dispersing agent, surfactant, lubricant, colorant, diluent, solubilizer, moistening agent, plasticizer, stabilizer, penetration enhancer, wetting agent, anti-foaming agent, antioxidant, preservative, or one or more combination thereof.

In still other aspects, using standard coating procedures, such as those described in Remington's Pharmaceutical Sciences, 20th Edition (2000), a film coating is provided around the compositions. In some embodiments, the compositions are formulated into particles (for example for administration by capsule) and some or all of the particles are coated. In some embodiments, the compositions are formulated into particles (for example for administration by capsule) and some or all of the particles are microencapsulated. In some embodiments, the compositions are formulated into particles (for example for administration by capsule) and some or all of the particles are not microencapsulated and are uncoated.

In certain embodiments, compositions provided herein can also include one or more preservatives to inhibit microbial activity. Suitable preservatives include mercury-containing substances such as merfen and thiomersal; stabilized chlorine dioxide; and quaternary ammonium compounds such as benzalkonium chloride, cetyltrimethylammonium bromide and cetylpyridinium chloride.

"Antifoaming agents" reduce foaming during processing which can result in coagulation of aqueous dispersions, bubbles in the finished film, or generally impair processing. Exemplary anti-foaming agents include silicon emulsions or sorbitan sesquoleate.

"Antioxidants" include, for example, butylated hydroxytoluene (BHT), sodium ascorbate, ascorbic acid, sodium metabisulfite and tocopherol. In certain embodiments, antioxidants enhance chemical stability where required.

Formulations described herein can benefit from antioxidants, metal chelating agents, thiol containing compounds and other general stabilizing agents. Examples of such stabilizing agents, include, but are not limited to: (a) about 0.5% to about 2% w/v glycerol, (b) about 0.1% to about 1% w/v methionine, (c) about 0.1% to about 2% w/v monothioglycerol, (d) about 1 mM to about 10 mM EDTA, (e) about 0.01% to about 2% w/v ascorbic acid, (f) 0.003% to about 0.02% w/v polysorbate 80, (g) 0.001% to about 0.05% w/v polysorbate 20, (h) arginine, (i) heparin, (j) dextran sulfate, (k) cyclodextrins, (l) pentosan polysulfate and other heparinoids, (m) divalent cations such as magnesium and zinc; or (n) combinations thereof.



"Binders" impart cohesive qualities and include, e.g., alginic acid and salts thereof; cellulose derivatives such as carboxymethylcellulose, methylcellulose (e.g., Methocel®), hydroxypropylmethylcellulose, hydroxyethylcellulose, hydroxypropylcellulose (e.g., Klucel®), ethylcellulose (e.g., Ethocel®), and microcrystalline cellulose (e.g., Avicel®); microcrystalline dextrose; amylose; magnesium aluminum silicate; polysaccharide acids; bentonites; gelatin; polyvinylpyrrolidone/vinyl acetate copolymer; crospovidone; povidone; starch; pregelatinized starch; tragacanth, dextrin, a sugar, such as sucrose (e.g., Dipac®), glucose, dextrose, molasses, mannitol, sorbitol, xylitol (e.g., Xylitab®), and lactose; a natural or synthetic gum such as acacia, tragacanth, ghatti gum, mucilage of isapol husks, polyvinylpyrrolidone (e.g., Polyvidone® CL, Kollidon® CL, Polyplasdone® XL-10), larch arabogalactan, Veegum®, polyethylene glycol, waxes, sodium alginate, and the like.

A "carrier" or "carrier materials" include any commonly used excipients in pharmaceuticals and should be selected on the basis of compatibility with compounds disclosed herein, such as, compounds of ibrutinib and An anticancer agent, and the release profile properties of the desired dosage form. Exemplary carrier materials include, e.g., binders, suspending agents, disintegration agents, filling agents, surfactants, solubilizers, stabilizers, lubricants, wetting agents, diluents, and the like. "Pharmaceutically compatible carrier materials" can include, but are not limited to, acacia, gelatin, colloidal silicon dioxide, calcium glycerophosphate, calcium lactate, maltodextrin, glycerine, magnesium silicate, polyvinylpyrrolidone (PVP), cholesterol, cholesterol esters, sodium caseinate, soy lecithin, taurocholic acid, phosphatidylcholine, sodium chloride, tricalcium phosphate, dipotassium phosphate, cellulose and cellulose conjugates, sugars sodium stearyl lactylate, carrageenan, monoglyceride, diglyceride, pregelatinized starch, and the like. See, e.g., Remington: The Science and Practice of Pharmacy, Nineteenth Ed (Easton, Pa.: Mack Publishing Company, 1995); Hoover, John E., Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pennsylvania 1975; Liberman, H. A. and Lachman, L., Eds., Pharmaceutical Dosage Forms, Marcel Decker, New York, N.Y., 1980; and Pharmaceutical Dosage Forms and Drug Delivery Systems, Seventh Ed. (Lippincott Williams & Wilkins 1999).

"Dispersing agents," and/or "viscosity modulating agents" include materials that control the diffusion and homogeneity of a drug through liquid media or a granulation method or blend method. In some embodiments, these agents also facilitate the effectiveness of a coating or eroding matrix. Exemplary diffusion facilitators/dispersing agents include, e.g., hydrophilic polymers, electrolytes, Tween® 60 or 80, PEG, polyvinylpyrrolidone (PVP; commercially known as Plasdane®), and the carbohydrate-based dispersing agents such as, for example, hydroxypropyl celluloses (e.g., HPC, HPC-SL, and HPC-L), hydroxypropyl methylcelluloses (e.g., HPMC K100, HPMC K4M, HPMC K15M, and HPMC K100M), carboxymethylcellulose sodium, methylcellulose, hydroxyethylcellulose, hydroxypropylcellulose, hydroxypropylmethylcellulose phthalate, hydroxypropylmethylcellulose acetate stearate (HPMCAS), non-crystalline cellulose, magnesium aluminum silicate, triethanolamine, polyvinyl alcohol (PVA), vinyl pyrrolidone/vinyl acetate copolymer (S630), 4-(1,1,3,3-tetramethylbutyl)-phenol polymer with ethylene oxide and formaldehyde (also known as tyloxapol), poloxamers (e.g., Pluronic F68®, F88®, and F108®, which are block copolymers of ethylene oxide and propylene oxide); and poloxamines (e.g., Tetronic 908®, also known as Poloxamine 908®, which is

a tetrafunctional block copolymer derived from sequential addition of propylene oxide and ethylene oxide to ethylenediamine (BASF Corporation, Parsippany, N.J.)), polyvinylpyrrolidone K12, polyvinylpyrrolidone K17, polyvinylpyrrolidone K25, or polyvinylpyrrolidone K30, polyvinylpyrrolidone/vinyl acetate copolymer (S-630), polyethylene glycol, e.g., the polyethylene glycol can have a molecular weight of about 300 to about 6000, or about 3350 to about 4000, or about 7000 to about 5400, sodium carboxymethylcellulose, methylcellulose, polysorbate-80, sodium alginate, gums, such as, e.g., gum tragacanth and gum acacia, guar gum, xanthans, including xanthan gum, sugars, cellulose, such as, e.g., sodium carboxymethylcellulose, methylcellulose, sodium carboxymethylcellulose, polysorbate-80, sodium alginate, polyethoxylated sorbitan monolaurate, polyethoxylated sorbitan monolaurate, povidone, carbomers, polyvinyl alcohol (PVA), alginates, chitosans and combinations thereof. Plasticizers such as cellulose or triethyl cellulose can also be used as dispersing agents. Dispersing agents particularly useful in liposomal dispersions and self-emulsifying dispersions are dimyristoyl phosphatidyl choline, natural phosphatidyl choline from eggs, natural phosphatidyl glycerol from eggs, cholesterol and isopropyl myristate.

Combinations of one or more erosion facilitator with one or more diffusion facilitator can also be used in the present compositions.

The term "diluent" refers to chemical compounds that are used to dilute the compound of interest prior to delivery. Diluents can also be used to stabilize compounds because they can provide a more stable environment. Salts dissolved in buffered solutions (which also can provide pH control or maintenance) are utilized as diluents in the art, including, but not limited to a phosphate buffered saline solution. In certain embodiments, diluents increase bulk of the composition to facilitate compression or create sufficient bulk for homogenous blend for capsule filling. Such compounds include e.g., lactose, starch, mannitol, sorbitol, dextrose, microcrystalline cellulose such as Avicel®; dibasic calcium phosphate, dicalcium phosphate dihydrate; tricalcium phosphate, calcium phosphate; anhydrous lactose, spray-dried lactose; pregelatinized starch, compressible sugar, such as Di-Pac® (Amstar); mannitol, hydroxypropylmethylcellulose, hydroxypropylmethylcellulose acetate stearate, sucrose-based diluents, confectioner's sugar; monobasic calcium sulfate monohydrate, calcium sulfate dihydrate; calcium lactate trihydrate, dextrates; hydrolyzed cereal solids, amylose; powdered cellulose, calcium carbonate; glycine, kaolin; mannitol, sodium chloride; inositol, bentonite, and the like.

"Filling agents" include compounds such as lactose, calcium carbonate, calcium phosphate, dibasic calcium phosphate, calcium sulfate, microcrystalline cellulose, cellulose powder, dextrose, dextrates, dextran, starches, pregelatinized starch, sucrose, xylitol, lactitol, mannitol, sorbitol, sodium chloride, polyethylene glycol, and the like.

"Lubricants" and "glidants" are compounds that prevent, reduce or inhibit adhesion or friction of materials. Exemplary lubricants include, e.g., stearic acid, calcium hydroxide, talc, sodium stearyl fumarate, a hydrocarbon such as mineral oil, or hydrogenated vegetable oil such as hydrogenated soybean oil (Sterotex®), higher fatty acids and their alkali-metal and alkaline earth metal salts, such as aluminum, calcium, magnesium, zinc, stearic acid, sodium stearates, glycerol, talc, waxes, Stearowet®, boric acid, sodium benzoate, sodium acetate, sodium chloride, leucine, a polyethylene glycol (e.g., PEG-4000) or a methoxypolyethylene

glycol such as Carbowax™, sodium oleate, sodium benzoate, glyceryl behenate, polyethylene glycol, magnesium or sodium lauryl sulfate, colloidal silica such as Syloid™, Cab-O-Sil®, a starch such as corn starch, silicone oil, a surfactant, and the like.

“Plasticizers” are compounds used to soften the micro-encapsulation material or film coatings to make them less brittle. Suitable plasticizers include, e.g., polyethylene glycols such as PEG 300, PEG 400, PEG 600, PEG 1450, PEG 3350, and PEG 800, stearic acid, propylene glycol, oleic acid, triethyl cellulose and triacetin. In some embodiments, plasticizers can also function as dispersing agents or wetting agents.

“Solubilizers” include compounds such as triacetin, triethylcitrate, ethyl oleate, ethyl caprylate, sodium lauryl sulfate, sodium docusate, vitamin E TPGS, dimethylacetamide, N-methylpyrrolidone, N-hydroxyethylpyrrolidone, polyvinylpyrrolidone, hydroxypropylmethyl cellulose, hydroxypropyl cyclodextrins, ethanol, n-butanol, isopropyl alcohol, cholesterol, bile salts, polyethylene glycol 200-600, glycofurol, transcitol, propylene glycol, and dimethyl isosorbide and the like.

“Stabilizers” include compounds such as any antioxidant agents, buffers, acids, preservatives and the like.

“Suspending agents” include compounds such as polyvinylpyrrolidone, e.g., polyvinylpyrrolidone K12, polyvinylpyrrolidone K17, polyvinylpyrrolidone K25, or polyvinylpyrrolidone K30, vinyl pyrrolidone/vinyl acetate copolymer (S630), polyethylene glycol, e.g., the polyethylene glycol can have a molecular weight of about 300 to about 6000, or about 3350 to about 4000, or about 7000 to about 5400, sodium carboxymethylcellulose, methylcellulose, hydroxypropylmethylcellulose, hydroxymethylcellulose acetate stearate, polysorbate-80, hydroxyethylcellulose, sodium alginate, gums, such as, e.g., gum tragacanth and gum acacia, guar gum, xanthans, including xanthan gum, sugars, celluloses, such as, e.g., sodium carboxymethylcellulose, methylcellulose, sodium carboxymethylcellulose, hydroxypropylmethylcellulose, hydroxyethylcellulose, polysorbate-80, sodium alginate, polyethoxylated sorbitan monolaurate, polyethoxylated sorbitan monolaurate, povidone and the like.

“Surfactants” include compounds such as sodium lauryl sulfate, sodium docusate, Tween 60 or 80, triacetin, vitamin E TPGS, sorbitan monooleate, polyoxyethylene sorbitan monooleate, polysorbates, polaxomers, bile salts, glyceryl monostearate, copolymers of ethylene oxide and propylene oxide, e.g., Pluronic® (BASF), and the like. Some other surfactants include polyoxyethylene fatty acid glycerides and vegetable oils, e.g., polyoxyethylene (60) hydrogenated castor oil; and polyoxyethylene alkylethers and alkylphenyl ethers, e.g., octoxynol 10, octoxynol 40. In some embodiments, surfactants can be included to enhance physical stability or for other purposes.

“Viscosity enhancing agents” include, e.g., methyl cellulose, xanthan gum, carboxymethyl cellulose, hydroxypropyl cellulose, hydroxypropylmethyl cellulose, hydroxypropylmethyl cellulose acetate stearate, hydroxypropylmethyl cellulose phthalate, carbomer, polyvinyl alcohol, alginates, acacia, chitosans and combinations thereof.

“Wetting agents” include compounds such as oleic acid, glyceryl monostearate, sorbitan monooleate, sorbitan monooleate, triethanolamine oleate, polyoxyethylene sorbitan monooleate, polyoxyethylene sorbitan monolaurate, sodium docusate, sodium oleate, sodium lauryl sulfate, sodium docusate, triacetin, Tween 80, vitamin E TPGS, ammonium salts and the like.

Kits/Article of Manufacture

Disclosed herein, in certain embodiments, are kits and articles of manufacture for use with one or more methods described herein. Such kits include a carrier, package, or container that is compartmentalized to receive one or more containers such as vials, tubes, and the like, each of the container(s) comprising one of the separate elements to be used in a method described herein. Suitable containers include, for example, bottles, vials, syringes, and test tubes. In one embodiment, the containers are formed from a variety of materials such as glass or plastic.

The articles of manufacture provided herein contain packaging materials. Examples of pharmaceutical packaging materials include, but are not limited to, blister packs, bottles, tubes, bags, containers, bottles, and any packaging material suitable for a selected formulation and intended mode of administration and treatment.

A kit typically includes labels listing contents and/or instructions for use, and package inserts with instructions for use. A set of instructions will also typically be included.

In some embodiments, a label is on or associated with the container. In one embodiment, a label is on a container when letters, numbers or other characters forming the label are attached, molded or etched into the container itself; a label is associated with a container when it is present within a receptacle or carrier that also holds the container, e.g., as a package insert. In one embodiment, a label is used to indicate that the contents are to be used for a specific therapeutic application. The label also indicates directions for use of the contents, such as in the methods described herein.

## EXAMPLES

These examples are provided for illustrative purposes only and not to limit the scope of the claims provided herein.

### Example 1

Antigenicity Bioinformatics Workflow for HPV Vaccine Designs  
Generating Consensus Sequences

Sample sequences for E5, E6, and E7 (for HPV16, HPV17, HPV31, HPV33, and HPV45, serotypes determined to have a higher predisposition for cancer) were downloaded, converted to FASTQ files, and imported into R statistical program. Individual AA sequences were read into R using the biostrings package (readAAStringSet function). Multiple sequence alignments were performed using the ClustalW algorithm in the msa R package. For each subgroup/subtype, consensus sequences were generated and output to PDF/FASTQ files using the msaPrettyPrint function.

Predicting Binding Affinity

NetMHC 4.0 was applied to each consensus sequence to predict binding affinity against all major MHC-I alleles (HLA-A0101, HLA-A0201, HLA-A0301, HLA-A2402, HLA-A2601, HLA-B0702, HLA-B0801, HLA-B2705, HLA-B3901, HLA-B4001, HLA-B5801, and HLA-B1501). NetMHC 4.0 uses artificial neural networks to predict the binding affinity of peptide sequences. This analysis was performed for HPV16, HPV18, HPV31, HPV33, and HPV45. Thresholds were arbitrarily established at 0.5% (strong binders) and 2% (weak binders) ranks. Peptides with predicted binding affinity greater than 99.5% were classified as strong binders, and peptides with predicted binding affinity greater than 98% were classified as weak binders. The position of each AA within the peptide sequences were

extracted and used to generate density curves (FIG. 7A and FIG. 7B). Using these density curves, first and second order differentials were calculated to determine peaks for strong and weak binders (FIG. 7C). Finally, the union of these positions was used to extract AA sequences likely to elicit a response.

Predicting Binding Affinity in HPV16, HPV18, HPV31, HPV33, and HPV45

Previous studies have shown that HPV strains 16, 18, 31, 33, and 45 were precursors for cervical carcinoma. To identify candidate peptides with broad coverage across these strains, binding affinity was predicted within each strain. The sequences with strong/weak binding affinity predictions were extracted. Using protein blast, these sequences were aligned against the consensus sequences across all five strains. Alignments were plotted for all five serotypes and coverage was assessed.

### Example 2

#### HPV Antigen Designs

Naturally existing sequence variations on HPV strains could potentially hinder the development of effective HPV vaccines. To address this concern, the present vaccine design approach utilized bioinformatics and protein engineering methods to select and design antigens with broader coverage of T cell epitopes, novel mutations, and enhancer agonist peptides. Drawing on available information of extended coverage of antigen regions with CTL-specific epitopes and in silico prediction results, the designed HPV vaccine antigens can induce robust HPV-16 and HPV-18 specific responses and potentially benefit patients at high risk of HPV-derived cancers.

The new HPV vaccines include the following engineered proteins, peptides and/or modifications:

- 1) truncated E5 regions that fused with E6 and E7 domains of HPV16 and HPV18;
- 2) newly designed point mutations that can potentially abrogate p53 binding with the known PTPN13 binding mutants (Weiking et al., *Cancer Gene Ther.* 19(10):667-74 (2012). doi: 10.1038/cgt.2012.55. Epub 2012 Aug. 24);
- 3) a novel truncation in HPV18 E6 non-immunogenic region at the C-terminus and application of p53 binding knockout mutants;
- 4) a novel truncation of HPV18 E7 non-immunogenic region at the N-terminus and incorporating pRb, Mi2B mutants (Weiking et al., 2012); and/or
- 5) agonist peptides from Tsang, et al., *Vaccine* 35(19): 2605-2611 (2017) doi: 10.1016/j.vaccine.2017.03.025. Epub 2017 Apr. 4 in the newly designed HPV vaccine constructs.

#### HPV Design 1

Consensus sequence information was utilized to select HPV 16/18 reference sequences for the design which included all major variants. The vaccine composition comprising different E6, E7 and E5 protein components with domain boundaries and mutation information is shown in FIG. 3. The reference sequence for E6 was obtained from the

Human Papillomavirus T cell Antigen Database (Bioinformatics Core at Cancer Vaccine Center, Dana-Farber Cancer Institute). Reference sequences for E5 and E7 were obtained from the Papilloma Virus Episteme (PaVE) database.

The design for HPV16 and HPV18 E5 were inspired via a combination of in silico predictions (IEDB and netMHC) and, for HPV16 also included findings of Chen et al., *J Virol.* 78(3):1333-43 (2004); whereas, the design for HPV18 included N terminal residues (1-26) and C terminal residues (41-53, 58-71) (see, FIGS. 4A and 4B).

Inactivating mutations for HPV16 and HPV18 were introduced into E6 and E7 peptide sequences in order prevent oncogenicity (Weiking et al., 2012 October; 19(10):667-74. doi: 10.1038/cgt.2012.55. Epub 2012 Aug. 24). For HPV16 E6, these mutations were E18A, L50G and alanine replacement from 148-151 (ETQL to AAAA). For HPV16 E7, four mutations (H2P, C24G, E46A, and L67R) were included (Weiking et al., 2012). For HPV18 E6, two mutations were included (E18A and L52G). For HPV18 E6, only the N terminus portion was predicted to contain MHC-I binding epitopes (IEDB and netMHC analysis), thus, residues starting at amino acid 121 were removed from these designs. An additional mutation was included in E6 in order to further abrogate its interaction with the p53 protein (Martinez-Zapien et al., *Nature* 529(7587):541-45 (2016)). For HPV18 E7, two mutations (Weiking et al., 2012) were included (E55A and L74R) and residues from (1-40) were removed because HPV-18 E7 contained majority of the predicted MHC-I binding epitopes (IEDB and netMHC predictions) in the C terminal region.

#### HPV Design 2

For designing recombinant multi-epitope proteins as HPV vaccine antigens, a total of 32 key immunogenic peptides, as listed in Table 2, were selected. These peptides included CTL specific peptides from E6 (HPV-16/-18), E7 (HPV-16/-18) and E5 (HPV16) genes. Most of these CTL epitopes were reported with multiple immunological assays, immuno-proteomics, and were included in IEDB resource and netMHC predictions. The novelty of this antigen design is to graft the CTL peptides on to a human ankyrin repeat protein scaffold for possible generation of new protein bearing T cell epitopes and or proteasome cleavage sites.

Human Ankyrin-like repeat ("ALR") protein (PDB code 1QYM) was selected as a scaffold on to which CTL peptides were assembled randomly, enabling different types of protein linker sequences embedded between the peptides. ALR proteins have generally high expression and high stability; the ALR protein provides a scaffold for the HPV peptides and could create novel CTLs. Caution was taken by random shuffling of the peptides to prevent any reformation of E6 and E7 oncogenic proteins.

A homology model (FIG. 4A) was used to assess the overall structural feature and to compare the design against native ankyrin repeats. This model suggested that the grafted CTL peptides on an ankyrin scaffold should retain their tertiary structure. It is expected that the scaffold amino acids connecting those peptides could serve as "linkers" and potentially create new CTL epitopes and or agonist peptide epitopes.

TABLE 2

Peptides from E5 (HPV16), E6 (HPV16 and HPV18), and E7 (HPV16 and HPV18) along with agonist peptides used to assemble multi-epitope vaccine constructs

| SEQ | HPV16-E6 (9) | SEQ | HPV16-E7 (8) | SEQ | HPV16-E5 (1) |
|-----|--------------|-----|--------------|-----|--------------|
|-----|--------------|-----|--------------|-----|--------------|

TABLE 2-continued

| Peptides from E5 (HPV16), E6 (HPV16 and HPV18), and E7 (HPV16 and HPV18)<br>along with agonist peptides used to assemble multi-epitope vaccine constructs |                 |     |                      |     |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|----------------------|-----|-------------------------|
| 113                                                                                                                                                       | HLDDKKQRFHNI    | 122 | LCVQSTHVDI           | 130 | YIIFVYIPL               |
| 114                                                                                                                                                       | RWTGRCMSCC      | 123 | RTLEDLLMGT           |     |                         |
| 115                                                                                                                                                       | TTLEQQYNKPLCDLL | 124 | TLGIVCPI             |     |                         |
| 116                                                                                                                                                       | ISEYRHYCY       | 125 | LLMGTLGIV            |     |                         |
| 117                                                                                                                                                       | VYDFAFRDL       | 126 | TLHEYMLDL            |     |                         |
| 118                                                                                                                                                       | TIHDIILECV      | 127 | AHYNIVTFCC           |     |                         |
| 119                                                                                                                                                       | KLPQLCTEL       | 128 | YMLDLQPETT           |     |                         |
| 120                                                                                                                                                       | FAFRDLCIVY      | 129 | CDSTLRLCV            |     |                         |
| 121                                                                                                                                                       | LCIVYRDGNPYAVCD |     |                      |     |                         |
| SEQ                                                                                                                                                       | HPV18-E6 (8)    | SEQ | HPV18-E7 (6)         | SEQ | Agonist<br>peptides (3) |
| 131                                                                                                                                                       | KLNTNTGLYNL     | 139 | DDLRAFQQFLNLTLS      | 145 | KLPQLCTEV               |
| 132                                                                                                                                                       | KCIDFYSRI       | 140 | FQQFLFLNTL           | 146 | QLYNKPLCDV              |
| 133                                                                                                                                                       | FAFKDLFVV       | 141 | QLFLNTLSFV           | 147 | RTLEDLLMGV              |
| 134                                                                                                                                                       | NLLIRCLRC       | 142 | LFLNTLSFVCPWCAS      |     |                         |
| 135                                                                                                                                                       | KLPDLCTEL       | 143 | TLQDIVLHL            |     |                         |
| 136                                                                                                                                                       | ELTEVFPEFA      | 144 | SEEENDEIDGVNHQHLPARR |     |                         |
| 137                                                                                                                                                       | SLQDIEITCV      |     |                      |     |                         |
| 138                                                                                                                                                       | KTVLELTEV       |     |                      |     |                         |

35

## HPV Design 3

Design 3 is similar to Design 1 with the addition of enhancer agonist peptides (FIG. 3). Overall, designs 1 and 3 contain all necessary sequences from each E5, E6, and E7 (FIGS. 4A and 4B). These were inspired via inventor-selected combinations using guidance from in silico epitope prediction analysis and available information on MHC-I binding and cytokine production following T cell activation.

For HPV16, this design includes: (1) peptides from Tsang et al., *Vaccine* 35(19):2605-2611 (2017) doi: 10.1016/j.vaccine.2017.03.025. Epub 2017 Apr. 4—three peptides that exhibit better MHC-I binding and elicit more robust cytotoxic T cell lymphocyte (CTL) response comprising mutations in E6 (L19V and Q91L/L99V) and mutation in E7 (T86V); (3) two enhancer peptides fused to the inactivated N and C terminus of E6; and (4) one enhancer peptide fused to the inactivated C terminus of E7.

For HPV18, this design includes: (1) three peptides with identical mutations to mimic enhancer agonist peptides similar to peptides of HPV16 (Tsang et al., 2017) comprising the mutations in E6 (L21V and L101V); and (2) two potential mimic enhancer agonists fused to N and C terminus regions of E6 and another agonist mimic fused to the C terminal region of E7. Since HPV18 naturally had the aforementioned mutations in E6 (Q91L) and E7 (T95V) naturally, no additional modifications were needed.

## HPV Design 4

This design was based on Design 2, utilizing the same ankyrin repeat approach. However, Design 4 includes an additional three unique agonist peptides from the peptides of Tsang et al., 2017 for a total of 35 key immunogenic

peptides (Table 2). The order of peptides grafted on the scaffold was again randomized in order to be different from Design 2 and for the potential to generate different CTL epitopes. A homology model of Design 4 was generated (FIG. 4B). The homology of this design was found similar to the Design 2, but indicated local structural variations.

## HPV Design 5

This design is a multi-epitope vaccine, designed by selecting all 35 key immunogenic peptides shown in Table 2. It was assembled with a charged dipeptide KK residue. Advantages of this design include potential for cleavage at “KK” residues and random modification with the “K” residue added at CTL epitopes.

## Example 3

## Assessing the Predicted Binding Affinities of HPV Vaccine Constructs

Bioinformatics predictions for the binding affinity for each design were carried out. Entire sequences were loaded into NetMHC, which was used to assess antigenicity and extent of coverage against various HPV genotypes. It should be noted that Designs 4 and 5 use the same 35 peptides. The main difference is that Design 4 peptides were grafted on Ankyrin protein and Design 5 peptides were connected by “KK” linkers (FIG. 5A). Ultimately, the binding affinity predictions on the matched regions of both designs were similar (FIG. 5B). The HPV designs were compared among themselves (Tables 3 and 4) and with reference sequences which are shown in FIG. 8, FIGS. 9A-9C, and FIGS. 10A-10C.

TABLE 3

| HPV16 Design Elements (HPV Designs 1 and 3) |                                 |                                                                                                                                                                                                                          |
|---------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MOD                                         | HPV type                        | Rationale                                                                                                                                                                                                                |
| 1755804                                     | HPV16 E6 non oncogenic          | Inclusion of p53 and PTPN13 binding site mutations [E18A, L50G, ETQL - AAAA]. These mutations do not allow E6/E7 to degrade p53, pRb, PTPN13, or activate telomerase: Martinez-Zapien et al., 2016; Weiking et al., 2012 |
| 1755805                                     | HPV16 E7 non oncogenic          | Removal of Rb and Mi2B binding sites; prior art in Etubics E7                                                                                                                                                            |
| 1755806                                     | HPV16 E5 (1-40, 58-75)          | Novel truncation of E5 with removal of 2 immuno-dominant regions                                                                                                                                                         |
| 1755814                                     | HPV16 E6 A1 peptide (L19V)      | Agonistic peptides                                                                                                                                                                                                       |
| 1755815                                     | HPV16 E6 A3 peptide (Q91L/L99V) | Agonistic peptides                                                                                                                                                                                                       |
| 1755816                                     | HPV16 E7 A3 peptide (T86V)      | Agonistic peptides                                                                                                                                                                                                       |

TABLE 4

| HPV18 Design Elements (HPV Designs 1 and 3) |                               |                                                                                                                                                            |
|---------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MOD                                         | HPV type                      | Notes                                                                                                                                                      |
| 1755807                                     | HPV18 E6 non oncogenic        | Novel C terminal truncated E6 domain (1-120) and inclusion of p53 binding sites mutations [E18A, L50G]; Martinez-Zapien et al., 2016; Weiking et al., 2012 |
| 1755808                                     | HPV18 E7 non oncogenic        | Novel N terminal truncated E7 domain (40-105), and inclusion of Rb and Mi2B binding mutants [E46A, L67R]; Weiking et al., 2012                             |
| 1755809                                     | HPV18 E5 (1-26; 41-53; 58-71) | Novel truncation of E5 with removal of 2 immuno-dominant regions                                                                                           |
| 1755817                                     | HPV18 E6 A1 peptide (L21V)    | Agonistic peptides                                                                                                                                         |
| 1755818                                     | HPV18 E6 A3 peptide (L101V)   | Agonistic peptides                                                                                                                                         |
| 1755819                                     | HPV18 E7 A3 peptide (T95V)    | Agonistic peptides                                                                                                                                         |

The foregoing description of the specific embodiments will so fully reveal the general nature of the invention that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present invention. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

Other embodiments of the invention will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

The present application claims benefit to U.S. Provisional Application No. 62/639,354 filed Mar. 6, 2018, which is incorporated herein by reference in its entirety.

## EMBODIMENTS

E1. A non-naturally occurring polynucleotide encoding a polypeptide comprising one or more immune response-inducing human papilloma virus (HPV) polypeptides.

E2. The polynucleotide of E1, wherein said non-naturally occurring polynucleotide encodes a polypeptide comprising two or more HPV polypeptides.

E3. The polynucleotide of E2, wherein said two or more HPV polypeptides comprise one or more HPV-16 immune response-inducing polypeptide sequences.

E4. The polynucleotide of E3, wherein said HPV-16 peptide comprises at least one of an E5 peptide, an E6 peptide or an E7 peptide.

E5. The polynucleotide of E3 or E4, wherein said HPV-16 peptide comprises an E5 peptide, and said E5 peptide has a sequence as shown in SEQ ID NO: 47.

E6. The polynucleotide of any one of E3 to E5, wherein said HPV-16 peptide comprises an E6 peptide, and said E6 peptide has a sequence as shown in SEQ ID NO: 45.

E7. The polynucleotide of any one of E3 to E6, wherein said HPV-16 peptide comprises an E7 peptide, and said E7 peptide has a sequence as shown in SEQ ID NO: 46.

E8. The polynucleotide of any one of E1 to E7, wherein said one or more HPV peptides comprises an HPV-18 peptide.

E9. The polynucleotide of E8, wherein said HPV-18 peptide comprises at least one of an E5 peptide, an E6 peptide or an E7 peptide.

E10. The polynucleotide of E8 or 9, wherein said HPV-18 peptide comprises an E5 peptide, and said E5 peptide has a sequence as shown in SEQ ID NO: 50.

E11. The polynucleotide of any one of E8 to 10, wherein said HPV-18 peptide comprises an E6 peptide, and said E6 peptide has a sequence as shown in SEQ ID NO: 48.

E12. The polynucleotide of any one of E8 to 10, wherein said HPV-18 peptide comprises an E7 peptide, and said E7 peptide has a sequence as shown in SEQ ID NO: 49.

E13. The polynucleotide of any one of E1 to E12, wherein said polypeptide has a sequence as shown in SEQ ID NO: 51.

E14. The polynucleotide of any one of E1 to E13, wherein at least one of said one or more HPV polypeptides is connected to an agonist peptide.

E15. The polynucleotide of E14, wherein said agonist peptide has a sequence comprising an agonist peptide sequence as shown in Table 2.

E16. The polynucleotide of E14 or E15, wherein said polypeptide has a sequence as shown in SEQ ID NO: 53.

E17. A polynucleotide comprising the polynucleotide of any one of E1 to E16, further comprising one or more polynucleotides encoding a gene switch system for inducible control of heterologous gene expression, wherein said heterologous gene expression is regulated by said gene switch system; and, wherein said heterologous gene comprises the polynucleotide of any one of E1 to E16.

E18. The polynucleotide of E17, wherein said gene switch system is an ecdysone receptor-based (EcR-based) gene switch system.

E19. The polynucleotide of any one of E1 to E18, wherein said one or more HPV polypeptides is for use in a vaccine.

E20. A vector comprising the polynucleotide of any one of E1 to E19.

E21. The vector of E20, wherein said vector is an adenoviral vector.

E22. The vector of E21, wherein said adenoviral vector is a gorilla adenoviral vector.

E23. A method of regulating the expression of a heterologous gene in a cell, the method comprising: introducing into said cell one or more polynucleotides that comprise (i) an repressible or inducible gene switch, and (ii) a heterologous immune response-inducing gene, wherein expression of said heterologous immune response-inducing gene is regulated by said gene switch, wherein said heterologous immune response-inducing gene encodes one or more HPV polypeptides; and exposing said cell to a compound in an amount sufficient to repress or induce expression of said heterologous immune response-inducing gene.

E24. The method of E23, wherein said target cell is a mammalian cell.

E25. The method of E23 or E24, wherein said gene switch comprises a ligand binding domain derived from at least one of an ecdysone receptor (EcR), a ubiquitous receptor, an orphan receptor 1, an NER-1, a steroid hormone nuclear receptor 1, a retinoid X receptor interacting protein-15, a liver X receptor  $\beta$ , a steroid hormone receptor like protein, a liver X receptor, a liver X receptor  $\alpha$ , a farnesoid X receptor, a receptor interacting protein 14, and a farnesol receptor.

E26. An E6 peptide, wherein said E6 peptide comprises an E18A amino acid substitution and at least one of an L50G, E148A, T149A, Q150A and L151A amino acid substitution as compared to a wildtype E6 peptide.

E27. The E6 peptide of E26, wherein said E6 peptide comprises said E18A amino acid substitution and said L50G, E148A, T149A, Q150A and L151A amino acid substitutions.

E28. The E6 peptide of E26, wherein said E6 peptide has a sequence as shown in SEQ ID NO: 45.

E29. The E6 peptide of any one of E26 to E28, wherein said E6 peptide is fused to an agonist peptide.

E30. The E6 peptide of E29, wherein said agonist peptide is fused to at least one of a C-terminus and an N-terminus of said E6 peptide.

E31. The E6 peptide of any one of E26 to E30, wherein said wildtype E6 peptide is from HPV-16.

E32. An E6 peptide, wherein said E6 peptide comprises a deletion as compared to a wildtype E6 peptide, wherein said deletion comprises a C-terminus of said wildtype E6 peptide.

E33. The E6 peptide of E32, wherein said deletion comprises amino acids from amino acid 121 to a C-terminus of said wildtype E6 peptide.

E34. The E6 peptide of E32 or E33, wherein said E6 peptide comprises at least one of an E18A and L50G substitution as compared to said wildtype E6 peptide.

E35. The E6 peptide of any one of E32 to E34, wherein said wildtype E6 peptide is from HPV-18.

E36. The E6 peptide of any one of E32 to E35, wherein said E6 peptide has a sequence as shown in SEQ ID NO: 48.

E37. An E7 peptide, wherein said E7 peptide comprises a deletion as compared to a wildtype E7 peptide, wherein said deletion comprises an N-terminus of said wildtype E7 peptide.

E38. The E7 peptide of E37, wherein said deletion comprises amino acids 1-39 of said wildtype E7 peptide.

E39. The E7 peptide of E37 or E38, wherein said E7 peptide comprises at least one of an E55A and L74R substitution as compared to said wildtype E7 peptide.

E40. The E7 peptide of any one of E37 to E39, wherein said wildtype E7 peptide is from HPV-18.

E41. The E7 peptide of any one of E37 to E40, wherein said E7 peptide has a sequence as shown in SEQ ID NO: 49.

E42. An E5 peptide, wherein said E5 peptide comprises a deletion as compared to a wildtype E5 peptide, wherein said deletion comprises amino acids 41-57 of said wildtype E5 peptide.

E43. The E5 peptide of E42, wherein said E5 peptide has a sequence as shown in SEQ ID NO: 47.

E44. The E5 peptide of E42 or E43, wherein said wildtype E5 peptide is from HPV-16.

E45. An E5 peptide, wherein said E5 peptide comprises a deletion as compared to a wildtype E5 peptide, wherein said deletion comprises at least one of amino acids 27-40 or amino acids 54-57 of said wildtype E5 peptide.

E46. The E5 peptide of E45, wherein said E5 peptide has a sequence as shown in SEQ ID NO: 50.

E47. The E5 peptide of E45 or E46, wherein said wildtype E5 peptide is from HPV-18.

E48. A polypeptide construct comprising the peptide of any one of E26 to E47.

E49. A polypeptide construct, wherein said polypeptide construct comprises an HPV-16 E6 peptide, wherein said HPV-16 E6 peptide comprises an E18A amino acid substitution and at least one of an L50G, E148A, T149A, Q150A and L151A amino acid substitution as compared to a wildtype HPV-16 E6 peptide.

E50. The polypeptide construct of E49, wherein said HPV-16 E6 peptide comprises said E18A amino acid substitution and said L50G, E148A, T149A, Q150A and L151A amino acid substitutions.

E51. The polypeptide construct of E49 or E50, wherein said HPV-16 E6 peptide has a sequence as shown in SEQ ID NO: 45.

E52. The polypeptide construct of any one of E49 to E51, wherein said polypeptide construct further comprises an HPV-16 E7 peptide, wherein said HPV-16 E7 peptide comprises at least one of an H2P, C24G, E46A and L67R amino acid substitution as compared to a wildtype HPV-16 E7 peptide.

E53. The polypeptide construct of E52, wherein said HPV-16 E7 peptide comprises said H2P, C24G, E46A and L67R amino acid substitutions.

E54. The polypeptide construct of E53, wherein said HPV-16 E7 peptide has a sequence as shown in SEQ ID NO: 46.

E55. The polypeptide construct of any one of E49 to E54, wherein said polypeptide construct further comprises an HPV-16 E5 peptide.

E56. The polypeptide construct of E55, wherein said HPV-16 E5 peptide comprises a deletion of one or more amino acids as compared to a wildtype HPV-16 E5 peptide.

E57. The polypeptide construct of E56, wherein said deletion comprises amino acids 41-57 of said wildtype HPV-16 E5 peptide.

E58. The polypeptide construct of any one of E55 to E57, wherein said HPV-16 E5 peptide has a sequence as shown in SEQ ID NO: 47.

E59. The polypeptide construct of any one of E49 to E58, wherein said polypeptide further comprises an HPV-18 E6 peptide.

E60. The polypeptide construct of E59, wherein said HPV-18 E6 peptide comprises an E18A and L50G substitution as compared to a wildtype HPV-18 E6 peptide.

E61. The polypeptide construct of E59 or E60, wherein said HPV-18 E6 peptide comprises a deletion of at least one C-terminus amino acid relative to said wildtype HPV-18 E6 peptide.

E62. The polypeptide construct of E61, wherein said deletion comprises amino acids from amino acid 121 to said C-terminus of said wildtype HPV-18 E6 peptide.

E63. The polypeptide construct of any one of E59 to E62, wherein said HPV-18 E6 peptide has a sequence as shown in SEQ ID NO: 48.

E64. The polypeptide construct of any one of E49 to E63, wherein said polypeptide construct further comprises an HPV-18 E7 peptide.

E65. The polypeptide construct of E64, wherein said HPV-18 E7 peptide comprises an E55A and L74R substitution as compared to a wildtype HPV-18 E7 peptide.

E66. The polypeptide construct of E64 or E65, wherein said HPV-18 E7 peptide comprises a deletion of at least one amino acid from an N-terminus of said HPV-18 E7 peptide.

E67. The polypeptide construct of E66, wherein said deletion comprises amino acids 1-40 of said wildtype HPV-18 E7 peptide.

E68. The polypeptide construct of any one of E64 to E67, wherein said HPV-18 E7 peptide has a sequence as shown in SEQ ID NO: 49.

E69. The polypeptide construct of any one of E59 to E68, wherein said polypeptide construct further comprises an HPV-18 E5 peptide.

E70. The polypeptide construct of E69, wherein said HPV-18 E5 peptide comprises a deletion of at least one amino acid as compared to a wildtype HPV-18 E5 peptide.

E71. The polypeptide construct of E70, wherein said deletion comprises amino acids 27-40 or 54-57 of said wildtype HPV-18 E5 peptide.

E72. The polypeptide construct of any one of E69 to E71, wherein said HPV-18 E5 peptide has a sequence as shown in SEQ ID NO: 50.

E73. The polypeptide construct of any one of E59 to E72, wherein said polypeptide construct has a sequence as shown in SEQ ID NO: 51.

E74. The polypeptide construct of any one of E5973, wherein said polypeptide construct further comprises at least one agonist peptide.

E75. The polypeptide construct of E74, wherein said at least one agonist peptide has a sequence comprising an agonist peptide sequence as shown in Table 2.

E76. The polypeptide construct of E74 or E75, wherein said polypeptide construct has a sequence as shown in SEQ ID NO: 53.

E77. A polypeptide construct comprising an ankyrin-like repeat domain and an HPV peptide.

E78. The polypeptide construct of E77, wherein said ankyrin-like repeat protein is a human ankyrin-like repeat protein.

E79. The polypeptide construct of E77 or E78, wherein said HPV peptide is linked to said ankyrin-like repeat protein by a linker.

E80. The polypeptide construct of any one of E77 to E79, wherein said HPV peptide comprises at least one of an HPV-16 peptide or an HPV-18 peptide.

E81. The polypeptide construct of any one of E77 to E80, wherein said HPV peptide comprises an HPV-16 peptide, and said HPV-16 peptide comprises at least one of an E5 peptide, an E6 peptide or an E7 peptide.

E82. The polypeptide construct of any one of E77 to E81, wherein said HPV peptide comprises an HPV-18 peptide, and said HPV-18 peptide comprises at least one of an E6 peptide or an E7 peptide.

E83. The polypeptide construct of any one of E77 to E82, wherein said HPV peptide comprises an HPV-16 E5

sequence, an HPV-16 E6 sequence, an HPV-16 E7 sequence, an HPV-18 E6 sequence or an HPV-18 E7 sequence as shown in Table 2.

E84. The polypeptide construct of any one of E77 to E83, wherein said polypeptide construct has a sequence as shown in SEQ ID NO: 52.

E85. The polypeptide construct of any one of E77 to E84, wherein said polypeptide construct further comprises at least one agonist peptide.

E86. The polypeptide construct of E85, wherein said polypeptide construct comprises three agonist peptides.

E87. The polypeptide construct of E86, wherein said polypeptide construct has a sequence as shown in SEQ ID NO: 54.

E88. A polypeptide construct, wherein said polypeptide construct comprises at least two HPV amino acid sequences as shown in Table 2, wherein said at least two HPV amino acid sequences are connected by a peptide linker, wherein said peptide linker is a KK linker.

E89. The polypeptide construct of E88, wherein said at least two HPV amino acid sequences comprise at least one of an HPV-16 peptide or an HPV-18 peptide as shown in Table 2.

E90. The polypeptide construct of E88 or E89, wherein said at least two HPV amino acid sequences comprise an HPV-16 peptide, and said HPV-16 peptide comprises at least one of an HPV-16 E5 peptide, an HPV-16 E6 peptide or an HPV-16 E7 peptide as shown in Table 2.

E91. The polypeptide construct of any one of E88 to E90, wherein said at least two HPV amino acid sequences comprise an HPV-18 peptide, and said HPV-18 peptide comprises at least one of an HPV-18 E6 peptide or an HPV-18 E7 peptide as shown in Table 2.

E92. The polypeptide construct of any one of E88 to E91, wherein said at least two HPV amino acid sequences comprise each of the amino acid sequences as shown in Table 2.

E93. The polypeptide construct of E92, wherein said each of the amino acid sequences is connected to another of said each of the amino acid sequences by said KK linker.

E94. The polypeptide construct of any one of E88 to E93, wherein said polypeptide construct has a sequence as shown in SEQ ID NO: 55.

E95. The polypeptide construct of any one of E48 to E94 for use in a vaccine.

E96. A polynucleotide encoding the polypeptide construct of any one of E58 to E95.

E97. A vector comprising the polynucleotide of E96.

E98. The vector of E97, wherein said vector is an adenoviral vector.

E99. The vector of E98, wherein said adenoviral vector is a gorilla adenoviral vector.

E100. A vector, wherein said vector comprises a polynucleotide that encodes at least one HPV peptide, wherein said vector is an adenoviral vector.

E101. A vector, wherein said vector comprises a polynucleotide that encodes at least one HPV peptide, wherein said vector is an adenoviral vector, wherein said adenoviral vector is a gorilla adenoviral vector.

Provided herein is a representative reference list of certain sequences included in embodiments provided herein (Table 5).

71

TABLE 5

| Polynucleotide/Amino Acid Sequences |                                                     |
|-------------------------------------|-----------------------------------------------------|
| SEQ ID NO:                          | Description                                         |
| 1                                   | Adenovirus pIX fragment nucleotides                 |
| 2                                   | Adenovirus DNA polymerase fragment nucleotides      |
| 3                                   | Adenovirus penton base protein fragment nucleotides |
| 4                                   | Adenovirus hexon protein fragment nucleotides       |
| 5                                   | Adenovirus fiber protein fragment nucleotides       |
| 6                                   | Adenovirus pIX nucleotides                          |
| 7                                   | Adenovirus DNA polymerase nucleotides               |
| 8                                   | Adenovirus penton base protein nucleotides          |
| 9                                   | Adenovirus hexon protein nucleotides                |
| 10                                  | Adenovirus fiber protein nucleotides                |
| 11                                  | Adenovirus pIX protein fragment                     |
| 12                                  | Adenovirus DNA polymerase fragment                  |
| 13                                  | Adenovirus penton base protein fragment             |
| 14                                  | Adenovirus hexon protein fragment                   |
| 15                                  | Adenovirus fiber protein fragment                   |
| 16                                  | Adenovirus pIX amino acids                          |
| 17                                  | Adenovirus DNA polymerase amino acids               |
| 18                                  | Adenovirus penton base protein                      |
| 19                                  | Adenovirus hexon protein                            |
| 20                                  | Adenovirus fiber protein                            |
| 21                                  | Adenovirus vector nucleotide sequences              |
| 22                                  | Adenovirus vector nucleotide sequences              |
| 23                                  | Adenovirus vector nucleotide sequences              |
| 24                                  | Adenovirus vector nucleotide sequences              |
| 25                                  | Adenovirus vector nucleotide sequences              |
| 26                                  | IL-2 core promoter                                  |
| 27                                  | IL-2 minimal promoter                               |
| 28                                  | IL-2 enhancer and promoter variant                  |
| 29                                  | L-2 enhancer and promoter variant                   |
| 30                                  | (NF- $\kappa$ B) <sub>1</sub> -IL2 promoter variant |
| 31                                  | (NF- $\kappa$ B) <sub>3</sub> -IL2 promoter variant |
| 32                                  | (NF- $\kappa$ B) <sub>6</sub> -IL2 promoter variant |
| 33                                  | 1X NFAT response elements-IL2 promoter variant      |
| 34                                  | 3X NFAT response elements-IL2 promoter variant      |
| 35                                  | 3X NFAT response elements-IL2 promoter variant      |

72

TABLE 5-continued

| Polynucleotide/Amino Acid Sequences |                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------|
| SEQ ID NO:                          | Description                                                                          |
| 36                                  | 6X NFAT response elements-IL2 promoter variant                                       |
| 37                                  | 6X NFAT response elements-IL2 promoter variant                                       |
| 38                                  | 6X NFAT response elements-IL2 promoter variant                                       |
| 39                                  | 6X NFAT response elements-IL2 promoter variant                                       |
| 40                                  | human EEF1A1 promoter variant                                                        |
| 41                                  | human EEF1A1 promoter variant                                                        |
| 42                                  | human EEF1A1 promoter and enhancer                                                   |
| 43                                  | human UBC promoter                                                                   |
| 44                                  | synthetic minimal promoter 1                                                         |
| 45                                  | HPV antigen design 1 HPV16 E6 amino acids                                            |
| 46                                  | HPV antigen design 1 HPV16 E7 amino acids                                            |
| 47                                  | HPV antigen design 1 HPV16 E5 amino acids                                            |
| 48                                  | HPV antigen design 1 HPV18 E6 amino acids                                            |
| 49                                  | HPV antigen design 1 HPV18 E7 amino acids                                            |
| 50                                  | HPV antigen design 1 HPV18 E5 amino acids                                            |
| 51                                  | HPV antigen design 1 amino acids                                                     |
| 52                                  | HPV antigen design 2 amino acids                                                     |
| 53                                  | HPV antigen design 3 amino acids                                                     |
| 54                                  | HPV antigen design 4 amino acids                                                     |
| 55                                  | HPV antigen design 5 amino acids                                                     |
| 56                                  | HPV antigen design 1 full nucleotide sequences of gorilla adenovirus shuttle plasmid |
| 57                                  | HPV antigen design 2 full nucleotide sequences of gorilla adenovirus shuttle plasmid |
| 58                                  | HPV antigen design 3 full nucleotide sequences of gorilla adenovirus shuttle plasmid |
| 59                                  | HPV antigen design 4 full nucleotide sequences of gorilla adenovirus shuttle plasmid |
| 60                                  | HPV antigen design 5 full nucleotide sequences of gorilla adenovirus shuttle plasmid |
| 61                                  | GCAd-RTS-IL12 design 1                                                               |
| 62                                  | GCAd-RTS-IL12 design 2                                                               |
| 63                                  | GCAd-RTS-IL12 design 3                                                               |

## SEQUENCE LISTING

&lt;160&gt; NUMBER OF SEQ ID NOS: 147

&lt;210&gt; SEQ ID NO 1

&lt;211&gt; LENGTH: 30

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Adenovirus pIX fragment nucleotides

&lt;400&gt; SEQUENCE: 1

agctcttttg tggcgagcgg cgcggcctct

30

&lt;210&gt; SEQ ID NO 2

&lt;211&gt; LENGTH: 439

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Adenovirus DNA polymerase fragment nucleotides

&lt;400&gt; SEQUENCE: 2

aacatcaata cctcaaagtc atggtcaggg acactttcgc cctcaccac acctccctcc 60

gcaaggcggc gcaggcctac gcgtgcccc tggagaaggg ctgttgcccc taccaggccg 120

tcaaccagtt ctacatgcta ggctcttacc gtcggacac ggacgggttt cccctccaag 180

agtactggaa agaccgcgaa gagttcgtcc tcaaccgcga gctgtggaaa aagaaggggg 240

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-continued

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ccgccgagct ggtgcacaag ctgcgcgagt cctacgcctc cttegtcagg gactcgggtg 360
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ccatcttcag gcagatcgc 439

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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus penton base protein fragment
nucleotides

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<400> SEQUENCE: 3

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```

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<210> SEQ ID NO 4
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus hexon protein fragment nucleotides

```

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<400> SEQUENCE: 4

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```

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<210> SEQ ID NO 5
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus fiber protein fragment nucleotides

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<400> SEQUENCE: 5

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gtagcaggcc ccctagctgt ggccaatggc 30

```

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<210> SEQ ID NO 6
<211> LENGTH: 399
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus pIX nucleotides

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<400> SEQUENCE: 6

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gttctgcctt caaatcgtc tacaatggcc tacgcgaccg tgggaggaac tccgctggac 180
gccgcgacct ccgcgcgcgc ctccgcgcgc gccgcgaccg cgcgcagcat ggctacggac 240
ctttacagct ctttggtggc gagcggcgcg gcctctcgcg cgtctgctcg ggatgagaaa 300
ctgaccgctc tgctgcttaa actggaagac ttgacccggg agctgggtca actgaccag 360
caggtctcca gcttgctga gagcagcctt gcctcccc 399

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```

<210> SEQ ID NO 7
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<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus DNA polymerase nucleotides

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<400> SEQUENCE: 7

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-continued

|            |             |            |            |             |             |      |
|------------|-------------|------------|------------|-------------|-------------|------|
| tggacctgcg | gctctcgcg   | cgtggtggtc | tgetccacca | tcgcccctcca | ggagacagat  | 120  |
| gtggcgggcc | agacaaccaa  | agtagaagac | caccagccac | acgggacccc  | aggcggggga  | 180  |
| cttagattcc | cgctcgctt   | cctcgtcaga | ggtcgccagg | ttcacctcgt  | gcaagatata  | 240  |
| caaccctgct | agcgtgcca   | gtactgcggt | cgtttttaca | aaagccagca  | cgagtgtctg  | 300  |
| gcccgcagac | gggacttcta  | ctttcaccac | atcaacagcc | aatcctccaa  | ctggtggcgg  | 360  |
| gagatccagt | tcttcccgat  | cggctctcat | cctcgacagg | agcgctctt   | tgtcacctac  | 420  |
| gatgtagaga | cctacacttg  | gatgggagcc | tttggaagc  | agctcgtgcc  | cttcagtctg  | 480  |
| gtcatgaaac | tggggggcga  | cgaggctctg | gtcgccgccc | cgcgcgacct  | cggccgagag  | 540  |
| ctcagatggg | acccttgagg  | gaaagacccc | ctcaccttct | actgcatcac  | ccccgaaaag  | 600  |
| atggccgtgg | ggcgacagtt  | cagaaccttc | cgcgacccgc | tgcagacct   | catggcccgc  | 660  |
| gacctctggc | gatccttct   | ggcgcccaac | cctcacttgc | aagactgggc  | cctggaggag  | 720  |
| cacggcctgg | aatcgccga   | ggagctcacc | tacgaggaac | tcaaaaagct  | ccccctccatc | 780  |
| aagggccagc | cccgttttt   | ggagctctac | atcgtgggcc | acaacataaa  | cggttttgac  | 840  |
| gagatcgctc | tggccgcccc  | ggtcatcaac | aaccgctcct | cggctccagg  | gccctttcgc  | 900  |
| atcaccagaa | acttcagtc   | tcgagcgggg | aagatcctct | tcaatgacct  | caccttctcc  | 960  |
| ctgcccaccc | cgcgtcccaa  | aaagcgcacg | gactacaccc | tgtgggaaca  | ggcggtgtgc  | 1020 |
| gatgacacag | acttcaaaac  | tcaatacctc | aaagtcatgg | tcagggacac  | tttcgcccctc | 1080 |
| acccacacct | ccctccgcaa  | ggcgccgcag | gcctacgcgc | tgcccgtgga  | gaagggtgt   | 1140 |
| tgcccctacc | agggcgccaa  | ccagttctac | atgctaggct | cttaccgttc  | ggacacggac  | 1200 |
| gggtttcccc | tccaagagta  | ctggaaagac | cgcgaagagt | tcgtcctcaa  | ccgcgagctg  | 1260 |
| tggaaaaaga | aggggggagga | taagtatgac | atcatccgcg | agacctcga   | ctactcgcg   | 1320 |
| ctcgacgtcc | aggtcacgcg  | cgagctggtg | cacaagctgc | gcgagtccta  | cgcctccttc  | 1380 |
| gtcagggact | cgggtggctt  | gcaagaagca | agcttcaacg | tcttcagcg   | gcccaccatc  | 1440 |
| tctccaact  | cccagccat   | cttcaggcag | atcgccctcc | gcgccgagcg  | ccccagcgc   | 1500 |
| accaacctcg | ggcccaacat  | gctggccccc | tcccacgagc | tctatgacta  | cgtgcgcgcc  | 1560 |
| agcatccgcg | ggggggcgtg  | ctaccccacc | tacctcgca  | tcctcaggga  | acccctgtac  | 1620 |
| gtgtatgaca | tctcgggcat  | gtacgcctcc | gcgtcacccc | accccatgcc  | ctggggcccc  | 1680 |
| ccctcaacc  | cctacgagcg  | cgcgtcgc   | gcccgcgaat | ggcagcgggc  | tctggacatg  | 1740 |
| caagcttgca | agatcgacta  | ctttgacccg | cgttgctcc  | cgggggtctt  | caccatcgac  | 1800 |
| gcggaccccc | caaacgagga  | ccagctggac | cccctacccc | ccttctgtct  | gcgcaagggc  | 1860 |
| ggccgcctct | gctggaccaa  | cgagcgctg  | cgcggcgagg | tcgccaccag  | cgtcgacatg  | 1920 |
| gtcaccctgc | acaaccgagg  | ctggagggtg | cgcctaattc | cagacgagcg  | caccaccgtc  | 1980 |
| ttccccagtg | ggaagtgcgt  | ggcccgcgag | tacgtgcaac | tcaacatcgc  | ggccaaggag  | 2040 |
| cgagccgacc | gcgacaaaaa  | ccagacctg  | cgtccatcg  | ccaagctgct  | ctccaacgcc  | 2100 |
| ctctacgggt | cgttcgccac  | caagcttgac | aacaaaaaaa | tagtgttttc  | tgaccagatg  | 2160 |
| gaccaggtga | ccctcaaagg  | tatcacctcc | ggacaggtga | acatcaaata  | ctctcatttt  | 2220 |
| ttagaaactg | acaacctgag  | cgtgagggtc | atgcccgct  | tcgagaggga  | atacttacct  | 2280 |
| cagcagctgg | ccctcgcaga  | cagcgatcg  | gaagagagt  | aagatgaaag  | ggcgccacc   | 2340 |
| cccttttata | cccccccgtc  | gggaaccccc | ggtcacgtgt | cctacacct   | caagccaata  | 2400 |
| acttttctgg | acgcggagga  | gggggacatg | tgctgcaca  | ccctggagaa  | ggtggacccg  | 2460 |

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ctagtggaca acgaccgeta cccctccac gtggcctcct tcgtcctggc ctggacgcgg 2520
gccttcgtct cagagtgggc agagtctctc tacgaggagg acagaggcac tccgctggaa 2580
gacaggcccc tgaagtcggt ctacggggac acggacagcc tcttcgtcac cgagaaggga 2640
caccgcctca tggagagcgc aggttaagaaa cgcatacaaa agcatggggg caacctgggt 2700
tttgaccctg accgcccggg gctcacttgg ctggtggaat gcgagacggt ctgcgcttcc 2760
tgcggcgcgg acgcctactc cccagagtcc gtgtttctcg ctcccaagct ctacgcctcg 2820
aagagcctgc agtgcccctc gtgcggcgcc acctccaagg gaaagctccg cgccaagggg 2880
cacgcgcgcg aggggtctcg ctacgagacc atggtcaaat gctacctggc cgacgcgcag 2940
ggcgaagagc ggcagcgatt cagcaccagc agaaccagcc tcaagcgcac cctggccagc 3000
gcccaccccg gagcgacccc cttcacctg acccagacca cctgacgag gacctgcgc 3060
ccatggaagg acatgactct ggcgccgctg gacgcccac ggctggtgcc ctacagcgaa 3120
agccgcccc aaccgcgaaa cgaggagatc tgctggatcg agatgccg 3168

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&lt;210&gt; SEQ ID NO 8

&lt;211&gt; LENGTH: 1974

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Adenovirus penton base protein nucleotides

&lt;400&gt; SEQUENCE: 8

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atgcggcgcg cgcgatggt cgaggagggg cctccccct cttacgagag cgcatgggg 60
atttctcctg cgcgccccc gcagcctccc tacgtgcctc ctcggtacct gcaacctaca 120
ggggggagaa atagcatctg ttactctgag ctgcagcccc tgtacgatac caccagactg 180
tacctgggtg acaacaagtc cgcggacgtg gcctccctga actaccagaa cgaccacagc 240
gattttttga ccacgggtgat ccaaaacaac gacttcaccc caaccgaggc cagcaccag 300
accataaacc tggataacag gtcgaactgg ggcggcgacc tgaagaccat cttgcacacc 360
aacatgcccc acgtgaacga gttcatgttc accaactctt ttaaggcgcg ggtgatgggtg 420
gcgcgcgagc aggggggaggc gaagtacgag tgggtggact tcacgctgcc cgagggcaac 480
tactcagaga ccatgactct cgacctgatg aacaatgcga tcgtggaaca ctatctgaaa 540
gtgggcaggc agaacggggt gaaggaaaac gatatcgggg tcaagtttga caccagaaac 600
ttcctgtcgg gctgggaccc cgtgaccggg ctggtcacgc cgggggtcta caccaacgag 660
gcctttcatc ccgacatagt gcttctgccc ggctgtgggg tggacttcac ccagagccgg 720
ctgagcaacc tgctgggcat tcgcaagcgg cagcctttcc aggagggttt caagatcacc 780
tatgaggatc tgaagggggg caacattccc gcgctccttg atctggacgc ctacgaggag 840
agcttgaaac ccgaggagag cgctggcgac agcggcgaga gtggcgagga gcaagccggc 900
ggcggtgggc gcgcgtcggt agaaaaagaa agtacgcccg cagtggcggc ggacgctgcg 960
gaggtcgagc cggaggccat gcagcaggac gcagaggagg gcgcacagga gggcgcgag 1020
aaggacatga acgatgggga gatcagggga gacacattcg ccacccgggg cgaagaaaaa 1080
gaggcagagg cggcgccggc ggcgacggcg gagggcgaac ccgaggttga ggcagaggca 1140
gagcccagag ccgaagtatt ggaagacatg aatgatggag aacgtagggg cgacacgttc 1200
gccacccggg gcgaagagaa ggcggcgagg gcagaagccg cggctgagga ggcggctgcg 1260
gctgcggcca agactgaggc tcgggctaag gctgagggtc aagccaatgt tcgggttgag 1320

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|                                                                     |      |
|---------------------------------------------------------------------|------|
| gctcaggctg aggaggaggg ggcggctgaa gcagttaagg aaaaggccca ggcagagcag   | 1380 |
| gaagagaaaa aacctgtcat tcaacctcta aaagaagata gcaaaaagcg cagttacaac   | 1440 |
| gtcatcgagg gcagcacctt taccagtag cgcagctggc acctggcgta caactacggc    | 1500 |
| gacccgggtca aggggggtgcg ctctgtggacc ctgctctgca cgccggagct cactcgggc | 1560 |
| tccgagcaga tgtactggtc gctgccgaac atgatgcaag acccggtgac cttccgctcc   | 1620 |
| acgcggcagg ttagcaactt cccgggtgtg ggcgcgaac tgcgtcccg gcactccaag     | 1680 |
| agtttttaca acgagcaggc cgtctactcc cagctgatcc gccaggccac ctctctgacc   | 1740 |
| cacgtgttca atcgctttcc cgagaaccag attttggcgc gcccgccggc cccaccatc    | 1800 |
| accaccgtga gtgaaaacgt tctgcccctc acagatcacg ggacgctacc gctgcgcaac   | 1860 |
| agcatctcag gagtccagcg agtgaccatt actgacgcca gacgcggac ctgcccctac    | 1920 |
| gtttacaagg ccttgggcat agtctcgccg cgcgtcctct ccagtcgcac tttt         | 1974 |

&lt;210&gt; SEQ ID NO 9

&lt;211&gt; LENGTH: 2877

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Adenovirus hexon protein nucleotides

&lt;400&gt; SEQUENCE: 9

|                                                                    |      |
|--------------------------------------------------------------------|------|
| atggcgaccc catcgatgat gcctcagtg tctacatgc acatctcggg ccaggacgct    | 60   |
| tccgagtacc tgagcccgg gctggtgcag ttcgcccgg ccacagacac ctacttcaac    | 120  |
| atgagtaaca agttcaggaa cccactgtg gcgcccacc acgatgtgac caggacccg     | 180  |
| tgcagcgcc tgacgctgcg gttcatcccc gtggatcggg aggacaccgc ctactcttac   | 240  |
| aaggcgggt tcacgctggc cgtggcgac aaccgctgc tggacatggc ctccacttac     | 300  |
| tttgacatca ggggggtgct ggacaggggc cccaccttca agccctactc gggactgccc  | 360  |
| tacaactccc tggccccc aa ggcgctccc aattcttgcg agtgggaaca agatgaacca  | 420  |
| gctcaggcag caatagtgta agatgaagaa gaactgaag aagaacaagc tcaggacgaa   | 480  |
| caggcgccca ctaagaaaac ccatgtatac gcccaggcac ctctttctgg tgaaaaaatt  | 540  |
| actaaggatg gtttgcaaat aggtgtggat gccacacagg cgggagataa ccctatatat  | 600  |
| gctgataaaa cattccaacc cgaacctcag ataggtgagt ctcagtggaa cgaggctgat  | 660  |
| gccacagtag caggaggcag agtcttaaaa aagaccacc ctatgagacc ttgctatgga   | 720  |
| tcctatgcca aacctactaa tgccaatggc ggtcaaggga tcatgggtgc caatgatcag  | 780  |
| ggagcgcttg aatctaaagt tgagatgcaa tttttctcca ccacaacgct tcttaatgta  | 840  |
| agggaaagtg aaaacaatct tcagccaaaa gtagtgctat acagcgaaga tgtaacttg   | 900  |
| gaatccccctg acactcatct gtcttcaaaa cctaaaaagg atgacaccaa ctctaaaatc | 960  |
| atgttgggtc agcaagccat gcccaacaga cccaacctca ttgcttttag ggacaacttt  | 1020 |
| attggactta tgtactacaa cagcacaggc aacatgggag tgcctggcagg acaggcctcc | 1080 |
| cagctaaacg ctgtggtaga cttgcaagac agaaacacag agctgtcata ccaactgatg  | 1140 |
| cttgattcca ttggagacag atcaagatac ttttccatgt ggaaccaggc agtggacagc  | 1200 |
| tatgaccag atgtcagaat cattgaaaac catgggggtg aagatgagct gcccaactat   | 1260 |
| tgctttcccc tggggggtat tggaattaca gacacatacc agtgcataaa accaaccgca  | 1320 |
| gctgctaata acactacatg gtctaaggat gaagaattta gtgatcgaa tgaaataggg   | 1380 |
| gtgggaaaca acttgcctat ggagatcaac atccaggcca acctctggag gaacttctc   | 1440 |

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|                                                                   |      |
|-------------------------------------------------------------------|------|
| tatgccaacg tggggctcta cctgccagac aagctcaagt acaacccac caacgtggac  | 1500 |
| atctctgaca accccaacac ctatgactac atgaacaagc gtgtggtggc tcccggcctg | 1560 |
| gtggactgct ttgtcaatgt gggagccagc tggccctgg actacatgga caacgtcaac  | 1620 |
| cccttcaacc accaccgcaa tgcgggtctg cgctaccgct ccatgatcct gggcaacggg | 1680 |
| cgctacgtgc ccttcacat tcaggtgccc cagaagttct ttgccatcaa gaacctctc   | 1740 |
| ctcctgccgg gctcctacac ttacgagtgg aacttcagga aggatgtcaa catggctctg | 1800 |
| cagagctctc tgggcaatga ccttaggtg gacggggcca gcatcaagtt tgacagcgtc  | 1860 |
| acctctatg ctaccttctt ccccatggct cacaacaccg cctccacgct cgaggccatg  | 1920 |
| ctgaggaacg acaccaacga ccagtccttc aatgactacc tctctggggc caacatgctc | 1980 |
| taccccatcc cgcacaagc caccaacgtg cccatctcca ttcctctcg caactgggc    | 2040 |
| gccttcagag gctgggcctt tacccgctt aagaccaagg aaacccctc cctgggctcg   | 2100 |
| ggttttgacc cctactttgt ctactcggga tccatccct acctggatgg cacctctac   | 2160 |
| ctcaaccaca cttttaagaa gatatccatc atgtatgact cctccgtcag ctggccgggc | 2220 |
| aatgaccgcc tgcaccccc caatgagttc gaggtcaagc gcgccgtgga cggcgagggc  | 2280 |
| tacaacgtgg ccagtgcaa catgaccaag gactggttcc tgggtcagat gctggccaac  | 2340 |
| tacaacatag gctaccaggg cttctacatc ccagagagct acaaggacag gatgtactcc | 2400 |
| ttcttcagaa atttcaacc catgagcagg caggtggtgg acgagacaa atacaaggac   | 2460 |
| tatcaggcca ttggcatcac tcaccagcac aacaactcgg gattcgtggg ctacctggct | 2520 |
| cccaccatgc gcgaggggca ggccatcccc gccaaactcc cctaccggt gataggcaaa  | 2580 |
| accgcggtcg acagcgtcac ccagaaaaag ttcctctcg accgcacct ctggcgcac    | 2640 |
| cccttctcta gcaacttcac gtccatgggt gcgctcacgg acctgggcca gaacctgctc | 2700 |
| tatgccaact ccgccatgc gctggacatg acttttgagg tggaccccat ggacgagccc  | 2760 |
| accttctct atattgtgt tgaagtgttc gacgtggtca gagtgcacca gccgcaccgc   | 2820 |
| ggtgtcatcg agaccgtgta cctgcgcacg cccttctcgg ccggcaacgc caccacc    | 2877 |

&lt;210&gt; SEQ ID NO 10

&lt;211&gt; LENGTH: 1749

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Adenovirus fiber protein nucleotides

&lt;400&gt; SEQUENCE: 10

|                                                                   |     |
|-------------------------------------------------------------------|-----|
| atgaaacgcg cgagatcgtc tgacgagacc ttcaaccccg tgtacccta cgataccgag  | 60  |
| atcgctccga cttctgtccc ttctctacc cctccctttg tgctatccgc aggaatgcaa  | 120 |
| gaaaatccag ctggggtgct gtccctgcac ttgtcagagc cccttaccac ccacaatggg | 180 |
| gccctgactc taaaaatggg gggcggcctg accctggaca aggaagggaa tctcacttcc | 240 |
| caaaacatca ccagtgtcga tcccccttc aaaaaagca agaacaacat cagccttcag   | 300 |
| accgcgcgac cctcgcgct cagctccggg gccctaacac tttttgccac tcccccccta  | 360 |
| gcggtcagtg gtgacaacct tactgtgcag tctcaggccc ctctcacttt ggaagactca | 420 |
| aaactaactc tggccaccaa aggaccccta actgtgtccg aaggcaact tgcctagaa   | 480 |
| acagaggctc cctgcgatgc aagtgcagc agcagcctgg gccttagcgt tacggcccca  | 540 |
| cttagcatta acaatgacag cctaggacta gatctgcagg caccattgt ctctcaaat   | 600 |

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ggaaaactgg ctctaaatgt agcaggcccc ctagctgtgg ccaatggcat taatgctttg 660
acagtaggca caggcaaagg tatttgtcta aatgaaacca gcaactcatt gcaagcaaag 720
ttggtcgccc ccctaggctt tgataccaat ggcaacatta agctaagcgt tgcaggaggc 780
atgagactaa ataatgacac acttatacta gatgtaaact acccatttga agtcaaggc 840
caactaagtc taagagtggg ccagggtccg ctgtatgtag attctagcag ccataacctg 900
accattagat gccttagagg attatacata acatcgtcta ataaccaaac cggcttagag 960
gccaacataa aactaacaaa aggccttgtc tatgatggaa atgccatagc agtcaatgtt 1020
ggtaaggat tgcaatacag cactactgcc acatcggaag gtgtgtatcc tatacagtct 1080
aagatagggt tgggaatgga atatgatacc aacggagcca tgatgacaaa actaggctct 1140
ggactaagct ttgacaattc aggagccatt gtagtgggaa acaaaaatga tgacaggctt 1200
actctgtgga ctacaccaga cccatctcct aactgtagaa tttattctga aaaagatact 1260
aaactaacct tgggtgtgac taagtgtggc agccaaatcc taggcacagt atctgccctt 1320
gctgtcagag gcagccttgc gcccatcact aatgcatcca gcatagtcca aatatttcta 1380
agatttgatg aaaatggact attgatgagc aactcatcgc tagacgggtga ttactggaat 1440
tacagaaatg gggactccac taatagcaca ccatatacaa atgcagtagg ctttatgcct 1500
aatctagcag cctatcctaa aggtcaggct acagctgcaa aaagcagtat tgtaagccag 1560
gtatacatgg atggtgacac tactaaacct ataacactaa aaataaactt caatggcatt 1620
gatgaaacaa cagaaaatac cctgttagt aaatattcca tgacattctc atggagctgg 1680
cccaccgcaa gctacatagg ccacactttt gcaacaaact cttttacttt ctctacatc 1740
gcccaagaa 1749

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<210> SEQ ID NO 11
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus pIX protein fragment

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<400> SEQUENCE: 11

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Ser Ser Leu Val Ala Ser Gly Ala Ala Ser
1          5          10

```

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<210> SEQ ID NO 12
<211> LENGTH: 147
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Adenovirus DNA polymerase fragment

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<400> SEQUENCE: 12

```

```

Lys His Gln Tyr Leu Lys Val Met Val Arg Asp Thr Phe Ala Leu Thr
1          5          10          15

```

```

His Thr Ser Leu Arg Lys Ala Ala Gln Ala Tyr Ala Leu Pro Val Glu
20          25          30

```

```

Lys Gly Cys Cys Pro Tyr Gln Ala Val Asn Gln Phe Tyr Met Leu Gly
35          40          45

```

```

Ser Tyr Arg Ser Asp Thr Asp Gly Phe Pro Leu Gln Glu Tyr Trp Lys
50          55          60

```

```

Asp Arg Glu Glu Phe Val Leu Asn Arg Glu Leu Trp Lys Lys Lys Gly
65          70          75          80

```

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Glu Asp Lys Tyr Asp Ile Ile Arg Glu Thr Leu Asp Tyr Cys Ala Leu

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|    |    |    |
|----|----|----|
| 85 | 90 | 95 |
|----|----|----|

Asp Val Gln Val Thr Ala Glu Leu Val His Lys Leu Arg Glu Ser Tyr  
           100                  105                  110

Ala Ser Phe Val Arg Asp Ser Val Gly Leu Gln Glu Ala Ser Phe Asn  
           115                  120                  125

Val Phe Gln Arg Pro Thr Ile Ser Ser Asn Ser His Ala Ile Phe Arg  
           130                  135                  140

Gln Ile Ala  
145

<210> SEQ ID NO 13  
 <211> LENGTH: 15  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Adenovirus penton base protein fragment

<400> SEQUENCE: 13

Lys Thr Glu Ala Ala Ala Lys Ala Glu Val Glu Ala Asn Val Ala  
 1                  5                  10                  15

<210> SEQ ID NO 14  
 <211> LENGTH: 15  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Adenovirus hexon protein fragment

<400> SEQUENCE: 14

Ile Gly Val Asp Ala Thr Gln Ala Gly Asp Asn Pro Ile Tyr Ala  
 1                  5                  10                  15

<210> SEQ ID NO 15  
 <211> LENGTH: 15  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Adenovirus fiber protein fragment

<400> SEQUENCE: 15

Leu Asn Val Ala Gly Pro Leu Ala Val Ala Asn Gly Ile Asn Ala  
 1                  5                  10                  15

<210> SEQ ID NO 16  
 <211> LENGTH: 133  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Adenovirus pIX amino acids

<400> SEQUENCE: 16

Met Ser Asp Thr Gly Asn Ser Phe Asp Gly Ser Ile Phe Ser Pro Tyr  
 1                  5                  10                  15

Leu Thr Val Arg Met Pro His Trp Ala Gly Val Arg Gln Asn Val Met  
           20                  25                  30

Gly Ser Asn Val Asp Gly Arg Pro Val Leu Pro Ser Asn Ser Ser Thr  
           35                  40                  45

Met Ala Tyr Ala Thr Val Gly Gly Thr Pro Leu Asp Ala Ala Thr Ser  
           50                  55                  60

Ala Ala Ala Ser Ala Ala Ala Thr Ala Arg Ser Met Ala Thr Asp  
 65                  70                  75                  80

Leu Tyr Ser Ser Leu Val Ala Ser Gly Ala Ala Ser Arg Ala Ser Ala

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| 85                                                              | 90  | 95  |
|-----------------------------------------------------------------|-----|-----|
| Arg Asp Glu Lys Leu Thr Ala Leu Leu Leu Lys Leu Glu Asp Leu Thr |     |     |
| 100                                                             | 105 | 110 |
| Arg Glu Leu Gly Gln Leu Thr Gln Gln Val Ser Ser Leu Arg Glu Ser |     |     |
| 115                                                             | 120 | 125 |
| Ser Leu Ala Ser Pro                                             |     |     |
| 130                                                             |     |     |
| <210> SEQ ID NO 17                                              |     |     |
| <211> LENGTH: 1056                                              |     |     |
| <212> TYPE: PRT                                                 |     |     |
| <213> ORGANISM: Artificial sequence                             |     |     |
| <220> FEATURE:                                                  |     |     |
| <223> OTHER INFORMATION: Adenovirus DNA polymerase amino acids  |     |     |
| <400> SEQUENCE: 17                                              |     |     |
| Met Asp Ser Ser Asn Val Arg Asp Val Val Ile Lys Leu Arg Pro Pro |     |     |
| 1                                                               | 5   | 10  |
| Ser Ala Glu Ile Trp Thr Cys Gly Ser Arg Gly Val Val Val Cys Ser |     |     |
| 20                                                              | 25  | 30  |
| Thr Ile Ala Leu Gln Glu Thr Asp Ala Gly Gly Gln Thr Thr Lys Val |     |     |
| 35                                                              | 40  | 45  |
| Glu Asp His Gln Pro His Gly Thr Pro Gly Gly Gly Leu Arg Phe Pro |     |     |
| 50                                                              | 55  | 60  |
| Leu Arg Phe Leu Val Arg Gly Arg Gln Val His Leu Val Gln Asp Ile |     |     |
| 65                                                              | 70  | 75  |
| Gln Pro Val Gln Arg Cys Gln Tyr Cys Gly Arg Phe Tyr Lys Ser Gln |     |     |
| 85                                                              | 90  | 95  |
| His Glu Cys Ser Ala Arg Arg Arg Asp Phe Tyr Phe His His Ile Asn |     |     |
| 100                                                             | 105 | 110 |
| Ser Gln Ser Ser Asn Trp Trp Arg Glu Ile Gln Phe Phe Pro Ile Gly |     |     |
| 115                                                             | 120 | 125 |
| Ser His Pro Arg Thr Glu Arg Leu Phe Val Thr Tyr Asp Val Glu Thr |     |     |
| 130                                                             | 135 | 140 |
| Tyr Thr Trp Met Gly Ala Phe Gly Lys Gln Leu Val Pro Phe Met Leu |     |     |
| 145                                                             | 150 | 155 |
| Val Met Lys Leu Gly Gly Asp Glu Ala Leu Val Ala Ala Ala Arg Asp |     |     |
| 165                                                             | 170 | 175 |
| Leu Ala Arg Glu Leu Arg Trp Asp Pro Trp Glu Lys Asp Pro Leu Thr |     |     |
| 180                                                             | 185 | 190 |
| Phe Tyr Cys Ile Thr Pro Glu Lys Met Ala Val Gly Arg Gln Phe Arg |     |     |
| 195                                                             | 200 | 205 |
| Thr Phe Arg Asp Arg Leu Gln Thr Leu Met Ala Arg Asp Leu Trp Arg |     |     |
| 210                                                             | 215 | 220 |
| Ser Phe Leu Ala Ala Asn Pro His Leu Gln Asp Trp Ala Leu Glu Glu |     |     |
| 225                                                             | 230 | 235 |
| His Gly Leu Glu Ser Pro Glu Glu Leu Thr Tyr Glu Glu Leu Lys Lys |     |     |
| 245                                                             | 250 | 255 |
| Leu Pro Ser Ile Lys Gly Gln Pro Arg Phe Leu Glu Leu Tyr Ile Val |     |     |
| 260                                                             | 265 | 270 |
| Gly His Asn Ile Asn Gly Phe Asp Glu Ile Val Leu Ala Ala Gln Val |     |     |
| 275                                                             | 280 | 285 |
| Ile Asn Asn Arg Ser Ser Val Pro Gly Pro Phe Arg Ile Thr Arg Asn |     |     |
| 290                                                             | 295 | 300 |
| Phe Met Pro Arg Ala Gly Lys Ile Leu Phe Asn Asp Leu Thr Phe Ser |     |     |



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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 305 |     |     |     |     | 310 |     |     |     |     |     | 315 |     |     |     |     | 320 |
| Leu | Pro | Asn | Pro | Arg | Ser | Lys | Lys | Arg | Thr | Asp | Tyr | Thr | Leu | Trp | Glu |     |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |     |
| Gln | Gly | Gly | Cys | Asp | Asp | Thr | Asp | Phe | Lys | His | Gln | Tyr | Leu | Lys | Val |     |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |     |
| Met | Val | Arg | Asp | Thr | Phe | Ala | Leu | Thr | His | Thr | Ser | Leu | Arg | Lys | Ala |     |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |     |
| Ala | Gln | Ala | Tyr | Ala | Leu | Pro | Val | Glu | Lys | Gly | Cys | Cys | Pro | Tyr | Gln |     |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |     |
| Ala | Val | Asn | Gln | Phe | Tyr | Met | Leu | Gly | Ser | Tyr | Arg | Ser | Asp | Thr | Asp |     |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |     |
| Gly | Phe | Pro | Leu | Gln | Glu | Tyr | Trp | Lys | Asp | Arg | Glu | Glu | Phe | Val | Leu |     |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |     |
| Asn | Arg | Glu | Leu | Trp | Lys | Lys | Lys | Gly | Glu | Asp | Lys | Tyr | Asp | Ile | Ile |     |
|     |     | 420 |     |     |     |     |     | 425 |     |     |     |     | 430 |     |     |     |
| Arg | Glu | Thr | Leu | Asp | Tyr | Cys | Ala | Leu | Asp | Val | Gln | Val | Thr | Ala | Glu |     |
|     | 435 |     |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |     |
| Leu | Val | His | Lys | Leu | Arg | Glu | Ser | Tyr | Ala | Ser | Phe | Val | Arg | Asp | Ser |     |
|     | 450 |     |     |     | 455 |     |     |     |     | 460 |     |     |     |     |     |     |
| Val | Gly | Leu | Gln | Glu | Ala | Ser | Phe | Asn | Val | Phe | Gln | Arg | Pro | Thr | Ile |     |
| 465 |     |     |     |     | 470 |     |     |     |     | 475 |     |     |     |     | 480 |     |
| Ser | Ser | Asn | Ser | His | Ala | Ile | Phe | Arg | Gln | Ile | Ala | Phe | Arg | Ala | Glu |     |
|     |     |     | 485 |     |     |     |     |     | 490 |     |     |     |     | 495 |     |     |
| Arg | Pro | Gln | Arg | Thr | Asn | Leu | Gly | Pro | Asn | Met | Leu | Ala | Pro | Ser | His |     |
|     |     | 500 |     |     |     |     |     | 505 |     |     |     |     | 510 |     |     |     |
| Glu | Leu | Tyr | Asp | Tyr | Val | Arg | Ala | Ser | Ile | Arg | Gly | Gly | Arg | Cys | Tyr |     |
|     | 515 |     |     |     |     | 520 |     |     |     |     | 525 |     |     |     |     |     |
| Pro | Thr | Tyr | Leu | Gly | Ile | Leu | Arg | Glu | Pro | Leu | Tyr | Val | Tyr | Asp | Ile |     |
|     | 530 |     |     |     |     | 535 |     |     |     |     | 540 |     |     |     |     |     |
| Cys | Gly | Met | Tyr | Ala | Ser | Ala | Leu | Thr | His | Pro | Met | Pro | Trp | Gly | Pro |     |
| 545 |     |     |     |     | 550 |     |     |     |     | 555 |     |     |     |     | 560 |     |
| Pro | Leu | Asn | Pro | Tyr | Glu | Arg | Ala | Leu | Ala | Ala | Arg | Glu | Trp | Gln | Arg |     |
|     |     |     | 565 |     |     |     |     |     | 570 |     |     |     |     | 575 |     |     |
| Ala | Leu | Asp | Met | Gln | Ala | Cys | Lys | Ile | Asp | Tyr | Phe | Asp | Pro | Arg | Leu |     |
|     |     | 580 |     |     |     |     |     | 585 |     |     |     |     | 590 |     |     |     |
| Leu | Pro | Gly | Val | Phe | Thr | Ile | Asp | Ala | Asp | Pro | Pro | Asn | Glu | Asp | Gln |     |
|     | 595 |     |     |     |     |     | 600 |     |     |     |     | 605 |     |     |     |     |
| Leu | Asp | Pro | Leu | Pro | Pro | Phe | Cys | Ser | Arg | Lys | Gly | Gly | Arg | Leu | Cys |     |
|     | 610 |     |     |     |     | 615 |     |     |     |     | 620 |     |     |     |     |     |
| Trp | Thr | Asn | Glu | Arg | Leu | Arg | Gly | Glu | Val | Ala | Thr | Ser | Val | Asp | Met |     |
| 625 |     |     |     | 630 |     |     |     |     |     | 635 |     |     |     |     | 640 |     |
| Val | Thr | Leu | His | Asn | Arg | Gly | Trp | Arg | Val | Arg | Leu | Ile | Pro | Asp | Glu |     |
|     |     |     | 645 |     |     |     |     |     | 650 |     |     |     |     | 655 |     |     |
| Arg | Thr | Thr | Val | Phe | Pro | Glu | Trp | Lys | Cys | Val | Ala | Arg | Glu | Tyr | Val |     |
|     |     |     | 660 |     |     |     |     | 665 |     |     |     |     | 670 |     |     |     |
| Gln | Leu | Asn | Ile | Ala | Ala | Lys | Glu | Arg | Ala | Asp | Arg | Asp | Lys | Asn | Gln |     |
|     | 675 |     |     |     |     |     | 680 |     |     |     |     | 685 |     |     |     |     |
| Thr | Leu | Arg | Ser | Ile | Ala | Lys | Leu | Leu | Ser | Asn | Ala | Leu | Tyr | Gly | Ser |     |
|     | 690 |     |     |     |     | 695 |     |     |     |     | 700 |     |     |     |     |     |
| Phe | Ala | Thr | Lys | Leu | Asp | Asn | Lys | Lys | Ile | Val | Phe | Ser | Asp | Gln | Met |     |
| 705 |     |     |     |     | 710 |     |     |     |     | 715 |     |     |     | 720 |     |     |
| Asp | Pro | Gly | Thr | Leu | Lys | Gly | Ile | Thr | Ser | Gly | Gln | Val | Asn | Ile | Lys |     |
|     |     |     | 725 |     |     |     |     |     | 730 |     |     |     |     | 735 |     |     |

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Ser Ser Ser Phe Leu Glu Thr Asp Asn Leu Ser Ala Glu Val Met Pro  
 740 745 750  
 Ala Phe Glu Arg Glu Tyr Leu Pro Gln Gln Leu Ala Leu Ala Asp Ser  
 755 760 765  
 Asp Ala Glu Glu Ser Glu Asp Glu Arg Ala Pro Thr Pro Phe Tyr Thr  
 770 775 780  
 Pro Pro Ser Gly Thr Pro Gly His Val Ser Tyr Thr Tyr Lys Pro Ile  
 785 790 795 800  
 Thr Phe Leu Asp Ala Glu Glu Gly Asp Met Cys Leu His Thr Leu Glu  
 805 810 815  
 Lys Val Asp Pro Leu Val Asp Asn Asp Arg Tyr Pro Ser His Val Ala  
 820 825 830  
 Ser Phe Val Leu Ala Trp Thr Arg Ala Phe Val Ser Glu Trp Ser Glu  
 835 840 845  
 Phe Leu Tyr Glu Glu Asp Arg Gly Thr Pro Leu Glu Asp Arg Pro Leu  
 850 855 860  
 Lys Ser Val Tyr Gly Asp Thr Asp Ser Leu Phe Val Thr Glu Lys Gly  
 865 870 875 880  
 His Arg Leu Met Glu Ser Arg Gly Lys Lys Arg Ile Lys Lys His Gly  
 885 890 895  
 Gly Asn Leu Val Phe Asp Pro Asp Arg Pro Glu Leu Thr Trp Leu Val  
 900 905 910  
 Glu Cys Glu Thr Val Cys Ala Ser Cys Gly Ala Asp Ala Tyr Ser Pro  
 915 920 925  
 Glu Ser Val Phe Leu Ala Pro Lys Leu Tyr Ala Leu Lys Ser Leu Gln  
 930 935 940  
 Cys Pro Ser Cys Gly Ala Thr Ser Lys Gly Lys Leu Arg Ala Lys Gly  
 945 950 955 960  
 His Ala Ala Glu Gly Leu Asp Tyr Glu Thr Met Val Lys Cys Tyr Leu  
 965 970 975  
 Ala Asp Ala Gln Gly Glu Glu Arg Gln Arg Phe Ser Thr Ser Arg Thr  
 980 985 990  
 Ser Leu Lys Arg Thr Leu Ala Ser Ala Gln Pro Gly Ala His Pro Phe  
 995 1000 1005  
 Thr Val Thr Gln Thr Thr Leu Thr Arg Thr Leu Arg Pro Trp Lys  
 1010 1015 1020  
 Asp Met Thr Leu Ala Pro Leu Asp Ala His Arg Leu Val Pro Tyr  
 1025 1030 1035  
 Ser Glu Ser Arg Pro Asn Pro Arg Asn Glu Glu Ile Cys Trp Ile  
 1040 1045 1050  
 Glu Met Pro  
 1055

<210> SEQ ID NO 18  
 <211> LENGTH: 658  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Adenovirus penton base protein

<400> SEQUENCE: 18

Met Arg Arg Ala Ala Met Phe Glu Glu Gly Pro Pro Pro Ser Tyr Glu  
 1 5 10 15  
 Ser Ala Met Gly Ile Ser Pro Ala Ala Pro Leu Gln Pro Pro Tyr Val  
 20 25 30

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Pro | Pro | Arg | Tyr | Leu | Gln | Pro | Thr | Gly | Gly | Arg | Asn | Ser | Ile | Cys | Tyr |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Ser | Glu | Leu | Gln | Pro | Leu | Tyr | Asp | Thr | Thr | Arg | Leu | Tyr | Leu | Val | Asp |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Asn | Lys | Ser | Ala | Asp | Val | Ala | Ser | Leu | Asn | Tyr | Gln | Asn | Asp | His | Ser |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     | 80  |     |
| Asp | Phe | Leu | Thr | Thr | Val | Ile | Gln | Asn | Asn | Asp | Phe | Thr | Pro | Thr | Glu |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Ala | Ser | Thr | Gln | Thr | Ile | Asn | Leu | Asp | Asn | Arg | Ser | Asn | Trp | Gly | Gly |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Asp | Leu | Lys | Thr | Ile | Leu | His | Thr | Asn | Met | Pro | Asn | Val | Asn | Glu | Phe |
|     | 115 |     |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Met | Phe | Thr | Asn | Ser | Phe | Lys | Ala | Arg | Val | Met | Val | Ala | Arg | Glu | Gln |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Gly | Glu | Ala | Lys | Tyr | Glu | Trp | Val | Asp | Phe | Thr | Leu | Pro | Glu | Gly | Asn |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Tyr | Ser | Glu | Thr | Met | Thr | Leu | Asp | Leu | Met | Asn | Asn | Ala | Ile | Val | Glu |
|     |     |     |     | 165 |     |     |     | 170 |     |     |     |     |     | 175 |     |
| His | Tyr | Leu | Lys | Val | Gly | Arg | Gln | Asn | Gly | Val | Lys | Glu | Ser | Asp | Ile |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Gly | Val | Lys | Phe | Asp | Thr | Arg | Asn | Phe | Arg | Leu | Gly | Trp | Asp | Pro | Val |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Thr | Gly | Leu | Val | Met | Pro | Gly | Val | Tyr | Thr | Asn | Glu | Ala | Phe | His | Pro |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Asp | Ile | Val | Leu | Leu | Pro | Gly | Cys | Gly | Val | Asp | Phe | Thr | Gln | Ser | Arg |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Leu | Ser | Asn | Leu | Leu | Gly | Ile | Arg | Lys | Arg | Gln | Pro | Phe | Gln | Glu | Gly |
|     |     |     | 245 |     |     |     |     | 250 |     |     |     |     |     | 255 |     |
| Phe | Lys | Ile | Thr | Tyr | Glu | Asp | Leu | Lys | Gly | Gly | Asn | Ile | Pro | Ala | Leu |
|     |     | 260 |     |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Leu | Asp | Leu | Asp | Ala | Tyr | Glu | Glu | Ser | Leu | Lys | Pro | Glu | Glu | Ser | Ala |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Gly | Asp | Ser | Gly | Glu | Ser | Gly | Glu | Glu | Gln | Ala | Gly | Gly | Gly | Gly | Gly |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Ala | Ser | Val | Glu | Asn | Glu | Ser | Thr | Pro | Ala | Val | Ala | Ala | Asp | Ala | Ala |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Glu | Val | Glu | Pro | Glu | Ala | Met | Gln | Gln | Asp | Ala | Glu | Glu | Gly | Ala | Gln |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Glu | Gly | Ala | Gln | Lys | Asp | Met | Asn | Asp | Gly | Glu | Ile | Arg | Gly | Asp | Thr |
|     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Phe | Ala | Thr | Arg | Gly | Glu | Glu | Lys | Glu | Ala | Glu | Ala | Ala | Ala | Ala | Ala |
|     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Thr | Ala | Glu | Ala | Glu | Thr | Glu | Val | Glu | Ala | Glu | Ala | Glu | Pro | Glu | Thr |
|     | 370 |     |     |     |     | 375 |     |     |     |     |     | 380 |     |     |     |
| Glu | Val | Met | Glu | Asp | Met | Asn | Asp | Gly | Glu | Arg | Arg | Gly | Asp | Thr | Phe |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Ala | Thr | Arg | Gly | Glu | Glu | Lys | Ala | Ala | Glu | Ala | Glu | Ala | Ala | Ala | Glu |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Glu | Ala | Ala | Ala | Ala | Ala | Ala | Lys | Thr | Glu | Ala | Ala | Ala | Lys | Ala | Glu |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Val | Glu | Ala | Asn | Val | Ala | Val | Glu | Ala | Gln | Ala | Glu | Glu | Glu | Ala | Ala |
|     | 435 |     |     |     |     |     | 440 |     |     |     |     |     | 445 |     |     |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Ala | Glu | Ala | Val | Lys | Glu | Lys | Ala | Gln | Ala | Glu | Gln | Glu | Glu | Lys | Lys |  |
| 450 |     |     |     |     |     | 455 |     |     |     |     | 460 |     |     |     |     |  |
| Pro | Val | Ile | Gln | Pro | Leu | Lys | Glu | Asp | Ser | Lys | Lys | Arg | Ser | Tyr | Asn |  |
| 465 |     |     |     |     | 470 |     |     |     |     | 475 |     |     |     |     | 480 |  |
| Val | Ile | Glu | Gly | Ser | Thr | Phe | Thr | Gln | Tyr | Arg | Ser | Trp | Tyr | Leu | Ala |  |
|     |     |     |     | 485 |     |     |     |     | 490 |     |     |     |     | 495 |     |  |
| Tyr | Asn | Tyr | Gly | Asp | Pro | Val | Lys | Gly | Val | Arg | Ser | Trp | Thr | Leu | Leu |  |
|     |     |     | 500 |     |     |     |     | 505 |     |     |     |     | 510 |     |     |  |
| Cys | Thr | Pro | Asp | Val | Thr | Cys | Gly | Ser | Glu | Gln | Met | Tyr | Trp | Ser | Leu |  |
|     |     | 515 |     |     |     |     | 520 |     |     |     |     | 525 |     |     |     |  |
| Pro | Asn | Met | Met | Gln | Asp | Pro | Val | Thr | Phe | Arg | Ser | Thr | Arg | Gln | Val |  |
|     |     | 530 |     |     |     | 535 |     |     |     |     | 540 |     |     |     |     |  |
| Ser | Asn | Phe | Pro | Val | Val | Gly | Ala | Glu | Leu | Leu | Pro | Val | His | Ser | Lys |  |
| 545 |     |     |     |     | 550 |     |     |     |     | 555 |     |     |     |     | 560 |  |
| Ser | Phe | Tyr | Asn | Glu | Gln | Ala | Val | Tyr | Ser | Gln | Leu | Ile | Arg | Gln | Ala |  |
|     |     |     | 565 |     |     |     |     |     | 570 |     |     |     |     | 575 |     |  |
| Thr | Ser | Leu | Thr | His | Val | Phe | Asn | Arg | Phe | Pro | Glu | Asn | Gln | Ile | Leu |  |
|     |     |     | 580 |     |     |     |     | 585 |     |     |     |     | 590 |     |     |  |
| Ala | Arg | Pro | Pro | Ala | Pro | Thr | Ile | Thr | Thr | Val | Ser | Glu | Asn | Val | Pro |  |
|     |     | 595 |     |     |     |     | 600 |     |     |     |     | 605 |     |     |     |  |
| Ala | Leu | Thr | Asp | His | Gly | Thr | Leu | Pro | Leu | Arg | Asn | Ser | Ile | Ser | Gly |  |
|     |     | 610 |     |     |     | 615 |     |     |     |     | 620 |     |     |     |     |  |
| Val | Gln | Arg | Val | Thr | Ile | Thr | Asp | Ala | Arg | Arg | Arg | Thr | Cys | Pro | Tyr |  |
| 625 |     |     |     |     | 630 |     |     |     |     | 635 |     |     |     |     | 640 |  |
| Val | Tyr | Lys | Ala | Leu | Gly | Ile | Val | Ser | Pro | Arg | Val | Leu | Ser | Ser | Arg |  |
|     |     |     |     | 645 |     |     |     |     | 650 |     |     |     |     | 655 |     |  |

Thr Phe

<210> SEQ ID NO 19  
 <211> LENGTH: 959  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Adenovirus hexon protein

<400> SEQUENCE: 19

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Met | Ala | Thr | Pro | Ser | Met | Met | Pro | Gln | Trp | Ser | Tyr | Met | His | Ile | Ser |  |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |  |
| Gly | Gln | Asp | Ala | Ser | Glu | Tyr | Leu | Ser | Pro | Gly | Leu | Val | Gln | Phe | Ala |  |
|     |     |     | 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |  |
| Arg | Ala | Thr | Asp | Thr | Tyr | Phe | Asn | Met | Ser | Asn | Lys | Phe | Arg | Asn | Pro |  |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |  |
| Thr | Val | Ala | Pro | Thr | His | Asp | Val | Thr | Thr | Asp | Arg | Ser | Gln | Arg | Leu |  |
|     |     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |  |
| Thr | Leu | Arg | Phe | Ile | Pro | Val | Asp | Arg | Glu | Asp | Thr | Ala | Tyr | Ser | Tyr |  |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     |     | 80  |  |
| Lys | Ala | Arg | Phe | Thr | Leu | Ala | Val | Gly | Asp | Asn | Arg | Val | Leu | Asp | Met |  |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     |     | 95  |     |  |
| Ala | Ser | Thr | Tyr | Phe | Asp | Ile | Arg | Gly | Val | Leu | Asp | Arg | Gly | Pro | Thr |  |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |  |
| Phe | Lys | Pro | Tyr | Ser | Gly | Thr | Ala | Tyr | Asn | Ser | Leu | Ala | Pro | Lys | Gly |  |
|     |     | 115 |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |  |
| Ala | Pro | Asn | Ser | Cys | Glu | Trp | Glu | Gln | Asp | Glu | Pro | Ala | Gln | Ala | Ala |  |
|     |     | 130 |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |  |

Ile Ala Glu Asp Glu Glu Glu Leu Glu Glu Glu Gln Ala Gln Asp Glu

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|                                                                 |     |     |     |
|-----------------------------------------------------------------|-----|-----|-----|
| 145                                                             | 150 | 155 | 160 |
| Gln Ala Pro Thr Lys Lys Thr His Val Tyr Ala Gln Ala Pro Leu Ser | 165 | 170 | 175 |
| Gly Glu Lys Ile Thr Lys Asp Gly Leu Gln Ile Gly Val Asp Ala Thr | 180 | 185 | 190 |
| Gln Ala Gly Asp Asn Pro Ile Tyr Ala Asp Lys Thr Phe Gln Pro Glu | 195 | 200 | 205 |
| Pro Gln Ile Gly Glu Ser Gln Trp Asn Glu Ala Asp Ala Thr Val Ala | 210 | 215 | 220 |
| Gly Gly Arg Val Leu Lys Lys Thr Thr Pro Met Arg Pro Cys Tyr Gly | 225 | 230 | 235 |
| Ser Tyr Ala Lys Pro Thr Asn Ala Asn Gly Gly Gln Gly Ile Met Val | 245 | 250 | 255 |
| Ala Asn Asp Gln Gly Ala Leu Glu Ser Lys Val Glu Met Gln Phe Phe | 260 | 265 | 270 |
| Ser Thr Thr Thr Ser Leu Asn Val Arg Glu Gly Glu Asn Asn Leu Gln | 275 | 280 | 285 |
| Pro Lys Val Val Leu Tyr Ser Glu Asp Val Asn Leu Glu Ser Pro Asp | 290 | 295 | 300 |
| Thr His Leu Ser Tyr Lys Pro Lys Lys Asp Asp Thr Asn Ser Lys Ile | 305 | 310 | 315 |
| Met Leu Gly Gln Gln Ala Met Pro Asn Arg Pro Asn Leu Ile Ala Phe | 325 | 330 | 335 |
| Arg Asp Asn Phe Ile Gly Leu Met Tyr Tyr Asn Ser Thr Gly Asn Met | 340 | 345 | 350 |
| Gly Val Leu Ala Gly Gln Ala Ser Gln Leu Asn Ala Val Val Asp Leu | 355 | 360 | 365 |
| Gln Asp Arg Asn Thr Glu Leu Ser Tyr Gln Leu Met Leu Asp Ser Ile | 370 | 375 | 380 |
| Gly Asp Arg Ser Arg Tyr Phe Ser Met Trp Asn Gln Ala Val Asp Ser | 385 | 390 | 395 |
| Tyr Asp Pro Asp Val Arg Ile Ile Glu Asn His Gly Val Glu Asp Glu | 405 | 410 | 415 |
| Leu Pro Asn Tyr Cys Phe Pro Leu Gly Gly Ile Gly Ile Thr Asp Thr | 420 | 425 | 430 |
| Tyr Gln Cys Ile Lys Pro Thr Ala Ala Ala Asn Asn Thr Thr Trp Ser | 435 | 440 | 445 |
| Lys Asp Glu Glu Phe Ser Asp Arg Asn Glu Ile Gly Val Gly Asn Asn | 450 | 455 | 460 |
| Phe Ala Met Glu Ile Asn Ile Gln Ala Asn Leu Trp Arg Asn Phe Leu | 465 | 470 | 475 |
| Tyr Ala Asn Val Gly Leu Tyr Leu Pro Asp Lys Leu Lys Tyr Asn Pro | 485 | 490 | 495 |
| Thr Asn Val Asp Ile Ser Asp Asn Pro Asn Thr Tyr Asp Tyr Met Asn | 500 | 505 | 510 |
| Lys Arg Val Val Ala Pro Gly Leu Val Asp Cys Phe Val Asn Val Gly | 515 | 520 | 525 |
| Ala Arg Trp Ser Leu Asp Tyr Met Asp Asn Val Asn Pro Phe Asn His | 530 | 535 | 540 |
| His Arg Asn Ala Gly Leu Arg Tyr Arg Ser Met Ile Leu Gly Asn Gly | 545 | 550 | 555 |
| Arg Tyr Val Pro Phe His Ile Gln Val Pro Gln Lys Phe Phe Ala Ile | 565 | 570 | 575 |

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Lys Asn Leu Leu Leu Leu Pro Gly Ser Tyr Thr Tyr Glu Trp Asn Phe
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Arg Lys Asp Val Asn Met Val Leu Gln Ser Ser Leu Gly Asn Asp Leu
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Arg Val Asp Gly Ala Ser Ile Lys Phe Asp Ser Val Thr Leu Tyr Ala
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Thr Phe Phe Pro Met Ala His Asn Thr Ala Ser Thr Leu Glu Ala Met
      625                      630                      635                      640
Leu Arg Asn Asp Thr Asn Asp Gln Ser Phe Asn Asp Tyr Leu Ser Gly
      645                      650                      655
Ala Asn Met Leu Tyr Pro Ile Pro Ala Lys Ala Thr Asn Val Pro Ile
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Ser Ile Pro Ser Arg Asn Trp Ala Ala Phe Arg Gly Trp Ala Phe Thr
      675                      680                      685
Arg Leu Lys Thr Lys Glu Thr Pro Ser Leu Gly Ser Gly Phe Asp Pro
      690                      695                      700
Tyr Phe Val Tyr Ser Gly Ser Ile Pro Tyr Leu Asp Gly Thr Phe Tyr
      705                      710                      715                      720
Leu Asn His Thr Phe Lys Lys Ile Ser Ile Met Tyr Asp Ser Ser Val
      725                      730                      735
Ser Trp Pro Gly Asn Asp Arg Leu Leu Thr Pro Asn Glu Phe Glu Val
      740                      745                      750
Lys Arg Ala Val Asp Gly Glu Gly Tyr Asn Val Ala Gln Cys Asn Met
      755                      760                      765
Thr Lys Asp Trp Phe Leu Val Gln Met Leu Ala Asn Tyr Asn Ile Gly
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Tyr Gln Gly Phe Tyr Ile Pro Glu Ser Tyr Lys Asp Arg Met Tyr Ser
      785                      790                      795                      800
Phe Phe Arg Asn Phe Gln Pro Met Ser Arg Gln Val Val Asp Glu Thr
      805                      810                      815
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      835                      840                      845
Tyr Pro Ala Asn Phe Pro Tyr Pro Leu Ile Gly Lys Thr Ala Val Asp
      850                      855                      860
Ser Val Thr Gln Lys Lys Phe Leu Cys Asp Arg Thr Leu Trp Arg Ile
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Pro Phe Ser Ser Asn Phe Met Ser Met Gly Ala Leu Thr Asp Leu Gly
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Gln Asn Leu Leu Tyr Ala Asn Ser Ala His Ala Leu Asp Met Thr Phe
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Glu Val Asp Pro Met Asp Glu Pro Thr Leu Leu Tyr Ile Val Phe Glu
      915                      920                      925
Val Phe Asp Val Val Arg Val His Gln Pro His Arg Gly Val Ile Glu
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<210> SEQ ID NO 20
<211> LENGTH: 583
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Adenovirus fiber protein

<400> SEQUENCE: 20

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Phe Val Ser Ser Ala Gly Met Gln Glu Asn Pro Ala Gly Val Leu Ser
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Leu His Leu Ser Glu Pro Leu Thr Thr His Asn Gly Ala Leu Thr Leu
      50           55           60

Lys Met Gly Gly Gly Leu Thr Leu Asp Lys Glu Gly Asn Leu Thr Ser
65           70           75           80

Gln Asn Ile Thr Ser Val Asp Pro Pro Leu Lys Lys Ser Lys Asn Asn
      85           90           95

Ile Ser Leu Gln Thr Ala Ala Pro Leu Ala Val Ser Ser Gly Ala Leu
      100          105          110

Thr Leu Phe Ala Thr Pro Pro Leu Ala Val Ser Gly Asp Asn Leu Thr
      115          120          125

Val Gln Ser Gln Ala Pro Leu Thr Leu Glu Asp Ser Lys Leu Thr Leu
      130          135          140

Ala Thr Lys Gly Pro Leu Thr Val Ser Glu Gly Lys Leu Val Leu Glu
145          150          155          160

Thr Glu Ala Pro Leu His Ala Ser Asp Ser Ser Ser Leu Gly Leu Ser
      165          170          175

Val Thr Ala Pro Leu Ser Ile Asn Asn Asp Ser Leu Gly Leu Asp Leu
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Gln Ala Pro Ile Val Ser Gln Asn Gly Lys Leu Ala Leu Asn Val Ala
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Gly Pro Leu Ala Val Ala Asn Gly Ile Asn Ala Leu Thr Val Gly Thr
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Gly Lys Gly Ile Gly Leu Asn Glu Thr Ser Thr His Leu Gln Ala Lys
225          230          235          240

Leu Val Ala Pro Leu Gly Phe Asp Thr Asn Gly Asn Ile Lys Leu Ser
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Val Ala Gly Gly Met Arg Leu Asn Asn Asp Thr Leu Ile Leu Asp Val
      260          265          270

Asn Tyr Pro Phe Glu Ala Gln Gly Gln Leu Ser Leu Arg Val Gly Gln
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Gly Pro Leu Tyr Val Asp Ser Ser Ser His Asn Leu Thr Ile Arg Cys
      290          295          300

Leu Arg Gly Leu Tyr Ile Thr Ser Ser Asn Asn Gln Thr Gly Leu Glu
305          310          315          320

Ala Asn Ile Lys Leu Thr Lys Gly Leu Val Tyr Asp Gly Asn Ala Ile
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Ala Val Asn Val Gly Gln Gly Leu Gln Tyr Ser Thr Thr Ala Thr Ser
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Glu Gly Val Tyr Pro Ile Gln Ser Lys Ile Gly Leu Gly Met Glu Tyr
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Asp Thr Asn Gly Ala Met Met Thr Lys Leu Gly Ser Gly Leu Ser Phe
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Asp Asn Ser Gly Ala Ile Val Val Gly Asn Lys Asn Asp Asp Arg Leu
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| caccgccgtc  | gccatctaga  | aatcgaaagg | gttctgccgc | gcgtcccggt  | gcgccacggg | 22680 |
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| tagacttgca | agacagaaac  | acagagctgt  | cataccaact | gatgcttgat | tccattggag |            | 20160 |
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atcctaggca cagtatctgc ccttgctgtc agaggcagcc ttgcgcccac cactaatgca 30660
tccagcatag tccaaatatt tctaagattt gatgaaaatg gactattgat gagcaactca 30720
tcgctagacg gtgattactg gaattacaga aatggggact cactaatag cacaccatat 30780
acaaatgcag taggctttat gcctaacta gcagcctatc ctaaaggta ggctacagct 30840
gcaaaaagca gtattgtaag ccaggatatac atggatgggtg acactactaa acctataaca 30900
ctaaaaataa acttcaatgg cattgatgaa acaacagaaa ataccctgt tagtaaatat 30960
tccatgacat tctcatggag ctggcccacc gcaagctaca taggccacac ttttgcaaca 31020
aactctttta ctttctccta catcgcccaa gaataaagaa agcacagaga tgcttgtttt 31080
gatttcaaaa ttgtgtgctt ttattttatt tcagcttaca gtatttccag tagtcattcg 31140
aataaagctt aatcaaatcg catgagaacc ctccacata gcttaaatga gcaccagtgc 31200
aaatggagaa aagcctcgag gtcgttgctc ggccgggac ggtgatcacc gatccagaca 31260
tgataagata cattgatgag tttggacaaa ccacaactag aatgcagtga aaaaaatgct 31320
ttatttgtga aatttgtgat gctattgctt tatttgtaac cattataagc tgcaataaac 31380
aagttcccg atcgcgatcc ggcccaggc tgtagccgac gatgggtgctc caggagagtt 31440
gttgattcat tgtttgcctc cctgctgctg ttttccaccg aagttcatgc cagtccagcg 31500
tttttgacg agaaaagcgg ccgacttcgg tttgcggtcg cgagtgaaga tccctttctt 31560
gttaccgcca acgcgcaata tgccctgcca ggtcgcaaaa tcggcgaaat tccatacctg 31620
ttaccgacg acggcgctga cgcgatcaaa gacgcggtga tacatatcca gccatgcaca 31680
ctgatactct tcactccaca tgcggtgta cattgagtgc agcccggcta acgtatccac 31740
ggcgtattcg gtgatgataa tcgggtgatg cagtttctcc tgccaggcca gaagttcttt 31800
ttccagtacc ttctctgccg ttccaaatc gccgctttgg acataccatc cgtaataacg 31860
gttcaggcac agcacatcaa agagatcgct gatggtatcg gtgtgagcgt cgcagaacat 31920
tacattgacg cagggtgatc gacgcgtcgg gtgcagttaa cgcgttgctt cgcacagtgg 31980
cgcgaaatat tcccgtagc cttgcggacg ggtatccggt tcgttggcaa tactccacat 32040
caccacgctt ggggtgtttt tgcacgcgc tatcagctct ttaatcgctt gtaagtgcgc 32100
ttgtgagatt tcccgttga ctgcctcttc gctgtacagt tctttcggtt tgttgccgc 32160
ttcgaaacca atgcctaaag agagggtaaa gccgacagca gcagtttcat caatcaccac 32220
gatgccatgt tcctctgcc agtcgagcat ctcttcagcg taagggtaat gcgaggtacg 32280

```

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```

gtaggagttg gccccaatcc agtcattaa tgcgtggctg tgcacccatca gcacgttatc 32340
gaatcctttg ccacgcaagt ccgcatcttc atgacgacca aagccagtaa agtagaacgg 32400
tttgtgggta atcaggaact gttcgccctt cactgccact gaccggatgc cgacgcgaag 32460
cgggtagata tcacactctg tctggctttt ggctgtgacg cacagttcat agagataacc 32520
ttcaccgggt tgccagaggt gcggattcac cacttgcaaa gtcccgttag tgccttgctc 32580
agttgcaacc acctgttgat ccgcatcagc cagttcaacg ctgacatcac cattggccac 32640
cacctgccag tcaacagacg cgtgggttaca gtcttgccgc acatgcgtca ccacggtgat 32700
atcgccacc caggtgttcg gcgtgggtg 32728

```

```

<210> SEQ ID NO 26
<211> LENGTH: 114
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: IL-2 core promoter

```

```

<400> SEQUENCE: 26

```

```

acattttgac acccccataa tatttttcca gaattaacag tataaattgc atctcttgtt 60
caagagttcc ctatcactct ctttaacac tactcacagt aacctcaact cctg 114

```

```

<210> SEQ ID NO 27
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: IL-2 minimal promoter

```

```

<400> SEQUENCE: 27

```

```

tcaagagttc cctatcactc tctttaatca ctactcacag taacctcaac tctg 55

```

```

<210> SEQ ID NO 28
<211> LENGTH: 380
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: IL-2 enhancer and promoter variant

```

```

<400> SEQUENCE: 28

```

```

tgatatcttt tctgagttac tttgtatcc ccacccctt aaagaaagga ggaaaaactg 60
tttcatacag aaggcggtta ttgcattgat tagagctatc acctaatgtt gggctaattg 120
aacaagagg gatttcacct acatccattc agtcagtctt tgggggttta aagaaattcc 180
aaagagtcac cagaagagga aaaatgaagg taatgttttt tcagactggg aaagtctttg 240
aaaatatgtg taatatgtaa aacattttga ccccccata atatttttcc agaattaaca 300
gtataaattg catctcttgt tcaagagttc cctatcactc tctttaatca ctactcacag 360
taacctcaac tctgcccaca 380

```

```

<210> SEQ ID NO 29
<211> LENGTH: 373
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: L-2 enhancer and promoter variant

```

```

<400> SEQUENCE: 29

```

```

ttttctgagt tacttttgta tccccacccc cttaaagaaa ggaggaaaaa ctgtttcata 60
cagaaggcgt taattgcatg aattagagct atcacctaag tgtgggctaa tgtaacaag 120

```

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```

agggatattca cctacatcca ttcagtcagt ctttgggggt ttaaagaaat tccaaagagt 180
catcagaaga ggaaaaatga aggtaatgtt ttttcagact ggtaaagtct ttgaaaatat 240
gtgtaatatg taaaacattt tgacaccccc ataatatattt tccagaatta acagtataaa 300
ttgcatctct tgttcaagag ttccctatca ctctctttaa tcaactactca cagtaacctc 360
aactcctgcc aca 373

```

```

<210> SEQ ID NO 30
<211> LENGTH: 161
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: (NF-?B)1-IL2 promoter variant

```

```

<400> SEQUENCE: 30

```

```

aattgggtccc atcgaagagg gatttcacct acataattgg tcccgggaca tttgacacc 60
cccataatat ttttcagaa ttaacagtat aaattgcac tcttgttcaa gagttcccta 120
tcaactctct taatcactac tcacagtaac ctcaactcct g 161

```

```

<210> SEQ ID NO 31
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: (NF-?B)3-IL2 promoter variant

```

```

<400> SEQUENCE: 31

```

```

aattgggtccc atcgaagagg gatttcacct acataagagg gatttcacct acataagagg 60
gatttcacct acataattgg tcccgggaca tttgacacc cccataatat ttttcagaa 120
ttaacagtat aaattgcac tcttgttcaa gagttcccta tcaactctct taatcactac 180
tcacagtaac ctcaactcct g 201

```

```

<210> SEQ ID NO 32
<211> LENGTH: 268
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: (NF-?B)6-IL2 promoter variant

```

```

<400> SEQUENCE: 32

```

```

aattgggtccc atcgaagagg gatttcacct acataagagg gatttcacct acataagagg 60
gatttcacct acataattgg taagagggat ttcacctaca taagagggat ttcacctaca 120
taagagggat ttcacctaca taattggtcc egggacattt tgacaccccc ataatatattt 180
tccagaatta acagtataaa ttgcatctct tgttcaagag ttccctatca ctctctttaa 240
tcaactactca cagtaacctc aactcctg 268

```

```

<210> SEQ ID NO 33
<211> LENGTH: 177
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 1X NFAT response elements-IL2 promoter variant

```

```

<400> SEQUENCE: 33

```

```

aattgggtccc atcgaattag gaggaaaaac tgtttcatat agaaggcgctc aattgggtccc 60
gggacatttt gacaccccc aatatatttt ccagaattaa cagtataaat tgcactctct 120

```

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gttcaagagt tccctatcac tctctttaat cactactcac agtaacctca actcctg 177

<210> SEQ ID NO 34  
 <211> LENGTH: 256  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: 3X NFAT response elements-IL2 promoter variant

&lt;400&gt; SEQUENCE: 34

|                                                                   |     |
|-------------------------------------------------------------------|-----|
| tgatatcaat tgggccatc gaattaggag gaaaaactgt ttcatacaga aggcgtcaat  | 60  |
| taggaggaaa aactgtttca tacagaaggc gtcaattagg aggaaaaact gtttcataca | 120 |
| gaaggcgtca attgggtccg ggacattttg acaccccat aatatttttc cagaattaac  | 180 |
| agtataaatt gcattctctg ttcaagagtt cctatcact ctctttaatc actactcaca  | 240 |
| gtaacctcaa ctctg                                                  | 256 |

<210> SEQ ID NO 35  
 <211> LENGTH: 249  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: 3X NFAT response elements-IL2 promoter variant

&lt;400&gt; SEQUENCE: 35

|                                                                     |     |
|---------------------------------------------------------------------|-----|
| aattgggtccc atcgatttag gaggaaaaac tgtttcatatc agaaggcgtc aattaggagg | 60  |
| aaaaactggt tcatacagaa ggcgtcaatt aggaggaaaa actgtttcat acagaaggcg   | 120 |
| tcaattggtc cggggacatt ttgacacccc cataatattt ttccagaatt aacagtataa   | 180 |
| attgcatctc ttgttcaaga gttccctatc actctcttta atcactactc acagtaacct   | 240 |
| caactctg                                                            | 249 |

<210> SEQ ID NO 36  
 <211> LENGTH: 358  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: 6X NFAT response elements-IL2 promoter variant

&lt;400&gt; SEQUENCE: 36

|                                                                   |     |
|-------------------------------------------------------------------|-----|
| gaattaggag gaaaaactgt ttcatacaga aggcgtcaat taggaggaaa aactgtttca | 60  |
| tacagaaggc gtcaattagg aggaaaaact gtttcataca gaaggcgtca attggtccca | 120 |
| tcgaattagg aggaaaaact gtttcataca gaaggcgtca attaggagga aaaactgttt | 180 |
| catacagaag gcgtcaatta ggaggaaaaa ctgtttcata cagaaggcgt caattggtcc | 240 |
| cgggacattt tgacaccccc ataatatttt tccagaatta acagtataaa ttgcatctct | 300 |
| tgttcaagag ttccctatca ctctccttaa tcactactca cagtaacctc aactcctg   | 358 |

<210> SEQ ID NO 37  
 <211> LENGTH: 374  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: 6X NFAT response elements-IL2 promoter variant

&lt;400&gt; SEQUENCE: 37

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| tgatatcgaa ttaggaggaa aaactgtttc atacagaagg cgtcaattag gagggaaaaa  | 60  |
| tgtttcatac agaaggcgtc aattaggagg aaaaactgtt tcatacagaa ggcgtcaatt  | 120 |
| gggtcccatcg aattaggagg aaaaactgtt tcatacagaa ggcgtcaatt aggaggaaaa | 180 |



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```

actgtttcat acagaaggcg tcaattagga ggaaaaactg tttcatacag aaggcgtaa 240
ttggtcccg gacattttga ccccccata atatttttcc agaattaaca gtataaattg 300
catctcttgt tcaagagttc cctatcactc tctttaatca ctactcacag taacctcaac 360
tcctgaattc catg 374

```

```

<210> SEQ ID NO 38
<211> LENGTH: 358
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 6X NFAT response elements-IL2 promoter variant

```

```

<400> SEQUENCE: 38
gaattaggag gaaaaactgt ttcatacaga aggcgtcaat taggaggaaa aactgtttca 60
tacagaaggc gtcaattagg aggaaaaact gtttcataca gaaggcgta attggtccca 120
tcgaattagg aggaaaaact gtttcataca gaaggcgta attaggagga aaaactgttt 180
catacagaag gcgtcaatta ggaggaaaaa ctgtttcata cagaaggcgt caattggtcc 240
cgggacattt tgacaccccc ataatatatt tccagaatta acagtataaa ttgcattctt 300
tgttcaagag ttccctatca ctctctttaa tcactactca cagtaacctc aactcctg 358

```

```

<210> SEQ ID NO 39
<211> LENGTH: 365
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: 6X NFAT response elements-IL2 promoter variant

```

```

<400> SEQUENCE: 39
tgatatcgaa ttaggaggaa aaactgtttc atacagaagg cgtcaattag gagggaaaaa 60
tgtttcatac agaaggcgtc aattaggagg aaaaactgtt tcatacagaa ggcgtcaatt 120
gggtcccatcg aattaggagg aaaaactgtt tcatacagaa ggcgtcaatt aggaggaaaa 180
actgtttcat acagaaggcg tcaattagga ggaaaaactg tttcatacag aaggcgtaa 240
ttggtcccg gacattttga ccccccata atatttttcc agaattaaca gtataaattg 300
catctcttgt tcaagagttc cctatcactc tctttaatca ctactcacag taacctcaac 360
tcctg 365

```

```

<210> SEQ ID NO 40
<211> LENGTH: 244
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: human EEFlA1 promoter variant

```

```

<400> SEQUENCE: 40
gagcgtcgt gaggtccgg tgcccgtag tgggcagagc gcacatcgcc cacagtcccc 60
gagaagtgg ggggggggg tcggcgattg aaccggtgcc tagagaagg ggcgcgggg 120
aaactgggaa agtgatgtcg tgtactggct ccgcctttt cccgagggtg ggggagaacc 180
gtatataagt gcagtagtcg ccgtgaacgt tctttttcgc aacgggtttg ccgcagaaac 240
acag 244

```

```

<210> SEQ ID NO 41
<211> LENGTH: 236
<212> TYPE: DNA

```

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```

<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: human EEFlA1 promoter variant

<400> SEQUENCE: 41

gcgtgaggct ccggtgcccc tcagtgggca gagcgacat cggccacagt ccccgagaag      60
ttgggggggag gggtcggcaa ttgaaccggt gcctagagaa ggtggcgcg ggtaaacctgg      120
gaaagtgatg tcgtgtactg gctccgcctt ttcccgagg gtgggggaga accgtatata      180
agtgcagtag tcgccgtgaa cgttcttttt cgcaacgggt ttgccgccag aacaca          236

```

```

<210> SEQ ID NO 42
<211> LENGTH: 1266
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: human EEFlA1 promoter and enhancer

```

```

<400> SEQUENCE: 42

gagctttgca aagatggata aagttttaa cagagaggaa tctttgcagc taatggacct      60
tctaggtctt gaaaggagtg ggaattggct ccggtgcccc tcagtgggca gagcgacat      120
cgccacacagt ccccgagaag ttgggggggag gggtcggcaa ttgaaccggt gcctagagaa      180
ggtggcgcg ggtaaacctgg gaaagtgatg tcgtgtactg gctccgcctt ttcccgagg      240
gtgggggaga accgtatata agtgcagtag tcgccgtgaa cgttcttttt cgcaacgggt      300
ttgccgccag aacacaggta agtgcctgtg ttggttcccg cgggcctggc ctctttacgg      360
gttatggccc ttgcgtgect tgaattactt ccacctggct gcagtacgtg attcttgatc      420
ccgagcttcg ggttggaagt gggtgaggaga gtccgaggcc ttgcgcttaa ggagcccctt      480
cgctcgtgc ttgagttgag gcctggcctg ggcgtgggg cgcgcgcgtg cgaatctggt      540
ggcaccttcg cgctgtctc gctgctttcg ataagtctct agccatttaa aatttttgat      600
gacctgctgc gacgtttttt ttctggcaag atagtcttgt aaatgcgggc caagatctgc      660
aacttggtat ttcggttttt ggggcccggg gcggcgacgg ggcccgtgcg tcccagcgca      720
catgttcgcg gagggggggc ctgcgagcgc ggccaccgag aatcggacgg gggtagtctc      780
aagctggccc gcctgctctg gtgcctggcc tcgcgcgcgc gtgtatgcgc ccgcccgtgg      840
cggcaaggct ggcccgtcg gcaccagttg cgtgagcgga aagatggccg cttcccggcc      900
ctgctgcagg gagctcaaaa tggaggacgc ggcgctcggg agagcgggcg ggtgagtcac      960
ccacacaaaag gaaaagggcc ttccgctcct cagccgtcgc ttcatgtgac tccacggagt      1020
accggggcgc gtccaggcac ctcgattagt tctcgagctt ttggagtacg tcgtctttag      1080
gttgggggga ggggttttat gcgatggagt ttccccacac tgagtgggtg gagactgaag      1140
ttaggccagc ttggcacttg atgtaattct ccttggaatt tgcccttttt gagtttggat      1200
cttggttcac tctcaagcct cagacagtgg ttcaaagttt tttcttcca tttcaggtgt      1260
cgtgag                                           1266

```

```

<210> SEQ ID NO 43
<211> LENGTH: 571
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: human UBC promoter

```

```

<400> SEQUENCE: 43

ggcctccgcg ccgggttttg gcgcctccc cgggcgcccc cctcctcacg gcgagcgtg      60

```

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```

ccacgtcaga cgaagggcgc agcgagcgtc ctgatocttc cgcccgacg ctcaggacag 120
cgcccgctg ctcataagac tcggccttag aaccccagta tcagcagaag gacattttag 180
gacgggactt ggggtactct agggcactgg ttttctttcc agagagcgga acaggcgagg 240
aaaagtagtc ccttctcggc gattctgcgg agggatctcc gtggggcggt gaacgccgat 300
gattatataa ggacgcgccg ggtgtggcac agctagtctc gtcgcagccg ggatttgggt 360
cgcggttctt gtttgtggat cgctgtgac gtcacttggt gactagcggg ctgctgggct 420
gggtacgtgc gctcgggggt ggcgagtgtg ttttgtgaag ttttttaggc accttttgaa 480
atgtaatcat ttgggtcaat atgtaatttt cagtgttaga ctagtaaatt gtccgctaaa 540
ttctggccgt ttttggcttt tttgttagac g 571

```

```

<210> SEQ ID NO 44
<211> LENGTH: 81
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthetic minimal promoter 1

```

```

<400> SEQUENCE: 44

```

```

aggtctatat aagcagagct cgtttagtga accctcattc tggagacgga tcccagaccg 60
agtgttttga cctccataga a 81

```

```

<210> SEQ ID NO 45
<211> LENGTH: 151
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: HPV antigen design 1 HPV16 E6 amino acids

```

```

<400> SEQUENCE: 45

```

```

Met Phe Gln Asp Pro Gln Glu Arg Pro Arg Lys Leu Pro Gln Leu Cys
1           5           10          15
Thr Ala Leu Gln Thr Thr Ile His Asp Ile Ile Leu Glu Cys Val Tyr
20          25          30
Cys Lys Gln Gln Leu Leu Arg Arg Glu Val Tyr Asp Phe Ala Phe Arg
35          40          45
Asp Gly Cys Ile Val Tyr Arg Asp Gly Asn Pro Tyr Ala Val Cys Asp
50          55          60
Lys Cys Leu Lys Phe Tyr Ser Lys Ile Ser Glu Tyr Arg His Tyr Cys
65          70          75          80
Tyr Ser Leu Tyr Gly Thr Thr Leu Glu Gln Gln Tyr Asn Lys Pro Leu
85          90          95
Cys Asp Leu Leu Ile Arg Cys Ile Asn Cys Gln Lys Pro Leu Cys Pro
100         105         110
Glu Glu Lys Gln Arg His Leu Asp Lys Lys Gln Arg Phe His Asn Ile
115         120         125
Arg Gly Arg Trp Thr Gly Arg Cys Met Ser Cys Cys Arg Ser Ser Arg
130         135         140
Thr Arg Arg Ala Ala Ala Ala
145         150

```

```

<210> SEQ ID NO 46
<211> LENGTH: 98
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:

```

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<223> OTHER INFORMATION: HPV antigen design 1 HPV16 E7 amino acids

&lt;400&gt; SEQUENCE: 46

Met Pro Gly Asp Thr Pro Thr Leu His Glu Tyr Met Leu Asp Leu Gln  
1 5 10 15

Pro Glu Thr Thr Asp Leu Tyr Gly Tyr Glu Gln Leu Asn Asp Ser Ser  
20 25 30

Glu Glu Glu Asp Glu Ile Asp Gly Pro Ala Gly Gln Ala Ala Pro Asp  
35 40 45

Arg Ala His Tyr Asn Ile Val Thr Phe Cys Cys Lys Cys Asp Ser Thr  
50 55 60

Leu Arg Arg Cys Val Gln Ser Thr His Val Asp Ile Arg Thr Leu Glu  
65 70 75 80

Asp Leu Leu Met Gly Thr Leu Gly Ile Val Cys Pro Ile Cys Ser Gln  
85 90 95

Lys Pro

&lt;210&gt; SEQ ID NO 47

&lt;211&gt; LENGTH: 58

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: HPV antigen design 1 HPV16 E5 amino acids

&lt;400&gt; SEQUENCE: 47

Met Thr Asn Leu Asp Thr Ala Ser Thr Thr Leu Leu Ala Cys Phe Leu  
1 5 10 15

Leu Cys Phe Cys Val Leu Leu Cys Val Cys Leu Leu Ile Arg Pro Leu  
20 25 30

Leu Leu Ser Val Ser Thr Tyr Thr Arg Cys Phe Ile Val Tyr Ile Ile  
35 40 45

Phe Val Tyr Ile Pro Leu Phe Leu Ile His  
50 55

&lt;210&gt; SEQ ID NO 48

&lt;211&gt; LENGTH: 120

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: HPV antigen design 1 HPV18 E6 amino acids

&lt;400&gt; SEQUENCE: 48

Met Ala Arg Phe Glu Asp Pro Thr Arg Arg Pro Tyr Lys Leu Pro Asp  
1 5 10 15

Leu Cys Thr Ala Leu Asn Thr Ser Leu Gln Asp Ile Glu Ile Thr Cys  
20 25 30

Val Tyr Cys Lys Thr Val Leu Glu Leu Thr Glu Val Phe Glu Phe Ala  
35 40 45

Phe Lys Asp Gly Phe Val Val Tyr Arg Asp Ser Ile Pro His Ala Ala  
50 55 60

Cys His Lys Cys Ile Asp Phe Tyr Ser Arg Ile Arg Glu Leu Arg His  
65 70 75 80

Tyr Ser Asp Ser Val Tyr Gly Asp Thr Leu Glu Lys Leu Thr Asn Thr  
85 90 95

Gly Leu Tyr Asn Leu Leu Ile Arg Cys Leu Arg Cys Gln Lys Pro Leu  
100 105 110

Asn Pro Ala Glu Lys Leu Arg His  
115 120

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<210> SEQ ID NO 49  
 <211> LENGTH: 66  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HPV antigen design 1 HPV18 E7 amino acids

<400> SEQUENCE: 49

Glu Ile Asp Gly Val Asn His Gln His Leu Pro Ala Arg Arg Ala Ala  
 1                    5                    10                    15  
 Pro Gln Arg His Thr Met Leu Cys Met Cys Cys Lys Cys Glu Ala Arg  
                   20                    25                    30  
 Ile Glu Arg Val Val Glu Ser Ser Ala Asp Asp Leu Arg Ala Phe Gln  
                   35                    40                    45  
 Gln Leu Phe Leu Asn Thr Leu Ser Phe Val Cys Pro Trp Cys Ala Ser  
                   50                    55                    60  
 Gln Gln  
 65

<210> SEQ ID NO 50  
 <211> LENGTH: 53  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HPV antigen design 1 HPV18 E5 amino acids

<400> SEQUENCE: 50

Met Leu Ser Leu Ile Phe Leu Phe Cys Phe Cys Val Cys Met Tyr Val  
 1                    5                    10                    15  
 Cys Cys His Val Pro Leu Leu Pro Ser Val Val Val Ile Thr Ser Pro  
                   20                    25                    30  
 Ala Thr Ala Phe Thr Val Tyr Leu Leu Pro Met Leu Leu Leu His Ile  
                   35                    40                    45  
 His Ala Ile Leu Ser  
 50

<210> SEQ ID NO 51  
 <211> LENGTH: 547  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HPV antigen design 1 amino acids

<400> SEQUENCE: 51

Met Met Phe Gln Asp Pro Gln Glu Arg Pro Arg Lys Leu Pro Gln Leu  
 1                    5                    10                    15  
 Cys Thr Ala Leu Gln Thr Thr Ile His Asp Ile Ile Leu Glu Cys Val  
                   20                    25                    30  
 Tyr Cys Lys Gln Gln Leu Leu Arg Arg Glu Val Tyr Asp Phe Ala Phe  
                   35                    40                    45  
 Arg Asp Gly Cys Ile Val Tyr Arg Asp Gly Asn Pro Tyr Ala Val Cys  
                   50                    55                    60  
 Asp Lys Cys Leu Lys Phe Tyr Ser Lys Ile Ser Glu Tyr Arg His Tyr  
 65                    70                    75                    80  
 Cys Tyr Ser Leu Tyr Gly Thr Thr Leu Glu Gln Gln Tyr Asn Lys Pro  
                   85                    90                    95  
 Leu Cys Asp Leu Leu Ile Arg Cys Ile Asn Cys Gln Lys Pro Leu Cys  
                   100                    105                    110

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Pro | Glu | Glu | Lys | Gln | Arg | His | Leu | Asp | Lys | Lys | Gln | Arg | Phe | His | Asn |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Ile | Arg | Gly | Arg | Trp | Thr | Gly | Arg | Cys | Met | Ser | Cys | Cys | Arg | Ser | Ser |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Arg | Thr | Arg | Arg | Ala | Ala | Ala | Ala | Met | Pro | Gly | Asp | Thr | Pro | Thr | Leu |
|     | 145 |     |     |     | 150 |     |     |     |     |     | 155 |     |     |     | 160 |
| His | Glu | Tyr | Met | Leu | Asp | Leu | Gln | Pro | Glu | Thr | Thr | Asp | Leu | Tyr | Gly |
|     |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Tyr | Glu | Gln | Leu | Asn | Asp | Ser | Ser | Glu | Glu | Glu | Asp | Glu | Ile | Asp | Gly |
|     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Pro | Ala | Gly | Gln | Ala | Ala | Pro | Asp | Arg | Ala | His | Tyr | Asn | Ile | Val | Thr |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Phe | Cys | Cys | Lys | Cys | Asp | Ser | Thr | Leu | Arg | Arg | Cys | Val | Gln | Ser | Thr |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| His | Val | Asp | Ile | Arg | Thr | Leu | Glu | Asp | Leu | Leu | Met | Gly | Thr | Leu | Gly |
|     |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Ile | Val | Cys | Pro | Ile | Cys | Ser | Gln | Lys | Pro | Met | Thr | Asn | Leu | Asp | Thr |
|     |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Ala | Ser | Thr | Thr | Leu | Leu | Ala | Cys | Phe | Leu | Leu | Cys | Phe | Cys | Val | Leu |
|     |     |     |     | 260 |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Leu | Cys | Val | Cys | Leu | Leu | Ile | Arg | Pro | Leu | Leu | Leu | Ser | Val | Ser | Thr |
|     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Tyr | Thr | Arg | Cys | Phe | Ile | Val | Tyr | Ile | Ile | Phe | Val | Tyr | Ile | Pro | Leu |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Phe | Leu | Ile | His | Met | Ala | Arg | Phe | Glu | Asp | Pro | Thr | Arg | Arg | Pro | Tyr |
|     | 305 |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Lys | Leu | Pro | Asp | Leu | Cys | Thr | Ala | Leu | Asn | Thr | Ser | Leu | Gln | Asp | Ile |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Glu | Ile | Thr | Cys | Val | Tyr | Cys | Lys | Thr | Val | Leu | Glu | Leu | Thr | Glu | Val |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Phe | Glu | Phe | Ala | Phe | Lys | Asp | Gly | Phe | Val | Val | Tyr | Arg | Asp | Ser | Ile |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Pro | His | Ala | Ala | Cys | His | Lys | Cys | Ile | Asp | Phe | Tyr | Ser | Arg | Ile | Arg |
|     |     | 370 |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Glu | Leu | Arg | His | Tyr | Ser | Asp | Ser | Val | Tyr | Gly | Asp | Thr | Leu | Glu | Lys |
|     |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Leu | Thr | Asn | Thr | Gly | Leu | Tyr | Asn | Leu | Leu | Ile | Arg | Cys | Leu | Arg | Cys |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Gln | Lys | Pro | Leu | Asn | Pro | Ala | Glu | Lys | Leu | Arg | His | Glu | Ile | Asp | Gly |
|     |     |     |     | 420 |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Val | Asn | His | Gln | His | Leu | Pro | Ala | Arg | Arg | Ala | Ala | Pro | Gln | Arg | His |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| Thr | Met | Leu | Cys | Met | Cys | Cys | Lys | Cys | Glu | Ala | Arg | Ile | Glu | Arg | Val |
|     | 450 |     |     |     |     | 455 |     |     |     |     | 460 |     |     |     |     |
| Val | Glu | Ser | Ser | Ala |     |     |     |     |     |     |     |     |     |     |     |

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| 530                                                                                                                                                                                                        | 535 | 540     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| Ile Leu Ser                                                                                                                                                                                                |     |         |
| 545                                                                                                                                                                                                        |     |         |
| <210> SEQ ID NO 52<br><211> LENGTH: 563<br><212> TYPE: PRT<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: HPV antigen design 2 amino acids<br><br><400> SEQUENCE: 52 |     |         |
| Met Gln Thr Asp Arg Thr Gly Glu Thr Ala Leu His Leu Ala Ala Arg                                                                                                                                            |     |         |
| 1                                                                                                                                                                                                          | 5   | 10 15   |
| Tyr Ser Arg Ser Asp His Leu Asp Lys Lys Gln Arg Phe His Asn Ile                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 20  | 25 30   |
| Ala Asp Ala Arg Trp Thr Gly Arg Cys Met Ser Cys Cys Asp Asn Met                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 35  | 40 45   |
| Thr Thr Leu Glu Gln Gln Tyr Asn Lys Pro Leu Cys Asp Leu Leu Pro                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 50  | 55 60   |
| Leu His Ala Ala Val Ser Ala Asp Ile Ser Glu Tyr Arg His Tyr Cys                                                                                                                                            |     |         |
| 65                                                                                                                                                                                                         | 70  | 75 80   |
| Tyr Arg Asn Arg Ala Thr Asp Leu Val Tyr Asp Phe Ala Phe Arg Asp                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 85  | 90 95   |
| Leu Met His Asp Thr Ile His Asp Ile Ile Leu Glu Cys Val Pro Leu                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 100 | 105 110 |
| Ile Leu Ala Ala Arg Leu Ala Val Lys Leu Pro Gln Leu Cys Thr Glu                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 115 | 120 125 |
| Leu Ser His Ala Asp Val Phe Ala Phe Arg Asp Leu Cys Ile Val Tyr                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 130 | 135 140 |
| Asp Asp Leu Leu Cys Ile Val Tyr Arg Asp Gly Asn Pro Tyr Ala Val                                                                                                                                            |     |         |
| 145                                                                                                                                                                                                        | 150 | 155 160 |
| Cys Asp Ala Leu His Trp Ala Ala Ala Val Asn Asn Val Leu Cys Val                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 165 | 170 175 |
| Gln Ser Thr His Val Asp Ile Gly Ala Asn Lys Arg Thr Leu Glu Asp                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 180 | 185 190 |
| Leu Leu Met Gly Thr Asn Asn Arg Thr Leu Gly Ile Val Cys Pro Ile                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 195 | 200 205 |
| Pro Leu Phe Leu Ala Ala Arg Glu Gly Ser Tyr Glu Leu Leu Met Gly                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 210 | 215 220 |
| Thr Leu Gly Ile Val Ala Asn Arg Thr Leu His Glu Tyr Met Leu Asp                                                                                                                                            |     |         |
| 225                                                                                                                                                                                                        | 230 | 235 240 |
| Leu Asp His Met Ala His Tyr Asn Ile Val Thr Phe Cys Cys Pro Arg                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 245 | 250 255 |
| Asp Tyr Met Leu Asp Leu Gln Pro Glu Thr Thr Ile Val Arg Leu Leu                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 260 | 265 270 |
| Asp Leu Glu Val Ser Gln Thr Ser Lys Leu Thr Arg Gln Thr Asp Arg                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 275 | 280 285 |
| Thr Gly Glu Thr Ala Leu His Leu Ala Ala Arg Tyr Ser Arg Ser Asp                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 290 | 295 300 |
| Cys Asp Ser Thr Leu Arg Leu Cys Val Ala Asp Ala Lys Leu Thr Asn                                                                                                                                            |     |         |
| 305                                                                                                                                                                                                        | 310 | 315 320 |
| Thr Gly Leu Tyr Asn Leu Asp Asn Met Lys Cys Ile Asp Phe Tyr Ser                                                                                                                                            |     |         |
|                                                                                                                                                                                                            | 325 | 330 335 |
| Arg Ile Pro Leu His Ala Ala Val Ser Ala Asp Phe Ala Phe Lys Asp                                                                                                                                            |     |         |

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| 340                         | 345                                 | 350 |
|-----------------------------|-------------------------------------|-----|
| Leu Phe Val Val Arg Asn Arg | Ala Thr Asp Leu Asn Leu Leu Ile Arg |     |
| 355                         | 360                                 | 365 |
| Cys Leu Arg Cys Met His Asp | Lys Leu Pro Asp Leu Cys Thr Glu Leu |     |
| 370                         | 375                                 | 380 |
| Pro Leu Ile Leu Ala Ala Arg | Leu Ala Val Glu Leu Thr Glu Val Phe |     |
| 385                         | 390                                 | 395 |
| Glu Phe Ala Ser His Ala Asp | Val Ser Leu Gln Asp Ile Glu Ile Thr |     |
| 405                         | 410                                 | 415 |
| Cys Val Asp Asp Leu Lys Thr | Val Leu Glu Leu Thr Glu Val Leu His |     |
| 420                         | 425                                 | 430 |
| Trp Ala Ala Ala Val Asn Asn | Val Asp Asp Leu Arg Ala Phe Gln Gln |     |
| 435                         | 440                                 | 445 |
| Leu Phe Leu Asn Thr Leu Ser | Gly Ala Asn Lys Phe Gln Gln Leu Phe |     |
| 450                         | 455                                 | 460 |
| Leu Asn Thr Leu Asn Asn Arg | Gln Leu Phe Leu Asn Thr Leu Ser Phe |     |
| 465                         | 470                                 | 475 |
| Val Pro Leu Phe Leu Ala Ala | Arg Glu Gly Ser Tyr Glu Leu Phe Leu |     |
| 485                         | 490                                 | 495 |
| Asn Thr Leu Ser Phe Val Cys | Pro Trp Cys Ala Ser Ala Asn Arg Thr |     |
| 500                         | 505                                 | 510 |
| Leu Gln Asp Ile Val Leu His | Leu Asp His Met Ser Glu Glu Glu Asn |     |
| 515                         | 520                                 | 525 |
| Asp Glu Ile Asp Gly Val Asn | His Gln His Leu Pro Ala Arg Arg Pro |     |
| 530                         | 535                                 | 540 |
| Arg Asp Tyr Ile Ile Phe Val | Tyr Ile Pro Leu Ile Val Arg Leu Leu |     |
| 545                         | 550                                 | 555 |
| 560                         |                                     |     |
| Asp Leu Glu                 |                                     |     |

&lt;210&gt; SEQ ID NO 53

&lt;211&gt; LENGTH: 605

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: HPV antigen design 3 amino acids

&lt;400&gt; SEQUENCE: 53

|                                 |                                     |
|---------------------------------|-------------------------------------|
| Met Lys Leu Pro Gln Leu Cys Thr | Glu Val Met Phe Gln Asp Pro Gln     |
| 1                               | 15                                  |
| Glu Arg Pro Arg Lys Leu Pro Gln | Leu Cys Thr Ala Leu Gln Thr Thr     |
| 20                              | 30                                  |
| Ile His Asp Ile Ile Leu Glu Cys | Val Tyr Cys Lys Gln Gln Leu Leu     |
| 35                              | 45                                  |
| Arg Arg Glu Val Tyr Asp Phe     | Ala Phe Arg Asp Gly Cys Ile Val Tyr |
| 50                              | 60                                  |
| Arg Asp Gly Asn Pro Tyr Ala     | Val Cys Asp Lys Cys Leu Lys Phe Tyr |
| 65                              | 80                                  |
| Ser Lys Ile Ser Glu Tyr Arg His | Tyr Cys Tyr Ser Leu Tyr Gly Thr     |
| 85                              | 95                                  |
| Thr Leu Glu Gln Gln Tyr Asn     | Lys Pro Leu Cys Asp Leu Leu Ile Arg |
| 100                             | 110                                 |
| Cys Ile Asn Cys Gln Lys Pro     | Leu Cys Pro Glu Glu Lys Gln Arg His |
| 115                             | 125                                 |
| Leu Asp Lys Lys Gln Arg Phe     | His Asn Ile Arg Gly Arg Trp Thr Gly |
| 130                             | 140                                 |



|            |            |            |            |            |            |            |            |            |            |            |            |            |     |            |            |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-----|------------|------------|
| Arg<br>145 | Cys        | Met        | Ser        | Cys        | Cys<br>150 | Arg        | Ser        | Ser        | Arg        | Thr<br>155 | Arg        | Arg        | Ala | Ala        | Ala<br>160 |
| Ala        | Gln        | Leu        | Tyr        | Asn<br>165 | Lys        | Pro        | Leu        | Cys        | Asp<br>170 | Val        | Met        | Pro        | Gly | Asp<br>175 | Thr        |
| Pro        | Thr        | Leu        | His<br>180 | Glu        | Tyr        | Met        | Leu        | Asp<br>185 | Leu        | Gln        | Pro        | Glu        | Thr | Thr        | Asp        |
| Leu        | Tyr        | Gly        | Tyr        | Glu        | Gln        | Leu        | Asn<br>200 | Asp        | Ser        | Ser        | Glu        | Glu        | Glu | Asp        | Glu        |
| Ile        | Asp<br>210 | Gly        | Pro        | Ala        | Gly        | Gln<br>215 | Ala        | Ala        | Pro        | Asp<br>220 | Arg        | Ala        | His | Tyr        | Asn        |
| Ile        | Val        | Thr        | Phe        | Cys        | Cys<br>230 | Lys        | Cys        | Asp        | Ser        | Thr<br>235 | Leu        | Arg        | Arg | Cys        | Val<br>240 |
| Gln        | Ser        | Thr        | His<br>245 | Val        | Asp        | Ile        | Arg        | Thr        | Leu<br>250 | Glu        | Asp        | Leu        | Leu | Met        | Gly        |
| Thr        | Leu        | Gly        | Ile<br>260 | Val        | Cys        | Pro        | Ile        | Cys        | Ser        | Gln        | Lys        | Pro        | Arg | Thr        | Leu        |
| Glu        | Asp        | Leu        | Leu        | Met        | Gly        | Val        | Met        | Thr        | Asn<br>280 | Leu        | Asp        | Thr        | Ala | Ser        | Thr        |
| Thr        | Leu        | Leu        | Ala        | Cys        | Phe        | Leu<br>295 | Leu        | Cys        | Phe        | Cys        | Val<br>300 | Leu        | Leu | Cys        | Val        |
| Cys<br>305 | Leu        | Leu        | Ile        | Arg        | Pro        | Leu        | Leu        | Leu        | Ser        | Val<br>315 | Ser        | Thr        | Tyr | Thr        | Arg<br>320 |
| Cys        | Phe        | Ile        | Val        | Tyr<br>325 | Ile        | Ile        | Phe        | Val        | Tyr<br>330 | Ile        | Pro        | Leu        | Phe | Leu        | Ile<br>335 |
| His        | Lys        | Leu        | Pro        | Asp        | Leu        | Cys        | Thr        | Glu        | Val<br>345 | Met        | Ala        | Arg        | Phe | Glu        | Asp        |
| Pro        | Thr        | Arg<br>355 | Arg        | Pro        | Tyr        | Lys        | Leu        | Pro        | Asp<br>360 | Leu        | Cys        | Thr        | Ala | Leu        | Asn        |
| Thr        | Ser        | Leu        | Gln        | Asp        | Ile        | Glu<br>375 | Ile        | Thr        | Cys        | Val        | Tyr<br>380 | Cys        | Lys | Thr        | Val        |
| Leu<br>385 | Glu        | Leu        | Thr        | Glu        | Val        | Phe        | Glu        | Phe        | Ala        | Phe<br>395 | Lys        | Asp        | Gly | Phe        | Val<br>400 |
| Val        | Tyr        | Arg        | Asp        | Ser        | Ile        | Pro        | His        | Ala        | Ala<br>410 | Cys        | His        | Lys        | Cys | Ile        | Asp<br>415 |
| Phe        | Tyr        | Ser        | Arg<br>420 | Ile        | Arg        | Glu        | Leu        | Arg        | His<br>425 | Tyr        | Ser        | Asp        | Ser | Val        | Tyr        |
| Gly        | Asp        | Thr<br>435 | Leu        | Glu        | Lys        | Leu        | Thr        | Asn        | Thr<br>440 | Gly        | Leu        | Tyr<br>445 | Asn | Leu        | Leu        |
| Ile        | Arg<br>450 | Cys        | Leu        | Arg        | Cys        | Gln<br>455 | Lys        | Pro        | Leu        | Asn<br>460 | Pro        | Ala        | Glu | Lys        | Leu        |
| Arg<br>465 | His        | Lys        | Leu        | Thr        | Asn        | Thr<br>470 | Gly        | Leu        | Tyr        | Asn<br>475 | Val        | Glu        | Ile | Asp        | Gly<br>480 |
| Val        | Asn        | His        | Gln        | His        | Leu        | Pro        | Ala        | Arg        | Arg<br>490 | Ala        | Ala        | Pro        | Gln | Arg        | His        |
| Thr        | Met        | Leu        | Cys        | Met        | Cys        | Cys        | Lys        | Cys        | Glu<br>505 | Ala        | Arg        | Ile        | Glu | Arg        | Val        |
| Val        | Glu        | Ser<br>515 | Ser        | Ala        | Asp        | Asp        | Leu        | Arg        | Ala<br>520 | Phe        | Gln        | Gln        | Leu | Phe        | Leu        |
| Asn        | Thr        | Leu        | Ser        | Phe        | Val        | Cys<br>535 | Pro        | Trp        | Cys        | Ala        | Ser<br>540 | Gln        | Gln | Arg        | Ala        |
| Phe<br>545 | Gln        | Gln        | Leu        | Phe        | Leu        | Asn<br>550 | Val        | Met        | Leu        | Ser        | Leu<br>555 | Ile        | Phe | Leu        | Phe<br>560 |

|                                                           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|-----------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Cys                                                       | Phe | Cys | Val | Cys | Met | Tyr | Val | Cys | Cys | His | Val | Pro | Leu | Leu | Pro |  |
|                                                           |     |     |     | 565 |     |     |     |     | 570 |     |     |     | 575 |     |     |  |
| Ser                                                       | Val | Val | Val | Ile | Thr | Ser | Pro | Ala | Thr | Ala | Phe | Thr | Val | Tyr | Leu |  |
|                                                           |     |     |     | 580 |     |     |     |     | 585 |     |     |     | 590 |     |     |  |
| Leu                                                       | Pro | Met | Leu | Leu | Leu | His | Ile | His | Ala | Ile | Leu | Ser |     |     |     |  |
|                                                           |     |     |     | 595 |     |     |     |     | 600 |     |     |     | 605 |     |     |  |
| <210> SEQ ID NO 54                                        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| <211> LENGTH: 572                                         |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| <212> TYPE: PRT                                           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| <213> ORGANISM: Artificial sequence                       |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| <220> FEATURE:                                            |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| <223> OTHER INFORMATION: HPV antigen design 4 amino acids |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| <400> SEQUENCE: 54                                        |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| Met                                                       | Gln | Thr | Asp | Arg | Glu | Leu | Thr | Glu | Val | Phe | Glu | Phe | Ala | Ala | Arg |  |
|                                                           |     |     |     | 5   |     |     |     |     | 10  |     |     |     | 15  |     |     |  |
| Tyr                                                       | Ser | Arg | Ser | Asp | Tyr | Ile | Ile | Phe | Val | Tyr | Ile | Pro | Leu | Ala | Asp |  |
|                                                           |     |     |     | 20  |     |     |     |     | 25  |     |     |     | 30  |     |     |  |
| Ala                                                       | Lys | Leu | Pro | Gln | Leu | Cys | Thr | Glu | Val | Asp | Asn | Met | Leu | Leu | Met |  |
|                                                           |     |     |     | 35  |     |     |     |     | 40  |     |     |     | 45  |     |     |  |
| Gly                                                       | Thr | Leu | Gly | Ile | Val | Pro | Leu | His | Ala | Ala | Val | Ser | Ala | Asp | Thr |  |
|                                                           |     |     |     | 50  |     |     |     |     | 55  |     |     |     | 60  |     |     |  |
| Leu                                                       | His | Glu | Tyr | Met | Leu | Asp | Leu | Arg | Asn | Arg | Ala | Thr | Asp | Leu | Ser |  |
|                                                           |     |     |     | 65  |     |     |     |     | 70  |     |     |     | 75  |     |     |  |
| Glu                                                       | Glu | Glu | Asn | Asp | Glu | Ile | Asp | Gly | Val | Asn | His | Gln | His | Leu | Pro |  |
|                                                           |     |     |     | 85  |     |     |     |     | 90  |     |     |     | 95  |     |     |  |
| Ala                                                       | Arg | Arg | Met | His | Asp | Asn | Leu | Leu | Ile | Arg | Cys | Leu | Arg | Cys | Pro |  |
|                                                           |     |     |     | 100 |     |     |     |     | 105 |     |     |     | 110 |     |     |  |
| Leu                                                       | Ile | Leu | Ala | Ala | Arg | Leu | Ala | Val | Leu | Phe | Leu | Asn | Thr | Leu | Ser |  |
|                                                           |     |     |     | 115 |     |     |     |     | 120 |     |     |     | 125 |     |     |  |
| Phe                                                       | Val | Cys | Pro | Trp | Cys | Ala | Ser | Ser | His | Ala | Asp | Val | Lys | Cys | Ile |  |
|                                                           |     |     |     | 130 |     |     |     |     | 135 |     |     |     | 140 |     |     |  |
| Asp                                                       | Phe | Tyr | Ser | Arg | Ile | Asp | Asp | Leu | Lys | Leu | Pro | Asp | Leu | Cys | Thr |  |
|                                                           |     |     |     | 145 |     |     |     |     | 150 |     |     |     | 155 |     |     |  |
| Glu                                                       | Leu | Ala | Leu | His | Trp | Ala | Ala | Ala | Val | Asn | Asn | Val | Gln | Leu | Phe |  |
|                                                           |     |     |     | 165 |     |     |     |     | 170 |     |     |     | 175 |     |     |  |
| Leu                                                       | Asn | Thr | Leu | Ser | Phe | Val | Gly | Ala | Asn | Lys | Phe | Gln | Gln | Leu | Phe |  |
|                                                           |     |     |     | 180 |     |     |     |     | 185 |     |     |     | 190 |     |     |  |
| Leu                                                       | Asn | Thr | Leu | Asn | Asn | Arg | Val | Tyr | Asp | Phe | Ala | Phe | Arg | Asp | Leu |  |
|                                                           |     |     |     | 195 |     |     |     |     | 200 |     |     |     | 205 |     |     |  |
| Pro                                                       | Leu | Phe | Leu | Ala | Ala | Arg | Glu | Gly | Ser | Tyr | Glu | Lys | Leu | Thr | Asn |  |
|                                                           |     |     |     | 210 |     |     |     |     | 215 |     |     |     | 220 |     |     |  |
| Thr                                                       | Gly | Leu | Tyr | Asn | Leu | Ala | Asn | Arg | Leu | Cys | Ile | Val | Tyr | Arg | Asp |  |
|                                                           |     |     |     | 225 |     |     |     |     | 230 |     |     |     | 235 |     |     |  |
| Gly                                                       | Asn | Pro | Tyr | Ala | Val | Cys | Asp | Asp | His | Met | Tyr | Met | Leu | Asp | Leu |  |
|                                                           |     |     |     | 245 |     |     |     |     | 250 |     |     |     | 255 |     |     |  |
| Gln                                                       | Pro | Glu | Thr | Thr | Pro | Arg | Asp | Phe | Ala | Phe | Arg | Asp | Leu | Cys | Ile |  |
|                                                           |     |     |     | 260 |     |     |     |     | 265 |     |     |     | 270 |     |     |  |
| Val                                                       | Tyr | Ile | Val | Arg | Leu | Leu | Asp | Leu | Glu | Val | Ser | Gln | Thr | Ser | Lys |  |
|                                                           |     |     |     | 275 |     |     |     |     | 280 |     |     |     | 285 |     |     |  |
| Leu                                                       | Thr | Arg | Gln | Thr | Asp | Arg | Thr | Leu | Gly | Ile | Val | Cys | Pro | Ile | Ala |  |
|                                                           |     |     |     | 290 |     |     |     |     | 295 |     |     |     | 300 |     |     |  |
| Ala                                                       | Arg | Tyr | Ser | Arg | Ser | Asp | Arg | Thr | Leu | Glu | Asp | Leu | Leu | Met | Gly |  |
|                                                           |     |     |     | 305 |     |     |     |     | 310 |     |     |     | 315 |     |     |  |
| 320                                                       |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |



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Phe Tyr Ser Arg Ile Lys Lys Lys Leu Pro Asp Leu Cys Thr Glu Leu  
                   115                  120                  125  
 Lys Lys Gln Leu Phe Leu Asn Thr Leu Ser Phe Val Lys Lys Phe Gln  
                   130                  135                  140  
 Gln Leu Phe Leu Asn Thr Leu Lys Lys Val Tyr Asp Phe Ala Phe Arg  
                   145                  150                  155                  160  
 Asp Leu Lys Lys Lys Leu Thr Asn Thr Gly Leu Tyr Asn Leu Lys Lys  
                   165                  170                  175  
 Leu Cys Ile Val Tyr Arg Asp Gly Asn Pro Tyr Ala Val Cys Asp Lys  
                   180                  185                  190  
 Lys Tyr Met Leu Asp Leu Gln Pro Glu Thr Thr Lys Lys Phe Ala Phe  
                   195                  200                  205  
 Arg Asp Leu Cys Ile Val Tyr Lys Lys Thr Leu Gly Ile Val Cys Pro  
                   210                  215                  220  
 Ile Lys Lys Arg Thr Leu Glu Asp Leu Leu Met Gly Val Lys Lys Thr  
                   225                  230                  235                  240  
 Ile His Asp Ile Ile Leu Glu Cys Val Lys Lys His Leu Asp Lys Lys  
                   245                  250                  255  
 Gln Arg Phe His Asn Ile Lys Lys Thr Thr Leu Glu Gln Gln Tyr Asn  
                   260                  265                  270  
 Lys Pro Leu Cys Asp Leu Leu Lys Lys Cys Asp Ser Thr Leu Arg Leu  
                   275                  280                  285  
 Cys Val Lys Lys Arg Trp Thr Gly Arg Cys Met Ser Cys Cys Lys Lys  
                   290                  295                  300  
 Ser Leu Gln Asp Ile Glu Ile Thr Cys Val Lys Lys Leu Cys Val Gln  
                   305                  310                  315                  320  
 Ser Thr His Val Asp Ile Lys Lys Ala His Tyr Asn Ile Val Thr Phe  
                   325                  330                  335  
 Cys Cys Lys Lys Asp Asp Leu Arg Ala Phe Gln Gln Leu Phe Leu Asn  
                   340                  345                  350  
 Thr Leu Ser Lys Lys Ile Ser Glu Tyr Arg His Tyr Cys Tyr Lys Lys  
                   355                  360                  365  
 Lys Leu Pro Gln Leu Cys Thr Glu Leu Lys Lys Lys Thr Val Leu Glu  
                   370                  375                  380  
 Leu Thr Glu Val Lys Lys Arg Thr Leu Glu Asp Leu Leu Met Gly Thr  
                   385                  390                  395                  400  
 Lys Lys Phe Ala Phe Lys Asp Leu Phe Val Val Lys Lys Gln Leu Tyr  
                   405                  410                  415  
 Asn Lys Pro Leu Cys Asp Val Lys Lys Thr Leu Gln Asp Ile Val Leu  
                   420                  425                  430  
 His Leu Lys Lys  
                   435

&lt;210&gt; SEQ ID NO 56

&lt;211&gt; LENGTH: 6380

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

 <223> OTHER INFORMATION: HPV antigen design 1 full nucleotide sequences  
 of gorilla adenovirus shuttle plasmid

&lt;400&gt; SEQUENCE: 56

gacgaaaggg cctcgtgata cgctatttt tatagggttaa tgtcatgata ataatggttt 60

cttagacgtc aggtggcact ttctggggaa atgtgcgcgg aaccttatt tgttatttt 120

tctaaataca ttcaaatatg tatccgctca tgagacaata accctgataa atgcttcaat 180

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|                                                                     |      |
|---------------------------------------------------------------------|------|
| aatattgaaa aaggaagagt atgagtattc aacattttccg tgtcgccctt attccctttt  | 240  |
| ttgcggcatt ttgccttcct gtttttgcgc acccagaaac gctggtgaaa gtaaaagatg   | 300  |
| ctgaagatca gttgggtgca cgagtgggtt acatcgaact ggatctcaac agcggtaaga   | 360  |
| tccttgagag ttttcgcccc gaagaacgtt ttccaatgat gagcactttt aaagttctgc   | 420  |
| tatgtggcgc ggtattatcc cgtattgacg ccgggcaaga gcaactcggc cgccgcatac   | 480  |
| actattctca gaatgacttg gttgagtact caccagtcac agaaaagcat cttacggatg   | 540  |
| gcatgacagt aagagaatta tgcagtgcgc ccataaccat gagtataaac actgcggcca   | 600  |
| acttacttct gacaacgacg ggaggaccga aggagctaac cgcttttttg cacaacatgg   | 660  |
| gggatcatgt aactcgccctt gatcggtggg aaccggagct gaatgaagcc ataccaaacg  | 720  |
| acgagcgtga caccacgatg cctgtagcaa tggcaacaac gttgcgcaaa ctattaactg   | 780  |
| gcgaactact tactctagct tcccggaac aattaataga ctggatggag gcggataaag    | 840  |
| ttgcaggacc actctgcgc tcggcccttc cggtcggctg gtttattgct gataaatctg    | 900  |
| gagccgggtga gcgtgggtct cgcgggtatca ttgcagcact ggggccagat ggtaagccct | 960  |
| cccgatcgt agttatctac acgacgggga gtcaggcaac tatggatgaa cgaaatagac    | 1020 |
| agatcgctga gatagggtcc tcaactgatta agcattggta actgtcagac caagtttact  | 1080 |
| catatatact ttagattgat ttaaaacttc atttttaatt taaaaggatc taggtgaaga   | 1140 |
| tcctttttga taatctcatg accaaaatcc cttaacgtga gttttcgctc cactgagcgt   | 1200 |
| cagaccccg agaaaagatc aaaggatctt cttgagatcc tttttttctg cgcgtaatct    | 1260 |
| gctgcttgca aacaaaaaaa ccaccgctac cagcgggtgtg ttgtttgccg gatcaagagc  | 1320 |
| taccaactct ttttcgaag gtaactggct tcagcagagc gcagatacca aatactgttc    | 1380 |
| ttctagtgtg gccgtagtta ggccaccact tcaagaactc tgtagcaccg cctacatacc   | 1440 |
| tcgctctgct aatcctgtta ccagtggctg ctgccagtgg cgataagtcg tgtcttaccg   | 1500 |
| ggttggactc aagacgatag ttaccggata aggcgcagcg gtcgggctga acggggggtt   | 1560 |
| cgtgcacaca gccacgcttg gagcgaacga cctacaccga actgagatac ctacagcgtg   | 1620 |
| agctatgaga aagcgccacg ctccccaag ggagaaaggg ggacaggat ccggtaagcg     | 1680 |
| gcagggtcgg aacaggagag cgcacgaggg agcttcacgg gggaaacgcc tggatatctt   | 1740 |
| atagtcctgt cgggtttcgc cacctctgac ttgagcgtcg atttttgtga tgctcgtcag   | 1800 |
| gggggaggag cctatggaaa aacgccagca acgcggcctt tttacggttc ctggcctttt   | 1860 |
| gctggccttt tgtcacatg ttctttctcg cgttatcccc tgattctgtg gataaccgta    | 1920 |
| ttaccgcctt tgagttagct gataccgctc gccgcagccg aacgaccgag cgcagcgagt   | 1980 |
| cagttagcga ggaagcggaa gagcgcccaa tacgcaaacc gcctctcccc gcgcgttggc   | 2040 |
| cgattcatta atgcagctgg cagcagaggt ttcccgaactg gaaagcgggc agtgagcgca  | 2100 |
| acgcaattaa tgtgagttag ctcaactcatt aggcacccca ggctttacac tttatgcttc  | 2160 |
| cggctcgtat gttgtgtgga attgtgagcg gataacaatt tcacacagga aacagctatg   | 2220 |
| accatgatta cgccaagctg ggtcaagtct tccagtttaa gcagcagagc ggtcagtttc   | 2280 |
| tcaccccgag cagacgcgcg agaggccgcg ccgctcgcca ccaaagagct gtaaaggctc   | 2340 |
| gtagccatgc tgcgcgcggg cgcggcgggc gcggaggcgg cggcggaggc cgcggcgctc   | 2400 |
| agcggagtto ctcacacggg cgcgtagggc attgtagacg aatttgaagg cagaacgggg   | 2460 |
| cgtccatcca cgttggaacc catcacattc tgacgcactc cagcccagtg aggcacgcgc   | 2520 |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| actgtcagat aggggctaaa gatgttcca tcaaagctgt tgccggtgtc gctcatggcg   | 2580 |
| gcggtctgtt caagacaaaa cagagagacc cttagacccc caattttatac acgccccacc | 2640 |
| cttctagcca cgcccacott acccacctca atcggtatcc tcatcgctag acccaaaactc | 2700 |
| ggccctgggt caggccagca ccagatggtc aggctctag ttattaatag taatcaatta   | 2760 |
| cggggtcatt agttcatagc ccatatatgg agttccgctg tacataaact acggtaaatg  | 2820 |
| gcccgcctgg ctgaccgccc aacgaccccc gccattgac gtcaataatg acgtatgttc   | 2880 |
| ccatagtaac gccaataggg actttccatt gacgtcaatg ggtggagtat ttacggtaaa  | 2940 |
| ctgcccactt ggcagtacat caagtgtatc atatgccaag tacgccccct attgacgtca  | 3000 |
| atgacggtaa atggcccgc tggcattatg ccagttacat gaccttatgg gactttccta   | 3060 |
| cttggcagta catctacgta ttagtcatcg ctattaccat ggtgatgcgg ttttggcagt  | 3120 |
| acatcaatgg cgtggatag cggtttgact cacggggatt tccaagtctc caccocattg   | 3180 |
| acgtcaatgg gagtttgtt tggcacaaaa atcaacggga ctttccaaaa tgtcgtaaca   | 3240 |
| actccgcccc attgacgcaa atgggcggtg ggcgtgtacg gtgggaggtc tatataagca  | 3300 |
| gagctctccc tatcagtgat agagatctcc ctatcagtga tagagatcgt cgacgagctc  | 3360 |
| gtttagttaa cgtcagatc gcttgagac gccatccacg ctgttttgac ctccatagaa    | 3420 |
| gacaccggga ccgatccagc ctccgcgcc gggaacggtg cattggaacg cggattcccc   | 3480 |
| gtgccaagag tgacgtaagt accgcctata gagtctatag gccaccccc ttggcttcac   | 3540 |
| ttgaggtgtg gcaggcttga gatctggcca tacacttgag tgacaatgac atccacttg   | 3600 |
| cctttctctc cacaggtgtc cactcccagg tccaactgca gccggatccg gtaccgccac  | 3660 |
| catgatgttc caggatctc aagaacggcc tcggaacta cctcagctgt gtacagcctt    | 3720 |
| acagacaaca atacatgata taatcctgga gtgctgtac tgtaaacagc aactgttgcg   | 3780 |
| tcgcgaggtc tatgatctcg ccttcaggga tggttgtatt gtatacagag acggcaatcc  | 3840 |
| gtatgcagtg tgtgataagt gtcctcaagt ctatagcaaa atcagtgaat accggcacta  | 3900 |
| ttgctatagc ctgtatggaa ctacactaga gcagcagtat aacaagccac tctgtgacct  | 3960 |
| gctaattaga tgtatcaact gccagaagcc tctgtgcccc gaagagaagc agcggcacct  | 4020 |
| tgacaagaag cagcggttc ataatatcag agggaggtgg accgggcggt gtatgtcctg   | 4080 |
| ctgccgcagc agcaggaccc gtcgggccgc ggcggcaatg ccaggggaca cgcccacact  | 4140 |
| ccacgagtac atgctcgacc ttcagccgga aaccaccgat ctctacggct atgaacagtt  | 4200 |
| gaacgacagc tccgaagagg aagatgagat cgacggccca gcaggacagg cggcccctga  | 4260 |
| cagagcacac tacaatatg ttacctctg ctgcaagtgc gacagcacgc tgcgccgatg    | 4320 |
| cgtacaatct acgcatgtgg acattaggac ccttgaggat ctgcttatgg ggacactcgg  | 4380 |
| aattgtttgt cccatatgca gtcagaaacc tatgaccaac ctacacacgg cgagcacgac  | 4440 |
| tcttctggct tgttctctgc tatgtttttg cgtgctgctt tgtgtttgcc tactcattcg  | 4500 |
| gcccctgctg ctgtctgtga gcacgtacac gaggtgtctt attgtctata ttatcttctg  | 4560 |
| ttacataccg ctgttctctga ttcacatggc aagatttgaa gatccacgc ggccggccata | 4620 |
| caaaactcct gatctctgca cggcactgaa cacttcgttg caggacatag aaatcacttg  | 4680 |
| cgtgtactgc aagactgtgc tagagctgac cgaggtgttt gagtttgctt tcaaggacgg  | 4740 |
| ttttgtgggt tatcgggata gcacccccca tgccgctgc cataaatgta ttgatttcta   | 4800 |
| ttcgagaatt agagaactca ggcattattc cgactccgta tacggcgaca ccctagaaaa  | 4860 |
| actgaccaac accggcttgt ataacctcct catccgctgc ctccgatgcc agaagccact  | 4920 |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| taaccacgccc gagaaattac gccacgagat cgacggggtc aaccatcagc acctgccggc | 4980 |
| cagacgagccc gccctcagc gccataccat gctgtgcatg tgttgcaagt gcgaagccag  | 5040 |
| gatcgaaagg gtggtggaat ctagcgctga cgacttaagg gcattccaac aactctttct  | 5100 |
| taacaccctg agcttcgtgt gtccttggtg tgcttccaa cagatgctta gtctgatttt   | 5160 |
| tctgttttgc ttttgctgtg gtatgtatgt ctgctgccac gtgcctctcc taccatctgt  | 5220 |
| agtgggtgatt acatcccccg cgactgcctt caccgtgtac ctgttgccaa tgctcctcct | 5280 |
| tcacatacac gccatcctga gctaagcggc cgctctcgag tctagctagt ctagactagc  | 5340 |
| tagaaagatc cggaacttg tttattgcag cttataatgg ttacaaataa agcaatagca   | 5400 |
| tcacaaattt cacaataaaa gcattttttt cactgcattc tagttgtggt ttgtccaaac  | 5460 |
| tcataaatgt atcttatcat gtctggatcg gtgatcaccg atcccgatct gtcaaatgac  | 5520 |
| ggtgacaata aaacggagac ttgaccccg aacgcggaaa ttcacgtaaa aaacacctgg   | 5580 |
| gcgagtcctc cagtaatcg gtcaaagtcc ctccgccctc ggtaaatatt acgcactatg   | 5640 |
| actaacgccc tattattcag ttttcacttc cccgtttcac ttttcgcgag aaaatggcca  | 5700 |
| aatcttacat ggtcccgccc aaaattacta cgatatccgg tgaaaagcgc gcgaaaattg  | 5760 |
| gcacttcctg aggtaggcgg cgctcatcaa aaacgtcaca ttttcgcga cggaagcttg   | 5820 |
| catgtgagct cctccactt gcaaatgcc aacttcgcc acacctccca acctactcg      | 5880 |
| cgcgtctac gtcacccgcc ccgctctctc ccgccacct cattatcata ttggccacaa    | 5940 |
| tccaaaaataa ggtatattat tgatgatggt ttaaacgccc aattcactgg ccgtcgtttt | 6000 |
| acaacgtcgt gactgggaaa accctggcgt tacccaactt aatgccttg cagcacatcc   | 6060 |
| ccctttcgcc agctggcgta atagcgaaga ggcccgacc gatcgccctt cccaacagtt   | 6120 |
| gcgcagcctg aatggcgaaat ggcgctgat gcggtatttt ctcttacgc atctgtgcgg   | 6180 |
| tatttcacac cgcataatgt gcactctcag tacaatctgc tctgatgcc catagttaag   | 6240 |
| ccagccccga cacccgccaa caccgctga cgcgcctga cgggcttgtc tgctcccgcc    | 6300 |
| atccgcttac agacaagctg tgaccgtctc cgggagctgc atgtgtcaga ggttttcacc  | 6360 |
| gtcatcacccg aaacgcgcga                                             | 6380 |

&lt;210&gt; SEQ ID NO 57

&lt;211&gt; LENGTH: 6428

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: HPV antigen design 2 full nucleotide sequences of gorilla adenovirus shuttle plasmid

&lt;400&gt; SEQUENCE: 57

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| gacgaaagg cctcgtgata cgcctatttt tatagggttaa tgcatagata ataattggttt | 60  |
| cttagacgtc aggtggcact tttcggggaa atgtgcgcgg aacctctatt tgtttatttt  | 120 |
| tctaaataca ttcaaatatg tatccgctca tgagacaata acctgataa atgettcaat   | 180 |
| aatattgaaa aaggaagagt atgagtattc aacatttccg tgctgcctct attccctttt  | 240 |
| ttgcggcatt ttgccttctc gtttttgctc acccagaaac gctggtgaaa gtaaaagatg  | 300 |
| ctgaagatca gttgggtgca cgagtgggtt acatcgaact ggatctcaac agcggtaaga  | 360 |
| tccttgagag ttttcgccc gaagaacgtt ttccaatgat gagcactttt aaagtctctgc  | 420 |
| tatgtggcgc ggtattatcc cgtattgacg ccgggcaaga gcaactcggc cgccgcatac  | 480 |
| actattctca gaatgacttg gttgagtact caccagtcac agaaaagcat cttacggatg  | 540 |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| gcatgacagt aagagaatta tgcagtgtctg ccataacat gagtgataac actgcggcca  | 600  |
| acttacttct gacaacgatc ggaggaccga aggagctaac cgcttttttg cacaacatgg  | 660  |
| gggatcatgt aactcgcctt gatcgttggg aaccggagct gaatgaagcc ataccaaacg  | 720  |
| acgagcgtga caccacgatg cctgtagcaa tggcaacaac gttgcgcaaa ctattaactg  | 780  |
| gcgaactact tactctagct tcccggcaac aattaataga ctggatggag gcggataaag  | 840  |
| ttgcaggacc acttctgcgc tcggcccttc cggtggctg gtttattgct gataaatctg   | 900  |
| gagccggtga gcgtgggtct cgcggtatca ttgcagcact ggggccagat ggtaagccct  | 960  |
| cccgatcgt agttatctac acgacgggga gtcaggcaac tatggatgaa cgaaatagac   | 1020 |
| agatcgctga gatagggtcc tactgatta agcatttgta actgtcagac caagtttact   | 1080 |
| catatatact ttagattgat ttaaaacttc atttttaatt taaaaggatc taggtgaaga  | 1140 |
| tcctttttga taatctcatg accaaaatcc cttaacgtga gtttcgttc cactgagcgt   | 1200 |
| cagaccccg agaaaagatc aaaggatctt cttgagatcc ttttttctg cgcgtaatct    | 1260 |
| gtgcttgca aacaaaaaaa ccaccgtac cagcgggtgtg ttgtttgccg gatcaagagc   | 1320 |
| taccaactct tttccgaag gtaactggct tcagcagagc gcagatacca aatactgttc   | 1380 |
| ttctagtgtg gccgtagtta ggccaccact tcaagaactc tgtagcaccg cctacatacc  | 1440 |
| tcgctctgct aatcctgtta ccagtggctg ctgccagtgg cgataagtcg tgtcttaccg  | 1500 |
| ggttggaactc aagacgatag ttaccggata aggcgcagcg gtcgggctga acggggggtt | 1560 |
| cgtgcacaca gccagcttg gagcgaacga cctacaccga actgagatac ctacagcgtg   | 1620 |
| agctatgaga aagcgcacg cttccgaag ggagaaaggc ggacaggtat ccggtaagcg    | 1680 |
| gcagggtcgg aacaggagag cgcacgagg agcttccagg gggaaacgcc tggatatctt   | 1740 |
| atagtcctgt cgggtttcgc cacctctgac ttgagcgtcg atttttgtga tgctcgtcag  | 1800 |
| gggggcgagg cctatggaaa aacgccagca acgcggcctt tttacggttc ctggcctttt  | 1860 |
| gctggccttt tgctcacatg ttctttctg cgttatcccc tgattctgtg gataaccgta   | 1920 |
| ttaccgcctt tgagttagct gataccgctc gccgcagccg aacgaccgag cgcagcgagt  | 1980 |
| cagttagcga ggaagcggaa gagcgcccaa tacgcaaacc gcctctcccc gcgcgttggc  | 2040 |
| cgattcatta atgcagctgg cagcagaggt ttcccgactg gaaagcgggc agtgagcgca  | 2100 |
| acgcaattaa tgtgagttag ctactcatt aggcacccca ggctttacac tttatgcttc   | 2160 |
| cggctcgat gttgtgtgga attgtgagcg gataacaatt tcacacagga aacagctatg   | 2220 |
| accatgatta cgccaagctg ggtcaagtct tccagtttaa gcagcagagc ggtaagtttc  | 2280 |
| tcaccccgag cagacgcgcg agaggccgcg ccgctcgcca ccaaagagct gtaaaggctc  | 2340 |
| gtagccatgc tgcgcgcggt cgcggcggcg gcggaggcg cggcggaggt cgcggcgctc   | 2400 |
| agcggagttc ctcccacggt cgcgtaggcc attgtagacg aatttgaagg cagaacgggg  | 2460 |
| cgtccatcca cgttggaacc catcacattc tgacgcactc cagcccagtg aggcattgcgc | 2520 |
| actgtcagat aggggctaaa gatgcttcca tcaaagctgt tgccggtgtc gctcatggcg  | 2580 |
| gcggctgttg caagacaaaa cagagagacc cttagacccc caatttatac acgccccacc  | 2640 |
| cttctagcca cgcccacctt acccacctca atcggtatcc tcacgctag acccaaacctc  | 2700 |
| ggccctgggt caggccagca ccagatggtc aggcctctag ttattaatag taatcaatta  | 2760 |
| cggggtcatt agttcatagc ccatatatgg agttccgcgt tacataactt acggtaaatg  | 2820 |
| gcccgcctgg ctgaccgccc aacgaccccc gccattgac gtcaataatg acgtatgttc   | 2880 |



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|                                                                    |      |
|--------------------------------------------------------------------|------|
| ccatagtaac gccaataggg actttccatt gacgtcaatg ggtggagtat ttacggtaaa  | 2940 |
| ctgcccactt ggcagtacat caagtgtatc atatgccaag tacgccccct attgacgtca  | 3000 |
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| acatcaatgg gcgtggatag cggtttgact cacggggatt tccaagtctc caccgccattg | 3180 |
| acgtcaatgg gagtttgttt tggcaccaaa atcaacggga ctttccaaaa tgtcgtaaca  | 3240 |
| actccgcccc attgacgcaa atgggcggta ggcgtgtacg gtgggaggtc tatataagca  | 3300 |
| gagctctccc tatcagtgat agagatctcc ctatcagtga tagagatcgt cgacgagctc  | 3360 |
| gtttagttaa cgcgcagatc gcttgagac gccatccacg ctgttttgac ctccatagaa   | 3420 |
| gacaccggga cgcgtccagc ctccgcgcc gggaacgggt cattggaacg cggattcccc   | 3480 |
| gtgccaagag tgacgtaagt accgcctata gagtctatag gccaccccc ttggcttcac   | 3540 |
| ttgaggtgtg gcaggcttga gatctggcca tacacttgag tgacaatgac atccacttg   | 3600 |
| cctttctctc cacagggtgc cactcccagg tccaactgca gccggatccg gtaccgccac  | 3660 |
| catgcagact gatagaaccg gcgaaaccgc tctacacctt gccgcccgtt atagtcgaa   | 3720 |
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| tgatttacag ccgaaacaa ccacgtgag actgctcgat ctggaggat ctacagacaag    | 4500 |
| taagctcaca agacaaacag acagaaccgg cgagactgct ttgcatctgg ctgcacgata  | 4560 |
| ttccaggfcc gactgtgata gcaccctgcg gttatgtgtg gcagacgcaa aacttacaa   | 4620 |
| taccggccta tataacctcg acaacatgaa gtgcacgac ttctacagca gaatccccct   | 4680 |
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| gtgtactgag ctgcctctca ttctggccgc acggctcgca gttgaactaa ccgaagtgtt  | 4860 |
| cgagttcgcc tcccacgccg acgtgtccct acaggacatc gagatcacat gcgtagacga  | 4920 |
| tctgaaaacc gttctcgaa tcacagaagt gcttcattgg gccgccgag tcaacaacgt    | 4980 |
| ggacgatctc cgggcattcc agcaactgtt cctgaataca ctgtctggcg ctaacaagtt  | 5040 |
| tcagcagtta tttttgaaca cacttaataa ccgccagcta ttcttgaata ccctgtcatt  | 5100 |
| cgtgccgctg ttcttggtg cgcgcgagg gtcctacgag ctgtttctga atacgtctc     | 5160 |
| cttcgtgtgt ccctgggtgc cctccgctaa taggacctg caagacatcg tacttcacct   | 5220 |
| ggatcacatg tcagaggagg agaacgatga gatcgacgtg gtcaaccacc agcacctccc  | 5280 |

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&lt;210&gt; SEQ ID NO 58

&lt;211&gt; LENGTH: 6554

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

<223> OTHER INFORMATION: HPV antigen design 3 full nucleotide sequences  
of gorilla adenovirus shuttle plasmid

&lt;400&gt; SEQUENCE: 58

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aatattgaaa aaggaagagt atgagtattc aacatttccg tgcgcctctt attccctfff 240
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| gagccggtga | gcgtgggtct | cgcggtatca  | ttgcagcact  | ggggccagat  | ggtaagccct | 960  |
| cccgatcgt  | agttatctac | acgacgggga  | gtcaggcaac  | tatggatgaa  | cgaaatagac | 1020 |
| agatcgctga | gatagggtcc | tactgatta   | agcattggta  | actgtcagac  | caagtttact | 1080 |
| catatatact | ttagattgat | ttaaaacttc  | atttttaatt  | taaaaggatc  | taggtgaaga | 1140 |
| tcctttttga | taatctcatg | acaaaaatcc  | cttaacgtga  | gttttcgttc  | cactgagcgt | 1200 |
| cagaccccg  | agaaaagatc | aaaggatctt  | cttgagatcc  | ttttttctg   | cgcgtaatct | 1260 |
| gctgcttgca | aacaaaaaaa | ccaccgctac  | cagcgggtgt  | ttgtttgccg  | gatcaagagc | 1320 |
| taccaactct | ttttccgaag | gtaactggct  | tcagcagagc  | gcagatacca  | aatactgttc | 1380 |
| ttctagtgt  | gccgtagtta | ggccaccact  | tcaagaactc  | tgtagcaccg  | cctacatacc | 1440 |
| tcgctctgct | aatcctgtta | ccagtggctg  | ctgccagtgg  | cgataagtcg  | tgtcttaccg | 1500 |
| ggttggaact | aagacgatag | ttaccggata  | aggcgcagcg  | gtcgggctga  | acgggggggt | 1560 |
| cgtgcacaca | gcccgcttg  | gagcgaacga  | cctacaccga  | actgagatac  | ctacagcgtg | 1620 |
| agctatgaga | aagcgccacg | cttcccgaag  | ggagaaaggc  | ggacaggtat  | ccggtaaagc | 1680 |
| gcagggtcgg | aacaggagag | cgcacgaggg  | agcttccagg  | gggaaacgcc  | tggtatcttt | 1740 |
| atagtctctg | cggtttctgc | cacctctgac  | ttgagcgtcg  | atttttgtga  | tgtctgtcag | 1800 |
| gggggcggag | cctatggaaa | aacgccagca  | acgcggcctt  | tttacggttc  | ctggcctttt | 1860 |
| gctggccttt | tgtccacatg | ttctttctctg | cgttatcccc  | tgattctgtg  | gataaccgta | 1920 |
| ttaccgcctt | tgagtgagct | gataccgctc  | gccgcagccg  | aacgaccgag  | cgcagcgagt | 1980 |
| cagtgagcga | ggaagcggaa | gagcgcccaa  | tacgcaaacc  | gcctctcccc  | gcgcgttgge | 2040 |
| cgattcatta | atgcagctgg | cacgacaggt  | ttcccgaactg | gaaagcgggc  | agtgaagcga | 2100 |
| acgcaattaa | tgtgagttag | ctcactcatt  | aggcacccca  | ggcttttacac | tttatgcttc | 2160 |
| cggctcgat  | gttgtgtgga | attgtgagcg  | gataacaatt  | tcacacagga  | aacagctatg | 2220 |
| accatgatta | cgccaagctg | ggteaagtct  | tccagtttaa  | gcagcagagc  | ggtcagtttc | 2280 |
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| cgggttcatt | agttcatagc | ccatatatgg  | agttccgcgt  | tacataactt  | acggtaaatg | 2820 |
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&lt;210&gt; SEQ ID NO 59

&lt;211&gt; LENGTH: 6455

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

<223> OTHER INFORMATION: HPV antigen design 4 full nucleotide sequences  
of gorilla adenovirus shuttle plasmid

&lt;400&gt; SEQUENCE: 59

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| gggggcggag cctatggaaa aacgccagca acgcggcctt ttacgggttc ctggcctttt  | 1860 |
| gctggccttt tgcacacatg ttcttctctg cgttatcccc tgattctgtg gataaccgta  | 1920 |
| ttaccgcctt tgagttagct gataccgtc gccgcagccg aacgaccgag cgcagcgagt   | 1980 |
| cagtgagcga ggaagcggaa gagcgcccaa tacgcaaacc gcctctcccc gcgcgttggc  | 2040 |
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| acgcaattaa tgtgagttag ctcaactcatt aggcacccca ggctttacac tttatgcttc | 2160 |
| cggctcgatg gttgtgtgga attgtgagcg gataacaatt tcacacagga aacagctatg  | 2220 |
| accatgatta cgccaagctg ggtcaagtct tccagtttaa gcagcagagc ggtcagtttc  | 2280 |
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| cgtccatcca cgttggaacc catcacattc tgacgcactc cagcccagtg aggcattgctc | 2520 |
| actgtcagat aggggctaaa gatgcttcca tcaaagctgt tgccggtgtc gctcatggcg  | 2580 |
| gcggctgttg caagacaaaa cagagagacc cttagacccc caattttatac acgccccacc | 2640 |
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| atgacggtaa atggcccgc tgccattatg ccagctacat gaccttatgg gactttccta   | 3060 |
| cttggcagta catctacgta ttagtcatcg ctattaccat ggtgatgcgg ttttggcagt  | 3120 |
| acatcaatgg gcgtggatag cggtttgact caccgggatt tccaagtctc caccctattg  | 3180 |
| acgtcaatgg gagtttggtt tggcaccaaa atcaacggga ctttccaaaa tgcgttaaca  | 3240 |
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| gagcgccgac accctgcatg aatacatgct ggacctgaga aatcgggcaa cgcacttacc  | 3900 |
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| acactacaac atcgttactt ttgtctgtct aactgggcc gcagccgtga ataactgtga   | 5040 |
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| ctagtctaga ctagctagaa agatccggga acttgtttat tgcagcttat aatgggtaca  | 5460 |
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| gtggtttgtc caaactcatc aatgtatctt atcatgtctg gatcggtgat caccgatccc  | 5580 |
| gatctgtcaa atgacgggtg caataaaacg gagactttga cccggaacgc ggaattcac   | 5640 |
| gtaaaaaaca cctggggcag tctccacgt aatcggtcaa agtccctcgg ccctcggtaa   | 5700 |
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&lt;210&gt; SEQ ID NO 60

&lt;211&gt; LENGTH: 6047

&lt;212&gt; TYPE: DNA

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: HPV antigen design 5 full nucleotide sequences of gorilla adenovirus shuttle plasmid

&lt;400&gt; SEQUENCE: 60

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| ttctagtgtg  | gccgtagtta | ggccaccact | tcaagaactc  | tgtagcaccg  | cctacatacc  | 1440 |
| tcgctctgct  | aatcctgtta | ccagtggtcg | ctgccagtgg  | cgataagtcg  | tgtcttaccg  | 1500 |
| ggttggaactc | aagacgatag | ttaccggata | aggcgcagcg  | gtcgggctga  | acgggggggtt | 1560 |
| cgtgcacaca  | gcccgacttg | gagcgaaacg | cctacaccga  | actgagatac  | ctacagcgtg  | 1620 |
| agctatgaga  | aagcgccacg | cttcccgaag | ggagaaagcg  | ggacaggtat  | ccggtaagcg  | 1680 |
| gcagggtcgg  | aacaggagag | cgcacgaggg | agcttccagg  | gggaaacgcc  | tggtatcttt  | 1740 |
| atagtctctg  | cgggtttctg | cacctctgac | ttgagcgtcg  | atttttgtga  | tgctcgtcag  | 1800 |
| gggggcggag  | cctatggaaa | aacgccagca | acgcggcctt  | tttacggttc  | ctggcctttt  | 1860 |
| gctggccttt  | tgtcacatg  | ttctttctcg | cgttatcccc  | tgattctgtg  | gataaccgta  | 1920 |
| ttaccgcctt  | tgagtgagct | gataccgctc | gccgcagccg  | aacgaccgag  | cgcagcgagt  | 1980 |
| cagtgagcga  | ggaagcggaa | gagcgcccaa | tacgcaaacc  | gcctctcccc  | gcgcgttggc  | 2040 |
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: GCAd-RTS-IL12 design 1

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| cgggtttcgc  | cacctctgac | ttgagcgtcg | atttttgtga  | tgtctctcag | gggggcggag | 1680 |
| cctatggaaa  | aacgccagca | acgcggcctt | tttacgggtc  | ctggcctttt | gctggccttt | 1740 |
| tgtctcacatg | ttctttctcg | cgttatcccc | tgattctgtg  | gataaccgta | ttaccgcctt | 1800 |
| tgagtgagct  | gataccgctc | gccgcagccg | aacgaccgag  | cgcagcaggt | cagtgcagca | 1860 |
| ggaaagcgaa  | gagcgcccaa | tacgcaaacc | gcctctcccc  | gcgcgttggc | cgattcatta | 1920 |
| atgcagctgg  | cacgacaggt | ttcccagctg | gaaagcgggc  | agtgagcgca | acgcaattaa | 1980 |
| tgtgagttag  | ctcactcatt | aggcacccca | ggctttacac  | tttatgcttc | cggctcgtat | 2040 |
| gttgtgtgga  | attgtgagcg | gataacaatt | tcacacagga  | aacagctatg | accatgatta | 2100 |
| cgccaaagctg | ggtaagctct | tccagtttaa | gcagcagagc  | ggtcagtttc | tcaccccag  | 2160 |
| cagacgcgcg  | agaggccgcg | ccgctcgcca | ccaaagagct  | gtaaagggtc | gtagccatgc | 2220 |
| tgcgcgcggt  | cgcgccggcg | gcggaggcgg | cgccggaggt  | cgccgcgtcc | agcggagttc | 2280 |
| ctcccacggt  | cgcgtaggcc | attgtagacg | aatttgaagg  | cagaacgggg | cgtccatcca | 2340 |
| cgttggaacc  | catcacatct | tgacgcactc | cagcccagtg  | aggcatgcgc | actgtcagat | 2400 |
| aggggctaaa  | gatgcttcca | tcaaagctgt | tgccggtgtc  | gctcatggcg | gcggctgttg | 2460 |
| caagacaaaa  | cagagagacc | cttagacccc | caatttatac  | acgccccacc | cttctagcca | 2520 |
| cgccacacct  | acccacctca | atcggtatcc | tcctcgtctg  | acccaaactc | ggccctgggt | 2580 |
| caggccagca  | ccagatggtc | aggcctgcag | gtacgtagcc  | gcaataaaat | atctttattt | 2640 |
| tcattacatc  | tgtgtgttgg | ttttttgtgt | gaatccatag  | tactaacata | cgtctcccat | 2700 |
| caaaacaaaa  | cgaacaaaaa | caaactagca | aaataggctg  | tccccagtcg | aagtcagggt | 2760 |
| gccagaacat  | ttctctatcc | ataatgcagg | ggtaccggaa  | ggaagagggg | cggggtcgat | 2820 |
| cgaccccgcc  | cctcttcctt | cgaagggaag | ggggcggggg  | ccaattgcgg | agtaactgtc | 2880 |
| tccgagcgga  | gtactgtcct | ccgagcggag | tactgtcctc  | cgagcggagt | actgtcctcc | 2940 |
| gagcggagta  | ctgtcctccg | agcggagtac | tgtcctccga  | gcggagagtc | cccggggacc | 3000 |
| tagaggggat  | ataatgggtg | ccttagctgg | tgtgtgacct  | catcttctgt | taegccctgt | 3060 |
| caggagatca  | cgcggtccac | catgggtcac | cagcagttgg  | tcactctctg | gttttccctg | 3120 |
| gtttttctgg  | catctccctt | cgtggccata | tgggaactga  | agaaagatgt | ttatgtcgta | 3180 |



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| acctgacca tccaagtaa agagtgtgga gatgctggc agtacacctg tcacaaagga     | 3360 |
| ggcgagggtc taagccattc gctcctgctg cttcacaaaa aggaagatgg aatttgggtc  | 3420 |
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| aagaattatt ctggacgttt cacctgctgg tggctgacga caatcagtac tgatttgaca  | 3540 |
| ttcagtgta aaagcagcag aggtctttct gacccccaag gggtgacgtg cggagctgct   | 3600 |
| acactctctg cagagagagt cagaggggac aacaaggagt atgagtactc agtggagtgc  | 3660 |
| caggaggaca gtgcctgccc agctgctgag gagagtctgc ccattgaggt catggtggat  | 3720 |
| gccgttcaca agctcaagta tgaaaactac accagcagct tcttcacag ggacatcac    | 3780 |
| aaacctgacc cacccaagaa cttgcagctg aagccattaa agaattctcg gcagggtggag | 3840 |
| gtcagctggg agtacctga cacctggagt actccacatt cctacttctc cctgacattc   | 3900 |
| tcggttcagg tccagggcaa gagcaagaga gaaaagaaag atagagtctt caccgacaag  | 3960 |
| acctcagcca cggctatctg ccgcaaaaat gccagcatta gcgtgcgggc ccaggaccgc  | 4020 |
| tactatagct catcttgag cgaatgggca tctgtgccct gcagtctcga gggcgcgga    | 4080 |
| gagggcagag gaagtcttct aacatgcggt gacgtggagg agaateccgg cctaggatg   | 4140 |
| ggtccagcgc gcagcctcct ccttgtggct acctgggtcc tctggacca cctcagtttg   | 4200 |
| gccagaaaac tcccgtggc cactccagac ccaggaatgt tccatgcct tcaccactcc    | 4260 |
| caaaacctgc tgagggcgt cagcaacatg ctcagaagg ccagacaaac tctagaattt    | 4320 |
| taccttgca cttctgaaga gattgatcat gaagatatca caaaagataa aaccagcaca   | 4380 |
| gtggaggcct gtttaccatt ggaattaacc aagaatgaga gttgcctaaa ttccagagag  | 4440 |
| acctctttca taactaatgg gagttgctg gcctccagaa agacctctt tatgatggcc    | 4500 |
| ctgtgcctta gtagtattta tgaagacttg aagatgtacc aggtggagt caagaccatg   | 4560 |
| aatgcaaagc ttctgatgga tctaagagg cagatcttc tagatcaaaa catgctggca    | 4620 |
| gttattgatg agctgatgca ggcctgaat ttcaacagtg agactgtgc acaaaaatcc    | 4680 |
| tccctgaag aaccggattt ttataaaact aaaatcaagc tctgcatact tcttcatgct   | 4740 |
| ttcagaatc gggcagtgac tattgataga gtgatgagct atctgaatgc ttctaaatc    | 4800 |
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| gtaaataaat gggtaaggg atgtaacatt gtttgttgtt tgggtggggg tggggcctcc   | 4920 |
| gcgcggggtt ttggcgctc ccgcggggc cccctcctc acggcgagcg ctgccacgtc     | 4980 |
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| cttgggtgac tctagggcac tggttttctt tccagagagc ggaacaggcg aggaaaagta  | 5160 |
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| catttgggtc aatatgtaat ttccagtgtt agactagtaa attgtccgct aaattctggc  | 5460 |
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|            |            |             |            |             |            |      |
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| ggattccccc | ggtcgggat  | ttacccccca  | cgactccgcc | ccctacggcg  | ctctggatat | 5700 |
| ggccgacttc | gagtttgagc | agatgtttac  | cgatgccctt | ggaattgacg  | agtacggtg  | 5760 |
| ggaattcgag | atgcctgtgg | acaggatcct  | ggaggcagag | cttgctgtgg  | aacagaagag | 5820 |
| tgaccagggc | gttgagggtc | ctgggggaac  | cgggggtagc | ggcagcagcc  | caaatgaccc | 5880 |
| tgtgactaac | atctgtcagg | cagctgacaa  | acagctattc | acgcttggtg  | agtgggcgaa | 5940 |
| gaggatccca | cacttttctc | ccttgccctc  | ggatgatcag | gtcatattgc  | tcggggcag  | 6000 |
| ctggaatgaa | ctcctcattg | cctccttttc  | acaccgatcc | attgatgttc  | gagatggcat | 6060 |
| cctccttgcc | acaggctctc | acgtgcaccg  | caactcagcc | cattcagcag  | gagtaggagc | 6120 |
| catctttgat | cgggtgtcga | cagagctagt  | gtccaaaatg | cgtgacatga  | ggatggacaa | 6180 |
| gacagagcct | ggctgcctga | gggcaatcat  | tctgtttaat | ccagagggtga | gggggttgaa | 6240 |
| atccgccccg | gaagttgaac | ttctacgtga  | aaaagtatat | gccgctttgg  | aagaatatac | 6300 |
| tagaacaaca | catcccgatg | aaccaggaag  | atgtgcaaaa | cttttgcttc  | gtctgccttc | 6360 |
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| ggtgcctctg | cggccaaaag | ccacgtgtat  | aagatacacc | tgcaaaggcg  | gcacaacccc | 6840 |
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| ctcggtgcac | atgctttaca | tgtgtttagt  | cgagggttaa | aaacgtctag  | gcccccgaa  | 7020 |
| ccacggggac | gtggttttcc | tttgaaaaac  | acgatctcta | ggcgccacca  | tgaagctact | 7080 |
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| ttctttacag | gatataaaag | cattgttaac  | aggattat   | gtacaagata  | atgtgaataa | 7380 |
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| gcatagaata | agtcgcacat | catcatcgga  | agagagtagt | aacaaaggtc  | aaagacagtt | 7500 |
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| cgccatgaag | cggaaagaga | agaaagcaca  | gaaggagaag | gacaaactgc  | ctgtcagcac | 7620 |
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| agcaaggatt | cacgaagtgg | tcccaagggt  | tctctccgac | aagctgttgg  | tgacaaaccg | 7740 |
| gcagaaaaac | atccccagct | tgacagccaa  | ccagcagttc | cttatcgcca  | ggctcatctg | 7800 |
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| ccgagtcgag | cgacgatacg  | atgcggcctc | agacagtatt | ctgttcgcga   | acaaccaagc  | 8100  |
| gtacactcgc | gacaactacc  | gcaaggctgg | catggccgag | gtcatcgagg   | atctactgca  | 8160  |
| cttctgcccg | tgcattgtact | ctatggcggt | ggacaacatc | cattacgcgc   | tgctcacggc  | 8220  |
| tgctgtcatc | ttttctgacc  | ggccagggtt | ggagcagccg | caactgggtg   | aagagatcca  | 8280  |
| gcggtactac | ctgaatacgc  | tccgcatcta | tatcctgaac | cagctgagcg   | ggtcggcgcg  | 8340  |
| ttcgtccgtc | atatacggca  | agatcctctc | aatcctctct | gagctacgca   | cgctcggcat  | 8400  |
| gcaaaactcc | aacatgtgca  | tctccctcaa | gctcaagaac | agaaagctgc   | cgcccttcct  | 8460  |
| cgaggagatc | tgggatgtgg  | cggacatgtc | gcacacccaa | ccgccgccta   | tctcagagtc  | 8520  |
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| ctgtcagaga | caagctgtgg  | tctgggaagg | ccttagcact | aaaagcacca   | caatgagaag  | 8880  |
| caaccgccag | aagcagggcc  | cgcaggcctt | tgttccagct | gcaaagagaa   | aggaaaaagt  | 8940  |
| ggggaataag | agttggggct  | gcggaggggg | tggggagcat | tgtgcagggt   | ccgtacttga  | 9000  |
| acagaaagca | gggaccaaca  | caaggaaggg | cgcgccacgc | gtcaaatgac   | ggtgacaata  | 9060  |
| aaacggagac | tttgaccggg  | aacgcggaaa | ttcacgtaaa | aaacacctgg   | gcgagtcctc  | 9120  |
| cacgtaatcg | gtcaaatgct  | ctcggccctc | ggtaaatatt | acgcactatg   | actaacgccc  | 9180  |
| tattattcag | ttttcacttc  | ccggtttcac | ttttcgcgcg | aaaatggcca   | aattcttacat | 9240  |
| ggtcccgccc | aaaattacta  | cgatatccgg | tgaaaagcgc | gcgaaaattg   | gcacttcggg  | 9300  |
| aggtaggcgg | cgctcatcaa  | aaacgtcaca | ttttccgcga | cggaagcttg   | catgtgagct  | 9360  |
| cctcccactt | gcaaatgcc   | cacttcggcc | acacctccca | accctactcg   | cgcgtcctac  | 9420  |
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| agctggcgta | atagegaaga  | ggcccgcacc | gatcgccctt | cccaacagtt   | gcgcagcctg  | 9660  |
| aatggcgaat | ggcgctgat   | gcgggtat   | ctccttacgc | atctgtgcgg   | tatttcacac  | 9720  |
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| ataatggttt | cttagacgtc  | aggtggcact | tttcggggaa | atgtgcgcgg   | aaccctatt   | 10020 |
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&lt;213&gt; ORGANISM: Artificial sequence

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&lt;223&gt; OTHER INFORMATION: GCAd-RTS-IL12 design 3

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| gaatgacttg gttgagtact caccagtcac agaaaagcat cttacggatg gcatgacagt    | 420  |
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| gacaacgacg ggaggaccga aggagctaac cgcttttttg cacaacatgg gggatcatgt    | 540  |
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| caccacgatg cctgtagcaa tggcaacaac gttgcgcaaa ctattaactg gcgaactact    | 660  |
| tactctagct tcccggaac aattaataga ctggatggag gcggataaag ttgcaggacc     | 720  |
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| taatctcatg accaaaatcc cttaacgtga gttttcgctc cactgagcgt cagaccccg     | 1080 |
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| aacaaaaaaa ccacgcgtac cagcgggtgt tttgttgccg gatcaagagc taccaactct    | 1200 |
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| gcggtagtta ggccaccact tcaagaactc tgtagcacgc cctacatacc tcgctctgct    | 1320 |
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| aagacgatag ttaccggata aggcgcagcg gtcgggctga acgggggggt cgtgcacaca    | 1440 |
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| tgctcacatg ttctttctcg cgttatcccc tgattctgtg gataaccgta ttaccgcctt    | 1800 |
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| ggaagcggaa gagcgcccaa tacgaaacc gcctctcccc gcgcgttggc cgattcatta     | 1920 |
| atgcagctgg cacgacaggt ttcccactg gaaagcgggc agtgagcga acgcaattaa      | 1980 |
| tgtgagttag ctactcatt aggcacccca ggctttacac tttatgcttc cggctcgat      | 2040 |
| gttgtgtgga attgtgagcg gataacaatt tcacacagga aacagctatg accatgatta    | 2100 |
| cgccaagctg ggtcaagtct tccagtttaa gcagcagagc ggtcagtttc tcatcccgag    | 2160 |
| cagacgcgcg agaggccgcg ccgctcgcca ccaaagagct gtaaaggctc gtagccatgc    | 2220 |
| tcgcgcgggt cgcggcggcg gcggaggcgg cggcggaggt cgcggcgtcc agcggagtcc    | 2280 |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| ctcccacggt cgcgtaggcc attgtagacg aatttgaagg cagaacgggg cgtccatcca  | 2340 |
| cgttggaacc catcacattc tgacgcactc cagcccagtg aggcattgcgc actgtcagat | 2400 |
| aggggctaaa gatgcttcca tcaaagctgt tgccgggtgc gctcatggcg gcggctgttg  | 2460 |
| caagacaaaa cagagagacc cttagacccc caatttatac acgccccacc cttctagcca  | 2520 |
| cgccacactt acccacctca atcggtatcc tcacgctag acccaaacctc ggccctgggtg | 2580 |
| caggccagca ccagatggtc aggcctgcag gtacgtagcc gcaataaaat atctttattt  | 2640 |
| tcattacatc tgtgtgttgg ttttttgtgt gaatccatag tactaacata cgtctccat   | 2700 |
| caaaacaaaa cgaaacaaaa caaactagca aaataggctg tccccagtc aagtccaggt   | 2760 |
| gccagaacat ttctctatcc ataatgcagg ggtaccgga ggaagagggg cggggctgat   | 2820 |
| cgaccccgcc cctcttcctt cgaaggaaga ggggcggggg ccaattgcgg agtactgtcc  | 2880 |
| tccgagcggg gtactgtcct ccgagcggag tactgtcctc cgagcggagt actgtcctcc  | 2940 |
| gagcggagta ctgtcctccg agcggagtac tgtcctccga gcggagagtc cccggggacc  | 3000 |
| tagaggggat ataatgggtg ccttagctgg tgtgtgacct catcttcctg tacgcccctg  | 3060 |
| caggcagccg ctaaatccaa ggtaaggta gaagagctag cgccaccatg tgtcaccagc   | 3120 |
| agttggtcat ctcttggttc agcctgggtt ttctggcatc tcccctcgtg gccatctggg  | 3180 |
| aactgaagaa agatgtttat gtcgtagaat tggattggta tcccgacgcc cctggagaaa  | 3240 |
| tgggtggtcct gacatgtgac acccctgaag aagatgggat cacctggacc ttggaccaga | 3300 |
| gcagtgaggt cttaggctct ggcaagaccc tgaccatcca agtcaaagag ttgggagatg  | 3360 |
| ctggccagta cacctgtcac aaaggaggcg aggttctaag ccattcgctc ctgctgcttc  | 3420 |
| acaaaaagga agatggaatt tggtcactg acattctgaa ggaccagaaa gaaccaaga    | 3480 |
| ataagacctt tctaagatgc gaggccaaaga attattctgg acgtttcacc tgctggtggc | 3540 |
| tgacgacaat cagtactgat ttgacattca gtgtcaaaag cagcagaggc tctctgacc   | 3600 |
| cccaaggggt gacgtgcgga gctgctacac tcagcgccga gagagtcaga ggggacaaca  | 3660 |
| aggagtatga gtactcagtg gagtgccagg aggacagtgc ctgcccagct gctgaggaga  | 3720 |
| gtctgcccat tgaggtcatg gtggatgccg ttcacaagct caagtatgaa aactacacca  | 3780 |
| gcagcttctt catcagggac atcatcaaac ctgaccacc caagaacttg cagctgaagc   | 3840 |
| ccctgaagaa cagcagacag gtggagggtca gctgggagta ccctgacacc tggagtactc | 3900 |
| cacattccta cttctccctg acattctgcg ttcagggtcca gggcaagagc aagagagaaa | 3960 |
| agaaagatag agtcttcacg gacaagacct cagccacggt catctgccgc aaaaatgcc   | 4020 |
| gcattagcgt gcgggccag gaccgctact atagctcatc ttggagcgaa tgggcatctg   | 4080 |
| tgccctgctc cggtggcggt ggcggcggtat ctagaaacct ccccgaggcc actccagacc | 4140 |
| caggaatgtt cccatgcctt caccacagcc agaacctgct gagggccgtc agcaacatgc  | 4200 |
| tccagaaggc cagacaaact ctagaatttt acccttgcac ttctgaagag attgatcatg  | 4260 |
| aagatatcac aaaagataaa accagcacag tggaggcctg tttaccattg gaattaacca  | 4320 |
| agaatgagag ttgcctaaat tccagagaga cctctttcat aactaatggg agttgcctgg  | 4380 |
| cctccagaaa gacctctttt atgatggccc tgtgccttag tagtatttat gaagacttga  | 4440 |
| agatgtacca ggtggagttc aagaccatga atgcaaagct gctgatggac cccaagaggc  | 4500 |
| agatctttct agatcaaac atgctggcag ttattgatga gctgatgcag gcctgaatt    | 4560 |
| tcaacagtga gactgtgcc aaaaaatcct ccctgaaga accggatttt tataaaacta    | 4620 |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| aaatcaagct ctgcatactt cttcatgctt tcagaatcag agcagtgact attgatagag  | 4680 |
| tgatgagcta tctgaatgct tcctaaatcg atttatttat cggcataaat aatttttttg  | 4740 |
| aagaagtaat actatttttc tttttttttg taaataaatg gggttaaggga tgtaacattg | 4800 |
| tttgttgttt ggtggggggt ggggctctcg cgcggggttt tggcgctctc cgcgggcgc   | 4860 |
| ccccctctca cggcgagcgc tggcacgtca gacgaagggc gcagcgagcg tcttgatcct  | 4920 |
| tccgcccga cgtcaggac agcggcccg tgctcataag actcggcctt agaaccocag     | 4980 |
| tatcagcaga aggacatttt aggacgggac ttgggtgact ctagggcact ggttttcttt  | 5040 |
| ccagagagcg gaacaggcga ggaagtag tcccttctcg gcgattctgc ggagggatct    | 5100 |
| ccgtggggcg gtgaacgcg atgattatat aaggacgcgc cgggtgtggc acagctagtt   | 5160 |
| ccgtcgcagc cgggatttgg gtcgcgggtt ttgtttgtgg atcgtgtga tcgtcacttg   | 5220 |
| gtgagtagcg ggctgctggg ctgggtacgt gcgctcgggg ttggcgagtg tgttttgtga  | 5280 |
| agtttttttag gcaccttttg aaatgtaatc atttgggtca atatgtaatt ttcagtgtta | 5340 |
| gactagtaaa ttgtccgcta aattctggcc gtttttggt tttttgttag acgagctagc   | 5400 |
| gccgccacca tgggccctaa aaagaagcgt aaagtgcgcc ccccgaccga tgtcagcctg  | 5460 |
| ggggacgagc tccacttaga cggcgaggac gtggcgatgg cgcagccga cgcgctagac   | 5520 |
| gatttcgac tggacatgtt gggggacggg gattccccgg gtccgggatt tacccccac    | 5580 |
| gactccgccc cctacggcgc tctggatatg gccgacttcg agtttgagca gatgtttacc  | 5640 |
| gatgcccctg gaattgacga gtacggtggg gaattcgaga tgctgtgga caggatcctg   | 5700 |
| gaggcagagc ttgctgtgga acagaagagt gaccagggcg ttgaggggtc tgggggaacc  | 5760 |
| gggggtagcg gcagcagccc aaatgaccct gtgactaaca tctgtcaggc agctgacaaa  | 5820 |
| cagctattca cgttgttga gtgggcgaag aggatccac acttttcctc cttgcctctg    | 5880 |
| gatgatcagg tcatattgct gcgggcaggc tggaatgaac tcctcattgc ctccctttca  | 5940 |
| caccgatcca ttgatgttcg agatggcatc ctccctgcca caggtcttca cgtgcaccgc  | 6000 |
| aactcagccc attcagcagg agtaggagcc atctttgatc ggggtgctgac agagctagtg | 6060 |
| tccaaaatgc gtgacatgag gatggacaag acagagcttg gctgcctgag ggcaatcatt  | 6120 |
| ctgtttaatc cagaggtgag gggtttgaaa tccgccagg aagttgaact tctacgtgaa   | 6180 |
| aaagtatatg ccgctttgga agaataact agaacaacac atcccgatga accaggaaga   | 6240 |
| tttgcaaac ttttgcttcg tctgccttct ttacgttcca taggccttaa gtgtttggag   | 6300 |
| catttgtttt tcttctgcct tattggagat gttccaattg atacgttctt gatggagatg  | 6360 |
| cttgaatcac cttctgattc ataacttagc ctagccccc tctccctccc cccccctaa    | 6420 |
| cgttactggc cgaagccgct tggaataagg ccggtgtgcg tttgtctata tgttattttc  | 6480 |
| caccatattg ccgtcttttg gcaatgtgag ggcccgaaa cctggccctg tcttcttgac   | 6540 |
| gagcattcct aggggtcttt cccctctcgc caaaggaatg caaggctctgt tgaatgtcgt | 6600 |
| gaaggaagca gttcctctgg aagcttcttg aagacaaaca acgtctgtag cgacccttg   | 6660 |
| caggcagcgg aacccccac ctggcgacag gtgcctctgc ggccaaaagc cacgtgtata   | 6720 |
| agatacacct gcaaaggcgg cacaacccca gtgccacgtt gtgagttgga tagttgtgga  | 6780 |
| aagagtcaaa tggctctcct caagcgtatt caacaagggg ctgaaggatg cccagaaggt  | 6840 |
| acccattgt atgggatctg atctggggcc tgggtgcaca tgctttacat gtgtttagt    | 6900 |
| gaggttaaaa aacgtctagg cccccgaac cacggggacg tggttttcct ttgaaaaaca   | 6960 |
| cgatctctag gcgccaccat gaagctactg tcttctatcg aacaagcatg cgatatttgc  | 7020 |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| cgacttaaaa agctcaagtg ctccaaagaa aaaccgaagt gcgccaagtg tctgaagaac  | 7080 |
| aactgggagt gtcgtacttc tcccaaaacc aaaagggtctc cgctgactag ggcacatctg | 7140 |
| acagaagtgg aatcaaggct agaagactg gaacagctat ttctactgat ttttcctcga   | 7200 |
| gaagaccttg acatgatttt gaaaatggat tctttacagg atataaaagc attgttaaca  | 7260 |
| ggattatttg tacaagataa tgtgaataaa gatgccgtca cagatagatt ggcttcagtg  | 7320 |
| gagactgata tgcctctaac attgagacag catagaataa gtgcgacatc atcatcgga   | 7380 |
| gagagtagta acaaagggtca aagacagttg actgtatcgc cggaattccc ggggatccgg | 7440 |
| cctgagtgcg tagtacccca gactcagtgc gccatgaagc ggaaagagaa gaaagcacag  | 7500 |
| aaggagaagg acaaaactgcc tgtcagcacg acgacgggtg acgaccacat gccgccatt  | 7560 |
| atgcagtgtg aacctccacc tctgaagca gcaaggattc acgaagtggc cccaaggttt   | 7620 |
| ctctccgaca agctgttggt gacaaaccgg cagaaaaaca tccccagtt gacagccaac   | 7680 |
| cagcagtttc ttatcgccag gctcatctgg taccaggacg ggtacgagca gccttctgat  | 7740 |
| gaagatttga agaggattac gcagacgtgg cagcaagcgg acgatgaaaa cgaagagtcg  | 7800 |
| gacactccct tccgccagat cacagagatg actatcctca cggccaact tatcgtggag   | 7860 |
| ttcggaagg gattgccagg gttcgccaag atctcgacgc ctgatcaaat tacgtgctt    | 7920 |
| aaggcttgct caagtgggt aatgatgtc cgagtcgcgc gacgatacga tgcggcctca    | 7980 |
| gacagtattc tgttcgcgaa caaccaagcg tactctcgcg acaactaccg caaggctggc  | 8040 |
| atggccgagg tcctcgagga tctactgcac ttctgccggt gcatgtactc tatggcgttg  | 8100 |
| gacaacatcc attacgcgt gctcacggct gtcgtcatct tttctgaccg gccaggggtg   | 8160 |
| gagcagccgc aactggtgga agagatccag cggtaactacc tgaatacgt ccgcattctat | 8220 |
| atcctgaacc agctgagcgg gtcggcgcggt tcgtccgtca tatacgga gatcctctca   | 8280 |
| atcctctctg agctacgcac gctcgcatg caaaactcca acatgtgcat ctccctcaag   | 8340 |
| ctcaagaaca gaaagctgcc gcctttcttc gaggagatct gggatgtggc ggacatgtcg  | 8400 |
| cacacccaac cgccgcctat cctcgagtcc cccacgaatc tctaaatcga ttacgctcct  | 8460 |
| ctactctttg agacatcact ggctataat aaatgggtta atttatgtaa caaaattgcc   | 8520 |
| ttggcttggt aactttatta gacattctga tgtttgcatt gtgtaatac tgttgtattg   | 8580 |
| gaaaagcgtg ccaagatgga ttattgtaat tcagtgtctt ttttagtagc gtcacgtgcc  | 8640 |
| aaacactggt agtcacagag ggcacagac agcctgtgct ggaacagctc agttcatagg   | 8700 |
| gctatggaga tggggagaaa ggggcgcttc gtgcagagac aagctgtggt ctgggaaggc  | 8760 |
| cttagcacta aaagcaccac aatgagaagc aaccgccaga agcagggccc gcaggccttt  | 8820 |
| gttcagctg caaagagaaa ggaaaaagtg gggaataaga gttggggctg cggagggggt   | 8880 |
| ggggagcatt gtgcaggttc cgtactgaa cagaaagcag ggaccaacac aagggaaggc   | 8940 |
| gcgccaccgg tcaaatgacg gtgacaataa aacggagact ttgaccgga acgcggaat    | 9000 |
| tcacgtaaaa aacacctggg cgagtcctcc acgtaatcgg tcaaagtc cggccctcg     | 9060 |
| gtaaatatta cgcactatga ctaacgccct attattcagt ttctacttcc ccgtttcact  | 9120 |
| tttcgcgcga aaatggccaa atcttacatg gtcccgccca aaattactac gatatccggt  | 9180 |
| gaaaagcgcg cgaatttg cacttccgga ggtaggcggc gctcatcaaa aacgtcacat    | 9240 |
| tttcgcgcac ggaagcttg atgtgagctc ctcccacttg caaatgccac acttccgcca   | 9300 |
| cacctcccaa ccctactcgc gcgtcctacg tcaccgccc cgccctctcc cgccacctc    | 9360 |

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attatcatat tggccacaat ccaaaataag gtatattatt gatgatggtt taaacgccca 9420
attcactggc cgctgcttta caacgtcgtg actgggaaaa cctggcggtt acccaactta 9480
atgccttgcc agcacatccc cctttcgcca gctggcgtaa tagcgaagag gcccgaccg 9540
atgcgccctc ccaacagttg cgcagcctga atggcgaatg gcgcctgatg cggtattttc 9600
tccttacgca tctgtgcggt atttcacacc gcatatggtg cactctcagt acaatctgct 9660
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tgtgtcagag gttttcaccg tcatacccca aacgcgcgag acgaaagggc ctcgtgatac 9840
gcctatTTTT ataggTTaTt gtcAtgataa taatggTttc ttagacgtca ggtggcactt 9900
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<210> SEQ ID NO 64
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Whitlow Linker

```

```

<400> SEQUENCE: 64

```

```

ggcagcacct cggcgagcgg caagcctggc agcggcgagg gcagcaccaa gggc 54

```

```

<210> SEQ ID NO 65
<211> LENGTH: 78
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

```

```

<400> SEQUENCE: 65

```

```

tctggcggag gatctggagg aggcggatct ggaggaggag gcagtggagg cggaggatct 60
ggcggaggat ctctgcag 78

```

```

<210> SEQ ID NO 66
<211> LENGTH: 9
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: GSG linker

```

```

<400> SEQUENCE: 66

```

```

ggaagcggc 9

```

```

<210> SEQ ID NO 67
<211> LENGTH: 12
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: SGSG linker

```

```

<400> SEQUENCE: 67

```

```

agtggcagcg gc 12

```

```

<210> SEQ ID NO 68
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: (G4S)3 linker

```

```

<400> SEQUENCE: 68

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|                                                                                                                                                                                                            |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| ggtagcggtg gctcgggagg tggtaggtcg gtaggcggcg gatct                                                                                                                                                          | 45 |
| <210> SEQ ID NO 69<br><211> LENGTH: 12<br><212> TYPE: DNA<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Furin cleavage site/ Furinlink1<br><400> SEQUENCE: 69       |    |
| cgtgcaaagc gt                                                                                                                                                                                              | 12 |
| <210> SEQ ID NO 70<br><211> LENGTH: 84<br><212> TYPE: DNA<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Fmdv<br><400> SEQUENCE: 70                                  |    |
| agagccaaga gggcaccggt gaaacagact ttgaattttg accttctgaa gttggcagga                                                                                                                                          | 60 |
| gacgttgagt ccaaccctgg gcc                                                                                                                                                                                  | 84 |
| <210> SEQ ID NO 71<br><211> LENGTH: 54<br><212> TYPE: DNA<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Thosea asigna virus 2A region (T2A)<br><400> SEQUENCE: 71   |    |
| gagggcagag gaagtctgct aacatgcggt gacgtcgagg agaatcctgg acct                                                                                                                                                | 54 |
| <210> SEQ ID NO 72<br><211> LENGTH: 75<br><212> TYPE: DNA<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Furin-GSG-T2A<br><400> SEQUENCE: 72                         |    |
| agagctaaga ggggaagcgg agagggcaga ggaagtctgc taacatgcgg tgacgtcgag                                                                                                                                          | 60 |
| gagaatcctg gacct                                                                                                                                                                                           | 75 |
| <210> SEQ ID NO 73<br><211> LENGTH: 78<br><212> TYPE: DNA<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Furin-SGSG-T2A<br><400> SEQUENCE: 73                        |    |
| agggccaaga ggagtggcag cggcgagggc agaggaagtc ttctaacatg cggtgacgtg                                                                                                                                          | 60 |
| gaggagaatc cggccct                                                                                                                                                                                         | 78 |
| <210> SEQ ID NO 74<br><211> LENGTH: 57<br><212> TYPE: DNA<br><213> ORGANISM: Artificial sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Porcine teschovirus-1 2A region (P2A)<br><400> SEQUENCE: 74 |    |

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gcaacgaact tctctctcct aaaacaggct ggtgatgtgg aggagaatcc tggcca 57

<210> SEQ ID NO 75  
 <211> LENGTH: 66  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: GSG-P2A

&lt;400&gt; SEQUENCE: 75

ggaagcggag ctactaactt cagcctgctg aagcaggctg gagacgtgga ggagaaccct 60

ggacct 66

<210> SEQ ID NO 76  
 <211> LENGTH: 60  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Equine rhinitis A virus 2A region (E2A)

&lt;400&gt; SEQUENCE: 76

cagtggtacta attatgctct cttgaaattg gctggagatg ttgagagcaa ccctggacct 60

<210> SEQ ID NO 77  
 <211> LENGTH: 66  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Foot-and-mouth disease virus 2A region (F2A)

&lt;400&gt; SEQUENCE: 77

gtcaaacaga ccctaaactt tgatctgcta aaactggccg gggatgtgga aagtaatccc 60

ggcccc 66

<210> SEQ ID NO 78  
 <211> LENGTH: 93  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: FP2A

&lt;400&gt; SEQUENCE: 78

cgtgcaaagc gtgcaccggt gaaacaggga agcggagcta ctaacttcag cctgctgaag 60

caggctggag acgtggagga gaaccctgga cct 93

<210> SEQ ID NO 79  
 <211> LENGTH: 24  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Linker-GSG

&lt;400&gt; SEQUENCE: 79

gcaccggtga aacagggaag cgga 24

<210> SEQ ID NO 80  
 <211> LENGTH: 15  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Linker

&lt;400&gt; SEQUENCE: 80

gcaccggtga aacag 15

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<210> SEQ ID NO 81  
<211> LENGTH: 18  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Whitlow Linker Amino Acid Sequence

<400> SEQUENCE: 81

Gly Ser Thr Ser Gly Ser Gly Lys Pro Gly Ser Gly Glu Gly Ser Thr  
1           5                   10                   15

Lys Gly

<210> SEQ ID NO 82  
<211> LENGTH: 26  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Linker Amino Acid Sequence

<400> SEQUENCE: 82

Ser Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly  
1           5                   10                   15

Gly Gly Gly Ser Gly Gly Gly Ser Leu Gln  
20                   25

<210> SEQ ID NO 83  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: GSG Linker Amino Acid Sequence

<400> SEQUENCE: 83

Gly Ser Gly  
1

<210> SEQ ID NO 84  
<211> LENGTH: 4  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: SGSG Linker Amino Acid Sequence

<400> SEQUENCE: 84

Ser Gly Ser Gly  
1

<210> SEQ ID NO 85  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: (G4S) 3 Linker Amino Acid sequence

<400> SEQUENCE: 85

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser  
1           5                   10                   15

<210> SEQ ID NO 86  
<211> LENGTH: 4  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Linker Polypeptide component

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<400> SEQUENCE: 86

Arg Ala Lys Arg  
1

<210> SEQ ID NO 87  
 <211> LENGTH: 28  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Fmdv Amino Acid Sequence

<400> SEQUENCE: 87

Arg Ala Lys Arg Ala Pro Val Lys Gln Thr Leu Asn Phe Asp Leu Leu  
 1 5 10 15

Lys Leu Ala Gly Asp Val Glu Ser Asn Pro Gly Pro  
 20 25

<210> SEQ ID NO 88  
 <211> LENGTH: 18  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Furin-GSG-T2A Amino Acid Sequence

<400> SEQUENCE: 88

Glu Gly Arg Gly Ser Leu Leu Thr Cys Gly Asp Val Glu Glu Asn Pro  
 1 5 10 15

Gly Pro

<210> SEQ ID NO 89  
 <211> LENGTH: 25  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Furin-SGSG-T2A Amino Acid Sequence

<400> SEQUENCE: 89

Arg Ala Lys Arg Gly Ser Gly Glu Gly Arg Gly Ser Leu Leu Thr Cys  
 1 5 10 15

Gly Asp Val Glu Glu Asn Pro Gly Pro  
 20 25

<210> SEQ ID NO 90  
 <211> LENGTH: 26  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Furin-SGSG-T2A

<400> SEQUENCE: 90

Arg Ala Lys Arg Ser Gly Ser Gly Glu Gly Arg Gly Ser Leu Leu Thr  
 1 5 10 15

Cys Gly Asp Val Glu Glu Asn Pro Gly Pro  
 20 25

<210> SEQ ID NO 91  
 <211> LENGTH: 19  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Porcine teschovirus-1 2A region (P2A)

<400> SEQUENCE: 91

Ala Thr Asn Phe Ser Leu Leu Lys Gln Ala Gly Asp Val Glu Glu Asn

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|   |   |    |    |
|---|---|----|----|
| 1 | 5 | 10 | 15 |
|---|---|----|----|

Pro Gly Pro

<210> SEQ ID NO 92  
 <211> LENGTH: 22  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: GSG-P2A Amino Acid Sequence

&lt;400&gt; SEQUENCE: 92

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gly | Ser | Gly | Ala | Thr | Asn | Phe | Ser | Leu | Leu | Lys | Gln | Ala | Gly | Asp | Val |
| 1   |     | 5   |     |     |     |     |     | 10  |     |     |     |     | 15  |     |     |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| Glu | Glu | Asn | Pro | Gly | Pro |
|     |     |     | 20  |     |     |

<210> SEQ ID NO 93  
 <211> LENGTH: 20  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Equine rhinitis A virus 2A region (E2A) Amino Acid Sequence

&lt;400&gt; SEQUENCE: 93

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gln | Cys | Thr | Asn | Tyr | Ala | Leu | Leu | Lys | Leu | Ala | Gly | Asp | Val | Glu | Ser |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     |     | 15  |     |

|     |     |     |     |
|-----|-----|-----|-----|
| Asn | Pro | Gly | Pro |
|     |     |     | 20  |

<210> SEQ ID NO 94  
 <211> LENGTH: 22  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Foot-and-mouth disease virus 2A region (F2A) Amino Acid Sequence

&lt;400&gt; SEQUENCE: 94

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Val | Lys | Gln | Thr | Leu | Asn | Phe | Asp | Leu | Leu | Lys | Leu | Ala | Gly | Asp | Val |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     |     | 15  |     |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| Glu | Ser | Asn | Pro | Gly | Pro |
|     |     |     |     |     | 20  |

<210> SEQ ID NO 95  
 <211> LENGTH: 31  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: FP2A Amino Acid Sequence

&lt;400&gt; SEQUENCE: 95

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Arg | Ala | Lys | Arg | Ala | Pro | Val | Lys | Gln | Gly | Ser | Gly | Ala | Thr | Asn | Phe |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Leu | Leu | Lys | Gln | Ala | Gly | Asp | Val | Glu | Glu | Asn | Pro | Gly | Pro |
|     |     |     |     | 20  |     |     |     |     | 25  |     |     |     | 30  |     |

<210> SEQ ID NO 96  
 <211> LENGTH: 8  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Linker-GSG Amino Acid

&lt;400&gt; SEQUENCE: 96

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Ala Pro Val Lys Gln Gly Ser Gly  
1 5

<210> SEQ ID NO 97  
<211> LENGTH: 5  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Linker Amino Acid

<400> SEQUENCE: 97

Ala Pro Val Lys Gln  
1 5

<210> SEQ ID NO 98  
<211> LENGTH: 154  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HBV HBx domain of HBV design 1

<400> SEQUENCE: 98

Met Ala Ala Arg Leu Cys Cys Gln Leu Asp Pro Ala Arg Asp Val Leu  
1 5 10 15

Cys Leu Arg Pro Val Gly Ala Glu Ser Arg Gly Arg Pro Phe Ser Gly  
20 25 30

Pro Leu Gly Ala Leu Ser Ser Ser Ser Pro Pro Ala Val Pro Thr Asp  
35 40 45

His Gly Ala His Leu Ser Leu Arg Gly Leu Pro Val Cys Ala Phe Ser  
50 55 60

Ser Ala Gly Pro Cys Ala Leu Arg Phe Thr Ser Ala Arg Arg Met Glu  
65 70 75 80

Thr Thr Val Asn Ala His Gln Phe Leu Pro Lys Val Leu His Lys Arg  
85 90 95

Thr Leu Gly Leu Ser Ala Met Ser Thr Thr Asp Leu Glu Ala Tyr Phe  
100 105 110

Lys Asp Cys Leu Phe Lys Asp Trp Glu Glu Leu Gly Glu Glu Leu Arg  
115 120 125

Leu Lys Val Phe Val Leu Gly Gly Cys Arg His Lys Leu Val Cys Ala  
130 135 140

Pro Ala Pro Cys Asn Phe Phe Thr Ser Ala  
145 150

<210> SEQ ID NO 99  
<211> LENGTH: 454  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HBV Pol domain of HBV design 1

<400> SEQUENCE: 99

Gly Pro Cys Ala Glu His Gly Glu His His Ile Arg Ile Pro Arg Thr  
1 5 10 15

Pro Ala Arg Val Thr Gly Gly Val Phe Leu Val Asp Lys Asn Pro His  
20 25 30

Asn Thr Ala Glu Ser Arg Leu Val Val Asp Phe Ser Gln Phe Ser Arg  
35 40 45

Gly Asn Tyr Arg Val Ser Trp Pro Lys Phe Ala Val Pro Asn Leu Gln  
50 55 60

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Leu | Thr | Asn | Leu | Leu | Ser | Ser | Asn | Leu | Cys | Trp | Leu | Ser | Leu | Asp | 65  | 70  | 75  | 80  |
| Val | Ser | Ala | Ala | Phe | Tyr | His | Leu | Pro | Leu | His | Pro | Ala | Ala | Met | Pro | 85  | 90  | 95  |     |
| His | Leu | Leu | Val | Gly | Ser | Ser | Gly | Leu | Ser | Arg | Tyr | Val | Ala | Arg | Leu | 100 | 105 | 110 |     |
| Ser | Ser | Asn | Ser | Arg | Ile | Ile | Asn | His | Gln | His | Gly | Thr | Leu | Gln | Asn | 115 | 120 | 125 |     |
| Leu | His | Asp | Ser | Cys | Ser | Arg | Asn | Leu | Tyr | Val | Ser | Leu | Leu | Leu | Leu | 130 | 135 | 140 |     |
| Tyr | Lys | Thr | Phe | Gly | Trp | Lys | Leu | His | Leu | Tyr | Ser | His | Pro | Ile | Ile | 145 | 150 | 155 | 160 |
| Leu | Gly | Phe | Arg | Lys | Ile | Pro | Met | Gly | Val | Gly | Leu | Ser | Pro | Phe | Leu | 165 | 170 | 175 |     |
| Leu | Ala | Gln | Phe | Thr | Ser | Ala | Ile | Cys | Ser | Val | Val | Arg | Arg | Ala | Phe | 180 | 185 | 190 |     |
| Pro | His | Cys | Leu | Ala | Phe | Ser | Gly | Ala | Lys | Ser | Val | Gln | His | Leu | Glu | 195 | 200 | 205 |     |
| Ser | Leu | Phe | Thr | Ala | Val | Thr | Asn | Phe | Leu | Leu | Ser | Leu | Gly | Ile | His | 210 | 215 | 220 |     |
| Leu | Asn | Pro | Asn | Lys | Thr | Lys | Arg | Trp | Gly | Tyr | Ser | Leu | Asn | Phe | Met | 225 | 230 | 235 | 240 |
| Gly | Tyr | Val | Ile | Gly | Ser | Trp | Gly | Ser | Leu | Pro | Gln | Asp | His | Ile | Arg | 245 | 250 | 255 |     |
| His | Lys | Ile | Lys | Glu | Cys | Phe | Arg | Lys | Leu | Pro | Val | His | Arg | Pro | Ile | 260 | 265 | 270 |     |
| Asp | Trp | Lys | Val | Cys | Gln | Arg | Ile | Val | Gly | Leu | Leu | Gly | Phe | Ala | Ala | 275 | 280 | 285 |     |
| Pro | Phe | Thr | Gln | Cys | Gly | Tyr | Pro | Ala | Leu | Met | Pro | Leu | Tyr | Ala | Cys | 290 | 295 | 300 |     |
| Ile | Gln | Ser | Lys | Gln | Ala | Phe | Thr | Phe | Ser | Pro | Thr | Tyr | Lys | Ala | Phe | 305 | 310 | 315 | 320 |
| Leu | Cys | Lys | Gln | Tyr | Leu | Asn | Leu | Tyr | Pro | Val | Ala | Arg | Gln | Arg | Pro | 325 | 330 | 335 |     |
| Gly | Leu | Cys | Gln | Val | Phe | Ala | Asp | Ala | Thr | Pro | Thr | Gly | Trp | Gly | Leu | 340 | 345 | 350 |     |
| Val | Met | Gly | His | Gln | Arg | Met | Arg | Gly | Thr | Phe | Ser | Ser | Arg | Lys | Tyr | 355 | 360 | 365 |     |
| Thr | Ser | Phe | Pro | Trp | Leu | Leu | Gly | Cys | Ala | Ala | Asn | Trp | Ile | Leu | Arg | 370 | 375 | 380 |     |
| Gly | Thr | Ser | Phe | Val | Tyr | Val | Pro | Ser | Ala | Leu | Asn | Pro | Ala | Asp | Asp | 385 | 390 | 395 | 400 |
| Pro | Ser | Arg | Gly | Arg | Leu | Gly | Pro | Cys | Arg | Pro | Leu | Leu | His | Leu | Pro | 405 | 410 | 415 |     |
| Phe | Arg | Pro | Thr | Thr | Gly | Arg | Thr | Ser | Leu | Tyr | Ala | Asp | Ser | Pro | Ser | 420 | 425 | 430 |     |
| Val | Pro | Ser | His | Leu | Pro | Asp | Arg | Val | His | Phe | Ala | Ser | Pro | Leu | His | 435 | 440 | 445 |     |
| Val | Ala | Trp | Arg | Pro | Pro |     |     |     |     |     |     |     |     |     |     | 450 |     |     |     |

&lt;210&gt; SEQ ID NO 100

&lt;211&gt; LENGTH: 226

&lt;212&gt; TYPE: PRT

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<213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HBV Surface (Env1) domain of HBV domain 1

<400> SEQUENCE: 100

```

Met Glu Asn Ile Thr Ser Gly Phe Leu Gly Pro Leu Leu Val Leu Gln
 1             5             10             15

Ala Gly Phe Phe Leu Leu Thr Arg Ile Leu Thr Ile Pro Gln Ser Leu
      20             25             30

Asp Ser Trp Trp Thr Ser Leu Ser Phe Leu Gly Gly Thr Thr Val Cys
      35             40             45

Leu Gly Gln Asn Ser Gln Ser Pro Thr Ser Asn His Ser Pro Thr Ser
      50             55             60

Cys Pro Pro Thr Cys Val Gly Tyr Arg Trp Met Cys Leu Arg Arg Phe
      65             70             75             80

Ile Ile Phe Leu Phe Ile Leu Leu Leu Cys Leu Ile Phe Leu Leu Val
      85             90             95

Leu Leu Asp Tyr Gln Gly Met Leu Pro Val Cys Pro Leu Ile Pro Gly
      100            105            110

Ser Ser Thr Thr Ser Thr Gly Pro Cys Arg Thr Cys Thr Thr Pro Ala
      115            120            125

Gln Gly Thr Ser Met Tyr Pro Ser Cys Cys Cys Thr Lys Pro Ser Asp
      130            135            140

Gly Asn Cys Thr Cys Ile Pro Ile Pro Ser Ser Trp Ala Phe Gly Lys
      145            150            155            160

Phe Leu Trp Glu Trp Ala Ser Ala Arg Phe Ser Trp Leu Ser Leu Leu
      165            170            175

Val Pro Phe Val Gln Trp Phe Val Gly Leu Ser Pro Thr Val Trp Leu
      180            185            190

Ser Val Ile Trp Met Met Trp Tyr Trp Gly Pro Ser Leu Tyr Asn Thr
      195            200            205

Leu Ser Pro Phe Leu Pro Leu Leu Pro Ile Phe Phe Tyr Leu Trp Val
      210            215            220

Tyr Ile
225

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<210> SEQ ID NO 101  
 <211> LENGTH: 180  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HBV Core domain of HBV design 1

<400> SEQUENCE: 101

```

Met Gln Leu Phe His Leu Cys Leu Ile Ile Ser Cys Ser Cys Pro Thr
 1             5             10             15

Val Gln Ala Ser Lys Leu Cys Leu Gly Trp Leu Trp Asp Met Asp Ile
      20             25             30

Asp Pro Tyr Lys Glu Phe Gly Ala Ser Val Glu Leu Leu Ser Phe Leu
      35             40             45

Pro Ser Asp Phe Phe Pro Ser Val Arg Asp Leu Leu Asp Thr Ala Thr
      50             55             60

Ala Leu Tyr Arg Asp Ala Leu Glu Ser Pro Glu His Cys Thr Pro His
      65             70             75             80

His Thr Ala Leu Arg His Val Cys Leu Cys Trp Gly Asp Leu Met Asn
      85             90             95

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Leu Ala Thr Trp Val Gly Thr Asn Leu Glu Asp Gln Ala Ser Arg Asp  
                   100                  105                  110  
 Leu Val Val Ser Tyr Val Asn Thr Asn Met Gly Leu Lys Phe Arg Gln  
                   115                  120                  125  
 Leu Leu Trp Phe His Ile Ser Cys Leu Thr Phe Gly Arg Asp Leu Val  
                   130                  135                  140  
 Leu Glu Tyr Leu Val Ser Phe Gly Val Trp Ile Arg Thr Pro Pro Ala  
                   145                  150                  155                  160  
 Tyr Arg Pro Ser Asn Ala Pro Ile Leu Ser Thr Leu Pro Glu Thr Thr  
                   165                  170                  175  
 Val Val Arg Gln  
                   180

<210> SEQ ID NO 102  
 <211> LENGTH: 226  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: SHB(Env) domain

<400> SEQUENCE: 102

Met Glu Asn Ile Thr Ser Gly Phe Leu Gly Pro Leu Leu Val Leu Gln  
 1                  5                  10                  15  
 Ala Gly Phe Phe Leu Leu Thr Arg Ile Leu Thr Ile Pro Gln Ser Leu  
                   20                  25                  30  
 Asp Ser Trp Trp Thr Ser Leu Ser Phe Leu Gly Gly Thr Thr Val Cys  
                   35                  40                  45  
 Leu Gly Gln Asn Ser Gln Ser Pro Thr Ser Asn His Ser Pro Thr Ser  
                   50                  55                  60  
 Cys Pro Pro Thr Cys Val Gly Tyr Arg Trp Met Cys Leu Arg Arg Phe  
                   65                  70                  75                  80  
 Ile Ile Phe Leu Phe Ile Leu Leu Leu Cys Leu Ile Phe Leu Leu Val  
                   85                  90                  95  
 Leu Leu Asp Tyr Gln Gly Met Leu Pro Val Cys Pro Leu Ile Pro Gly  
                   100                  105                  110  
 Ser Ser Thr Thr Ser Thr Gly Pro Cys Arg Thr Cys Thr Thr Pro Ala  
                   115                  120                  125  
 Gln Gly Thr Ser Met Tyr Pro Ser Cys Cys Cys Thr Lys Pro Ser Asp  
                   130                  135                  140  
 Gly Asn Cys Thr Cys Ile Pro Ile Pro Ser Ser Trp Ala Phe Gly Lys  
                   145                  150                  155                  160  
 Phe Leu Trp Glu Trp Ala Ser Ala Arg Phe Ser Trp Leu Ser Leu Leu  
                   165                  170                  175  
 Val Pro Phe Val Gln Trp Phe Val Gly Leu Ser Pro Thr Val Trp Leu  
                   180                  185                  190  
 Ser Val Ile Trp Met Met Trp Tyr Trp Gly Pro Ser Leu Tyr Asn Thr  
                   195                  200                  205  
 Leu Ser Pro Phe Leu Pro Leu Leu Pro Ile Phe Phe Tyr Leu Trp Val  
                   210                  215                  220  
 Tyr Ile  
 225

<210> SEQ ID NO 103  
 <211> LENGTH: 180  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:

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&lt;223&gt; OTHER INFORMATION: HBeAg domain

&lt;400&gt; SEQUENCE: 103

```

Met Gln Leu Phe His Leu Cys Leu Ile Ile Ser Cys Ser Cys Pro Thr
 1             5             10             15

Val Gln Ala Ser Lys Leu Cys Leu Gly Trp Leu Trp Asp Met Asp Ile
      20             25             30

Asp Pro Tyr Lys Glu Phe Gly Ala Ser Val Glu Leu Leu Ser Phe Leu
      35             40             45

Pro Ser Asp Phe Phe Pro Ser Val Arg Asp Leu Leu Asp Thr Ala Thr
      50             55             60

Ala Leu Tyr Arg Asp Ala Leu Glu Ser Pro Glu His Cys Thr Pro His
 65             70             75             80

His Thr Ala Leu Arg His Val Cys Leu Cys Trp Gly Asp Leu Met Asn
      85             90             95

Leu Ala Thr Trp Val Gly Thr Asn Leu Glu Asp Gln Ala Ser Arg Asp
      100            105            110

Leu Val Val Ser Tyr Val Asn Thr Asn Met Gly Leu Lys Phe Arg Gln
      115            120            125

Leu Leu Trp Phe His Ile Ser Cys Leu Thr Phe Gly Arg Asp Leu Val
      130            135            140

Leu Glu Tyr Leu Val Ser Phe Gly Val Trp Ile Arg Thr Pro Pro Ala
 145            150            155            160

Tyr Arg Pro Ser Asn Ala Pro Ile Leu Ser Thr Leu Pro Glu Thr Thr
      165            170            175

Val Val Arg Gln
      180

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&lt;210&gt; SEQ ID NO 104

&lt;211&gt; LENGTH: 154

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: HBx domain

&lt;400&gt; SEQUENCE: 104

```

Met Ala Ala Arg Leu Cys Cys Gln Leu Asp Pro Ala Arg Asp Val Leu
 1             5             10             15

Cys Leu Arg Pro Val Gly Ala Glu Ser Arg Gly Arg Pro Phe Ser Gly
      20             25             30

Pro Leu Gly Ala Leu Ser Ser Ser Pro Pro Ala Val Pro Thr Asp
      35             40             45

His Gly Ala His Leu Ser Leu Arg Gly Leu Pro Val Cys Ala Phe Ser
      50             55             60

Ser Ala Gly Pro Cys Ala Leu Arg Phe Thr Ser Ala Arg Arg Met Glu
 65             70             75             80

Thr Thr Val Asn Ala His Gln Phe Leu Pro Lys Val Leu His Lys Arg
      85             90             95

Thr Leu Gly Leu Ser Ala Met Ser Thr Thr Asp Leu Glu Ala Tyr Phe
      100            105            110

Lys Asp Cys Leu Phe Lys Asp Trp Glu Glu Leu Gly Glu Glu Leu Arg
      115            120            125

Leu Lys Val Phe Val Leu Gly Gly Cys Arg His Lys Leu Val Cys Ala
      130            135            140

Pro Ala Pro Cys Asn Phe Phe Thr Ser Ala
 145            150

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<210> SEQ ID NO 105
<211> LENGTH: 454
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Pol domain

<400> SEQUENCE: 105

Gly Pro Cys Ala Glu His Gly Glu His His Ile Arg Ile Pro Arg Thr
1      5      10      15
Pro Ala Arg Val Thr Gly Gly Val Phe Leu Val Asp Lys Asn Pro His
20     25     30
Asn Thr Ala Glu Ser Arg Leu Val Val Asp Phe Ser Gln Phe Ser Arg
35     40     45
Gly Asn Tyr Arg Val Ser Trp Pro Lys Phe Ala Val Pro Asn Leu Gln
50     55     60
Ser Leu Thr Asn Leu Leu Ser Ser Asn Leu Cys Trp Leu Ser Leu Asp
65     70     75     80
Val Ser Ala Ala Phe Tyr His Leu Pro Leu His Pro Ala Ala Met Pro
85     90     95
His Leu Leu Val Gly Ser Ser Gly Leu Ser Arg Tyr Val Ala Arg Leu
100    105    110
Ser Ser Asn Ser Arg Ile Ile Asn His Gln His Gly Thr Leu Gln Asn
115    120    125
Leu His Asp Ser Cys Ser Arg Asn Leu Tyr Val Ser Leu Leu Leu Leu
130    135    140
Tyr Lys Thr Phe Gly Trp Lys Leu His Leu Tyr Ser His Pro Ile Ile
145    150    155    160
Leu Gly Phe Arg Lys Ile Pro Met Gly Val Gly Leu Ser Pro Phe Leu
165    170    175
Leu Ala Gln Phe Thr Ser Ala Ile Cys Ser Val Val Arg Arg Ala Phe
180    185    190
Pro His Cys Leu Ala Phe Ser Gly Ala Lys Ser Val Gln His Leu Glu
195    200    205
Ser Leu Phe Thr Ala Val Thr Asn Phe Leu Leu Ser Leu Gly Ile His
210    215    220
Leu Asn Pro Asn Lys Thr Lys Arg Trp Gly Tyr Ser Leu Asn Phe Met
225    230    235    240
Gly Tyr Val Ile Gly Ser Trp Gly Ser Leu Pro Gln Asp His Ile Arg
245    250    255
His Lys Ile Lys Glu Cys Phe Arg Lys Leu Pro Val His Arg Pro Ile
260    265    270
Asp Trp Lys Val Cys Gln Arg Ile Val Gly Leu Leu Gly Phe Ala Ala
275    280    285
Pro Phe Thr Gln Cys Gly Tyr Pro Ala Leu Met Pro Leu Tyr Ala Cys
290    295    300
Ile Gln Ser Lys Gln Ala Phe Thr Phe Ser Pro Thr Tyr Lys Ala Phe
305    310    315    320
Leu Cys Lys Gln Tyr Leu Asn Leu Tyr Pro Val Ala Arg Gln Arg Pro
325    330    335
Gly Leu Cys Gln Val Phe Ala Asp Ala Thr Pro Thr Gly Trp Gly Leu
340    345    350
Val Met Gly His Gln Arg Met Arg Gly Thr Phe Ser Ser Arg Lys Tyr
355    360    365

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Thr Ser Phe Pro Trp Leu Leu Gly Cys Ala Ala Asn Trp Ile Leu Arg  
 370 375 380

Gly Thr Ser Phe Val Tyr Val Pro Ser Ala Leu Asn Pro Ala Asp Asp  
 385 390 395 400

Pro Ser Arg Gly Arg Leu Gly Pro Cys Arg Pro Leu Leu His Leu Pro  
 405 410 415

Phe Arg Pro Thr Thr Gly Arg Thr Ser Leu Tyr Ala Asp Ser Pro Ser  
 420 425 430

Val Pro Ser His Leu Pro Asp Arg Val His Phe Ala Ser Pro Leu His  
 435 440 445

Val Ala Trp Arg Pro Pro  
 450

<210> SEQ ID NO 106  
 <211> LENGTH: 1014  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HBV design 1

<400> SEQUENCE: 106

Met Glu Asn Ile Thr Ser Gly Phe Leu Gly Pro Leu Leu Val Leu Gln  
 1 5 10 15

Ala Gly Phe Phe Leu Leu Thr Arg Ile Leu Thr Ile Pro Gln Ser Leu  
 20 25 30

Asp Ser Trp Trp Thr Ser Leu Ser Phe Leu Gly Gly Thr Thr Val Cys  
 35 40 45

Leu Gly Gln Asn Ser Gln Ser Pro Thr Ser Asn His Ser Pro Thr Ser  
 50 55 60

Cys Pro Pro Thr Cys Val Gly Tyr Arg Trp Met Cys Leu Arg Arg Phe  
 65 70 75 80

Ile Ile Phe Leu Phe Ile Leu Leu Leu Cys Leu Ile Phe Leu Leu Val  
 85 90 95

Leu Leu Asp Tyr Gln Gly Met Leu Pro Val Cys Pro Leu Ile Pro Gly  
 100 105 110

Ser Ser Thr Thr Ser Thr Gly Pro Cys Arg Thr Cys Thr Thr Pro Ala  
 115 120 125

Gln Gly Thr Ser Met Tyr Pro Ser Cys Cys Cys Thr Lys Pro Ser Asp  
 130 135 140

Gly Asn Cys Thr Cys Ile Pro Ile Pro Ser Ser Trp Ala Phe Gly Lys  
 145 150 155 160

Phe Leu Trp Glu Trp Ala Ser Ala Arg Phe Ser Trp Leu Ser Leu Leu  
 165 170 175

Val Pro Phe Val Gln Trp Phe Val Gly Leu Ser Pro Thr Val Trp Leu  
 180 185 190

Ser Val Ile Trp Met Met Trp Tyr Trp Gly Pro Ser Leu Tyr Asn Thr  
 195 200 205

Leu Ser Pro Phe Leu Pro Leu Leu Pro Ile Phe Phe Tyr Leu Trp Val  
 210 215 220

Tyr Ile Met Gln Leu Phe His Leu Cys Leu Ile Ile Ser Cys Ser Cys  
 225 230 235 240

Pro Thr Val Gln Ala Ser Lys Leu Cys Leu Gly Trp Leu Trp Asp Met  
 245 250 255

Asp Ile Asp Pro Tyr Lys Glu Phe Gly Ala Ser Val Glu Leu Leu Ser  
 260 265 270

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Phe | Leu | Pro | Ser | Asp | Phe | Phe | Pro | Ser | Val | Arg | Asp | Leu | Leu | Asp | Thr |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Ala | Thr | Ala | Leu | Tyr | Arg | Asp | Ala | Leu | Glu | Ser | Pro | Glu | His | Cys | Thr |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Pro | His | His | Thr | Ala | Leu | Arg | His | Val | Cys | Leu | Cys | Trp | Gly | Asp | Leu |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Met | Asn | Leu | Ala | Thr | Trp | Val | Gly | Thr | Asn | Leu | Glu | Asp | Gln | Ala | Ser |
|     |     |     | 325 |     |     |     |     | 330 |     |     |     |     |     | 335 |     |
| Arg | Asp | Leu | Val | Val | Ser | Tyr | Val | Asn | Thr | Asn | Met | Gly | Leu | Lys | Phe |
|     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Arg | Gln | Leu | Leu | Trp | Phe | His | Ile | Ser | Cys | Leu | Thr | Phe | Gly | Arg | Asp |
|     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Leu | Val | Leu | Glu | Tyr | Leu | Val | Ser | Phe | Gly | Val | Trp | Ile | Arg | Thr | Pro |
| 370 |     |     |     |     | 375 |     |     |     |     |     | 380 |     |     |     |     |
| Pro | Ala | Tyr | Arg | Pro | Ser | Asn | Ala | Pro | Ile | Leu | Ser | Thr | Leu | Pro | Glu |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Thr | Thr | Val | Val | Arg | Gln | Met | Ala | Ala | Arg | Leu | Cys | Cys | Gln | Leu | Asp |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Pro | Ala | Arg | Asp | Val | Leu | Cys | Leu | Arg | Pro | Val | Gly | Ala | Glu | Ser | Arg |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Gly | Arg | Pro | Phe | Ser | Gly | Pro | Leu | Gly | Ala | Leu | Ser | Ser | Ser | Ser | Pro |
|     | 435 |     |     |     |     | 440 |     |     |     |     |     | 445 |     |     |     |
| Pro | Ala | Val | Pro | Thr | Asp | His | Gly | Ala | His | Leu | Ser | Leu | Arg | Gly | Leu |
| 450 |     |     |     |     | 455 |     |     |     |     | 460 |     |     |     |     |     |
| Pro | Val | Cys | Ala | Phe | Ser | Ser | Ala | Gly | Pro | Cys | Ala | Leu | Arg | Phe | Thr |
| 465 |     |     |     |     | 470 |     |     |     |     | 475 |     |     |     |     | 480 |
| Ser | Ala | Arg | Arg | Met | Glu | Thr | Thr | Val | Asn | Ala | His | Gln | Phe | Leu | Pro |
|     |     |     |     | 485 |     |     |     |     | 490 |     |     |     |     | 495 |     |
| Lys | Val | Leu | His | Lys | Arg | Thr | Leu | Gly | Leu | Ser | Ala | Met | Ser | Thr | Thr |
|     |     | 500 |     |     |     |     |     | 505 |     |     |     |     | 510 |     |     |
| Asp | Leu | Glu | Ala | Tyr | Phe | Lys | Asp | Cys | Leu | Phe | Lys | Asp | Trp | Glu | Glu |
|     | 515 |     |     |     |     |     | 520 |     |     |     |     | 525 |     |     |     |
| Leu | Gly | Glu | Glu | Leu | Arg | Leu | Lys | Val | Phe | Val | Leu | Gly | Gly | Cys | Arg |
| 530 |     |     |     |     | 535 |     |     |     |     |     | 540 |     |     |     |     |
| His | Lys | Leu | Val | Cys | Ala | Pro | Ala | Pro | Cys | Asn | Phe | Phe | Thr | Ser | Ala |
| 545 |     |     |     |     | 550 |     |     |     |     | 555 |     |     |     |     | 560 |
| Gly | Pro | Cys | Ala | Glu | His | Gly | Glu | His | His | Ile | Arg | Ile | Pro | Arg | Thr |
|     |     |     | 565 |     |     |     |     | 570 |     |     |     |     | 575 |     |     |
| Pro | Ala | Arg | Val | Thr | Gly | Gly | Val | Phe | Leu | Val | Asp | Lys | Asn | Pro | His |
|     |     |     | 580 |     |     |     |     | 585 |     |     |     |     | 590 |     |     |
| Asn | Thr | Ala | Glu | Ser | Arg | Leu | Val | Val | Asp | Phe | Ser | Gln | Phe | Ser | Arg |
|     | 595 |     |     |     |     | 600 |     |     |     |     |     | 605 |     |     |     |
| Gly | Asn | Tyr | Arg | Val | Ser | Trp | Pro | Lys | Phe | Ala | Val | Pro | Asn | Leu | Gln |
| 610 |     |     |     |     | 615 |     |     |     |     |     | 620 |     |     |     |     |
| Ser | Leu | Thr | Asn | Leu | Leu | Ser | Ser | Asn | Leu | Cys | Trp | Leu | Ser | Leu | Asp |
| 625 |     |     |     | 630 |     |     |     |     |     | 635 |     |     |     |     | 640 |
| Val | Ser | Ala | Ala | Phe | Tyr | His | Leu | Pro | Leu | His | Pro | Ala | Ala | Met | Pro |
|     |     |     | 645 |     |     |     |     | 650 |     |     |     |     |     | 655 |     |
| His | Leu | Leu | Val | Gly | Ser | Ser | Gly | Leu | Ser | Arg | Tyr | Val | Ala | Arg | Leu |
|     | 660 |     |     |     |     |     | 665 |     |     |     |     | 670 |     |     |     |
| Ser | Ser | Asn | Ser | Arg | Ile | Ile | Asn | His | Gln | His | Gly | Thr | Leu | Gln | Asn |
|     | 675 |     |     |     |     |     | 680 |     |     |     |     | 685 |     |     |     |

[illegible]

<400> SEQUENCE: 107

Ser Val Glu Leu Leu Ser Phe Leu Pro Ser Asp Phe Phe Pro Ser Val  
1 5 10 15

Arg Asp Leu Leu Asp Thr Ala Thr Ala Leu Tyr Arg Asp Ala Leu Glu  
20 25 30

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Pro | Glu | His | Cys | Thr | Pro | His | Thr | Ala | Leu | Arg | His | Val | Cys |     |
| 35  |     |     |     |     |     | 40  |     |     |     | 45  |     |     |     |     |     |
| Leu | Cys | Trp | Gly | Asp | Leu | Met | Asn | Leu | Ala | Thr | Trp | Val | Gly | Thr | Asn |
| 50  |     |     |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |
| Leu | Glu | Asp | Gln | Ala | Ser | Arg | Asp | Leu | Val | Val | Ser | Tyr | Val | Asn | Thr |
| 65  |     |     |     | 70  |     |     |     |     |     | 75  |     |     |     | 80  |     |
| Asn | Met | Gly | Leu | Lys | Phe | Arg | Gln | Leu | Leu | Trp | Phe | His | Ile | Ser | Cys |
|     |     |     |     | 85  |     |     |     | 90  |     |     |     |     |     | 95  |     |
| Leu | Thr | Phe | Gly | Arg | Asp | Leu | Val | Leu | Glu | Tyr | Leu | Val | Ser | Phe | Gly |
|     |     | 100 |     |     |     |     |     | 105 |     |     |     | 110 |     |     |     |
| Val | Trp | Ile | Arg | Thr | Pro | Pro | Ala | Tyr | Arg | Pro | Ser | Asn | Ala | Pro | Ile |
| 115 |     |     |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
| Leu | Ser | Thr | Leu | Pro | Glu | Thr | Thr | Val | Val | Arg | Gln | Arg | Gly | Arg | Thr |
| 130 |     |     |     |     |     | 135 |     |     |     |     |     | 140 |     |     |     |
| Ile | Val | Leu | His | Lys | Arg | Thr | Leu | Gly | Leu | Met | Gly | Gln | Asn | Leu | Ser |
| 145 |     |     |     | 150 |     |     |     |     |     | 155 |     |     |     | 160 |     |
| Thr | Ser | Asn | Pro | Leu | Gly | Phe | Phe | Pro | Asp | His | Gln | Leu | Asp | Pro | Ala |
|     |     |     |     | 165 |     |     |     | 170 |     |     |     | 175 |     |     |     |
| Phe | Arg | Ala | Asn | Thr | Asn | Asn | Pro | Asp | Trp | Asp | Phe | Asn | Pro | Asn | Lys |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     | 190 |     |     |     |
| Asp | Thr | Trp | Pro | Asp | Ala | Asn | Lys | Val | Gly | Ala | Gly | Ala | Phe | Gly | Leu |
| 195 |     |     |     |     |     | 200 |     |     |     |     |     | 205 |     |     |     |
| Gly | Phe | Thr | Pro | Pro | His | Gly | Gly | Leu | Leu | Gly | Trp | Ser | Pro | Gln | Ala |
| 210 |     |     |     |     |     | 215 |     |     |     |     |     | 220 |     |     |     |
| Gln | Gly | Ile | Met | Gln | Thr | Leu | Pro | Ala | Asn | Pro | Pro | Pro | Ala | Ser | Thr |
| 225 |     |     |     | 230 |     |     |     |     |     | 235 |     |     |     | 240 |     |
| Asn | Arg | Gln | Ser | Gly | Arg | Gln | Pro | Thr | Pro | Leu | Leu | Pro | Lys | Val | Leu |
|     |     |     |     | 245 |     |     |     | 250 |     |     |     | 255 |     |     |     |
| His | Lys | Arg | Thr | Leu | Met | Pro | Leu | Ser | Tyr | Gln | His | Phe | Arg | Lys | Leu |
|     |     | 260 |     |     |     |     |     | 265 |     |     |     | 270 |     |     |     |
| Leu | Leu | Leu | Asp | Asn | Glu | Ala | Gly | Pro | Leu | Glu | Glu | Glu | Leu | Pro | Arg |
| 275 |     |     |     |     |     | 280 |     |     |     |     |     | 285 |     |     |     |
| Leu | Ala | Asp | Glu | Asp | Leu | Asn | Arg | Arg | Val | Ala | Glu | Asp | Leu | Asn | Leu |
| 290 |     |     |     |     |     | 295 |     |     |     |     |     | 300 |     |     |     |
| Gly | Asn | Leu | Asn | Val | Ser | Ile | Pro | Trp | Thr | His | Lys | Val | Gly | Asn | Phe |
| 305 |     |     |     | 310 |     |     |     |     |     | 315 |     |     |     | 320 |     |
| Thr | Gly | Leu | Tyr | Ser | Ser | Ser | Val | Pro | Val | Phe | Asn | Pro | His | Trp | Lys |
|     |     |     |     | 325 |     |     |     | 330 |     |     |     | 335 |     |     |     |
| Thr | Pro | Ser | Phe | Pro | Asn | Ile | His | Leu | His | Gln | Asp | Ile | Ile | Lys | Lys |
|     |     | 340 |     |     |     | 345 |     |     |     |     |     | 350 |     |     |     |
| Cys | Glu | Gln | Phe | Val | Gly | Pro | Leu | Thr | Val | Asn | Glu | Lys | Arg | Arg | Leu |
| 355 |     |     |     |     |     | 360 |     |     |     |     |     | 365 |     |     |     |
| Lys | Leu | Ile | Met | Pro | Ala | Arg | Phe | Tyr | Pro | Asn | Phe | Thr | Lys | Tyr | Leu |
| 370 |     |     |     |     |     | 375 |     |     |     |     |     | 380 |     |     |     |
| Pro | Leu | Asp | Lys | Gly | Ile | Lys | Pro | Tyr | Tyr | Pro | Glu | His | Leu | Val | Asn |
| 385 |     |     |     | 390 |     |     |     |     |     | 395 |     |     |     | 400 |     |
| His | Tyr | Phe | His |     |     |     |     |     |     |     |     |     |     |     |     |

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| 450        |            |            |            | 455        |            |            |            | 460        |            |            |            |            |            |            |            |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Gln<br>465 | Ser        | Lys        | His        | Gln<br>470 | Gln        | Ser        | Arg        | Leu        | Gly<br>475 | Leu        | Gln        | Ser        | Gln        | Gln        | Gly<br>480 |
| His        | Leu        | Ala        | Arg        | Arg<br>485 | Gln        | Gln        | Gly        | Arg        | Ser<br>490 | Trp        | Ser        | Ile        | Arg        | Thr<br>495 | Arg        |
| Val        | His        | Pro        | Thr        | Ala<br>500 | Arg        | Arg        | Pro        | Ser<br>505 | Gly        | Val        | Glu        | Pro        | Ser        | Gly        | Ser        |
| Gly        | His        | Asn<br>515 | Ala        | Asn        | Leu        | Ala        | Ser<br>520 | Lys        | Ser        | Ala        | Ser        | Cys<br>525 | Leu        | Tyr        | Gln        |
| Ser        | Thr<br>530 | Val        | Arg        | Thr        | Ala<br>535 | Ala        | Tyr        | Pro        | Ala        | Val        | Ser<br>540 | Thr        | Ser        | Glu        | Asn        |
| His<br>545 | Ser        | Ser        | Ser        | Gly        | His<br>550 | Ala        | Val        | Glu        | Leu        | His<br>555 | Asn        | Leu        | Pro        | Pro        | Asn<br>560 |
| Ser        | Ala        | Arg        | Ser        | Gln<br>565 | Ser        | Glu        | Arg        | Pro        | Val<br>570 | Ser        | Pro        | Cys        | Trp        | Trp<br>575 | Leu        |
| Gln        | Phe        | Arg        | Asn<br>580 | Ser        | Lys        | Pro        | Cys        | Ser<br>585 | Asp        | Tyr        | Cys        | Leu        | Ser        | His        | Ile        |
| Val        | Asn        | Leu<br>595 | Leu        | Glu        | Asp        | Trp        | Gly<br>600 | Pro        | Cys        | His        | Lys        | Arg<br>605 | Thr        | Leu        | Gly        |
| Leu<br>610 | Ser        | Ala        | Met        | Ser        | Pro        | Pro<br>615 | Leu        | Arg        | Thr        | Thr        | His<br>620 | Pro        | Gln        | Ala        | Met        |
| Gln<br>625 | Trp        | Asn        | Ser        | Thr        | Thr<br>630 | Phe        | His        | Gln        | Thr        | Leu<br>635 | Gln        | Asp        | Pro        | Arg        | Val<br>640 |
| Arg        | Gly        | Leu        | Tyr        | Leu<br>645 | Pro        | Ala        | Gly        | Gly        | Ser<br>650 | Ser        | Ser        | Gly        | Thr        | Val<br>655 | Asn        |
| Pro        | Val        | Pro        | Thr<br>660 | Thr        | Ala        | Ser        | Pro        | Thr<br>665 | Leu        | Ser        | Thr        | Ser        | Ser        | Arg        | Ile        |
| Gly        | Asp        | Pro<br>675 | Ala        | Leu        | Asn        | Gln        | Phe<br>680 | Leu        | Pro        | Lys        | Val        | Leu<br>685 | His        | Lys        | Arg        |
| Ser<br>690 | Arg        | Gly        | Asn        | Tyr        | Arg        | Val<br>695 | Ser        | Trp        | Pro        | Lys        | Phe<br>700 | Ala        | Val        | Pro        | Asn        |
| Leu<br>705 | Gln        | Ser        | Leu        | Thr        | Asn<br>710 | Leu        | Leu        | Ser        | Ser        | Asn<br>715 | Leu        | Cys        | Trp        | Leu        | Ser<br>720 |
| Leu        | Asp        | Val        | Ser        | Ala<br>725 | Ala        | Phe        | Tyr        | His        | Leu<br>730 | Pro        | Leu        | His        | Pro        | Ala<br>735 | Ala        |
| Met        | Pro        | His        | Leu<br>740 | Leu        | Val        | Gly        | Ser        | Ser<br>745 | Gly        | Leu        | Ser        | Arg        | Tyr<br>750 | Val        | Ala        |
| Arg        | Leu<br>755 | Ser        | Ser        | Asn        | Ser        | Arg        | Ile<br>760 | Ile        | Asn        | His        | Gln        | His<br>765 | Gly        | Thr        | Leu        |
| Gln<br>770 | Asn        | Leu        | His        | Asp        | Ser        | Cys<br>775 | Ser        | Arg        | Asn        | Leu        | Tyr<br>780 | Val        | Ser        | Leu        | Leu        |
| Leu<br>785 | Leu        | Tyr        | Lys        | Thr        | Phe<br>790 | Gly        | Trp        | Lys        | Leu        | His<br>795 | Leu        | Tyr        | Ser        | His        | Pro<br>800 |
| Ile        | Ile        | Leu        | Gly        | Phe<br>805 | Arg        | Lys        | Ile        | Pro        | Met<br>810 | Gly        | Val        | Gly        | Leu        | Ser<br>815 | Pro        |
| Phe        | Leu        | Leu        | Ala<br>820 | Gln        | Phe        | Thr        | Ser        | Ala<br>825 | Ile        | Thr        | Val        | Asn        | Ala<br>830 | His        | Gln        |
| Phe        | Leu<br>835 | Pro        | Lys        | Gln        | Asn        | Ser        | Gln<br>840 | Ser        | Pro        | Thr        | Ser        | Asn<br>845 | His        | Ser        | Pro        |
| Thr<br>850 | Ser        | Cys        | Pro        | Pro        | Thr        | Cys<br>855 | Val        | Gly        | Tyr        | Arg        | Trp<br>860 | Met        | Cys        | Leu        | Arg        |
| Arg<br>865 | Phe        | Ile        | Ile        | Phe<br>870 | Leu        | Phe        | Ile        | Leu        | Leu        | Leu<br>875 | Cys        | Leu        | Ile        | Phe        | Leu<br>880 |



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Leu Val Leu Leu Asp Tyr Gln Gly Met Leu Pro Val Cys Pro Leu Ile  
                   885                                  890                                  895  
 Pro Gly Ser Ser Thr Thr Ser Thr Gly Pro Cys Arg Thr Cys Thr Thr  
                   900                                  905                                  910  
 Pro Ala Gln Gly Thr Ser Met Tyr Pro Ser Cys Cys Cys Thr Lys Pro  
                   915                                  920                                  925  
 Ser Asp Gly Asn Cys Thr Cys Ile Pro Ile Pro Ser Ser Trp Ala Phe  
                   930                                  935                                  940  
 Gly Lys Phe Leu Trp Glu Trp Ala Ser Ala Arg Phe Ser Trp Leu Ser  
                   945                                  950                                  955                                  960  
 Leu Leu Val Pro Phe Val Gln Trp Phe Val Gly Leu Ser Pro Thr Val  
                   965                                  970                                  975  
 Trp Leu Ser Val Ile Trp Met Met Trp Tyr Trp Gly Pro Ser Leu Tyr  
                   980                                  985                                  990  
 Asn Thr Leu Ser Pro Phe Leu Pro Leu Leu Pro Ile Phe Phe Tyr Leu  
                   995                                  1000                                  1005  
 Trp Val Tyr Ile Leu Ser Ala Met Ser Thr Thr Asp Leu  
                   1010                                  1015                                  1020

<210> SEQ ID NO 108  
 <211> LENGTH: 532  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: HBV design 3

<400> SEQUENCE: 108

Met Gln Thr Asp Arg Thr Gly Glu Thr Ala Leu His Leu Ala Ala Arg  
 1                  5                                  10                                  15  
 Tyr Ser Arg Ser Asp Phe Leu Pro Ser Asp Phe Phe Pro Ser Val Ala  
                   20                                  25                                  30  
 Asp Ala Gly Val Trp Ile Arg Thr Pro Pro Ala Asp Asn Met Glu Tyr  
                   35                                  40                                  45  
 Leu Val Ser Phe Gly Val Trp Pro Leu His Ala Ala Val Ser Ala Asp  
                   50                                  55                                  60  
 Cys Trp Gly Glu Leu Met Thr Leu Arg Asn Arg Ala Thr Asp Leu Gly  
                   65                                  70                                  75                                  80  
 Gly Pro Asn Leu Asp Asn Ile Leu Met His Asp Ile Leu Arg Ser Phe  
                   85                                  90                                  95  
 Ile Pro Leu Leu Pro Leu Ile Leu Ala Ala Arg Leu Ala Val Ser Thr  
                   100                                  105                                  110  
 Leu Pro Glu Thr Thr Val Val Arg Ser His Ala Asp Val Phe Leu  
                   115                                  120                                  125  
 Gly Gly Pro Pro Val Cys Leu Asp Asp Leu Phe Leu Leu Thr Arg Ile  
                   130                                  135                                  140  
 Leu Thr Ile Ala Leu His Trp Ala Ala Ala Val Asn Asn Val Leu Thr  
                   145                                  150                                  155                                  160  
 Phe Gly Arg Glu Thr Val Leu Glu Tyr Gly Ala Asn Lys Trp Leu Ser  
                   165                                  170                                  175  
 Leu Leu Val Pro Phe Val Asn Asn Arg Phe Leu Lys Gln Gln Tyr Met  
                   180                                  185                                  190  
 Asn Leu Pro Leu Phe Leu Ala Ala Arg Glu Gly Ser Tyr Glu Asp Leu  
                   195                                  200                                  205  
 Leu Asp Thr Ala Ser Ala Leu Tyr Ala Asn Arg Phe Leu Ser Lys Gln  
                   210                                  215                                  220

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Tyr Met Asp Leu Asp His Met Thr Val Ser Thr Lys Leu Cys Lys Ile
225                230                235                240

Pro Arg Asp Leu Trp Phe His Ile Ser Cys Leu Thr Phe Ile Val Arg
                245                250                255

Leu Leu Asp Leu Glu Val Ser Gln Thr Ser Lys Leu Thr Arg Gln Thr
                260                265                270

Asp Arg Thr Gly Glu Thr Ala Leu His Leu Ala Ala Arg Tyr Ser Arg
                275                280                285

Ser Asp Leu Thr Thr Val Pro Ala Ala Ser Leu Leu Ala Ala Asp Ala
                290                295                300

Gly Leu Ser Arg Tyr Val Ala Arg Leu Asp Asn Met Lys Leu His Leu
305                310                315                320

Tyr Ser His Pro Ile Pro Leu His Ala Ala Val Ser Ala Asp Gly Leu
                325                330                335

Ser Pro Thr Val Trp Leu Ser Val Arg Asn Arg Ala Thr Asp Leu Phe
                340                345                350

Leu Leu Ser Leu Gly Ile His Leu Met His Asp Ser Leu Tyr Ala Asp
                355                360                365

Ser Pro Ser Val Pro Leu Ile Leu Ala Ala Arg Leu Ala Val His Lys
                370                375                380

Arg Thr Leu Gly Leu Ser Ala Met Ser His Ala Asp Val Thr Leu Cys
385                390                395                400

Ile Pro His Val Ala Val Asp Asp Leu Leu Leu Leu Lys Ala Thr Leu
                405                410                415

Cys Ile Ala Leu His Trp Ala Ala Ala Val Asn Asn Val Gln Phe Leu
                420                425                430

Pro Lys Val Leu His Lys Arg Gly Ala Asn Lys Ala Leu Met Pro Leu
                435                440                445

Tyr Ala Cys Ile Asn Asn Arg Thr Val Asn Ala His Gln Phe Leu Pro
                450                455                460

Lys Pro Leu Phe Leu Ala Ala Arg Glu Gly Ser Tyr Glu Leu Pro Lys
465                470                475                480

Val Leu His Lys Arg Thr Leu Ala Asn Arg Val Leu His Lys Arg Thr
                485                490                495

Leu Gly Leu Asp His Met Leu Ser Ala Met Ser Thr Thr Asp Leu Pro
                500                505                510

Arg Asp Leu Leu Val Pro Phe Val Gln Trp Phe Val Ile Val Arg Leu
                515                520                525

Leu Asp Leu Glu
530

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<210> SEQ ID NO 109
<211> LENGTH: 368
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: HBV design 4

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<400> SEQUENCE: 109

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Lys Lys Phe Leu Pro Ser Asp Phe Phe Pro Ser Val Lys Lys Cys Trp
1                5                10                15

Gly Glu Leu Met Thr Leu Lys Lys Gly Val Trp Ile Arg Thr Pro Pro
                20                25                30

Ala Lys Lys Ser Thr Leu Pro Glu Thr Thr Val Val Arg Arg Lys Lys
35                40                45

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Leu Thr Phe Gly Arg Glu Thr Val Leu Glu Tyr Lys Lys Asp Leu Leu  
 50 55 60  
 Asp Thr Ala Ser Ala Leu Tyr Lys Lys Leu Trp Phe His Ile Ser Cys  
 65 70 75 80  
 Leu Thr Phe Lys Lys Glu Tyr Leu Val Ser Phe Gly Val Trp Lys Lys  
 85 90 95  
 Gly Gly Pro Asn Leu Asp Asn Ile Leu Lys Lys Leu Thr Thr Val Pro  
 100 105 110  
 Ala Ala Ser Leu Leu Ala Lys Lys Ile Leu Arg Ser Phe Ile Pro Leu  
 115 120 125  
 Leu Lys Lys Phe Leu Gly Gly Pro Pro Val Cys Leu Lys Lys Phe Leu  
 130 135 140  
 Leu Thr Arg Ile Leu Thr Ile Lys Lys Trp Leu Ser Leu Leu Val Pro  
 145 150 155 160  
 Phe Val Lys Lys Gly Leu Ser Pro Thr Val Trp Leu Ser Val Lys Lys  
 165 170 175  
 Leu Leu Val Pro Phe Val Gln Trp Phe Val Lys Lys Phe Leu Lys Gln  
 180 185 190  
 Gln Tyr Met Asn Leu Lys Lys Phe Leu Ser Lys Gln Tyr Met Asp Leu  
 195 200 205  
 Lys Lys Thr Val Ser Thr Lys Leu Cys Lys Ile Lys Lys Gly Leu Ser  
 210 215 220  
 Arg Tyr Val Ala Arg Leu Lys Lys Lys Leu His Leu Tyr Ser His Pro  
 225 230 235 240  
 Ile Lys Lys Phe Leu Leu Ser Leu Gly Ile His Leu Lys Lys Ser Leu  
 245 250 255  
 Tyr Ala Asp Ser Pro Ser Val Lys Lys Ala Leu Met Pro Leu Tyr Ala  
 260 265 270  
 Cys Ile Lys Lys Leu Leu Leu Lys Ala Thr Leu Cys Ile Lys Lys Thr  
 275 280 285  
 Leu Cys Ile Pro His Val Ala Val Lys Lys Val Leu His Lys Arg Thr  
 290 295 300  
 Leu Gly Leu Lys Lys Leu Pro Lys Val Leu His Lys Arg Thr Leu Lys  
 305 310 315 320  
 Lys His Lys Arg Thr Leu Gly Leu Ser Ala Met Lys Lys Gln Phe Leu  
 325 330 335  
 Pro Lys Val Leu His Lys Arg Lys Lys Thr Val Asn Ala His Gln Phe  
 340 345 350  
 Leu Pro Lys Lys Lys Leu Ser Ala Met Ser Thr Thr Asp Leu Lys Lys  
 355 360 365

<210> SEQ ID NO 110  
 <211> LENGTH: 20  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: splice primer

<400> SEQUENCE: 110

tgccaagagt gacgtgtcca

20

<210> SEQ ID NO 111  
 <211> LENGTH: 21  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial sequence  
 <220> FEATURE:

-continued

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<223> OTHER INFORMATION: splice probe

<400> SEQUENCE: 111

cccaggtcca actgcagccg g 21

<210> SEQ ID NO 112

<211> LENGTH: 12

<212> TYPE: DNA

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Furin linker polypeptide

<400> SEQUENCE: 112

agagctaaga gg 12

<210> SEQ ID NO 113

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 113

His Leu Asp Lys Lys Gln Arg Phe His Asn Ile  
1 5 10

<210> SEQ ID NO 114

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 114

Arg Trp Thr Gly Arg Cys Met Ser Cys Cys  
1 5 10

<210> SEQ ID NO 115

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 115

Thr Thr Leu Glu Gln Gln Tyr Asn Lys Pro Leu Cys Asp Leu Leu  
1 5 10 15

<210> SEQ ID NO 116

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 116

Ile Ser Glu Tyr Arg His Tyr Cys Tyr  
1 5

<210> SEQ ID NO 117

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 117

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Val Tyr Asp Phe Ala Phe Arg Asp Leu  
1 5

<210> SEQ ID NO 118  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 118

Thr Ile His Asp Ile Ile Leu Glu Cys Val  
1 5 10

<210> SEQ ID NO 119  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 119

Lys Leu Pro Gln Leu Cys Thr Glu Leu  
1 5

<210> SEQ ID NO 120  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 120

Phe Ala Phe Arg Asp Leu Cys Ile Val Tyr  
1 5 10

<210> SEQ ID NO 121  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E6

<400> SEQUENCE: 121

Leu Cys Ile Val Tyr Arg Asp Gly Asn Pro Tyr Ala Val Cys Asp  
1 5 10 15

<210> SEQ ID NO 122  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 122

Leu Cys Val Gln Ser Thr His Val Asp Ile  
1 5 10

<210> SEQ ID NO 123  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 123

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Arg Thr Leu Glu Asp Leu Leu Met Gly Thr  
1 5 10

<210> SEQ ID NO 124  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 124

Thr Leu Gly Ile Val Cys Pro Ile  
1 5

<210> SEQ ID NO 125  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 125

Leu Leu Met Gly Thr Leu Gly Ile Val  
1 5

<210> SEQ ID NO 126  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 126

Thr Leu His Glu Tyr Met Leu Asp Leu  
1 5

<210> SEQ ID NO 127  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 127

Ala His Tyr Asn Ile Val Thr Phe Cys Cys  
1 5 10

<210> SEQ ID NO 128  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 128

Tyr Met Leu Asp Leu Gln Pro Glu Thr Thr  
1 5 10

<210> SEQ ID NO 129  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E7

<400> SEQUENCE: 129

Cys Asp Ser Thr Leu Arg Leu Cys Val

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1                    5

<210> SEQ ID NO 130  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 130

Tyr Ile Ile Phe Val Tyr Ile Pro Leu  
1                    5

<210> SEQ ID NO 131  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 131

Lys Leu Thr Asn Thr Gly Leu Tyr Asn Leu  
1                    5                    10

<210> SEQ ID NO 132  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 132

Lys Cys Ile Asp Phe Tyr Ser Arg Ile  
1                    5

<210> SEQ ID NO 133  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 133

Phe Ala Phe Lys Asp Leu Phe Val Val  
1                    5

<210> SEQ ID NO 134  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 134

Asn Leu Leu Ile Arg Cys Leu Arg Cys  
1                    5

<210> SEQ ID NO 135  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 135

Lys Leu Pro Asp Leu Cys Thr Glu Leu  
1                    5

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<210> SEQ ID NO 136  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 136

Glu Leu Thr Glu Val Phe Glu Phe Ala  
1 5

<210> SEQ ID NO 137  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 137

Ser Leu Gln Asp Ile Glu Ile Thr Cys Val  
1 5 10

<210> SEQ ID NO 138  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV16-E5

<400> SEQUENCE: 138

Lys Thr Val Leu Glu Leu Thr Glu Val  
1 5

<210> SEQ ID NO 139  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV18-E7

<400> SEQUENCE: 139

Asp Asp Leu Arg Ala Phe Gln Gln Leu Phe Leu Asn Thr Leu Ser  
1 5 10 15

<210> SEQ ID NO 140  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV18-E7

<400> SEQUENCE: 140

Phe Gln Gln Leu Phe Leu Asn Thr Leu  
1 5

<210> SEQ ID NO 141  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV18-E7

<400> SEQUENCE: 141

Gln Leu Phe Leu Asn Thr Leu Ser Phe Val  
1 5 10



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<210> SEQ ID NO 142  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV18-E7

<400> SEQUENCE: 142

Leu Phe Leu Asn Thr Leu Ser Phe Val Cys Pro Trp Cys Ala Ser  
1 5 10 15

<210> SEQ ID NO 143  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV18-E7

<400> SEQUENCE: 143

Thr Leu Gln Asp Ile Val Leu His Leu  
1 5

<210> SEQ ID NO 144  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: HPV18-E7

<400> SEQUENCE: 144

Ser Glu Glu Glu Asn Asp Glu Ile Asp Gly Val Asn His Gln His Leu  
1 5 10 15

Pro Ala Arg Arg  
20

<210> SEQ ID NO 145  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Agonist peptides

<400> SEQUENCE: 145

Lys Leu Pro Gln Leu Cys Thr Glu Val  
1 5

<210> SEQ ID NO 146  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Agonist peptides

<400> SEQUENCE: 146

Gln Leu Tyr Asn Lys Pro Leu Cys Asp Val  
1 5 10

<210> SEQ ID NO 147  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial sequence  
<220> FEATURE:

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<223> OTHER INFORMATION: Agonist peptides

<400> SEQUENCE: 147

Arg Thr Leu Glu Asp Leu Leu Met Gly Val  
1                   5                   10

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What is claimed is:

1. A polynucleotide encoding a polypeptide construct comprising the amino acid sequence of SEQ ID NO: 52 or 54.

2. A vector comprising the polynucleotide of claim 1.

3. The vector of claim 2, wherein the vector is an adenoviral vector.

4. The polynucleotide of claim 1, wherein the polypeptide construct comprises the amino acid sequence of SEQ ID NO: 54.

5. The polynucleotide of claim 1, wherein the polypeptide construct comprises the amino acid sequence of SEQ ID NO: 52.

6. The vector of claim 3, wherein the adenoviral vector is a gorilla adenoviral vector.

10 7. The vector of claim 6, wherein the gorilla adenoviral vector does not comprise an E1 region.

8. The vector of claim 6, wherein the gorilla adenoviral vector does not comprise an E4 region.

15 9. The vector of claim 6, wherein the gorilla adenoviral vector does not comprise an E1 region or an E4 region.

10. The vector of claim 6, wherein the gorilla adenoviral vector is a GC46 vector.

20 11. The vector of claim 6, wherein the polynucleotide is under the control of an immediate early cytomegalovirus (CMV) promoter.

12. A method of vaccinating a subject against human papilloma virus, the method comprising administering to the subject the vector of claim 6.

\* \* \* \* \*